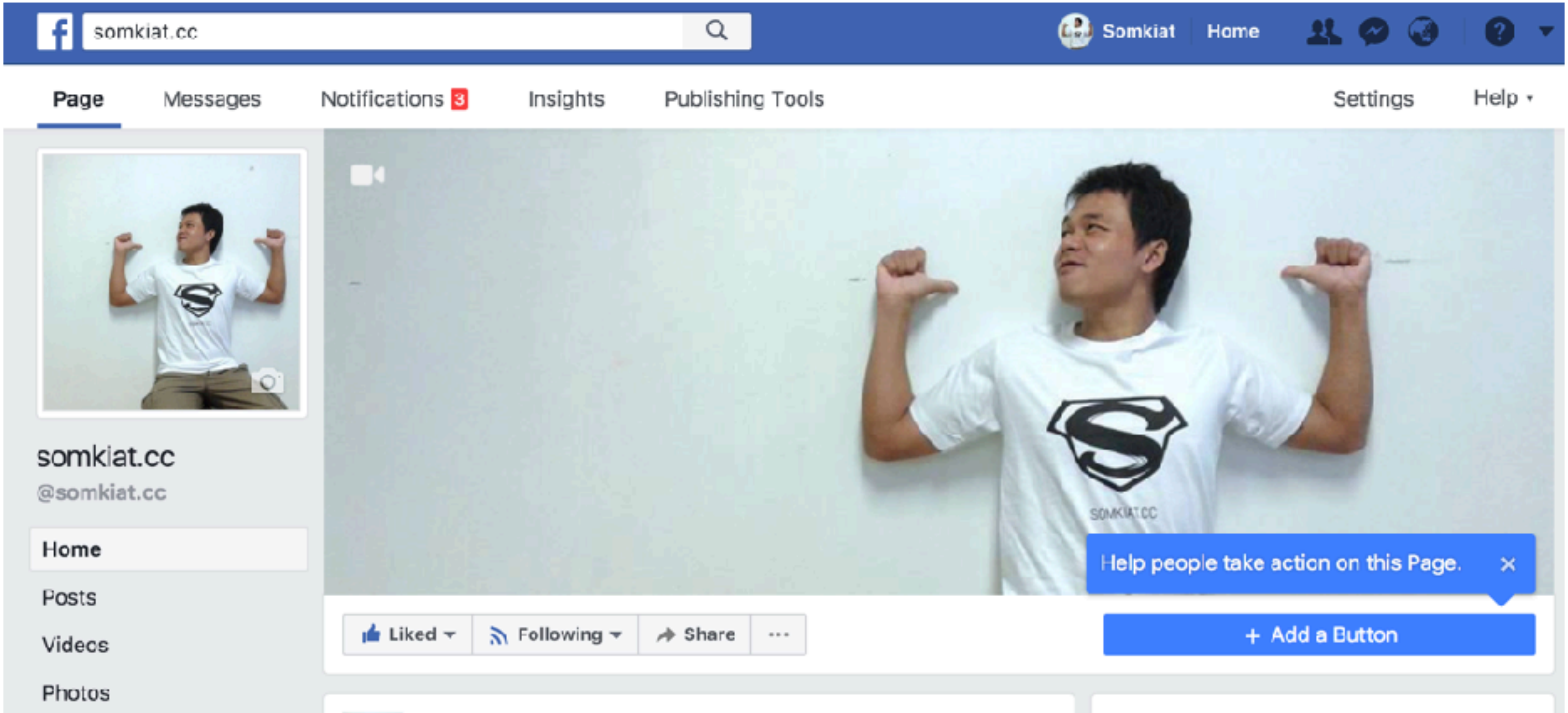
Generative AI for Software Development 2026

สยามชำนาญกิจ
SIAM CHAMNAN KIT

SHU HA RI
SOFTWARE DEVELOPMENT SCHOOL


somkiat.cc
Somkiat
Home
Page
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Insights
Publishing Tools
Settings
Help
somkiat.cc
@somkiat.cc
Home
Posts
Videos
Photos
Help people take action on this Page.
Liked
Following
Share
+ Add a Button

https://github.com/up1/workshop-ai-with-technical-team

Goals

Integrate Generative AI in Software Development
Optimize code quality
Team up with AI on coding tasks
Develop innovative solutions

Learning Path

AI/ML

AI Model

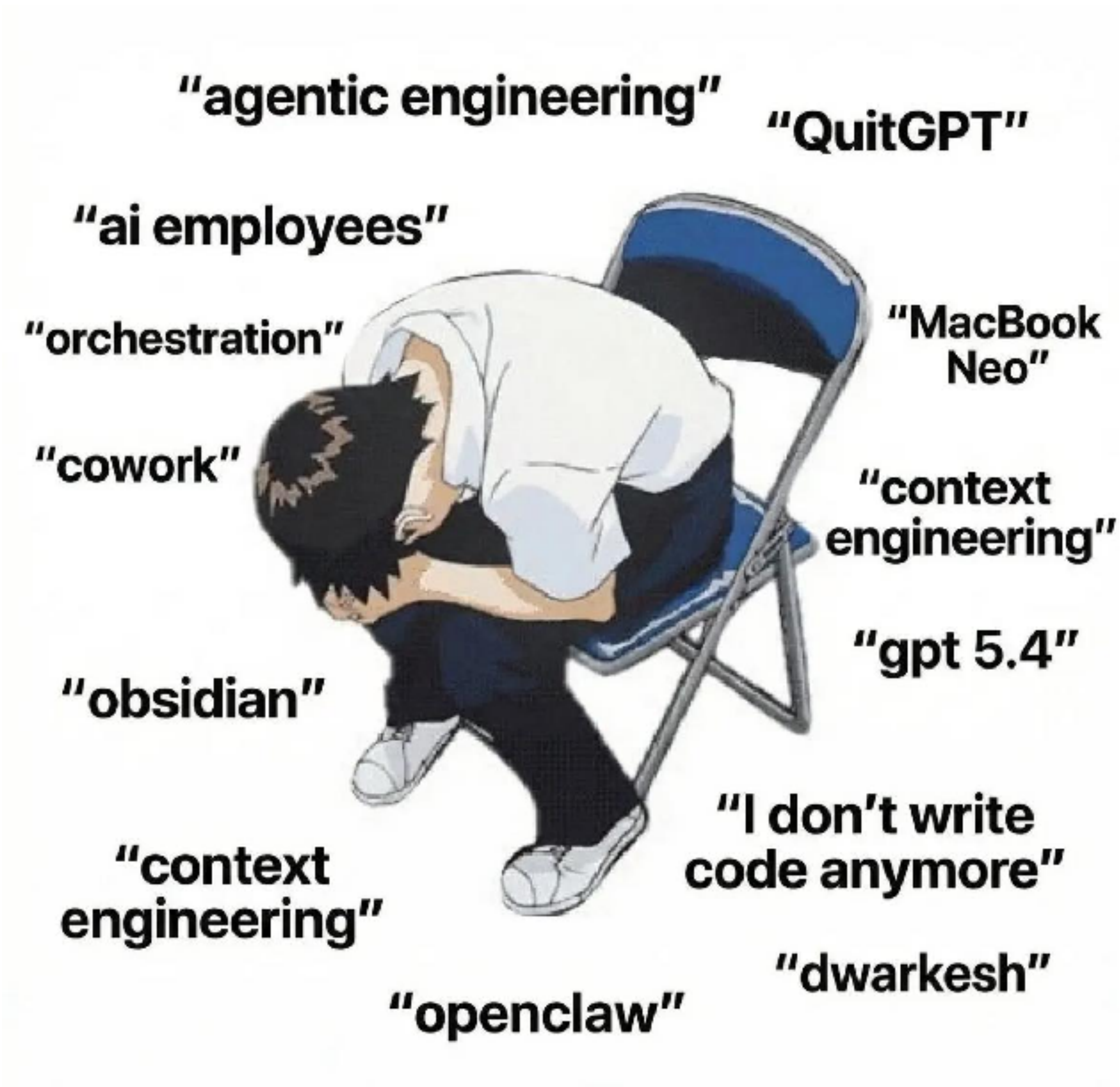
Prompt Engineer

AI in Software development

Develop AI/LLM app

RAG

Use cases and workshops


"agentic engineering"
"QuitGPT"
"ai employees"
"orchestration"
"MacBook Neo"
"cowork"
"context engineering"
"gpt 5.4"
"obsidian"
"I don't write code anymore"
"context engineering"
"dwarkesh"
"openclaw"

Steps to use AI ?

Q/A Chat-based → Assistant → Workflow → System Autonomous

Q/A Chat-based	Assistant	Workflow	System Autonomous
	Summarize Explain Research	Define steps n8n Agent Skills	AI Agent Sub-agent

Steps to use AI ?

Q/A
Chat-based → Assistant → Workflow → System
Autonomous

Summarize
Explain
Research

Define steps
n8n
Agent Skills

AI Agent
Sub-agent

Prompt engineering

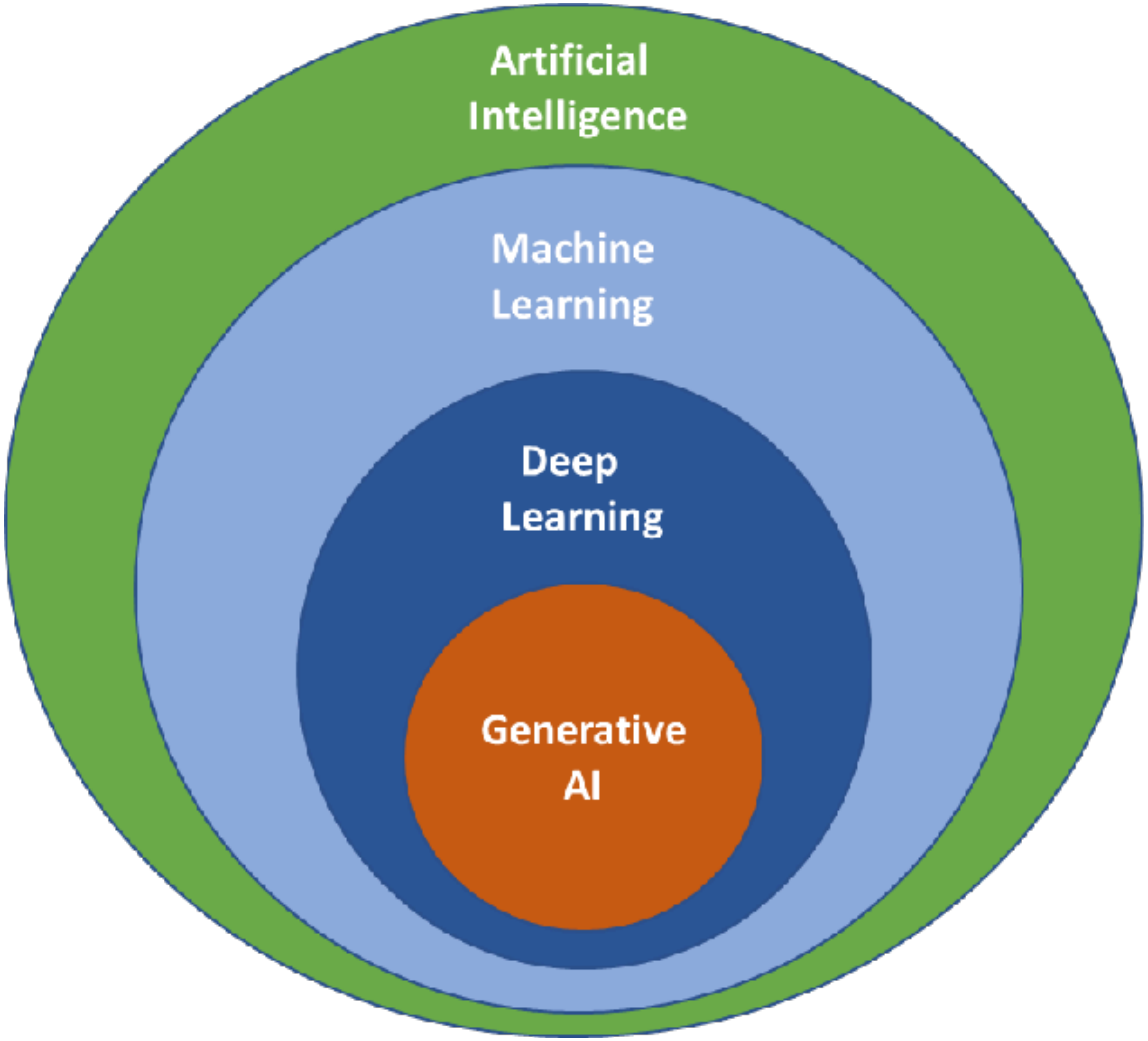
Manage context and knowledge for AI (Context engineering)
Human in the Loop (HITL)

Software Development

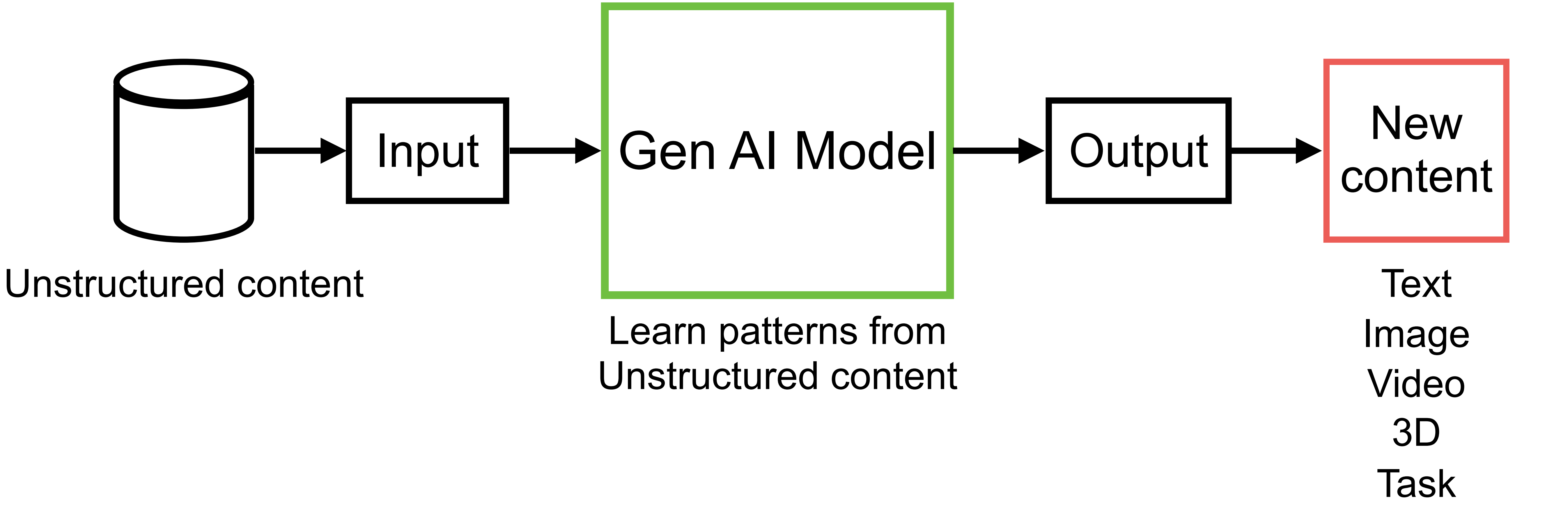
Requirement | Design | Develop | Testing | Deploy

Generative AI

Improve Productivity … (Replace human !!)


Artificial Intelligence
Machine Learning
Deep Learning
Generative AI

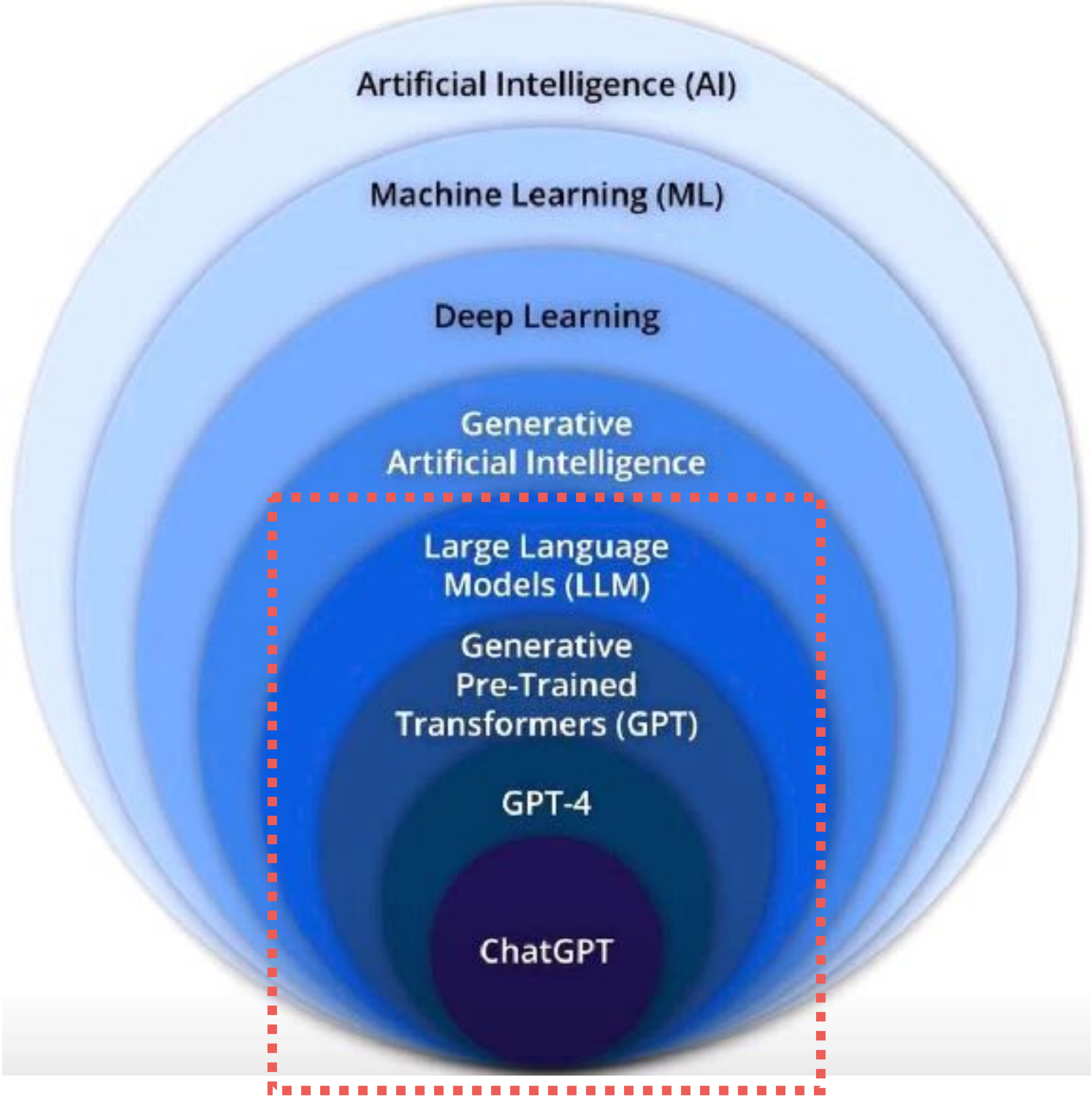
Generative AI



https://grow.google/ai-essentials/

Generative AI

LLMs	GANs	VAEs
Large Language Models	**Generative Adversarial Network**	**Variational Autoencoders**
Text generation	Image generation	Data compression
Code generation	Deep fake	Synthetic data generation
Chatbot	Art creation	Image reconstruction
Conversation AI	Simulate financial market	


Artificial Intelligence (AI)
Machine Learning (ML)
Deep Learning
Generative
Artificial Intelligence
Large Language
Models (LLM)
Generative
Pre-Trained
Transformers (GPT)
GPT-4
ChatGPT

Large Language Model (LLM)

Type of AI model
Process, understand
Generate human readable data

LLM

Training data !!

LLMs in industry

Model name	Company
Bidirectional Encoder Representation from Transformers (BERT)	Google AI
Generative Pre-trained transformer-5 (GPT-5)	OpenAI
Pathways Language Model-E (PaLM-E)	Google AI
BLOOM	NVIDIA AI
Llama 4	Facebook
Claude 4.5 Sonnet/Opus	Anthropic

LLM Development timeline

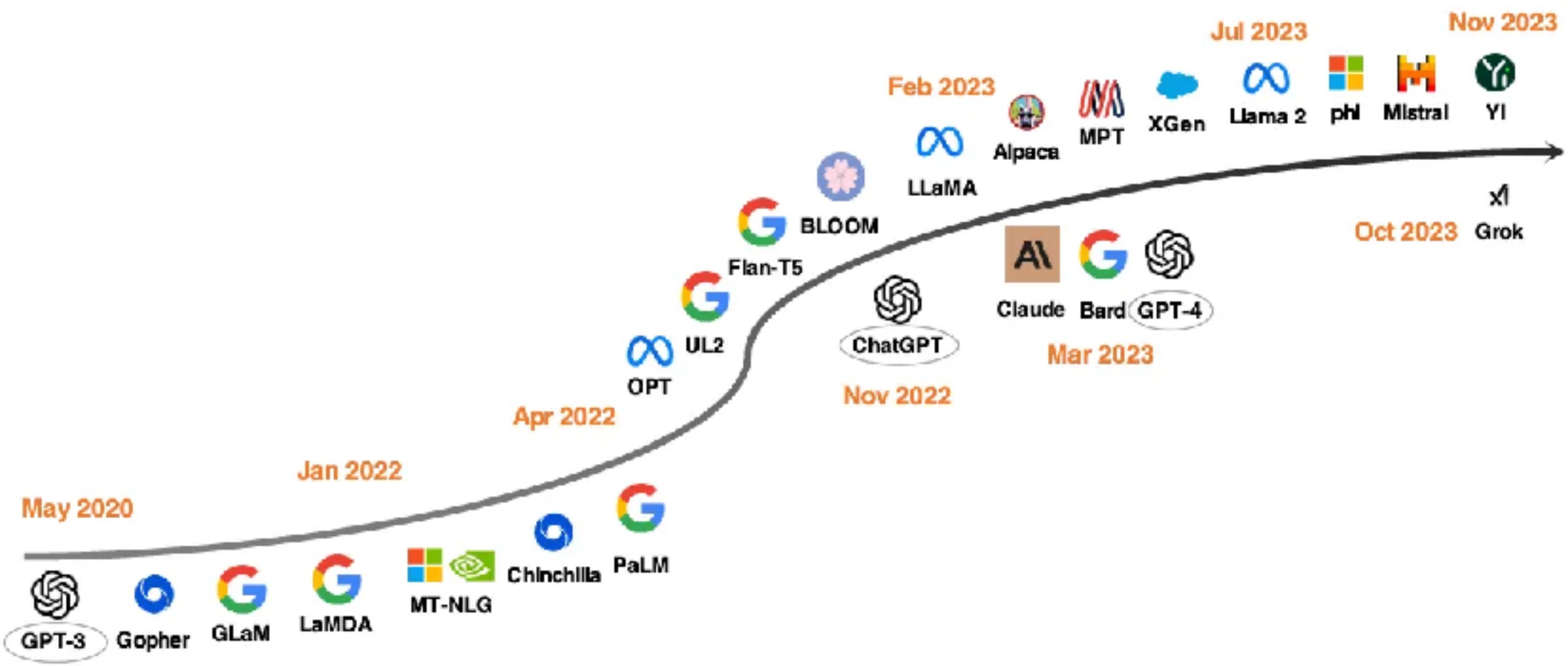


Figure 3: LLM development timeline. The models below the arrow are closed-source while those above the arrow are open-source.

2026
GPT
Gemini
DeepSeek
Llama
Mistral
GLM
Kimi

https://arxiv.org/abs/2311.16989

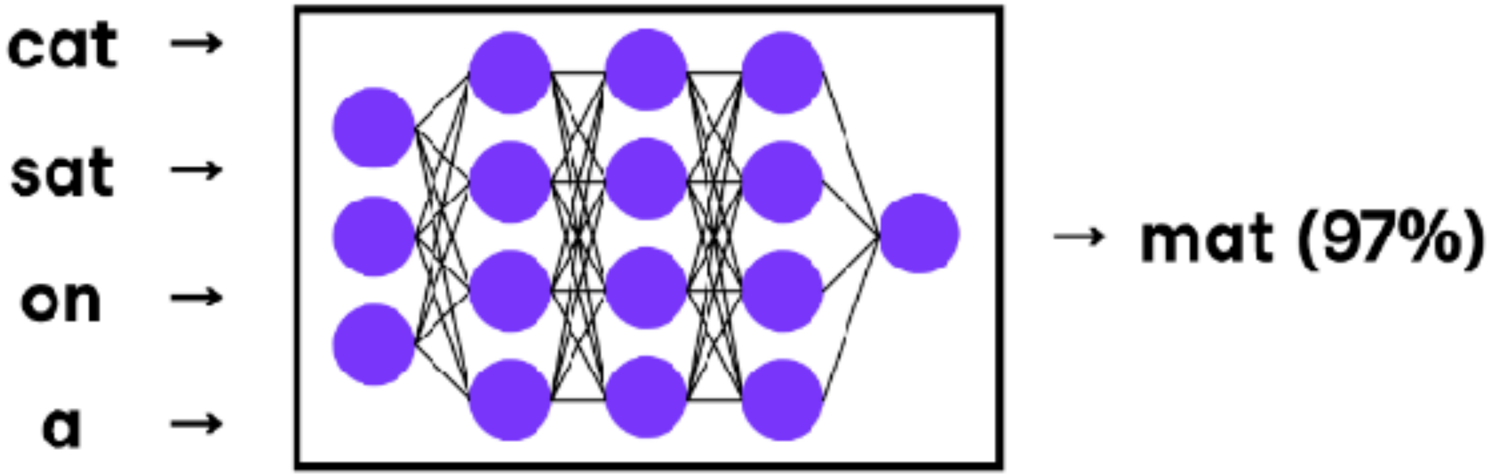
Let’s go !!

สีเหลือง

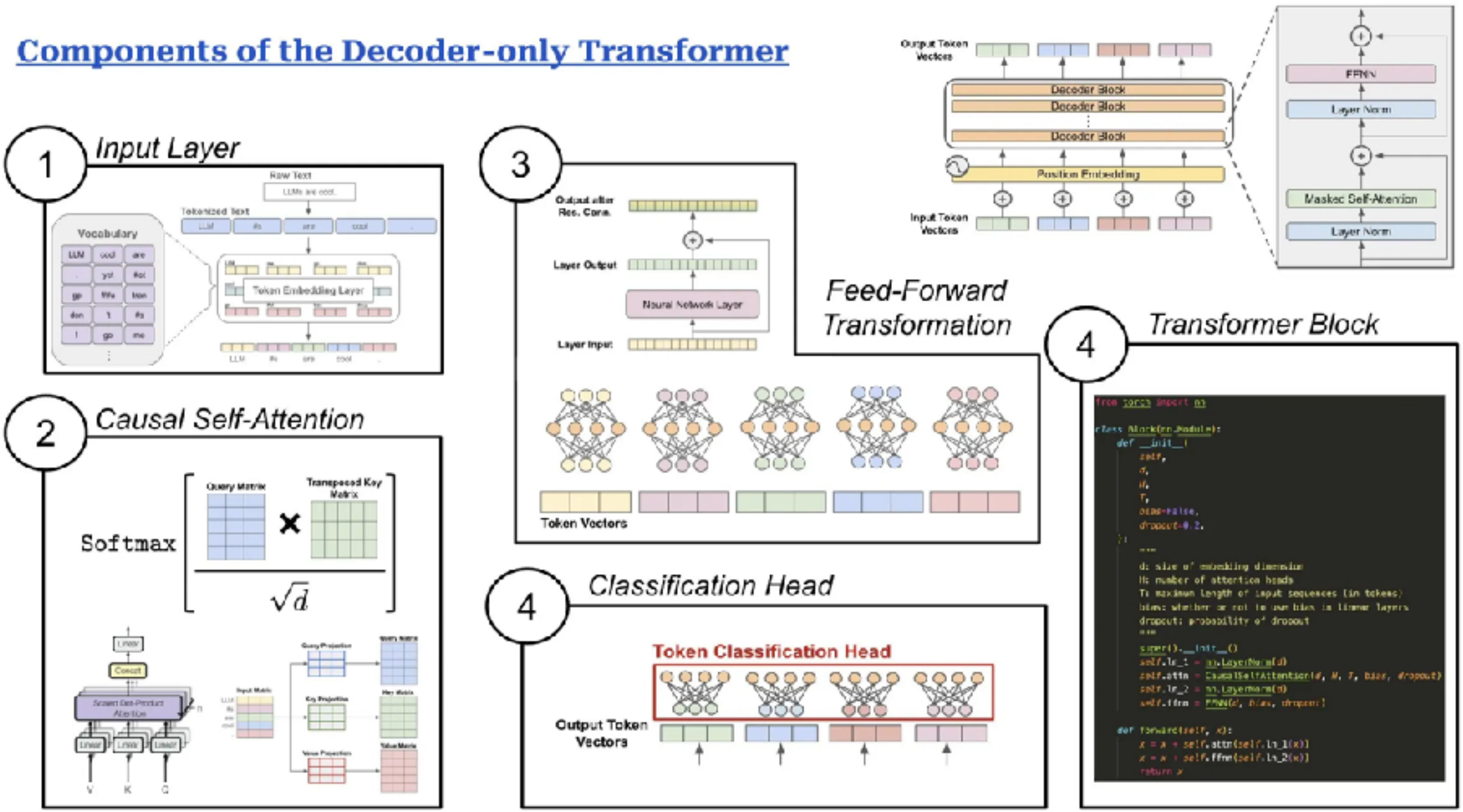
Large Language Model (LLM)

Neural network

Predicts the next word in a sequence

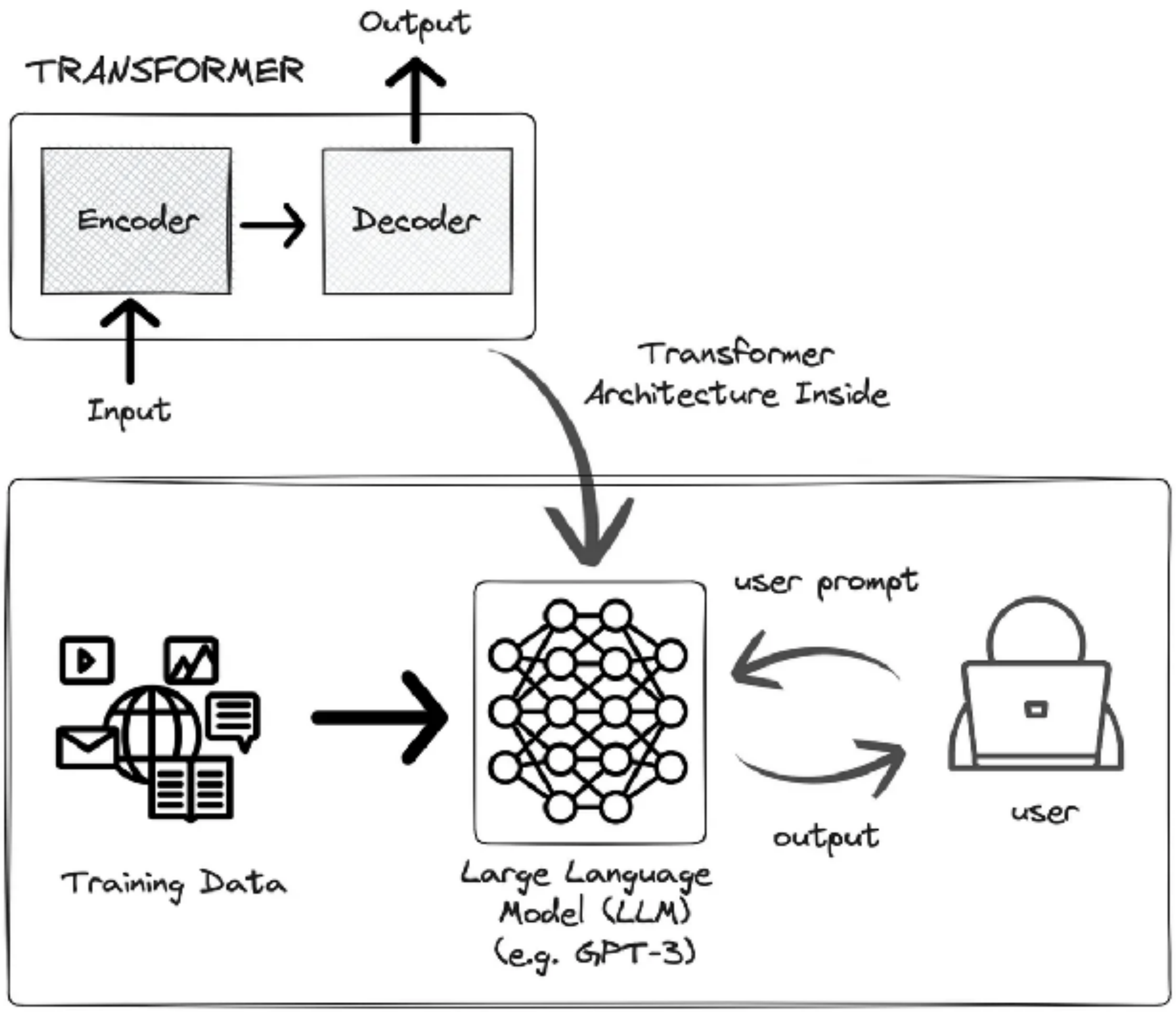


LLM components !!



https://stackoverflow.blog/2024/08/22/llms-evolve-quickly-their-underlying-architecture-not-so-much

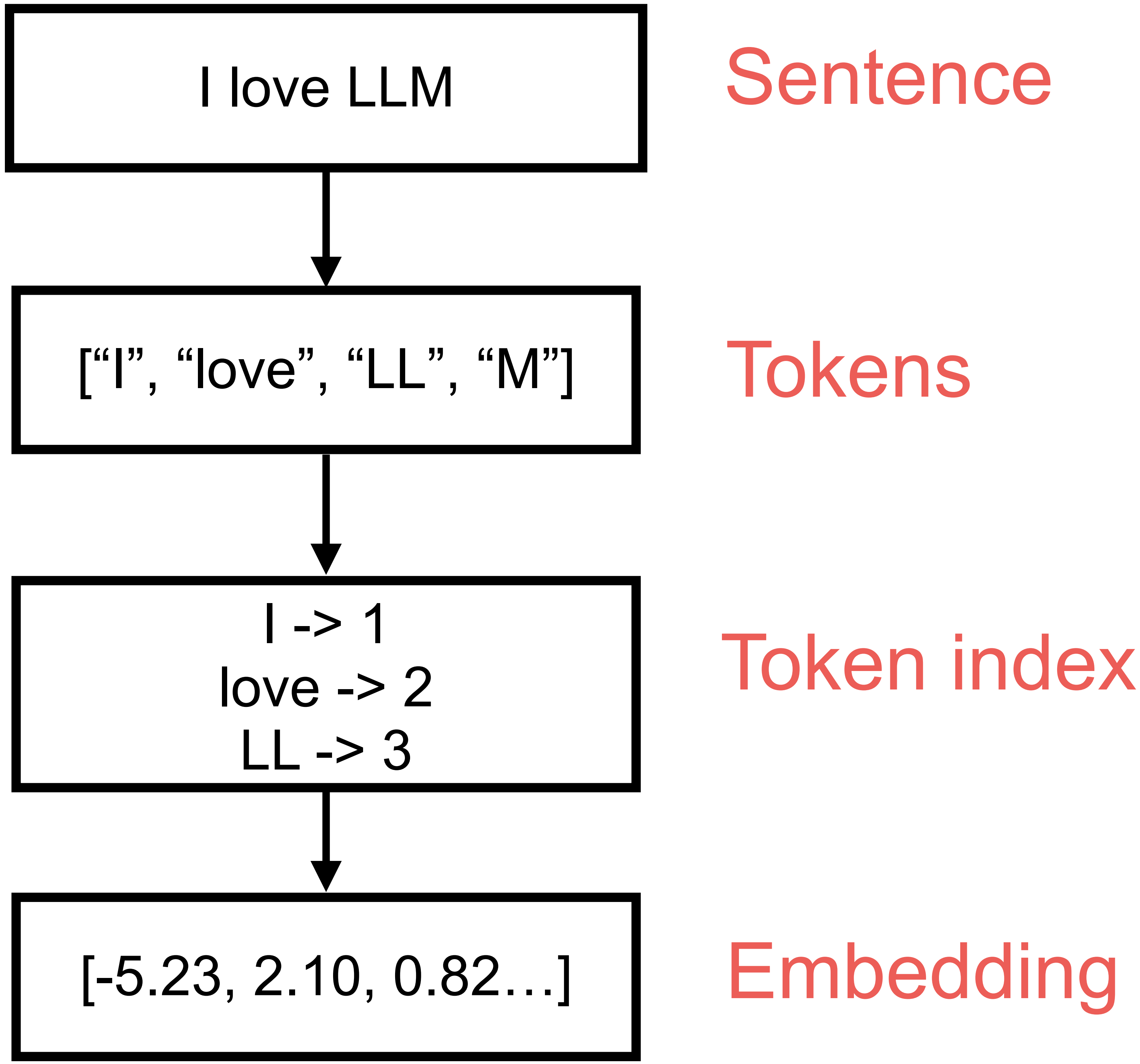
Transformer inside



Transformer process

- I love LLM — Sentence
- ↓
- ["I", "love", "LL", "M"] — Tokens
- ↓
- [-5.23, 2.10, 0.82…] — Embedding

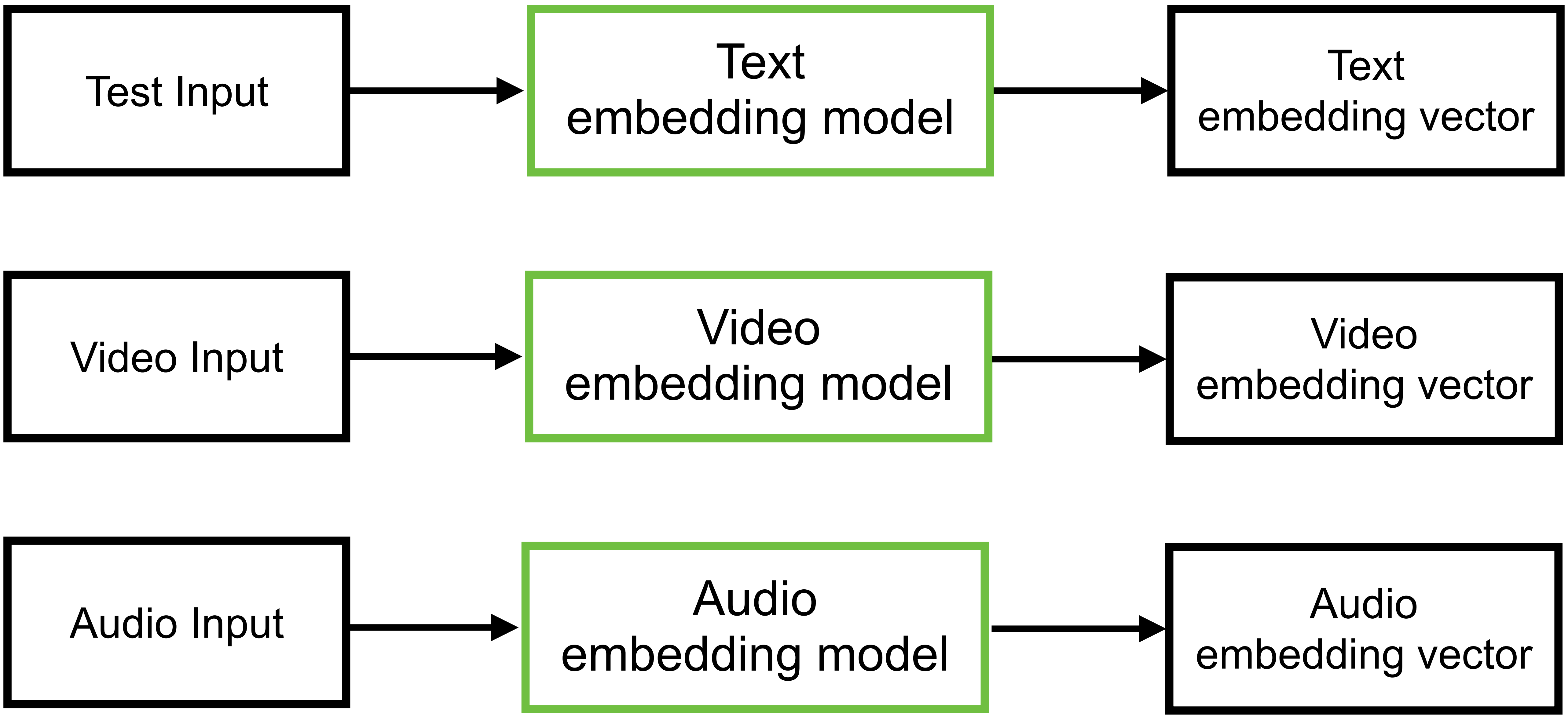
Transformer process



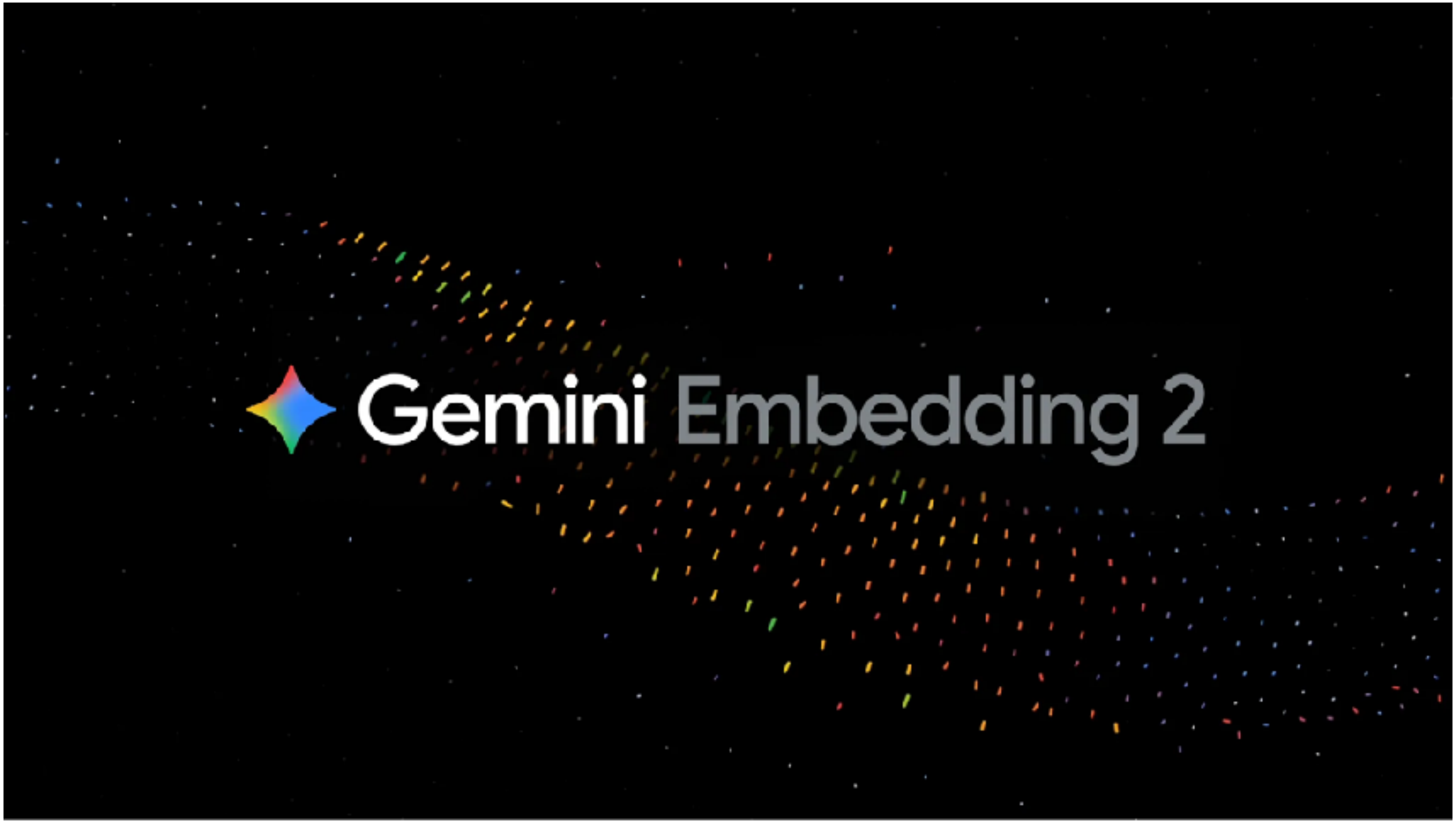
OpenAI Tokenizer

https://platform.openai.com/tokenizer

Embedding

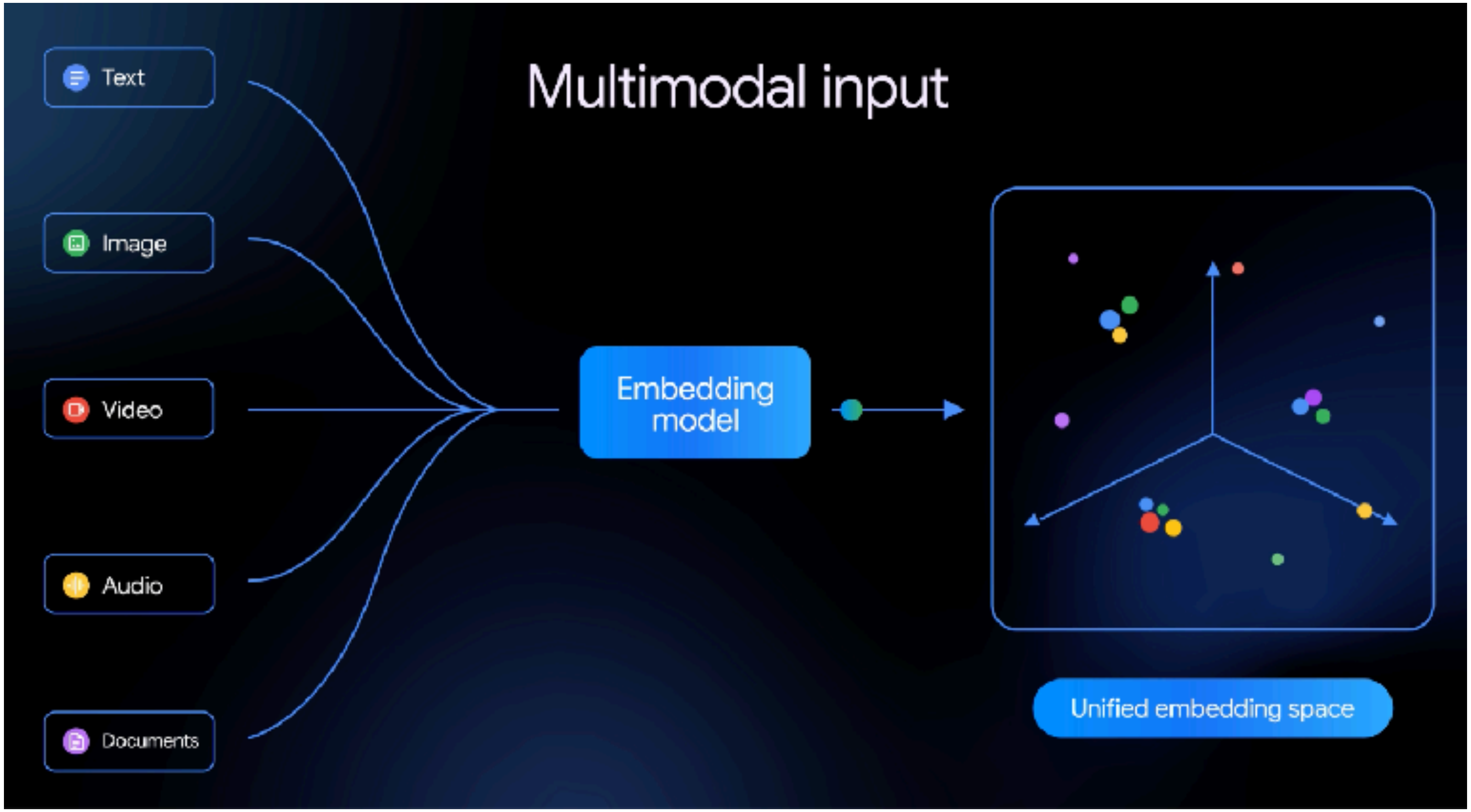

Test Input
Text embedding model
Text embedding vector
Video Input
Video embedding model
Video embedding vector
Audio Input
Audio embedding model
Audio embedding vector

Multimodal Embedding Model



https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/

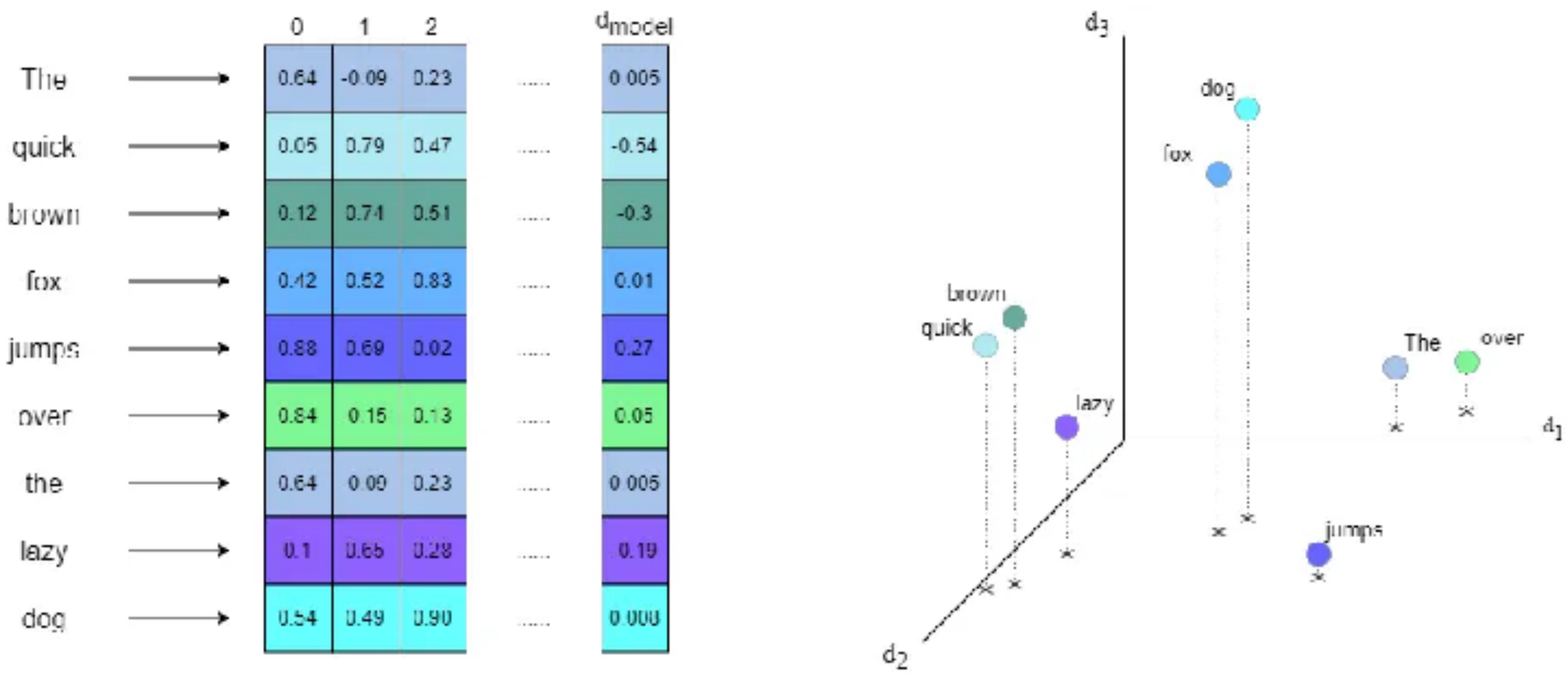
Multimodal Embedding Model



https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/

Embedding ?

Map items of **unstructured data** to high-dimensional real vectors



	0	1	2		d_{model}
The	0.64	-0.09	0.23	......	0.005
quick	0.05	0.79	0.47	......	-0.54
brown	0.12	0.74	0.51	......	-0.3
fox	0.42	0.52	0.83	......	0.01
jumps	0.88	0.69	0.02	......	0.27
over	0.84	0.15	0.13	......	0.05
the	0.64	0.09	0.23	......	0.005
lazy	0.1	0.65	0.28	......	0.19
dog	0.54	0.49	0.90	......	0.008

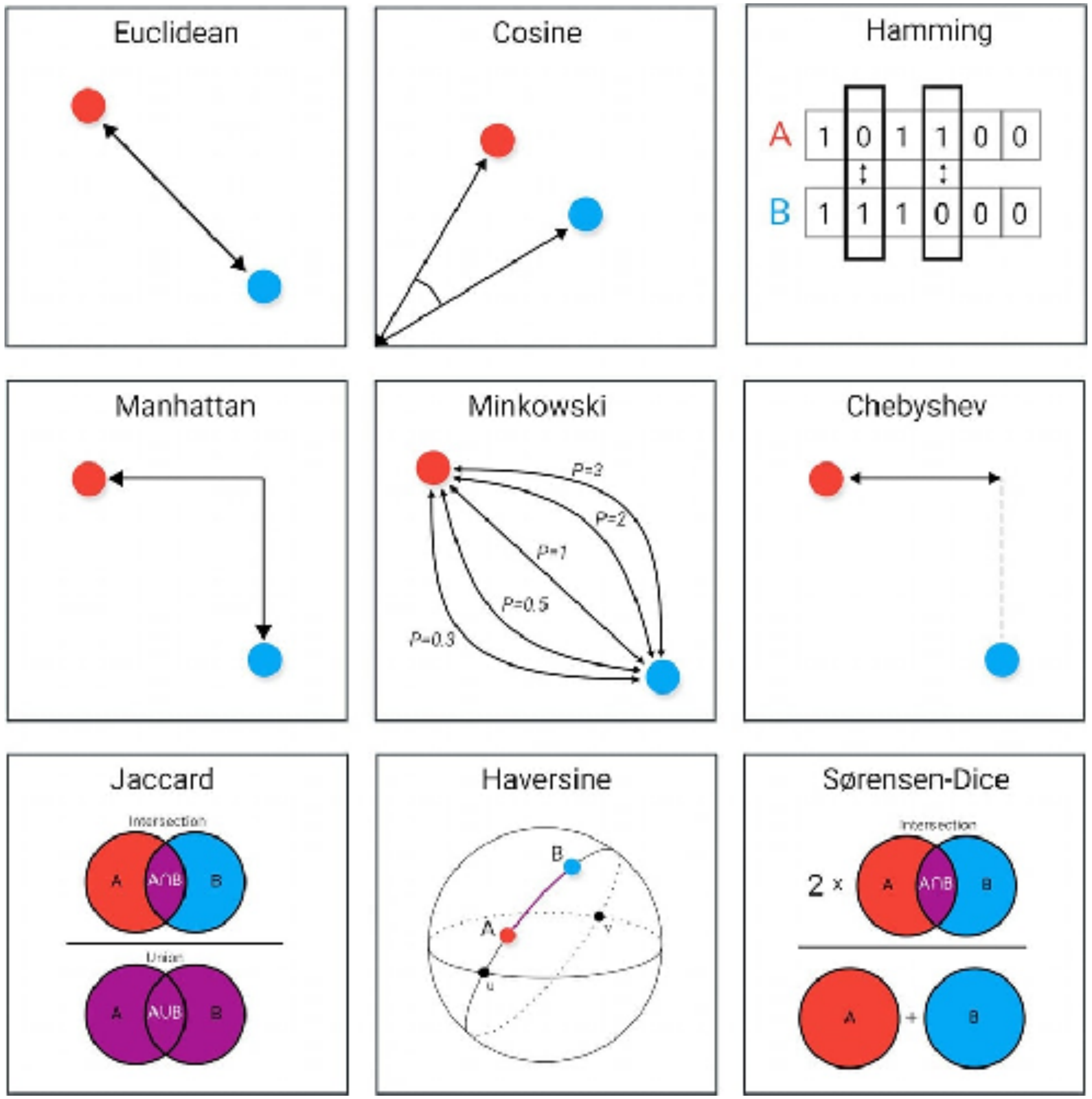
https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca

Embedding Leaderboard

Rank (Bor…	Model	Embedding D…	Max Tokens	Mean (T…	Mean (TaskT…	Bitext …	Classification	Clustering	Instructi
1	KaLM-Embedding-Gemma3-12B-2511	3840	32768	**72.32**	**62.51**	**83.76**	**77.88**	55.77	5.49
2	llama-embed-nemotron-8b	4096	32768	69.46	61.09	81.72	73.21	54.35	10.82
3	Qwen3-Embedding-8B	4096	32768	70.58	61.69	80.89	74.00	57.65	10.06
4	gemini-embedding-001	3072	2048	68.37	59.59	79.28	71.82	54.59	5.18
5	Qwen3-Embedding-4B	2560	32768	69.45	60.86	79.36	72.33	57.15	11.56
6	Octen-Embedding-8B	4096	32768	67.84	60.28	80.35	66.68	55.68	8.98
7	F2LLM-v2-14B	5120	40960	68.74	59.45	77.15	73.60	**60.91**	0.62
8	F2LLM-v2-8B	4096	40960	68.09	58.99	75.96	71.93	60.62	0.85
9	Seed1.6-embedding-1215	2048	32768	70.26	61.34	78.68	76.75	56.78	-0.82
10	F2LLM-v2-4B	2560	40960	67.06	58.25	74.49	70.73	59.53	2.25
11	jina-embeddings-v5-text-small	1024	32768	67.00	58.90	69.71	71.32	53.41	1.35

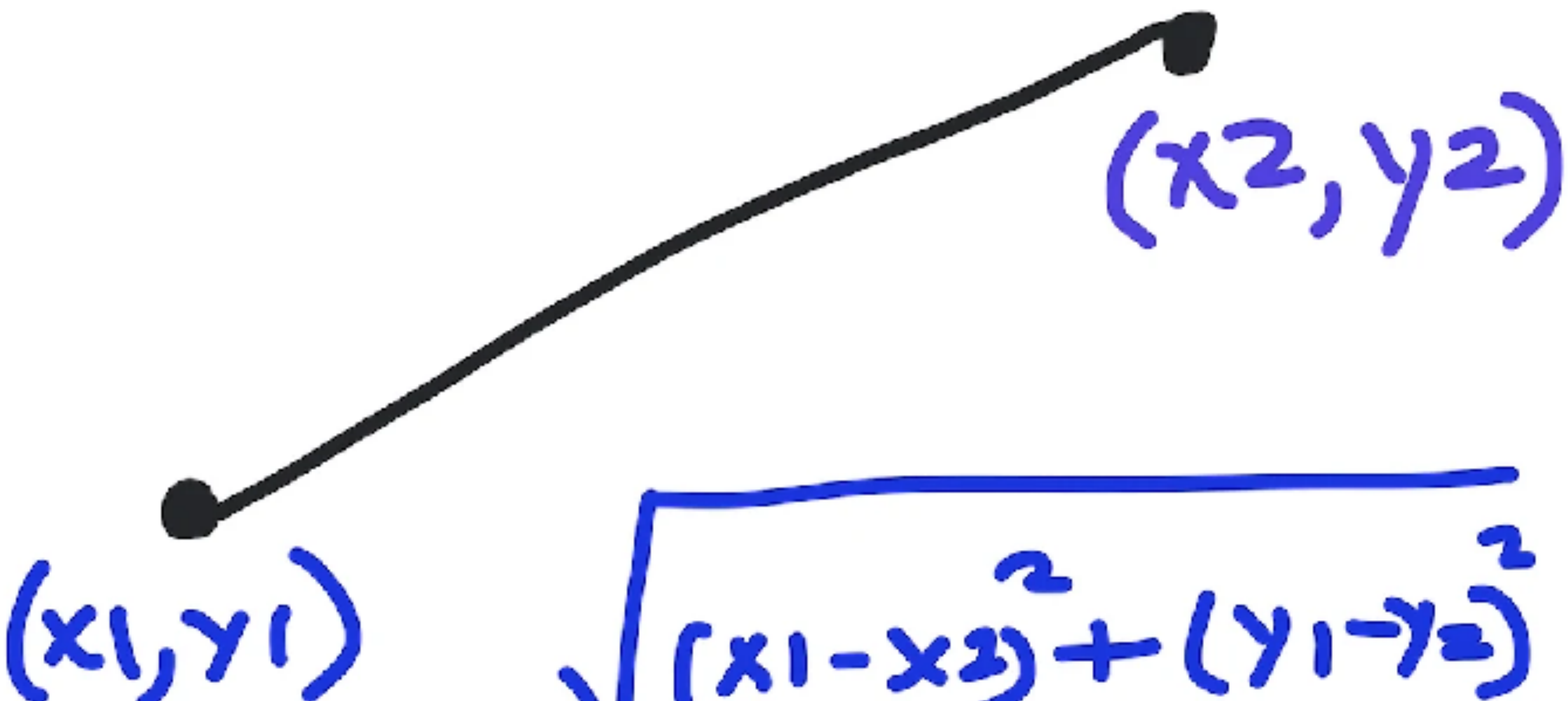
https://huggingface.co/spaces/mteb/leaderboard

Distance measure in Data Science



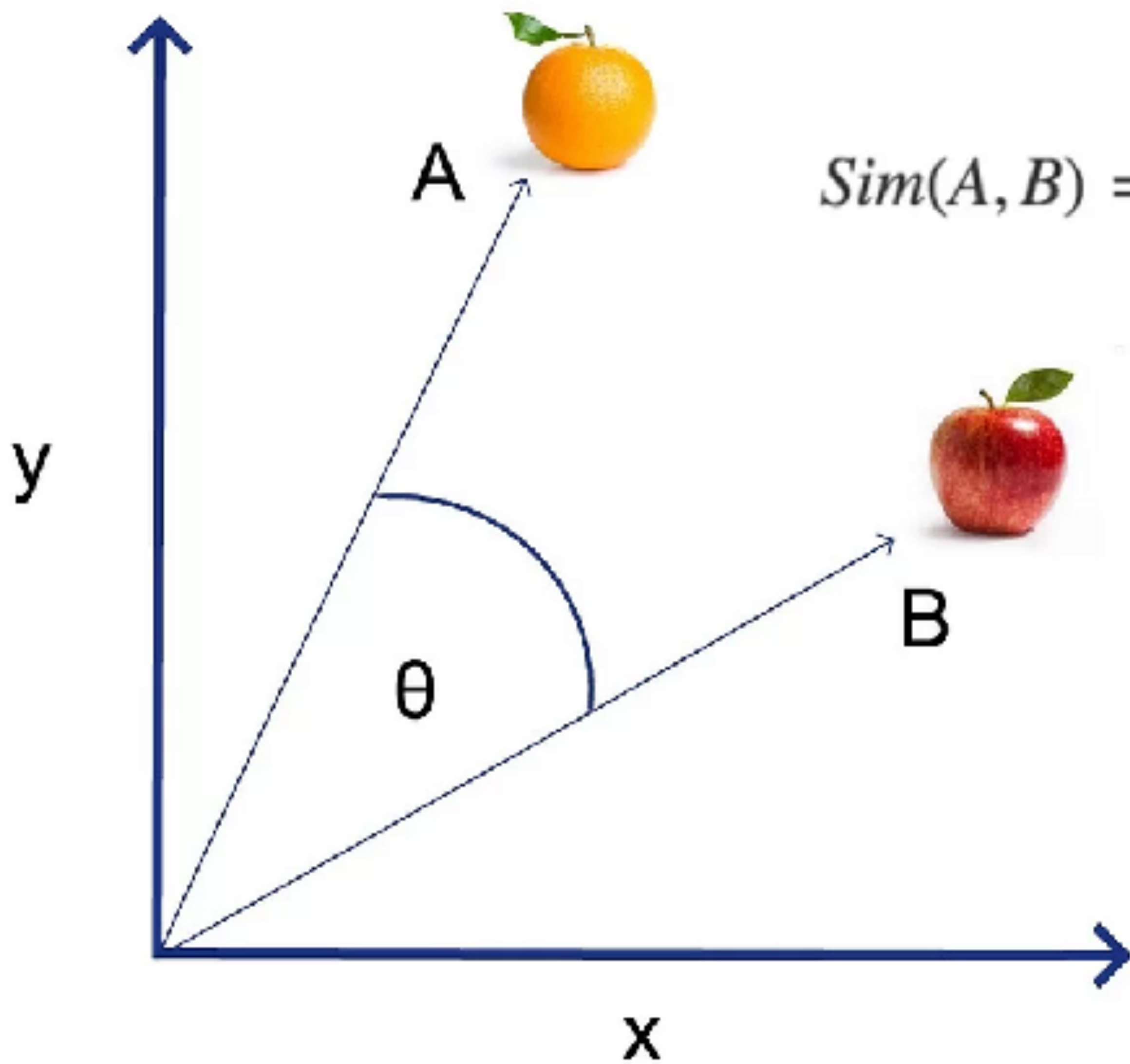
https://www.maartengrootendorst.com/blog/distances/

Euclidean Distance



$$\sqrt{(x1-x2)^2+(y1-y2)^2}$$

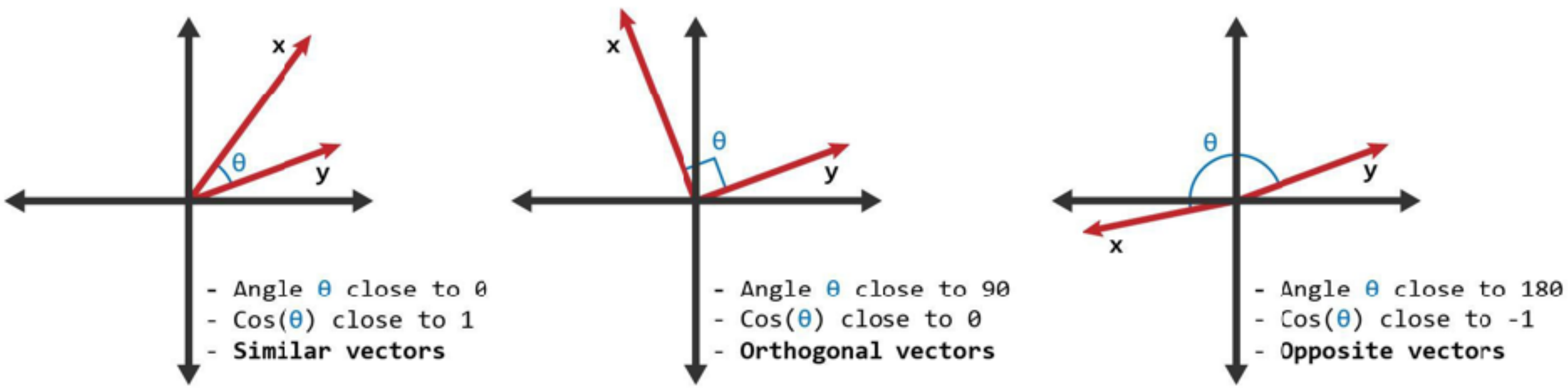
Cosine Similarity



$$Sim(A,B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

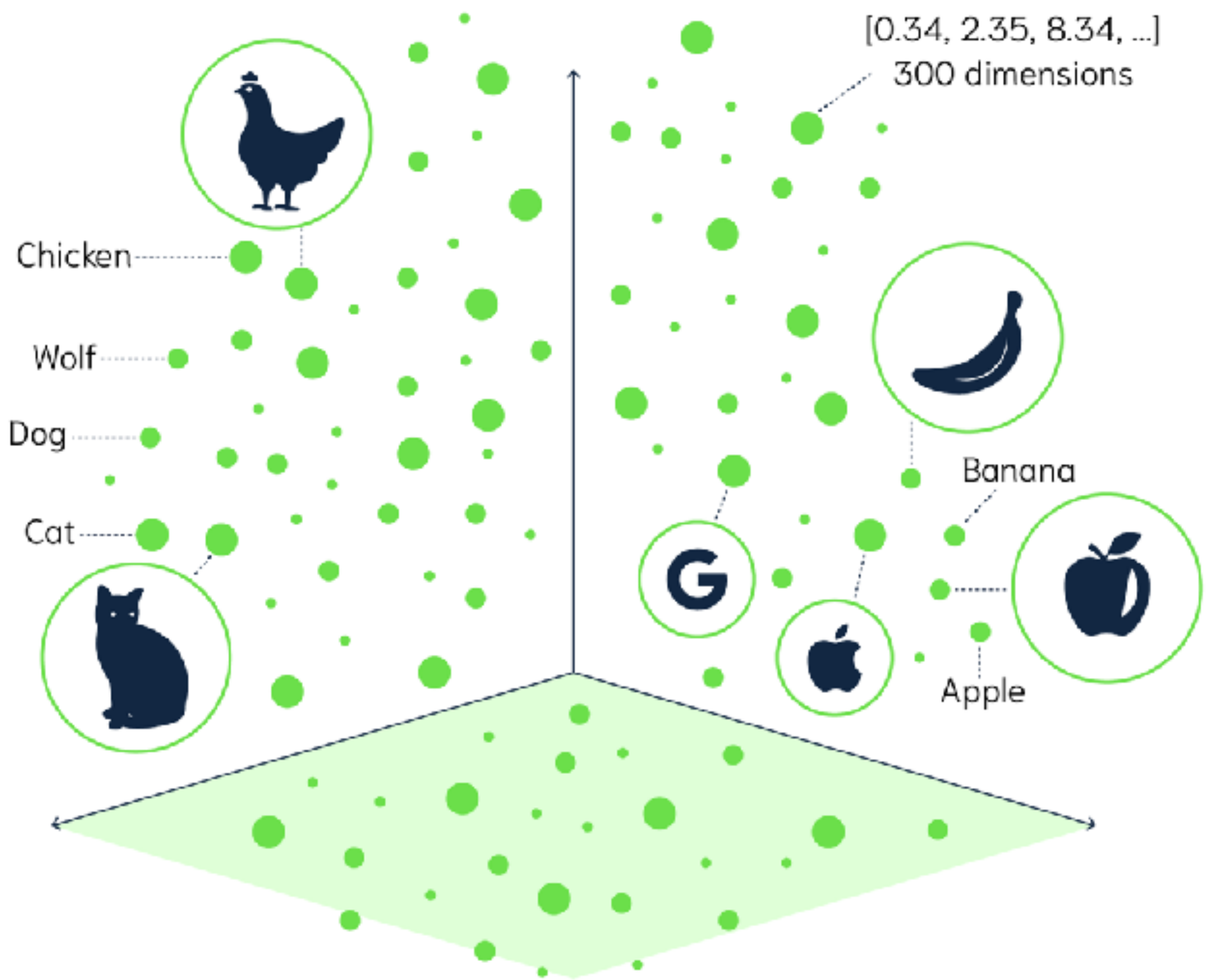
https://python.plainenglish.io/mastering-cosine-metrics-a-data-scientists-essential-toolkit-3ce95eda91f9

Cosign Similarity

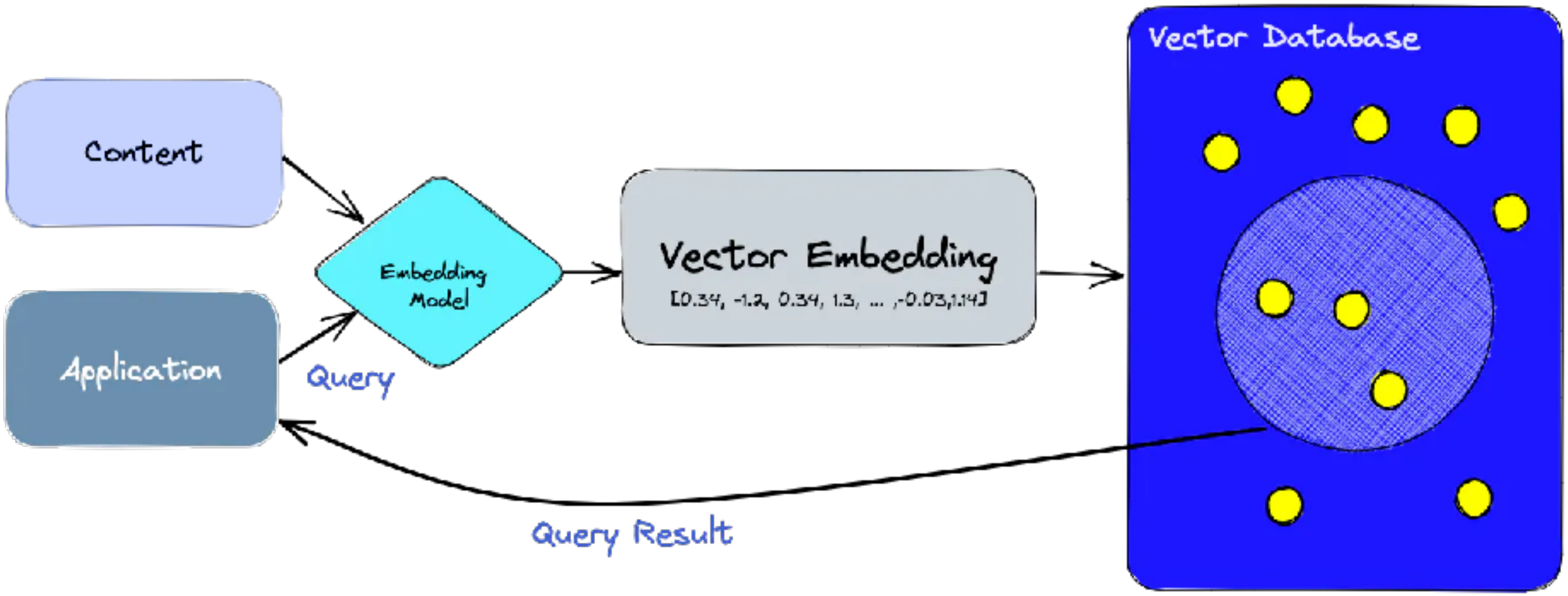


https://www.learndatasci.com/glossary/cosine-similarity/

Visual of Vector space

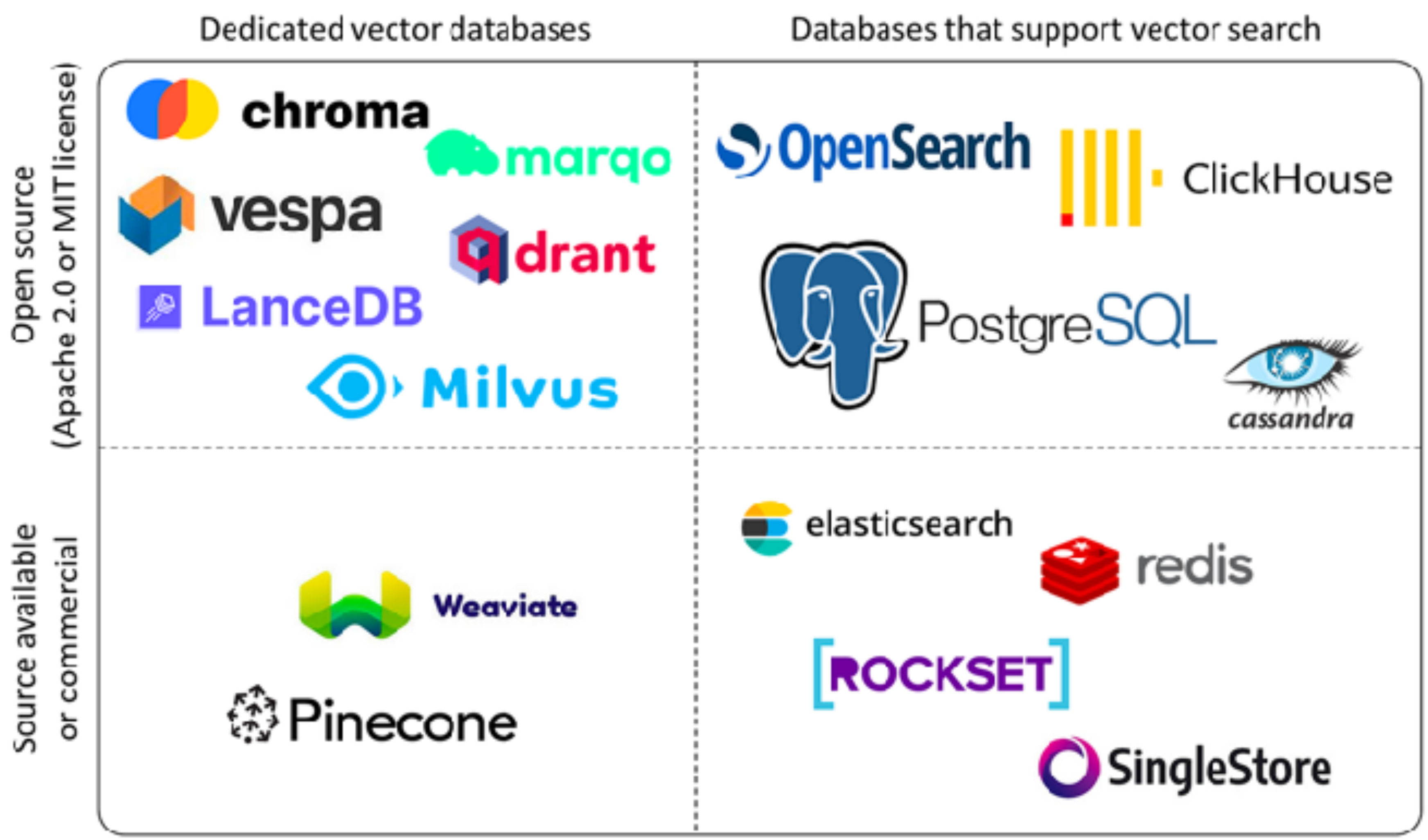


Store data in Vector Database

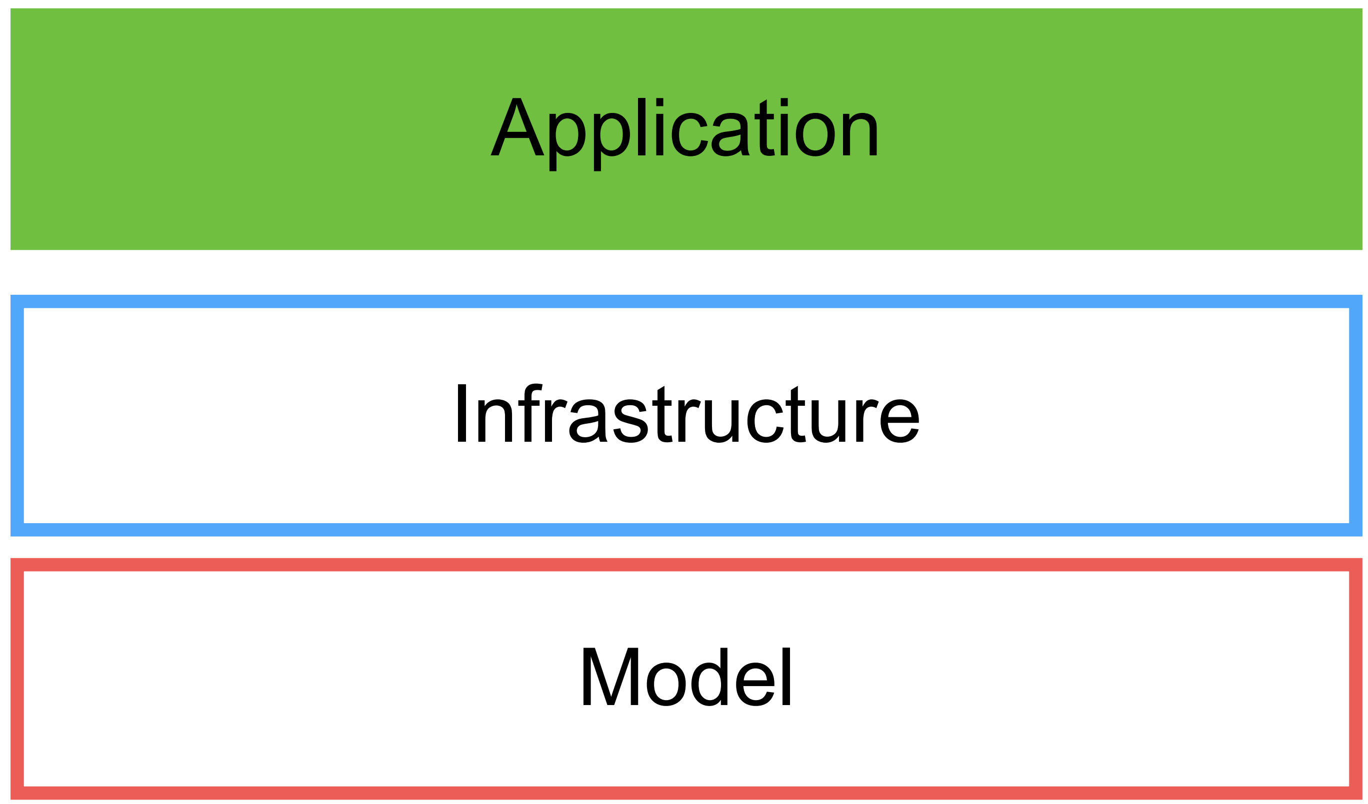


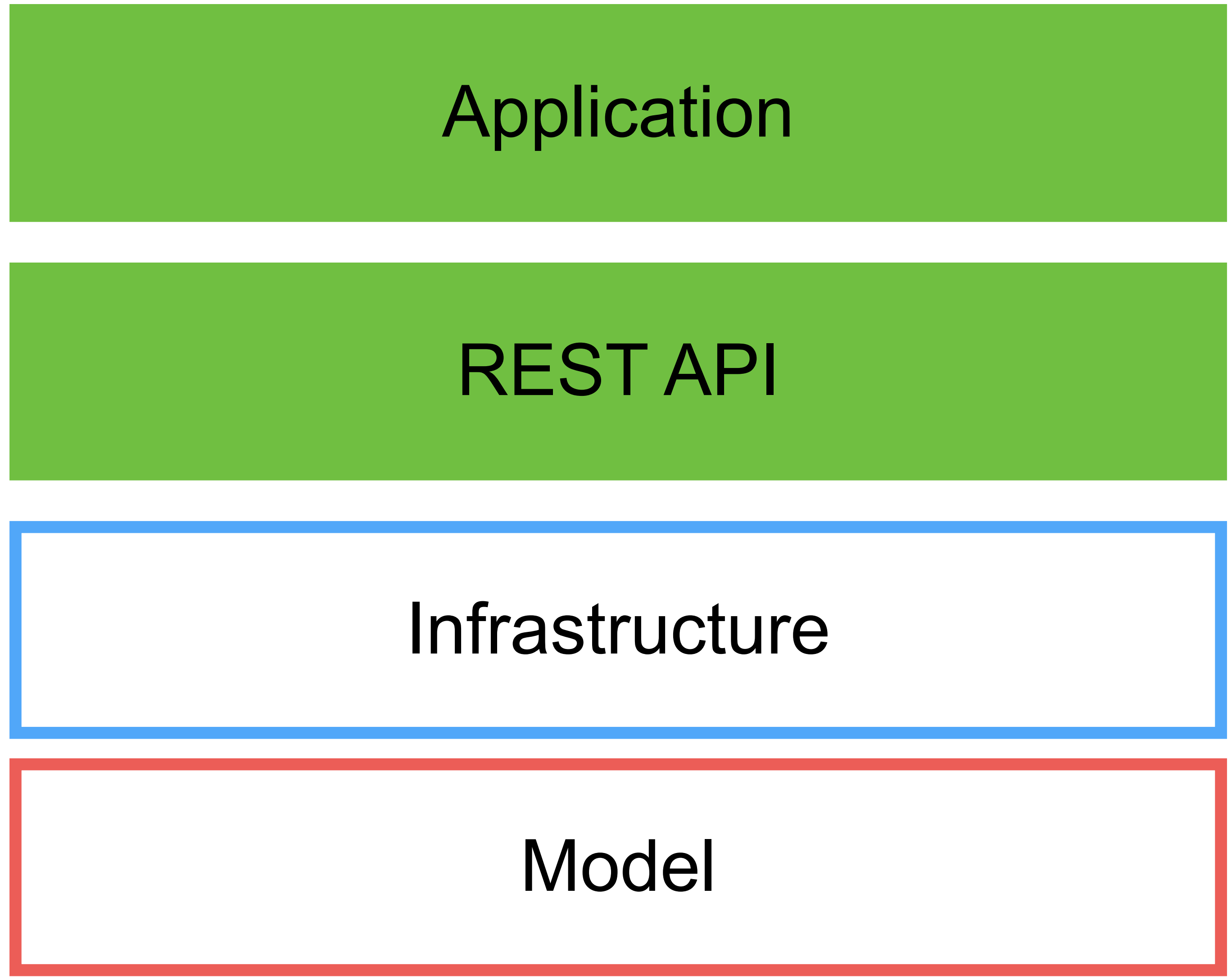
Vector Database ?

Map items of unstructured data to high-dimensional real **vectors**

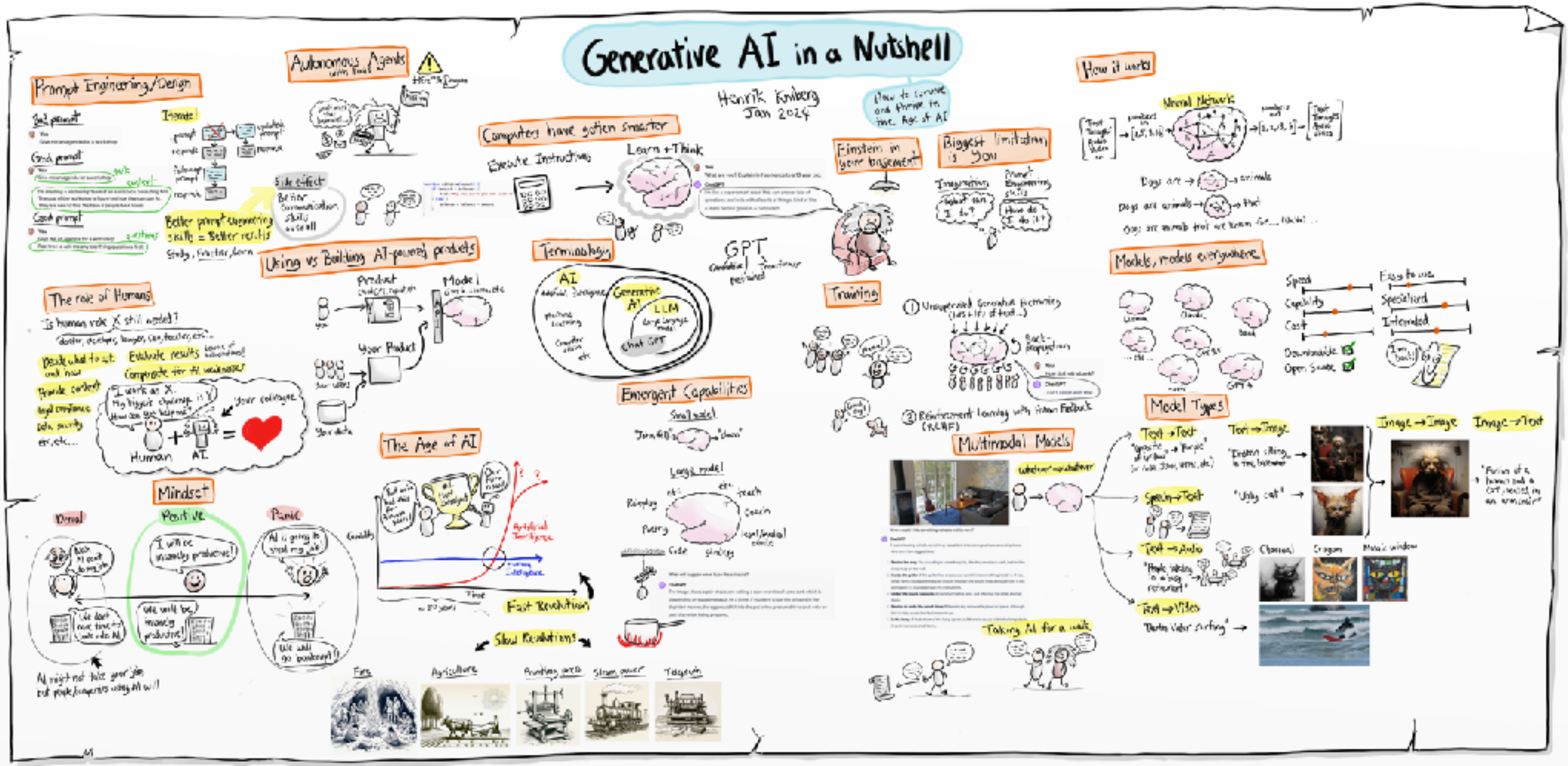


https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca


Application
Infrastructure
Model

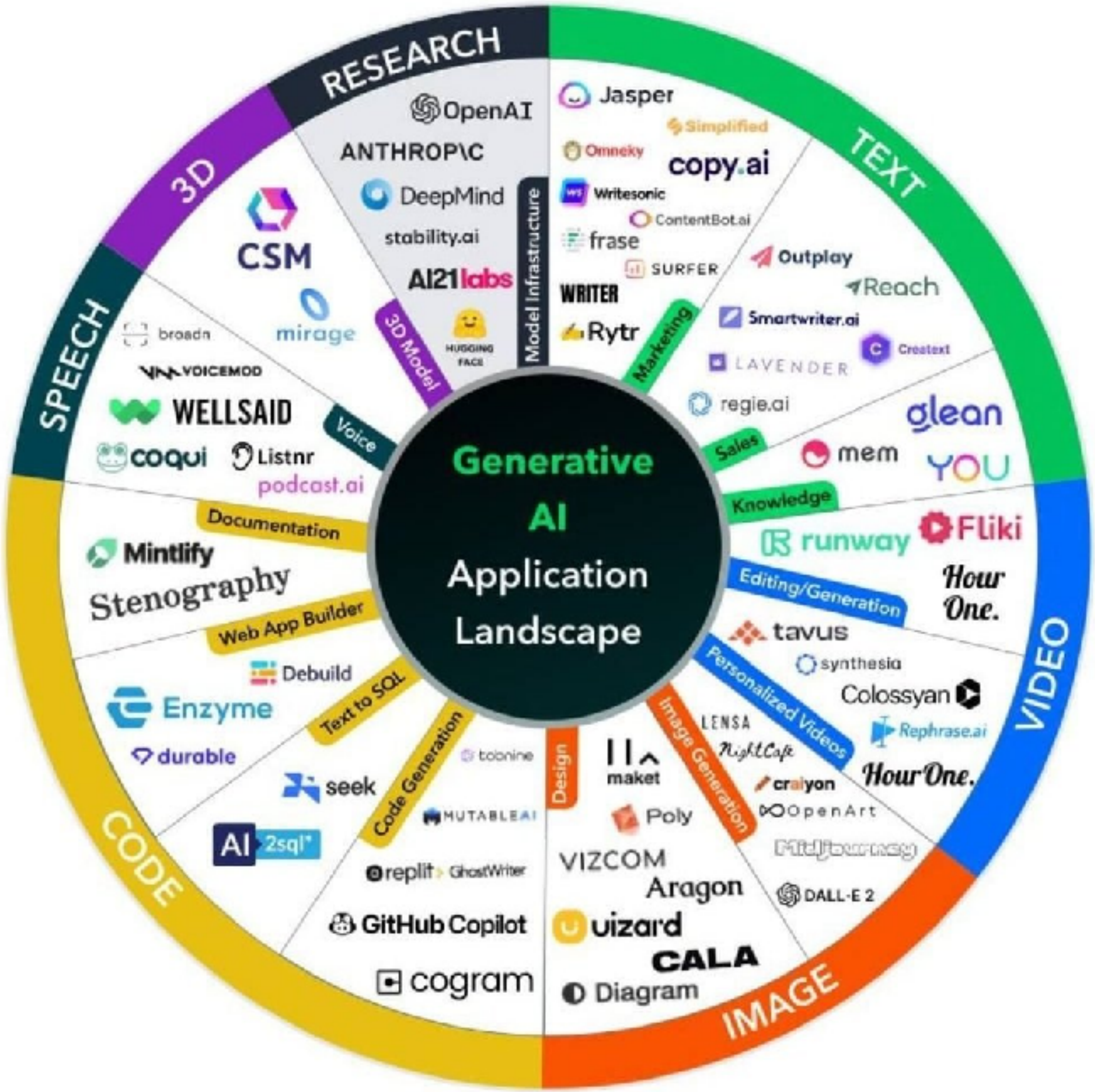

Application
REST API
Infrastructure
Model

Generative AI in Nutshell



https://www.youtube.com/watch?v=2IK3DFHRFfw

Application Tool chains



Generative AI Application Landscape
RESEARCH
OpenAI
ANTHROP\C
DeepMind
stability.ai
AI21labs
HUGGING FACE
Model Infrastructure
TEXT
Jasper
Simplified
Omneky
copy.ai
Writesonic
ContentBot.ai
frase
SURFER
WRITER
Rytr
Marketing
Outplay
Reach
Smartwriter.ai
Crostext
LAVENDER
regie.ai
Sales
glean
mem
YOU
Knowledge
VIDEO
runway
Fliki
Hour One.
Editing/Generation
tavus
synthesia
Colossyan
Rephrase.ai
Personalized Videos
Hour One.
IMAGE
LENSA
NightCafe
craiyon
OpenArt
Midjourney
DALL-E 2
Image Generation
maket
Design
Poly
VIZCOM
Aragon
uizard
CALA
Diagram
CODE
tabnine
Code Generation
MUTABLEAI
seek
replit GhostWriter
GitHub Copilot
cogram
AI2sql
Text to SQL
durable
Enzyme
Debuild
Web App Builder
Stenography
Mintlify
Documentation
SPEECH
broadn
VOICEMOD
WELLSAID
coqui
Listnr
podcast.ai
Voice
3D
CSM
mirage
3D Model

Tool chains category

Assist tasks

Interaction modes

Prompt composition

Properties of model

LLM models

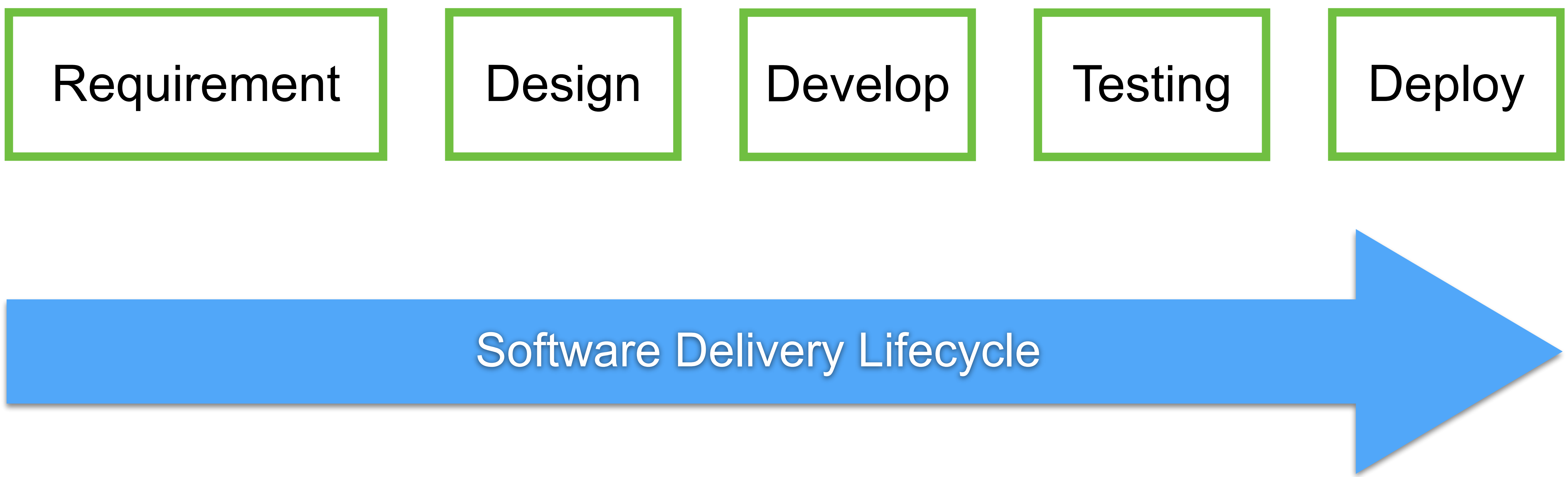
Assist tasks

Finding information faster in context

Generating code

Reasoning about code

Transforming code into something ..



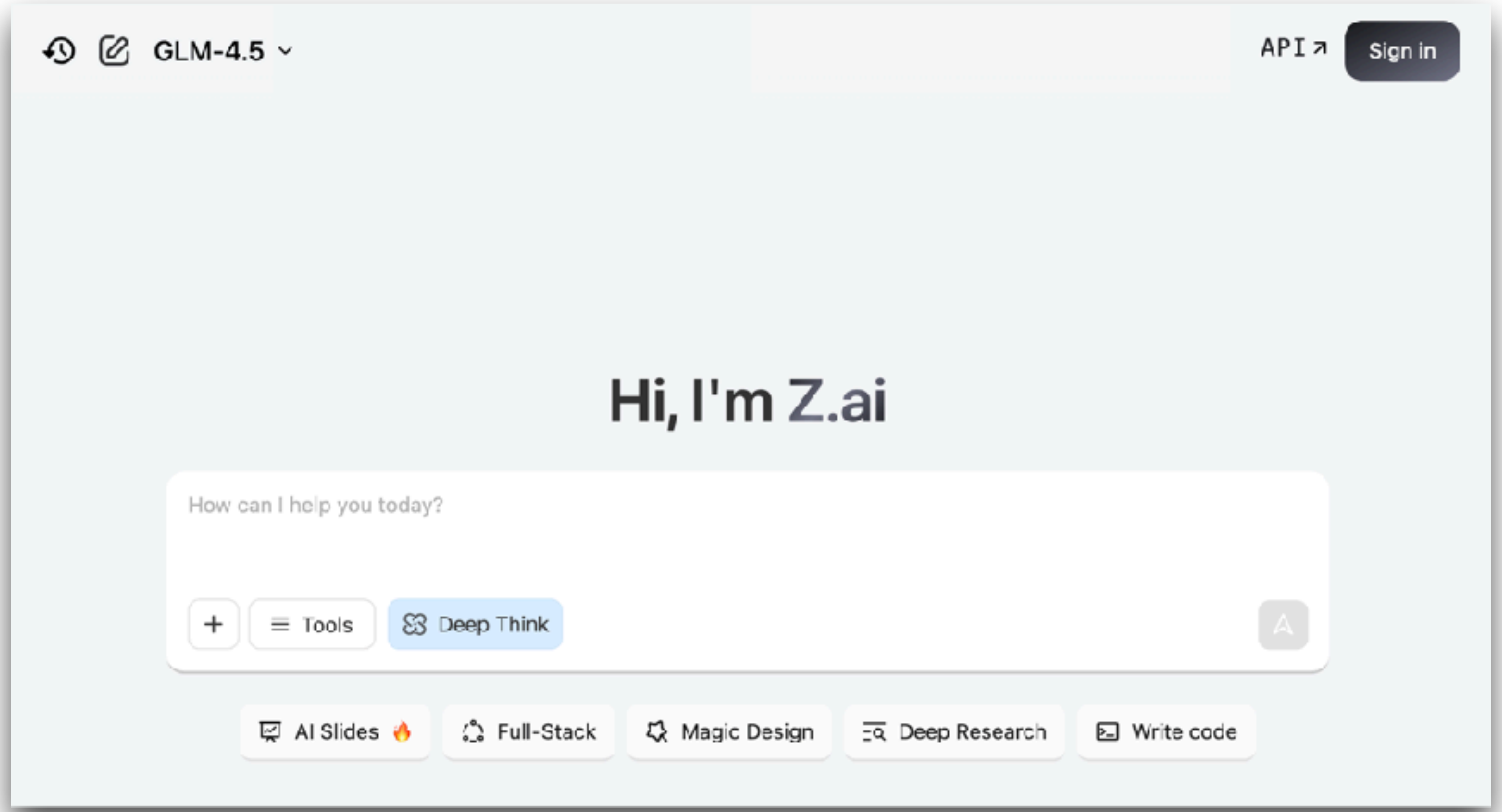
Interaction modes

Chat interfaces

In-line assistance (typing in code editor)

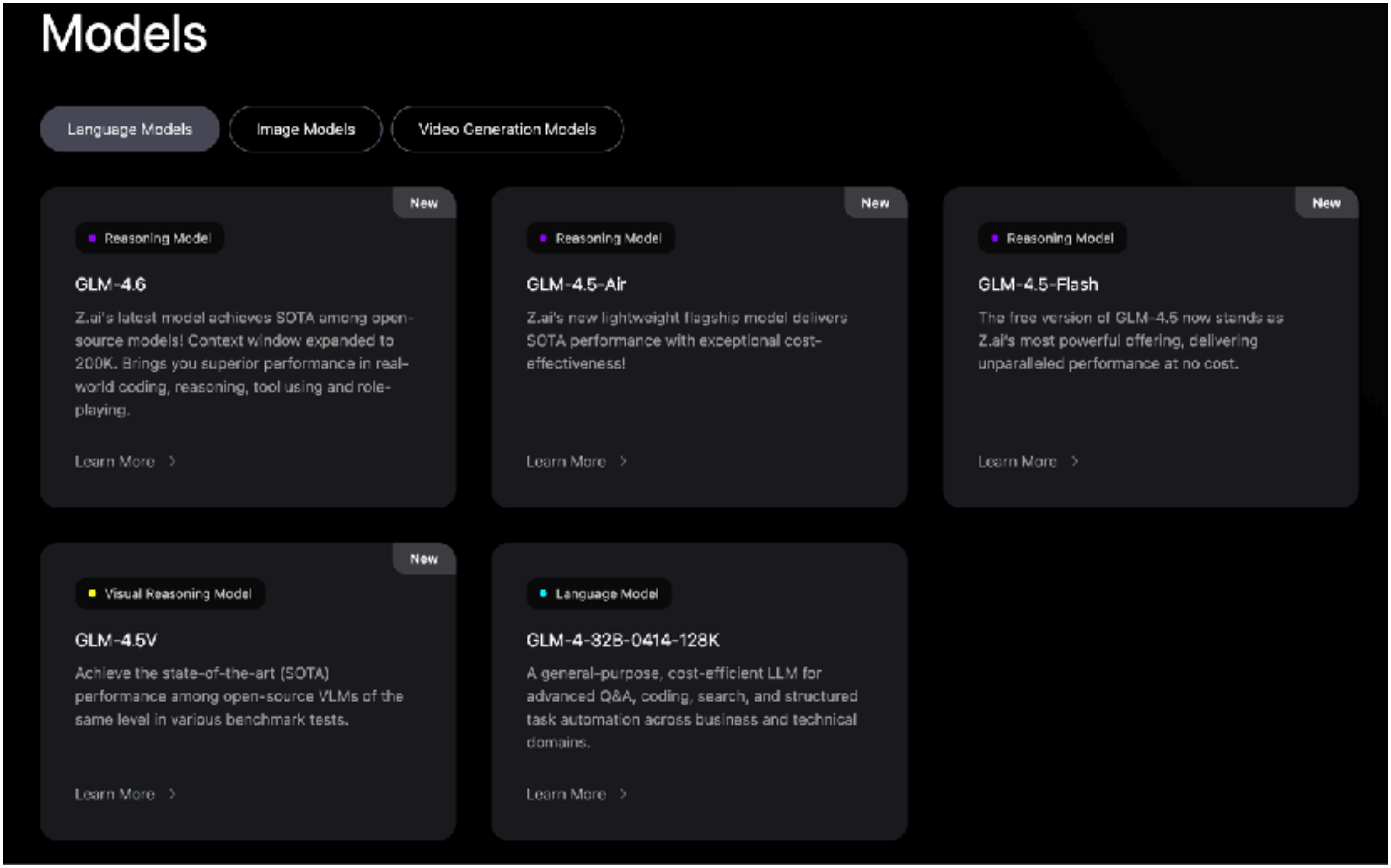
CLI (command-line interface)

Z.ai



https://z.ai/

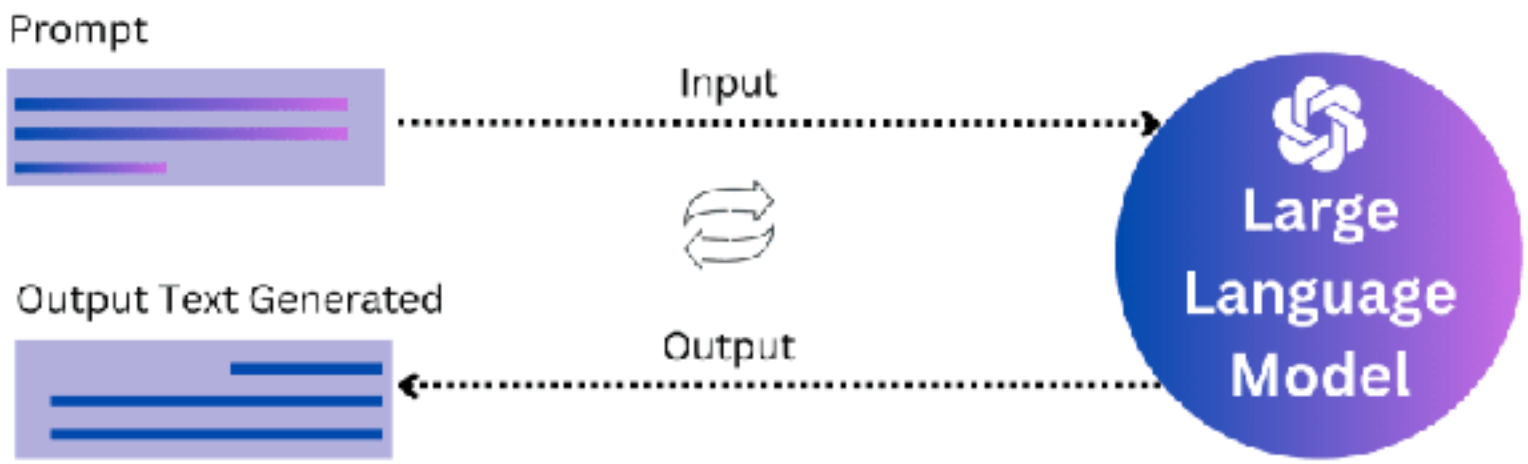
Models



https://z.ai/

Prompt composition

Prompt engineering
Compose prompts from user inputs and context



https://platform.openai.com/docs/guides/prompt-engineering

Better Prompt

Write clear instructions

Provide reference text

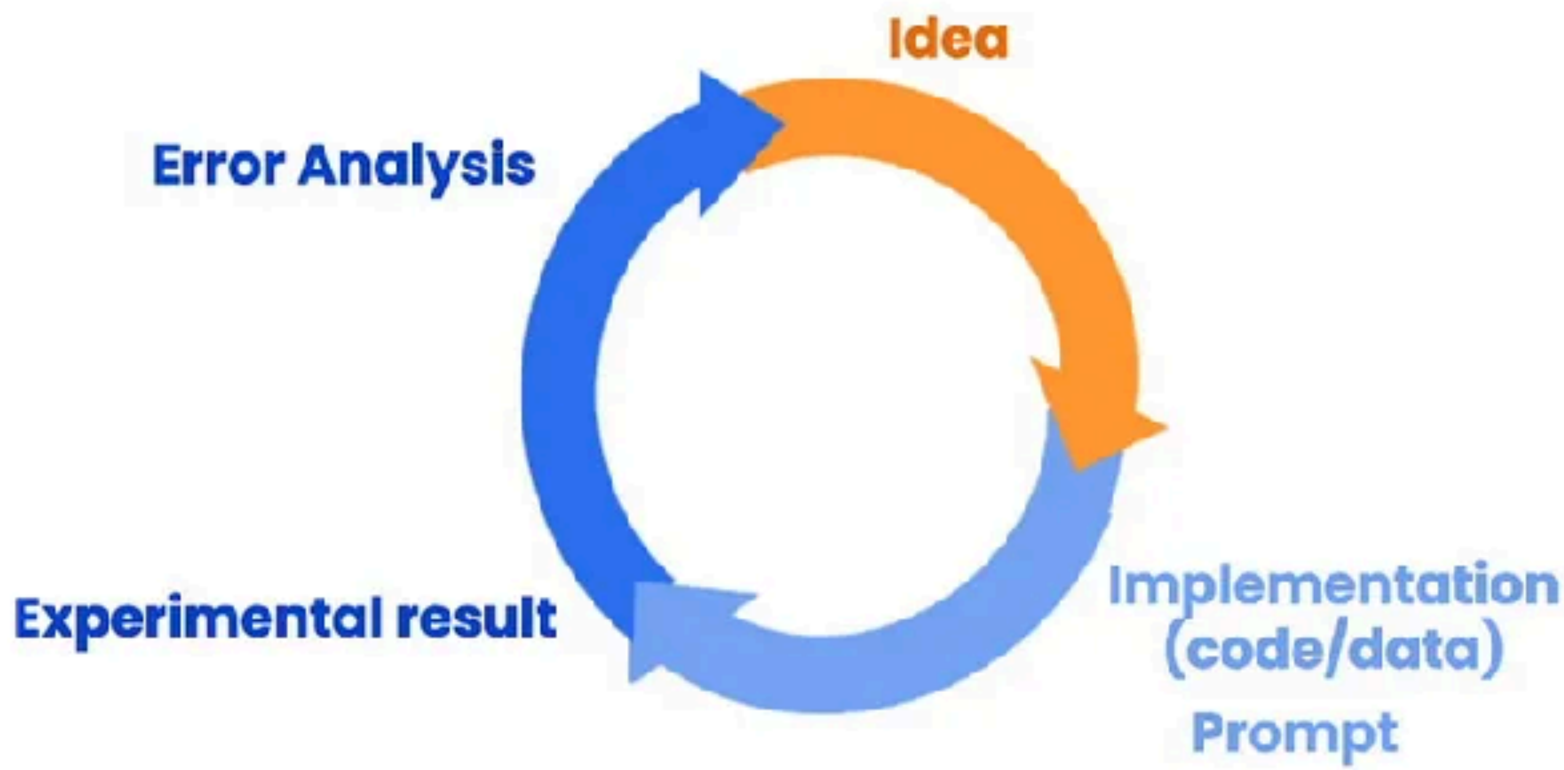
Split complex tasks into simpler subtasks

Give the model time to think

Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering

Iterative Prompt Development

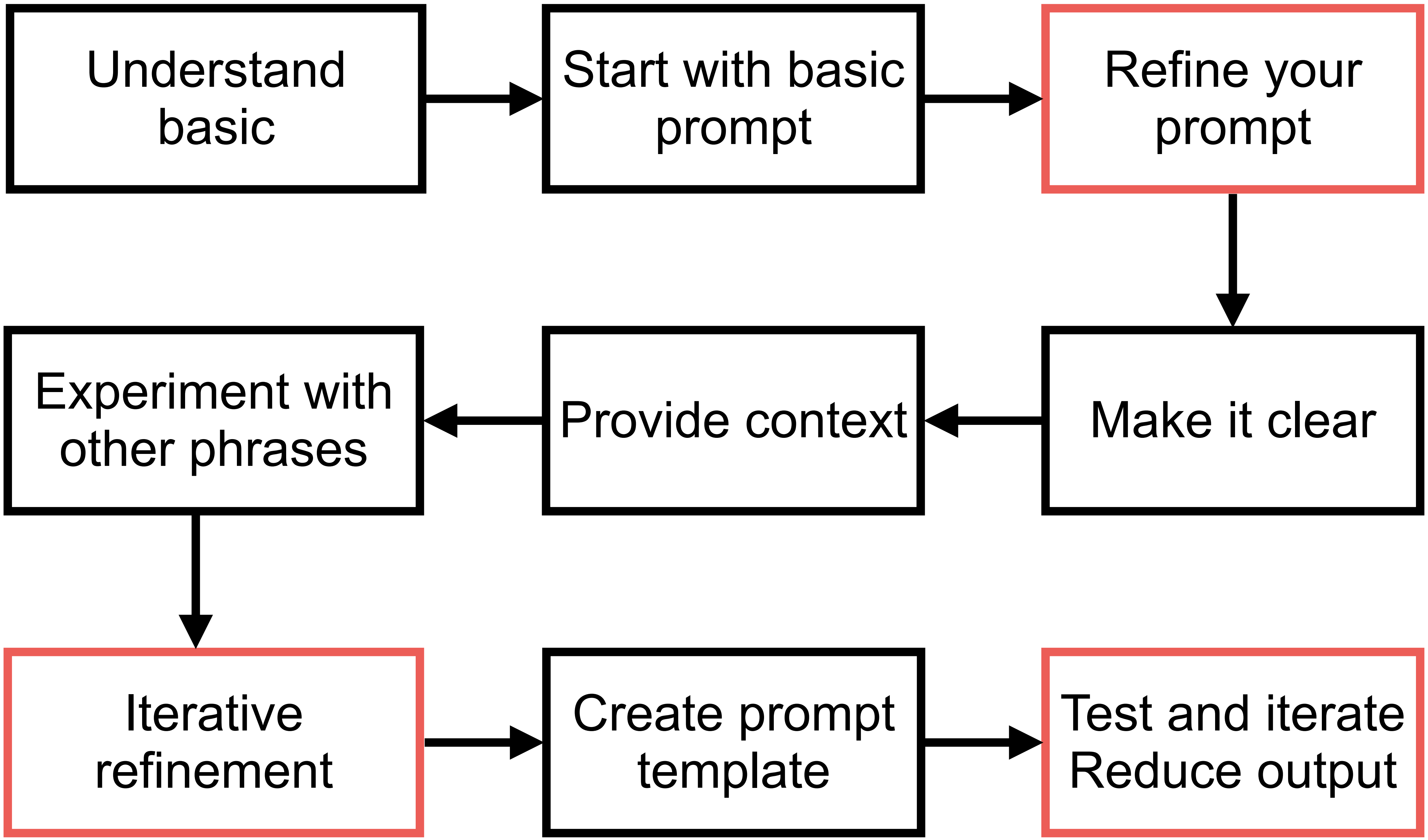


Iterative Process

- Try something
- Analyze where the result does not give what you want
- Clarify instructions, give more time to think
- Refine prompts with a batch of examples

https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/

Basic of Prompt Engineer



Better Prompt

Write clear instructions

Provide reference text

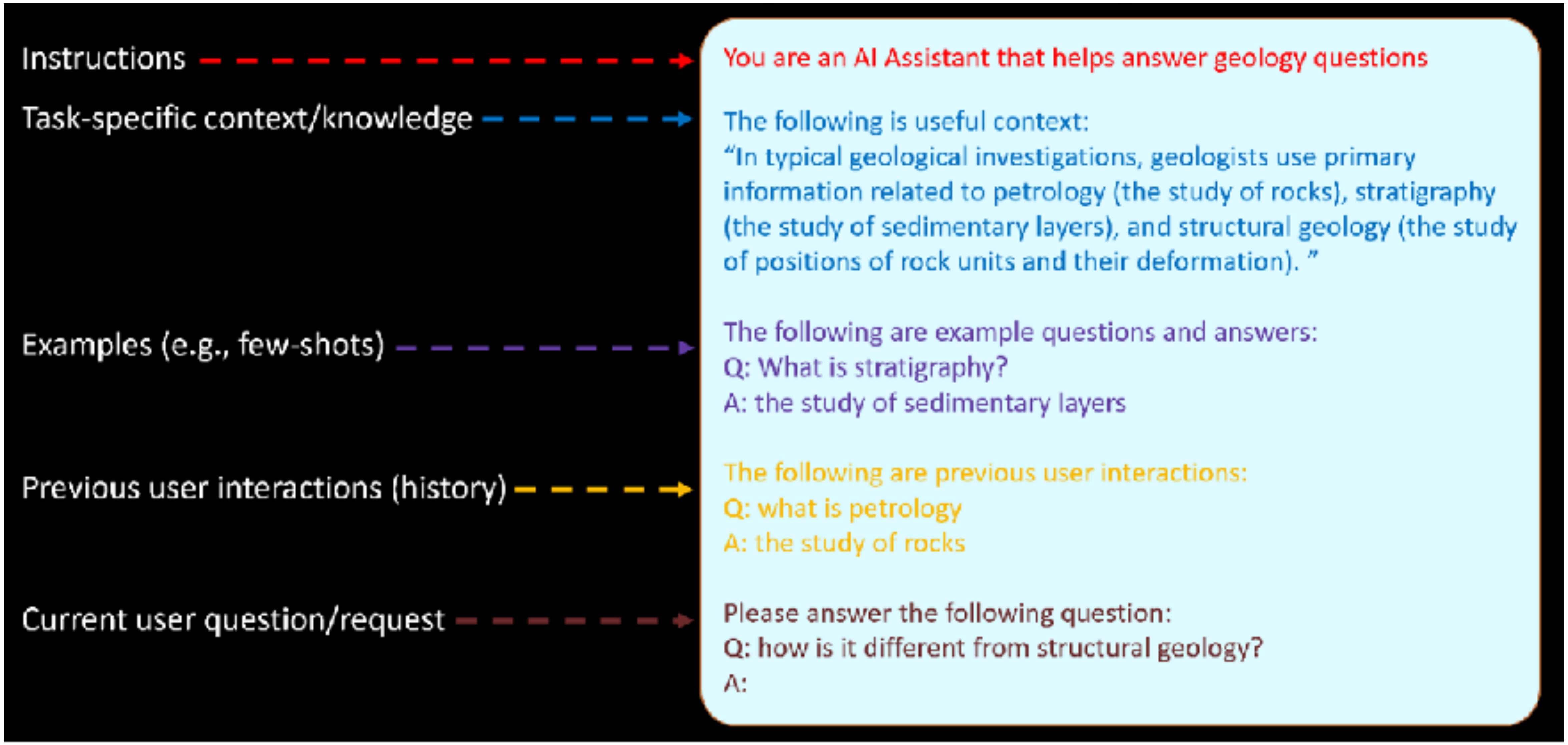
Split complex tasks into simpler subtasks

Give the model time to think

Testing and improve …

https://platform.openai.com/docs/guides/prompt-engineering/six-strategies-for-getting-better-results

Prompt Structure



https://learn.microsoft.com/en-us/ai/playbook/technology-guidance/generative-ai/working-with-llms/prompt-engineering

Structure of Prompt

Context

Instruction

Input data

Output format

https://platform.openai.com/docs/guides/prompt-engineering

Prompting Guide

Prompt Engineering

Prompt Engineering Guide

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning. Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Prompt engineering is not just about designing and developing prompts. It encompasses a wide range of skills and techniques that are useful for interacting and developing with LLMs. It's an important skill to interface, build with, and understand capabilities of LLMs. You can use prompt engineering to improve safety of LLMs and build new capabilities like augmenting LLMs with domain knowledge and external tools.

https://www.promptingguide.ai/

Structure of Prompt !!

APE
Action, Purpose, Expectation

RACE
Role, Action, Context, Expectation

TAG
Task, Action, Goal

COAST
Context, Objective, Action, Scenario, Task

RISE
Role, Input, Step, Expectation

https://twitter.com/pradeepeth/status/1673271866696544257

Prompt Techniques

Zero-shot	Chain-of Thought (CoT)
Few-shot	Meta or structure

https://www.promptingguide.ai/techniques

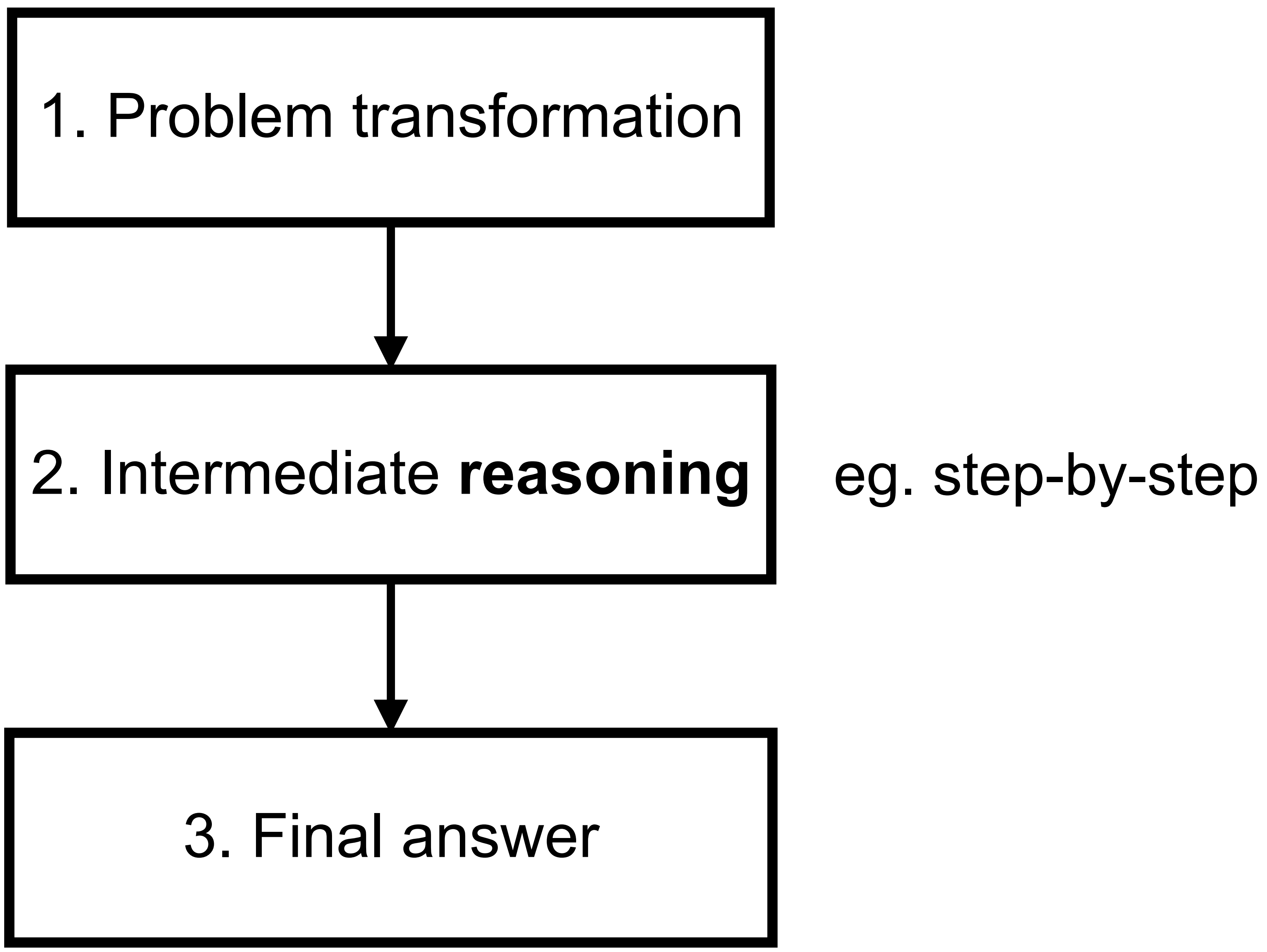
Chain of Thought Prompting (CoT)

Technique used to improve the reasoning ability of LLM

Try to break down a complex problem into smaller, More manageable steps, lead to final answer

Reasoning model !!

Chain of Thought Prompting (CoT)



Chain of Thought Prompting (CoT)

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The answer is 8. X

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) 8 X

(d) Zero-shot-CoT (Ours)

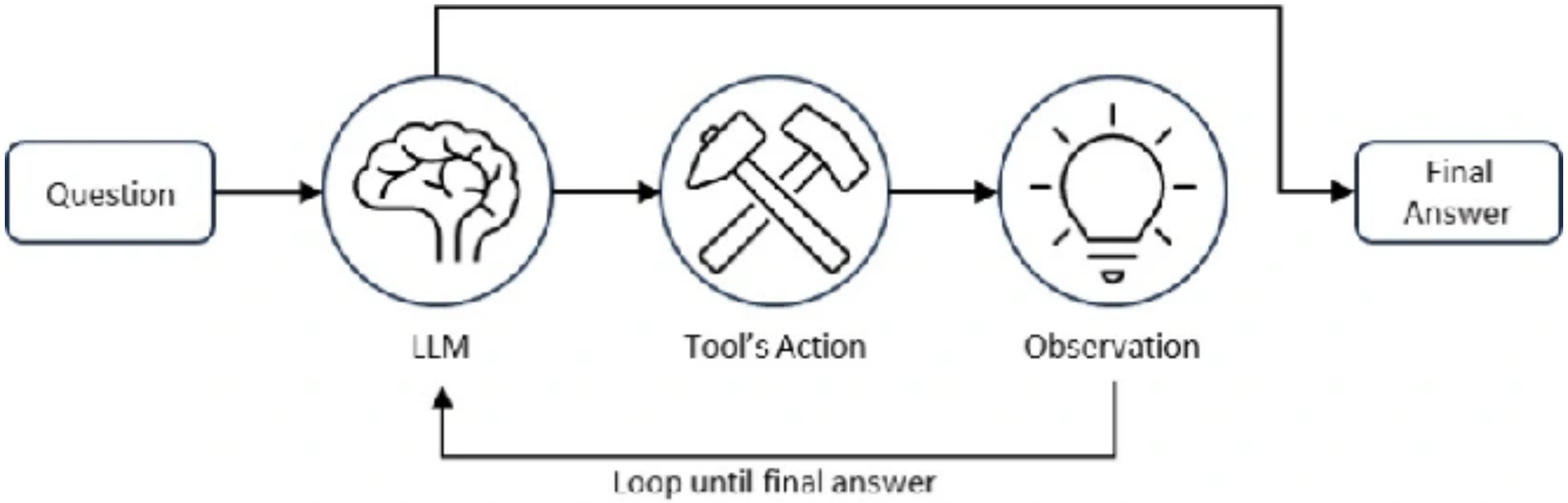
Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

https://www.promptingguide.ai/techniques/cot

ReAct Prompt

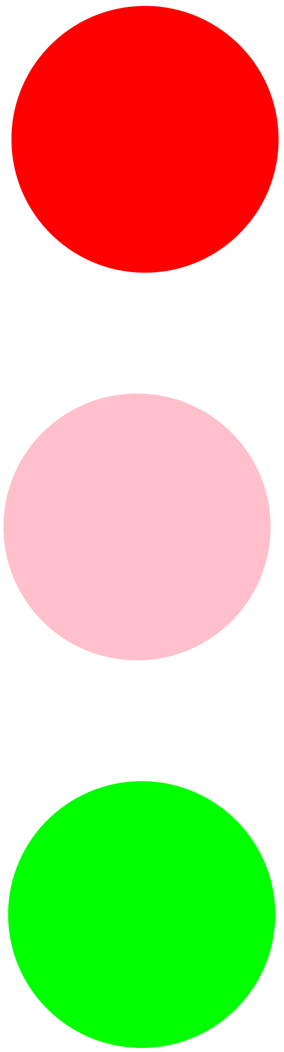
LLM reasoning and additional tools (expert)
Improve better answer



https://www.promptingguide.ai/techniques/react

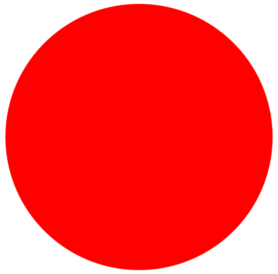
Workshop

RGB ?



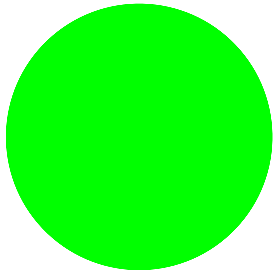
https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb

RGB ?



[255, 0, 0]

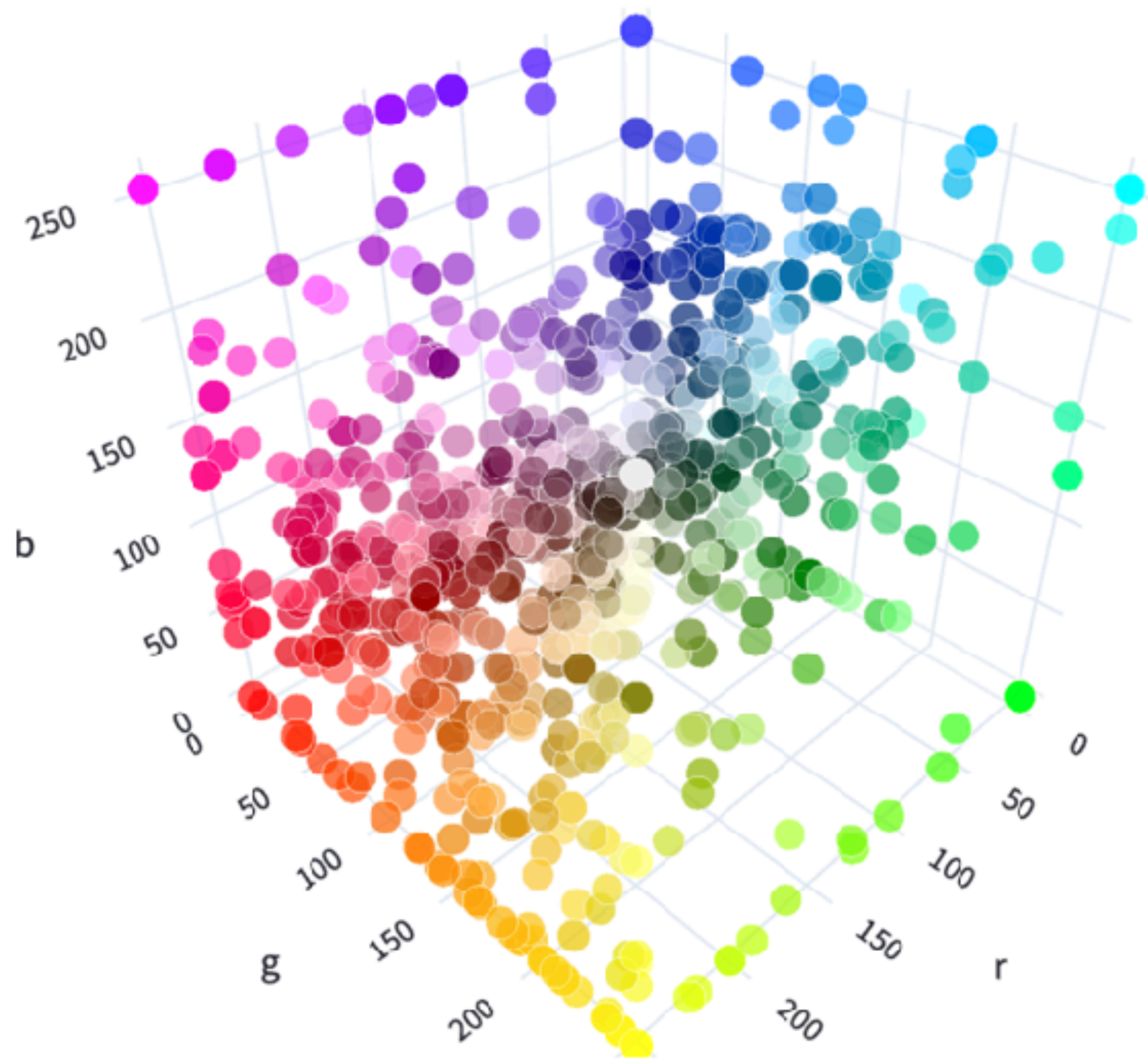
[255, 192, 203]



[0, 255, 0]

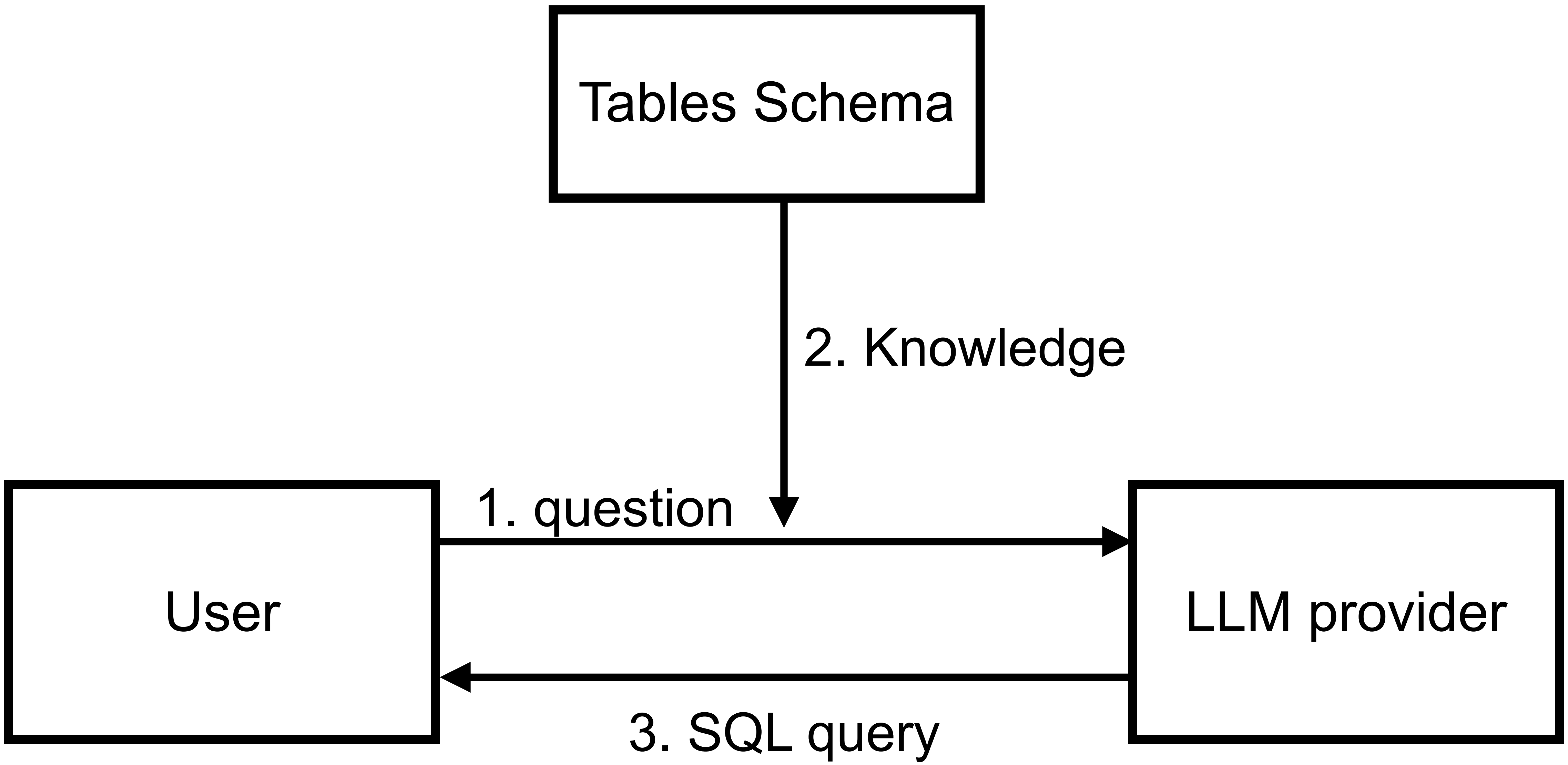
https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb

Visualize RGB



https://huggingface.co/spaces/jphwang/colorful_vectors

Text-to-SQL



https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-sql

Software Development Life Cycle

SDLC

Requirement | Design | Develop | Testing | Deploy

Planning

- Written planning process
- Arch diagrams

Testing

- Automated testing
- TDD
- Testing environments
- Testing in prod
- Performance testing
- Load testing
- Generate test data

Shipping

- FF & experimentation
- Logging
- Monitoring & alerting
- Staged rollouts

Development

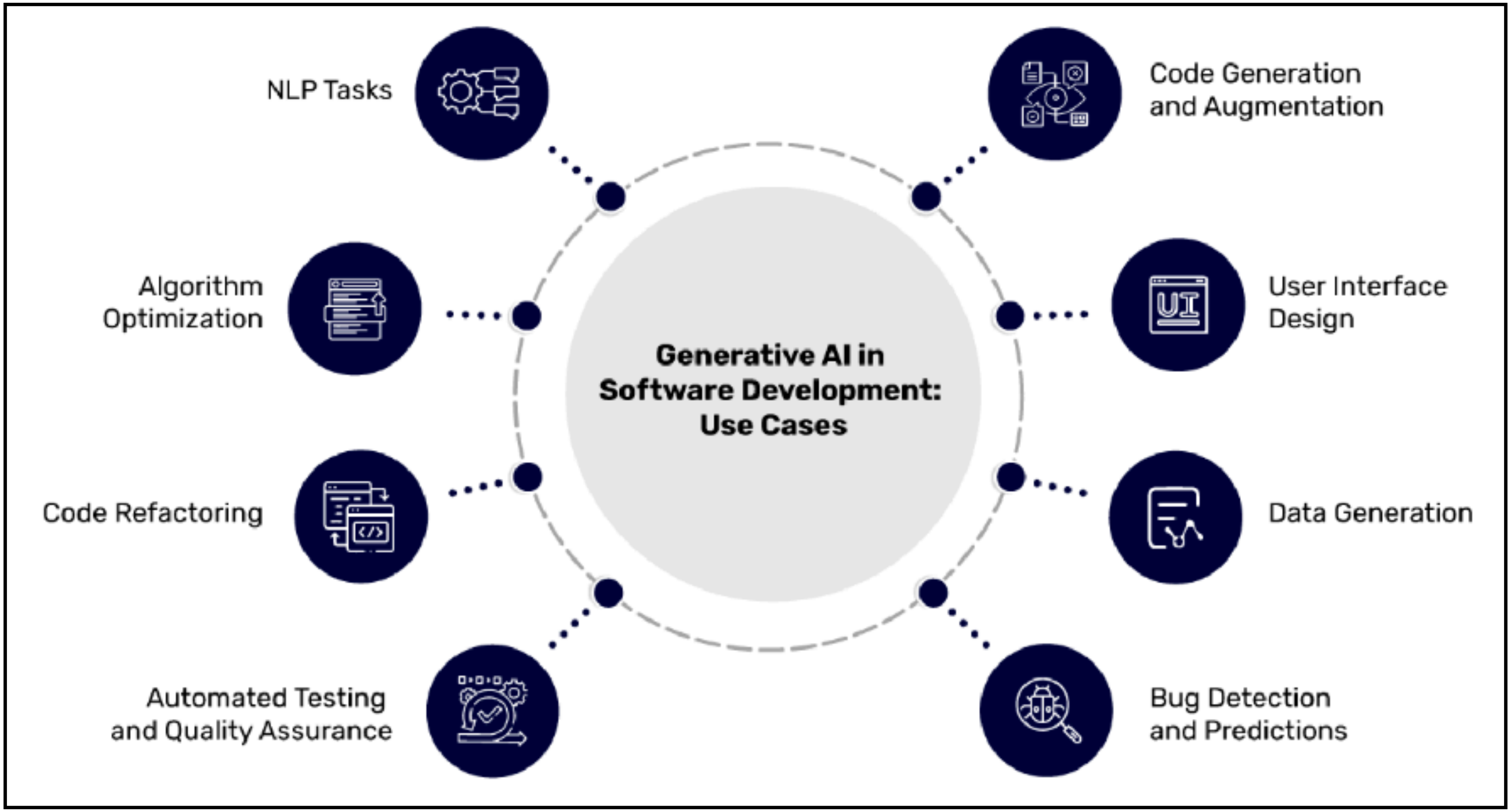
- Automated dev env
- CI/CD
- Prototyping
- Code review
- Code generation
- Templates
- Cross-platform dev
- Preview env
- Post-commit code review
- Linting
- Static code analysis
- Project mgmt

Maintenance

- Debug production
- Documentation
- Runbook examples
- Mitigation runbook

pragmaticengineer.com

SDLC


NLP Tasks
Code Generation and Augmentation
Algorithm Optimization
User Interface Design
Generative AI in Software Development: Use Cases
Code Refactoring
Data Generation
Automated Testing and Quality Assurance
Bug Detection and Predictions

Impacts with productivity ?

Automated simple tasks

Improve quality and reliability

Improve communication

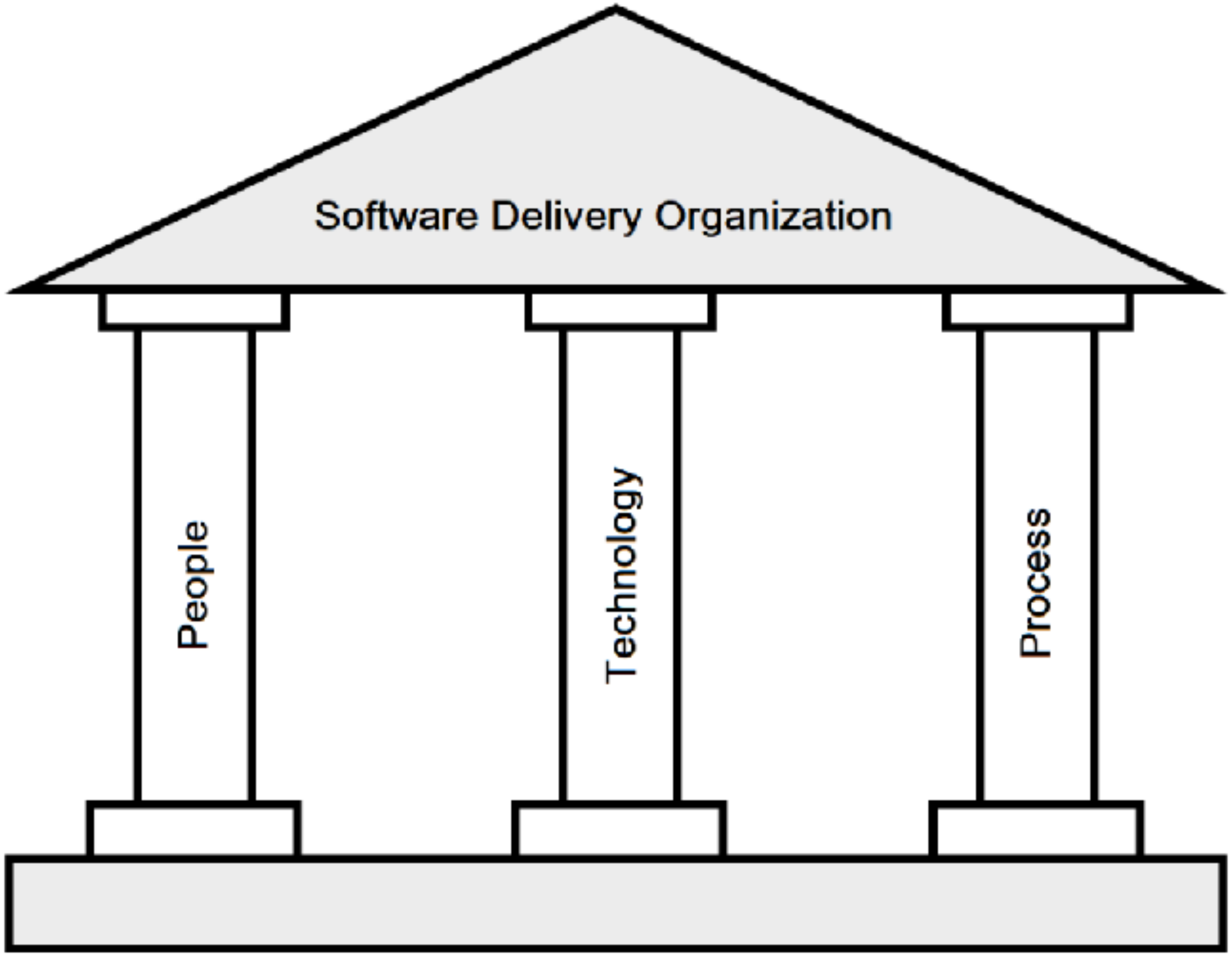
Faster prototype

Generative AI isn't just a tool

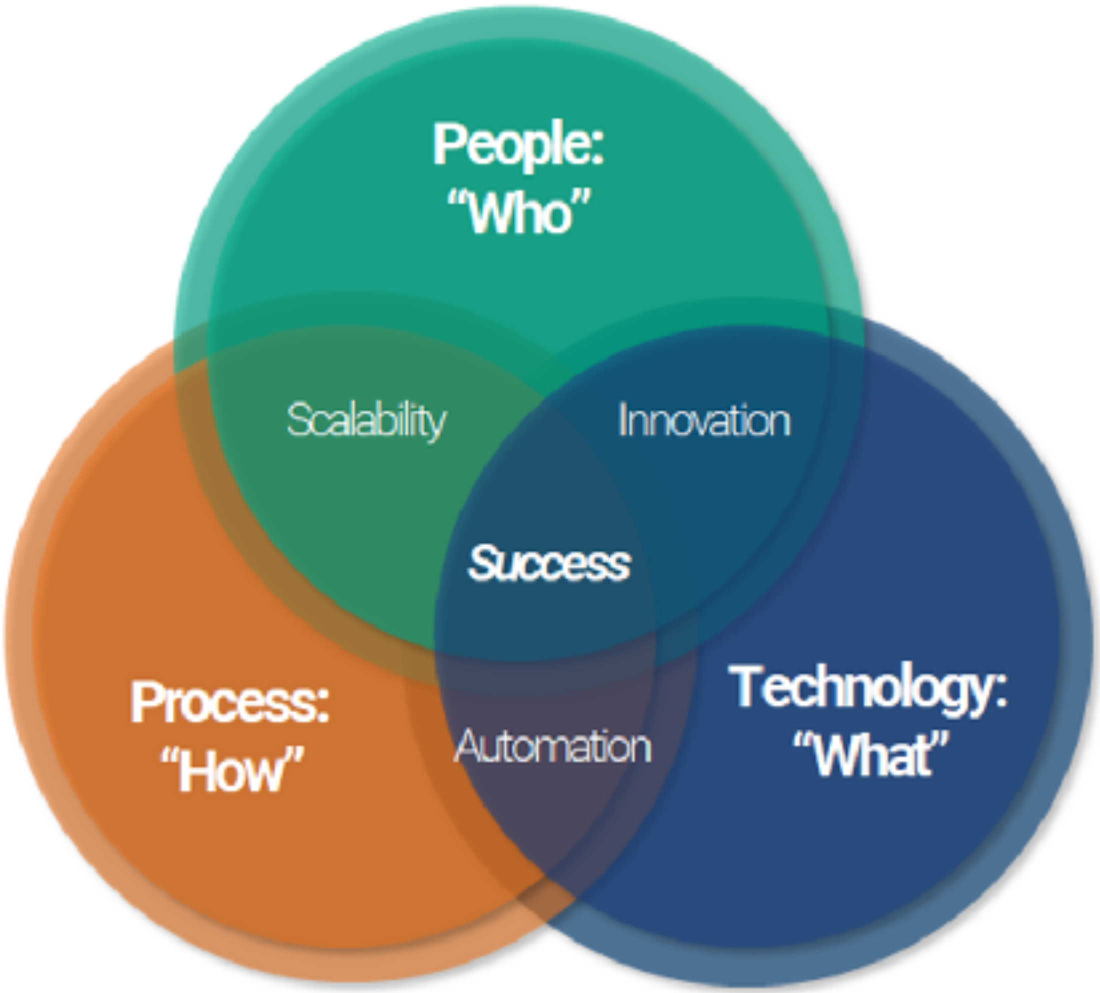
it's your **team member**

Trust, but verify output !!

3 Pillar of Software Development



3 Pillar of Software Development



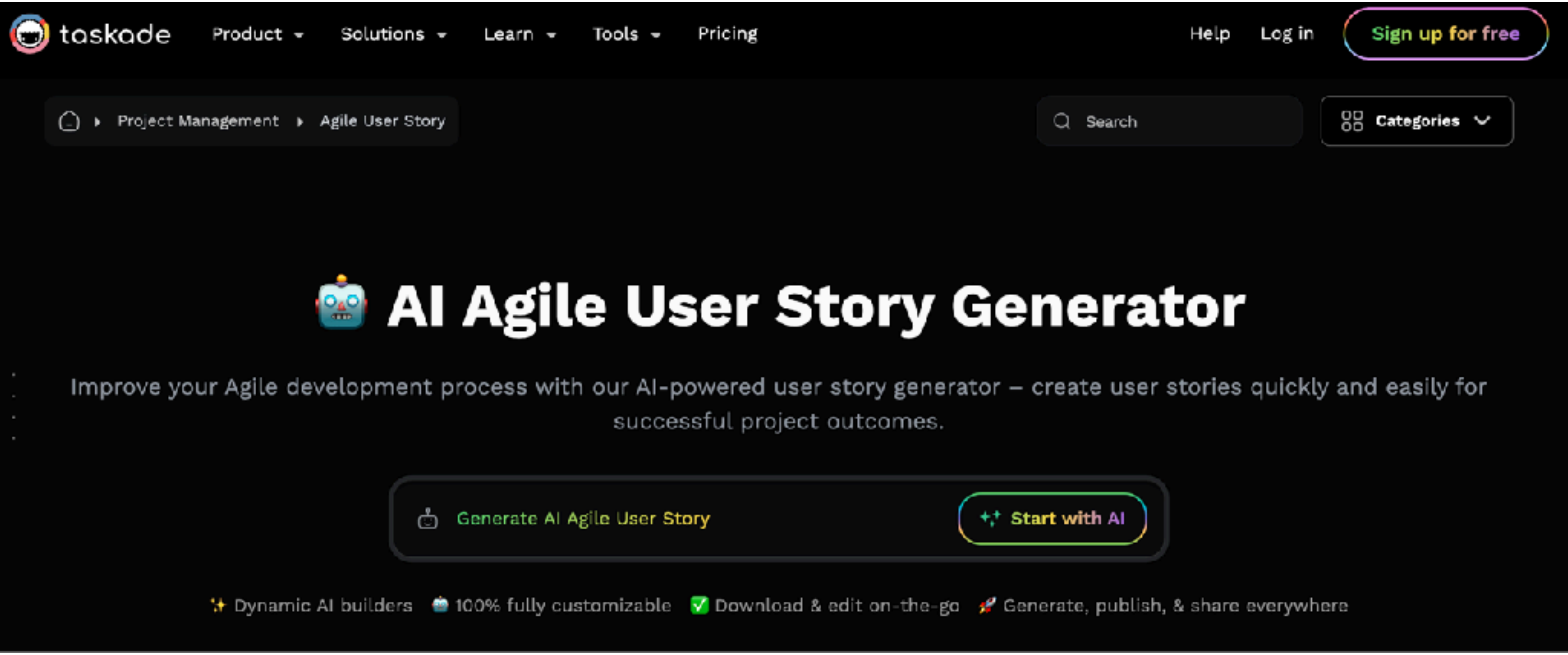
Requirement and Analysis

Requirement and Analysis

Requirement	Design	Develop	Testing	Deploy

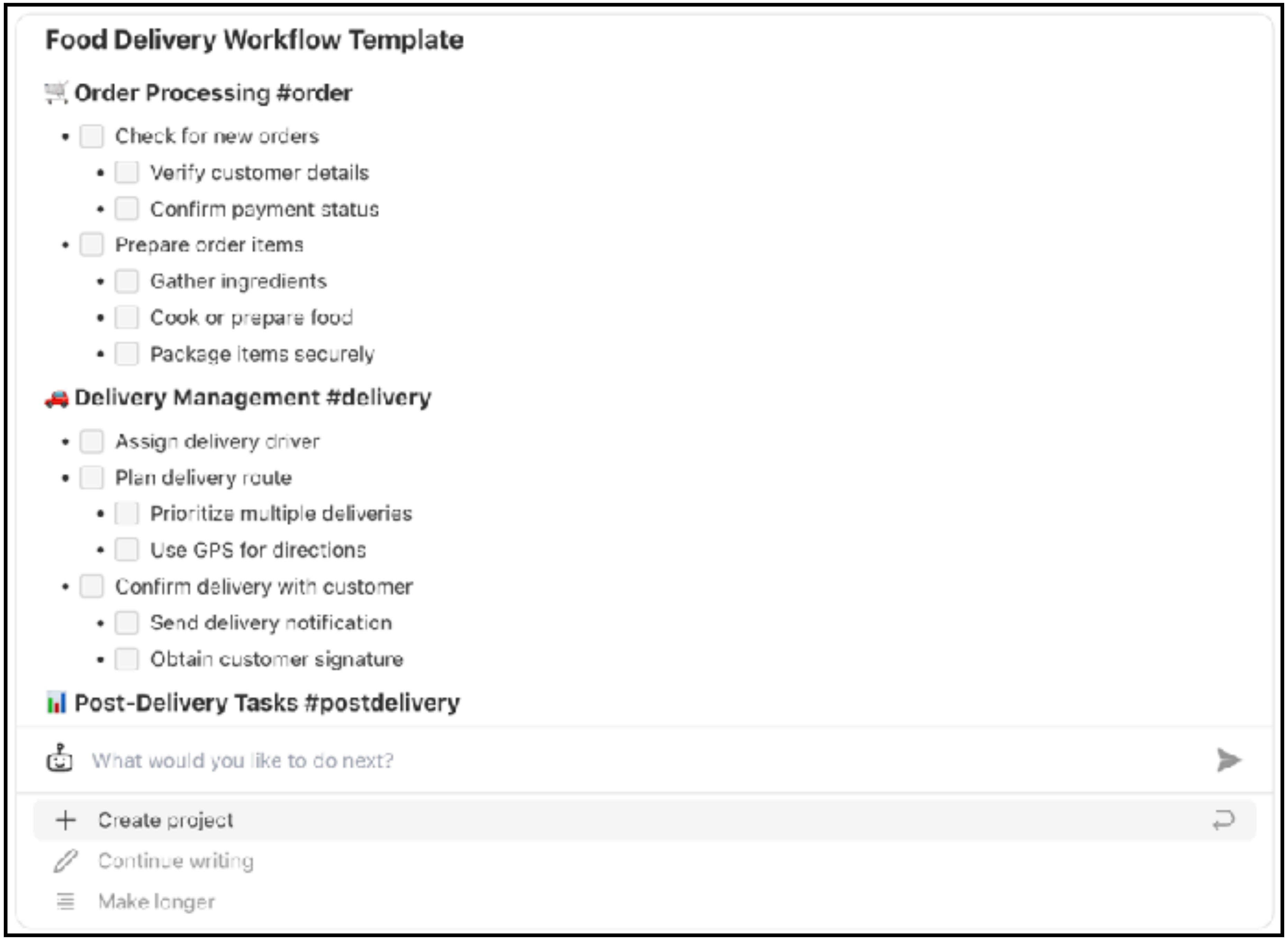
Requirements writing and analysis
User story generation

Agents AI for Automated tasks



https://www.taskade.com/generate/project-management/agile-user-story

Example with food delivery



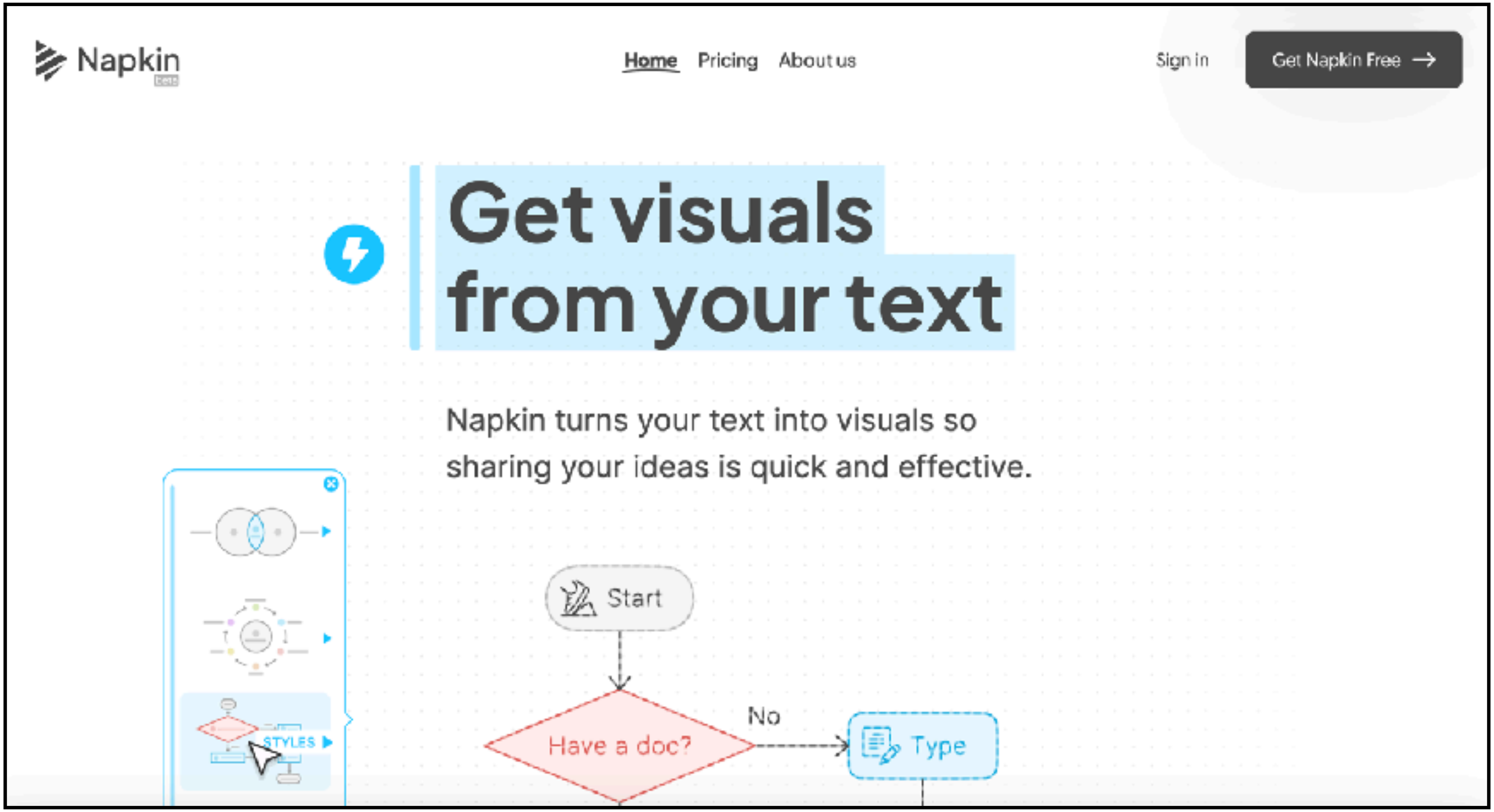
https://www.taskade.com/generate/project-management/agile-user-story

User Story Generator

Idea → Features → Roles → User stories

https://userstorygenerator.ai/

Napkin



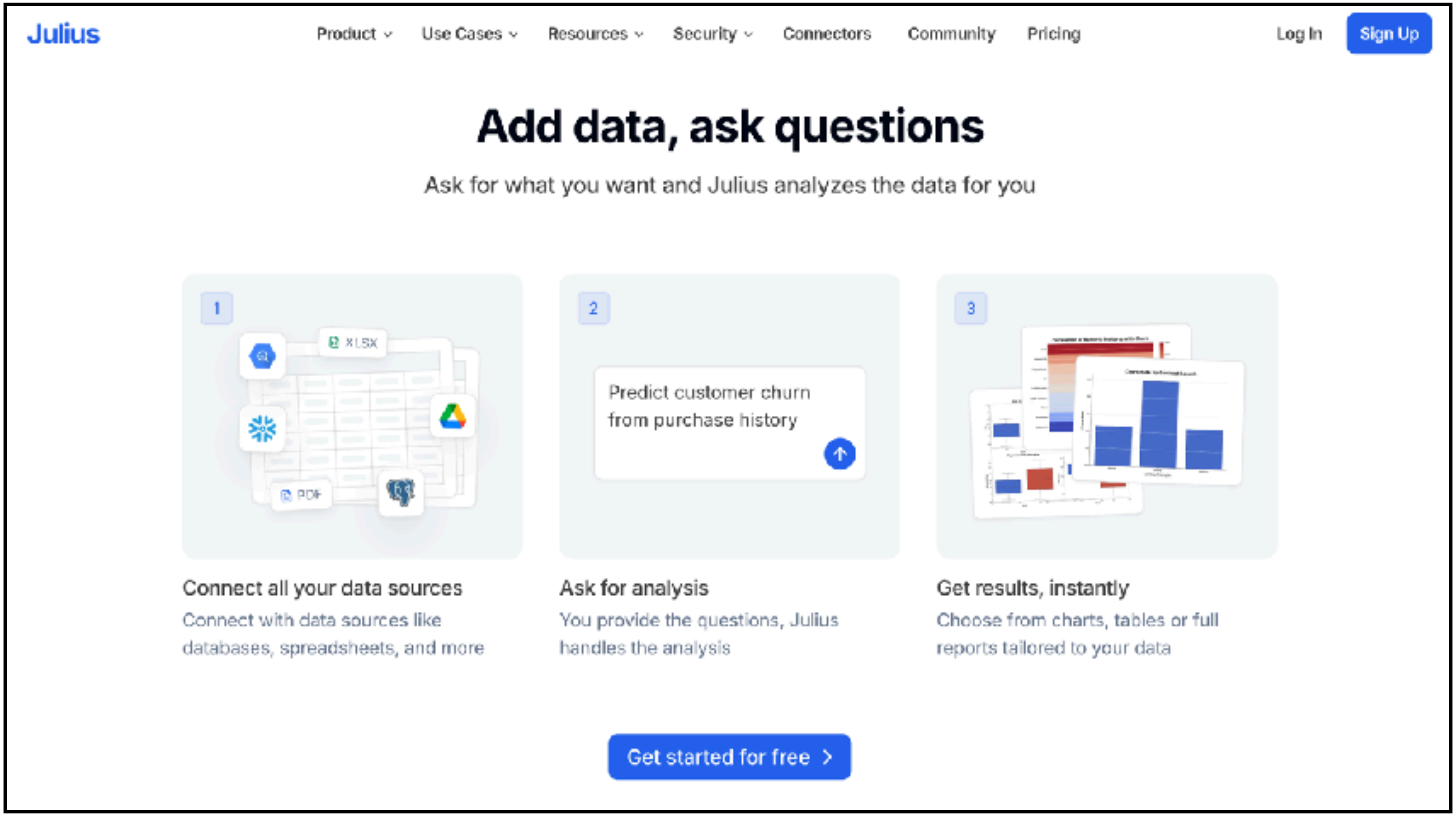
https://www.napkin.ai/

Requirement analysis

Clarify of User requirement ?

https://github.com/up1/workshop-ai-with-technical-team/wiki/Requirement-analysis

Data Analysis



https://julius.ai/

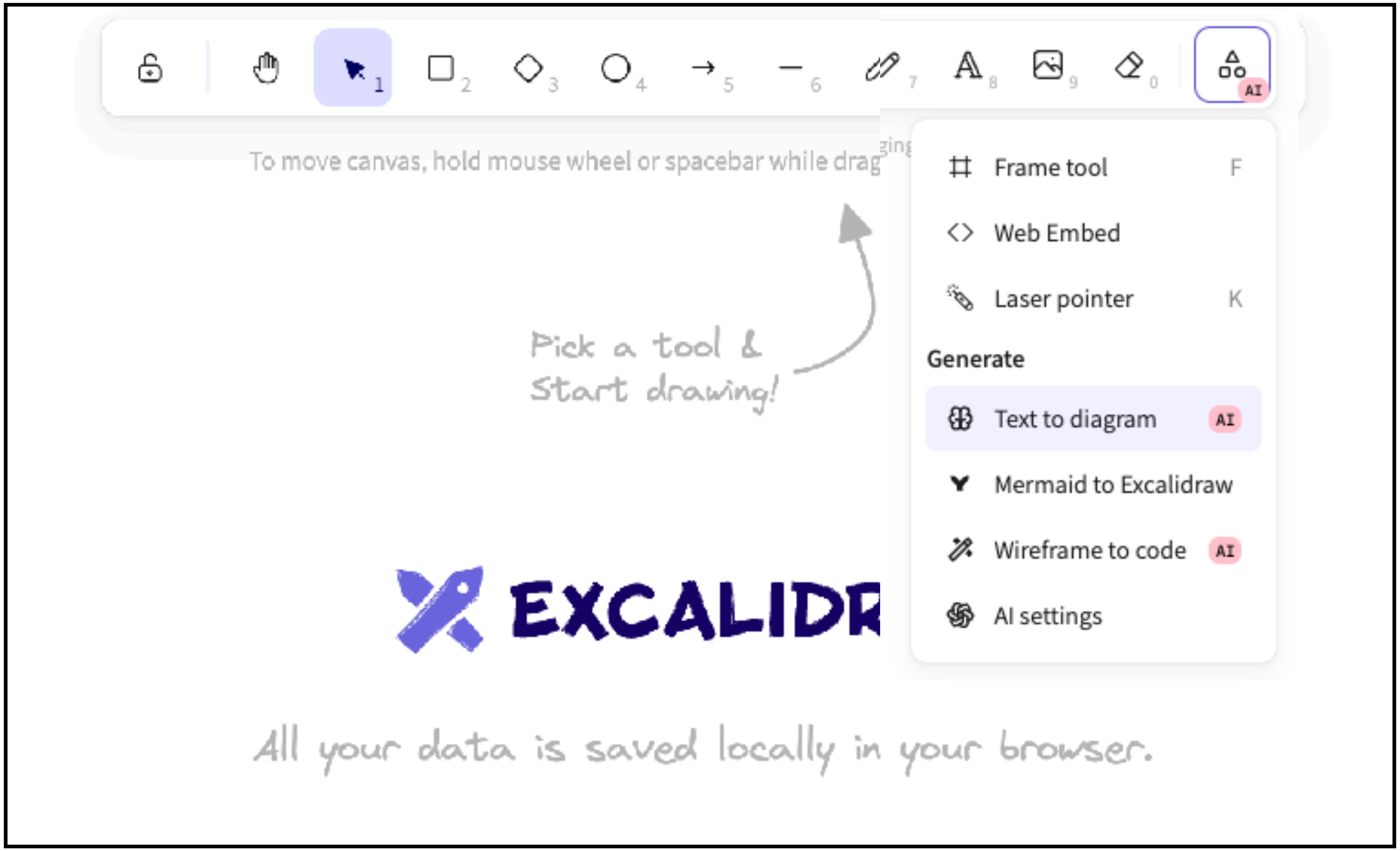
Design Process

Design

Requirement	Design	Develop	Testing	Deploy

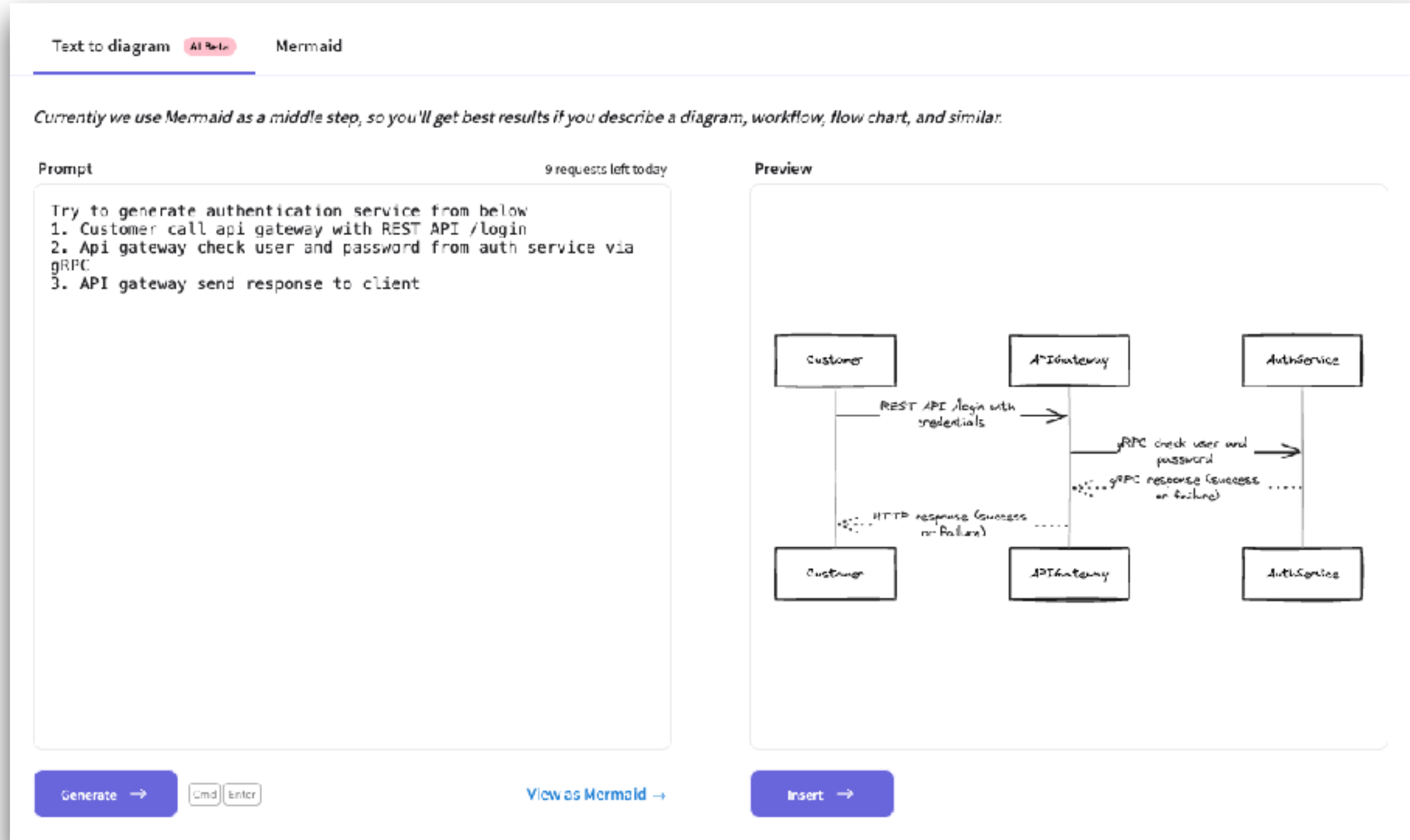
Architecture writing assistance
Sequence flow diagram generation
Data modeling
UX/UI design assistance

Excalidraw with AI

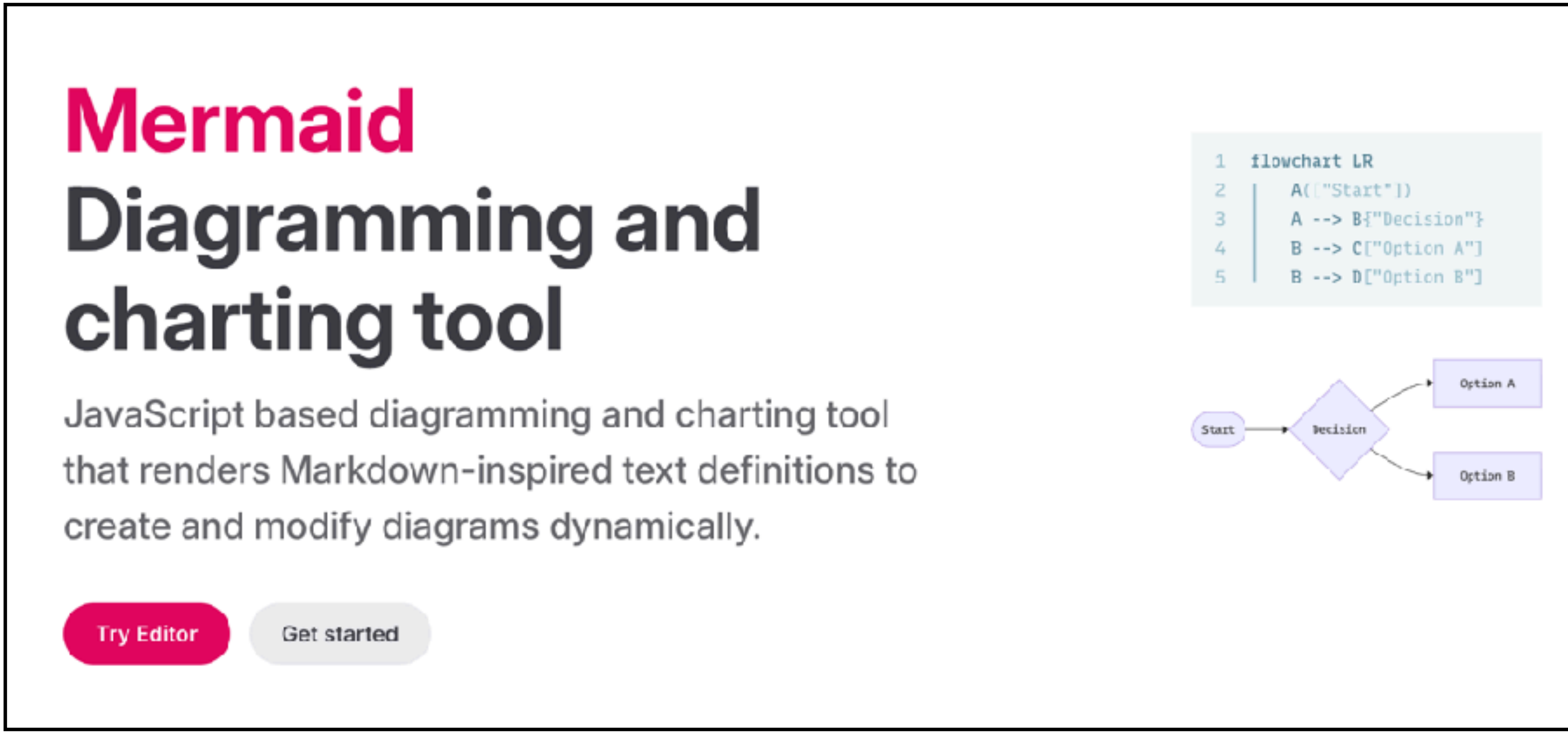


https://excalidraw.com/

Demo

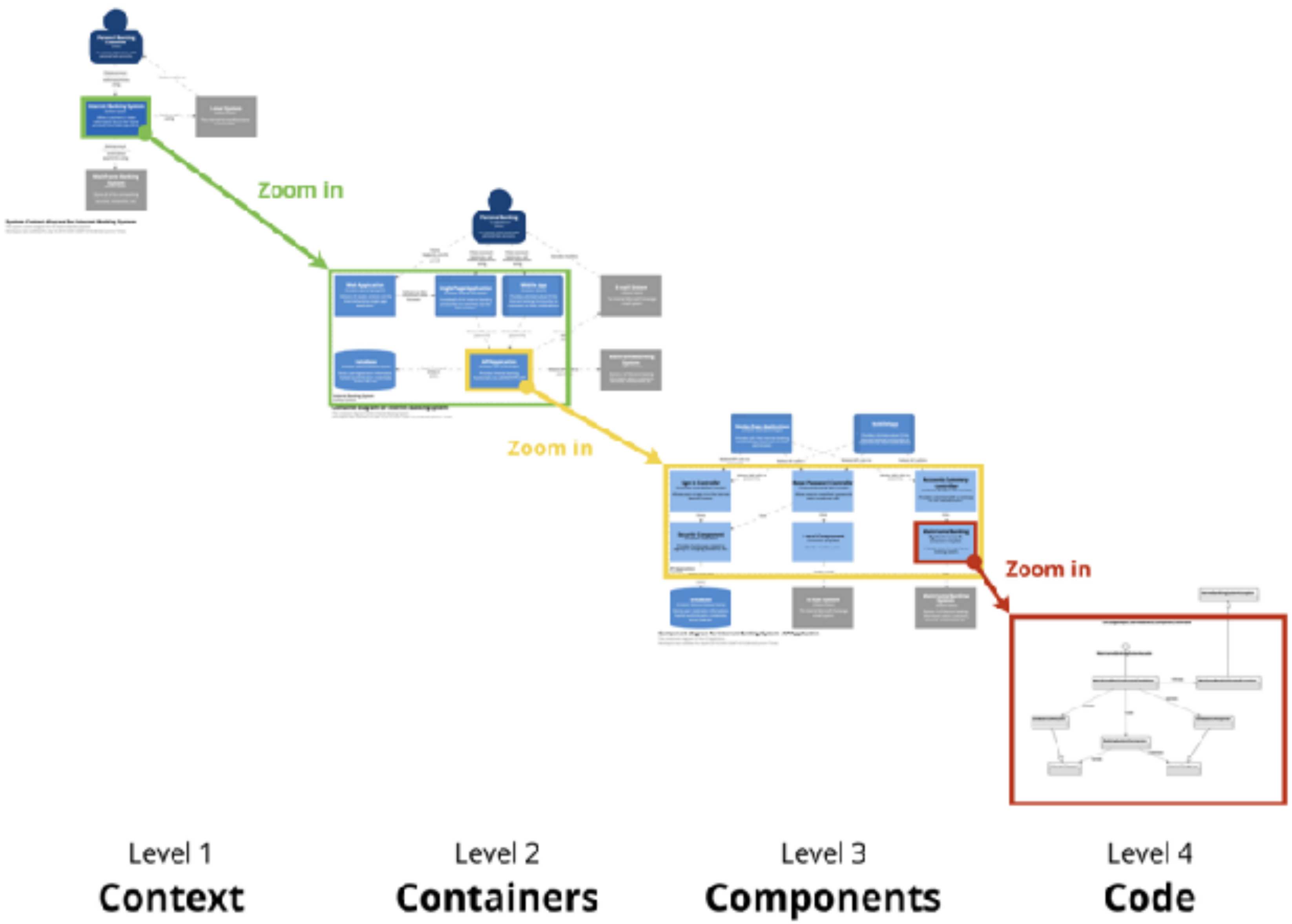

Text to diagram
AI Beta
Mermaid
Currently we use Mermaid as a middle step, so you'll get best results if you describe a diagram, workflow, flow chart, and similar.
Prompt
9 requests left today
Try to generate authentication service from below
1. Customer call api gateway with REST API /login
2. Api gateway check user and password from auth service via gRPC
3. API gateway send response to client
Preview
Customer
APIGateway
AuthService
REST API /login with credentials
gRPC check user and password
gRPC response (success or failure)
HTTP response (success or failure)
Generate
Cmd
Enter
View as Mermaid →
Insert

Mermaid Diagram



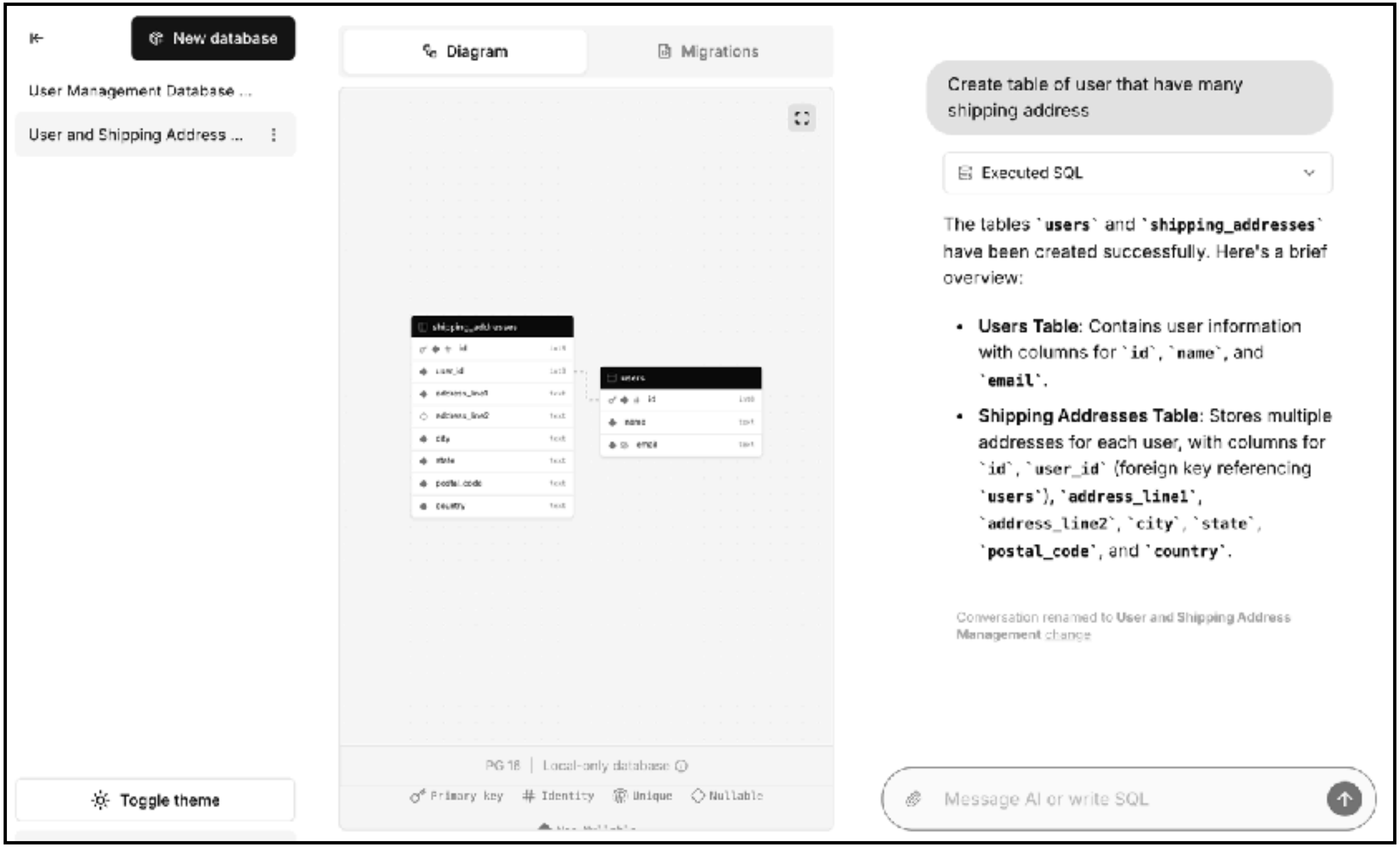
https://mermaid.js.org/

Software Architecture



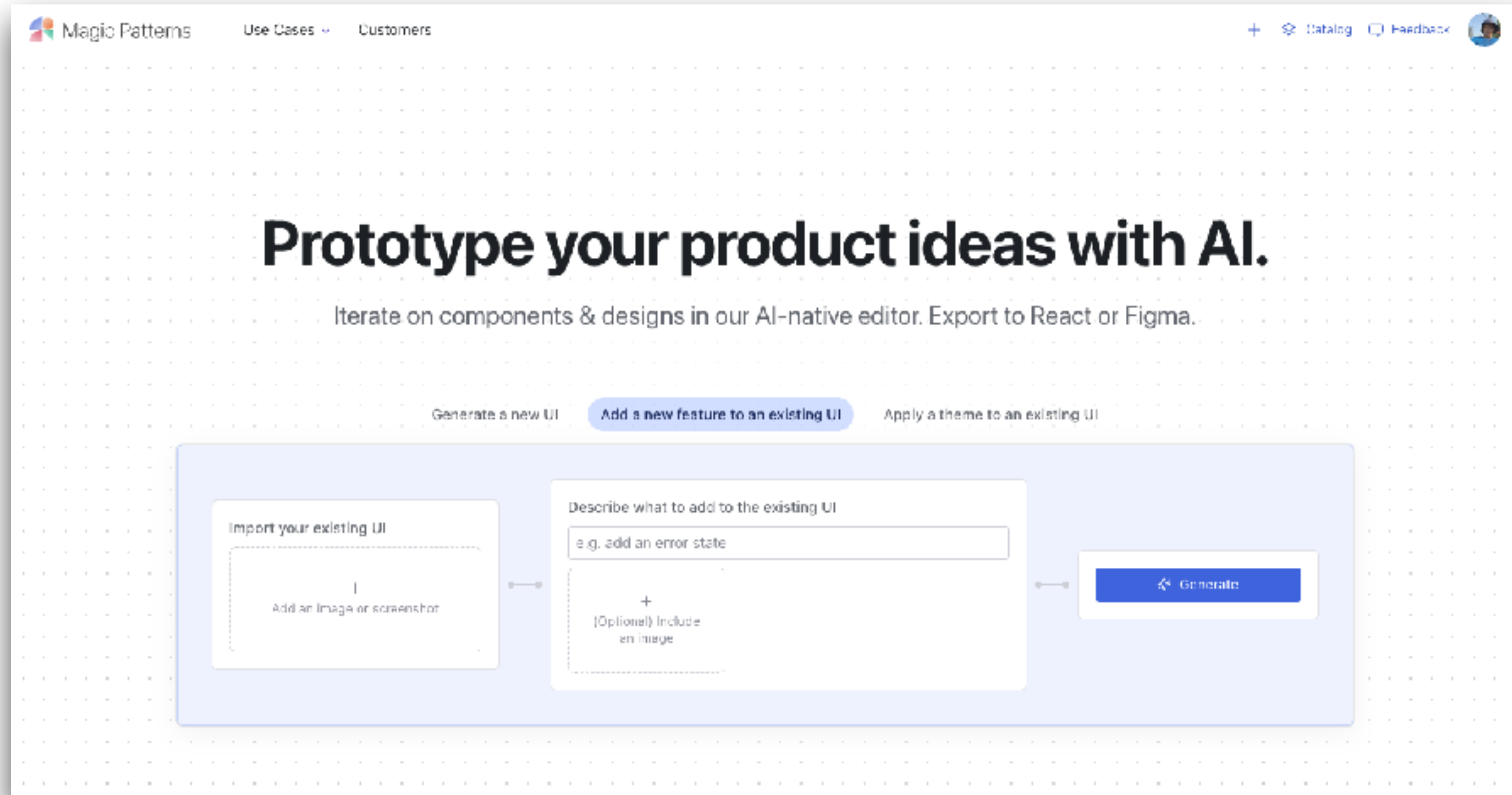
https://c4model.com/

Database



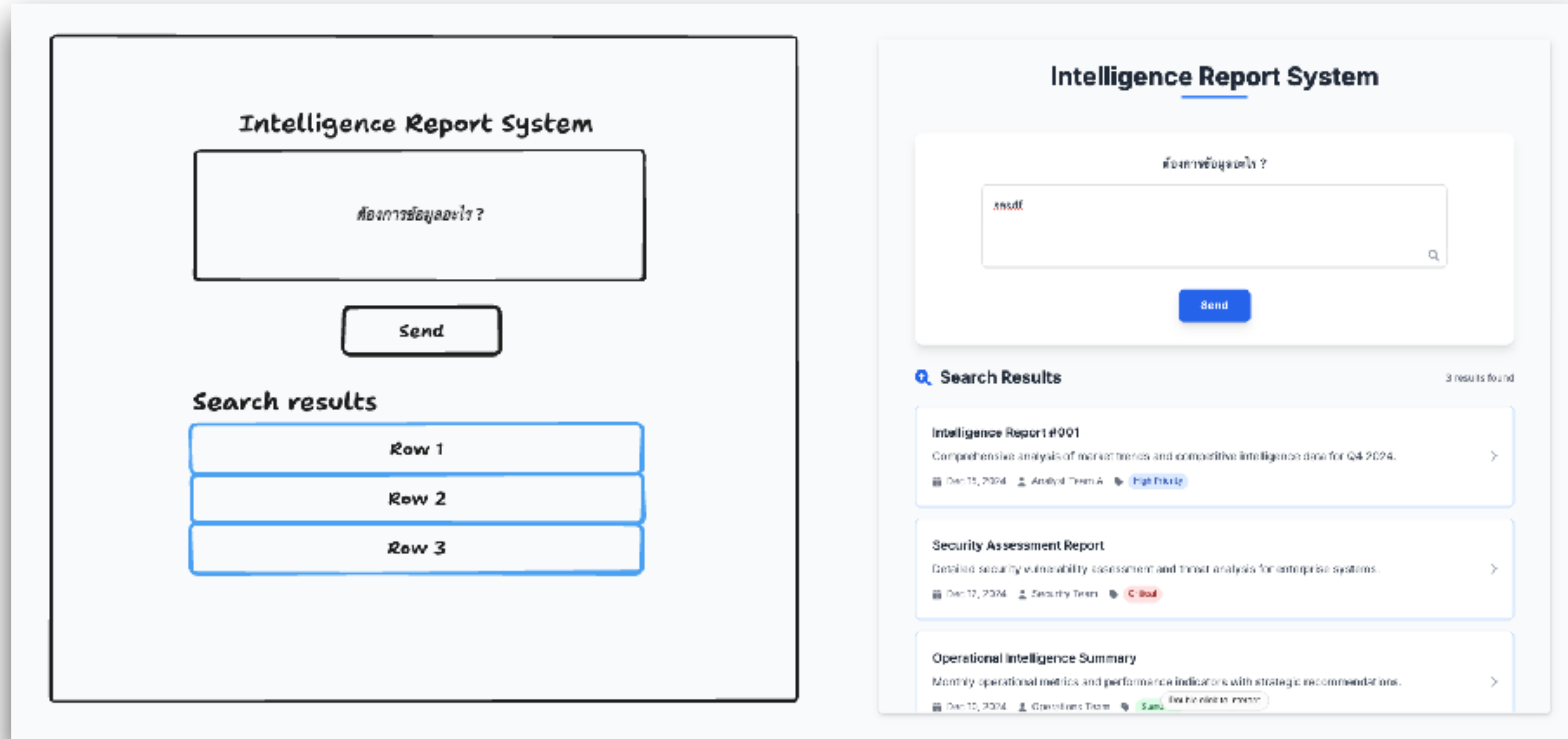
https://database.build/

Magic Pattern



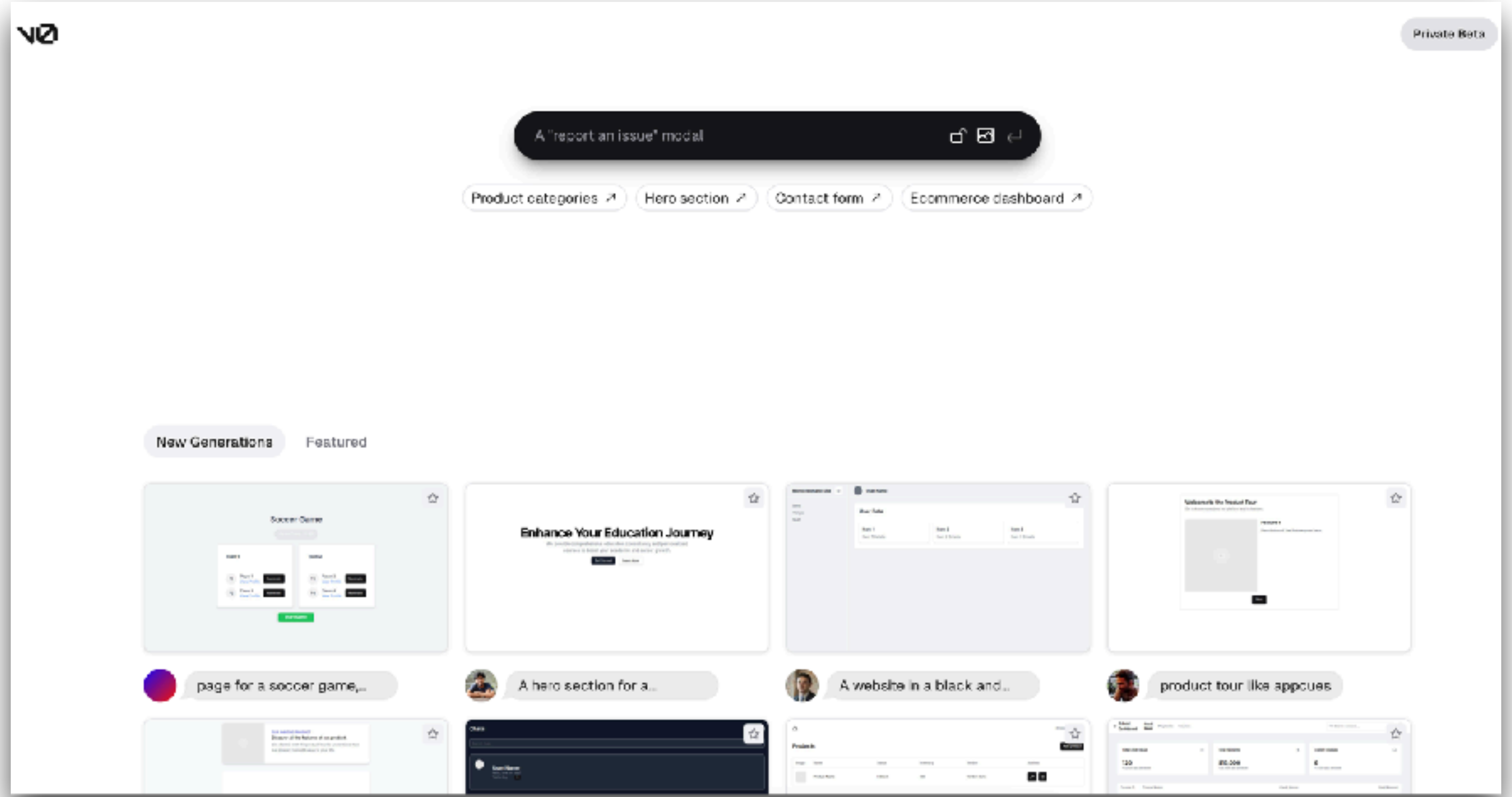
https://www.magicpatterns.com/

Make Real



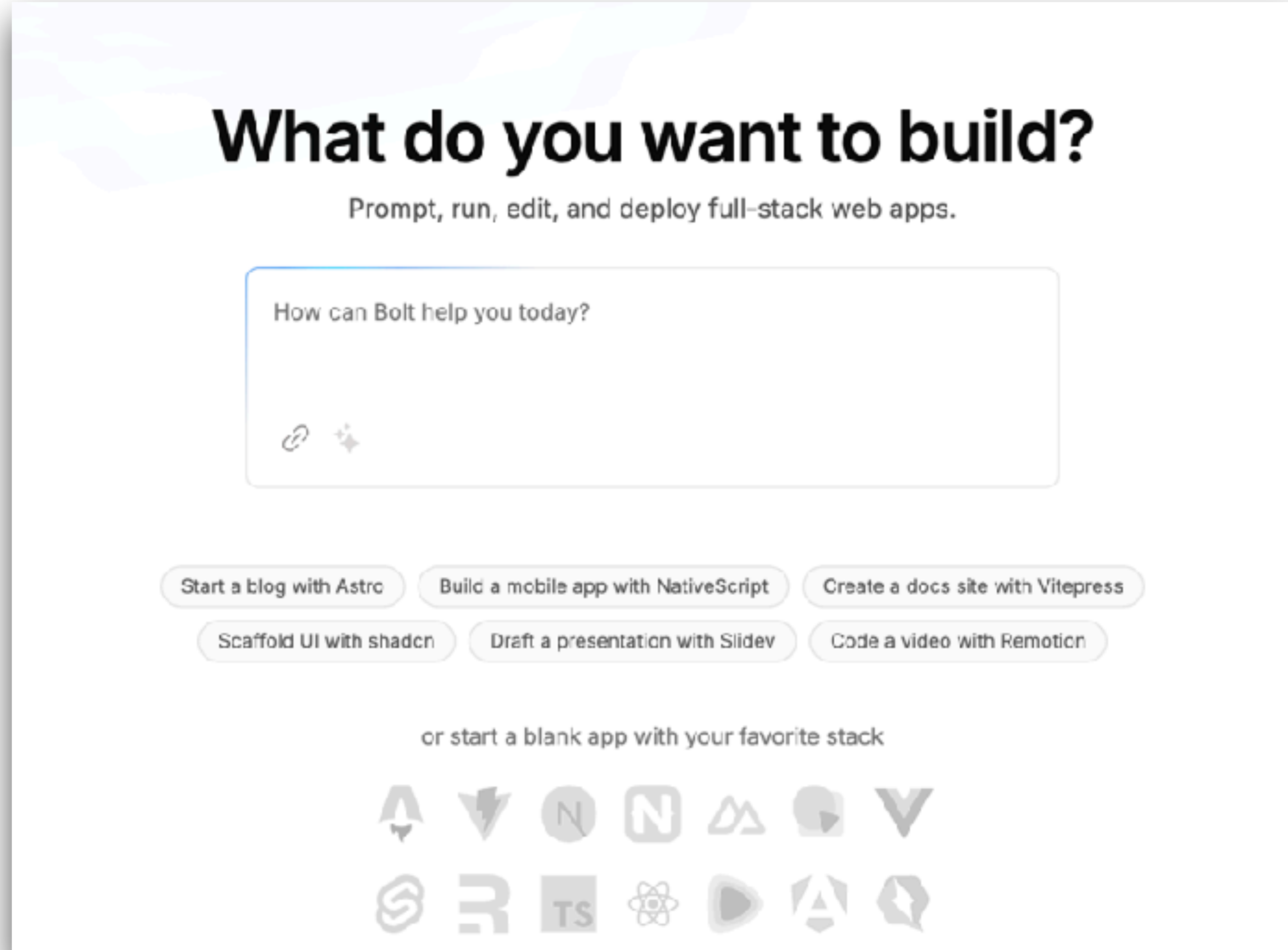
https://github.com/tldraw/make-real

V0.dev



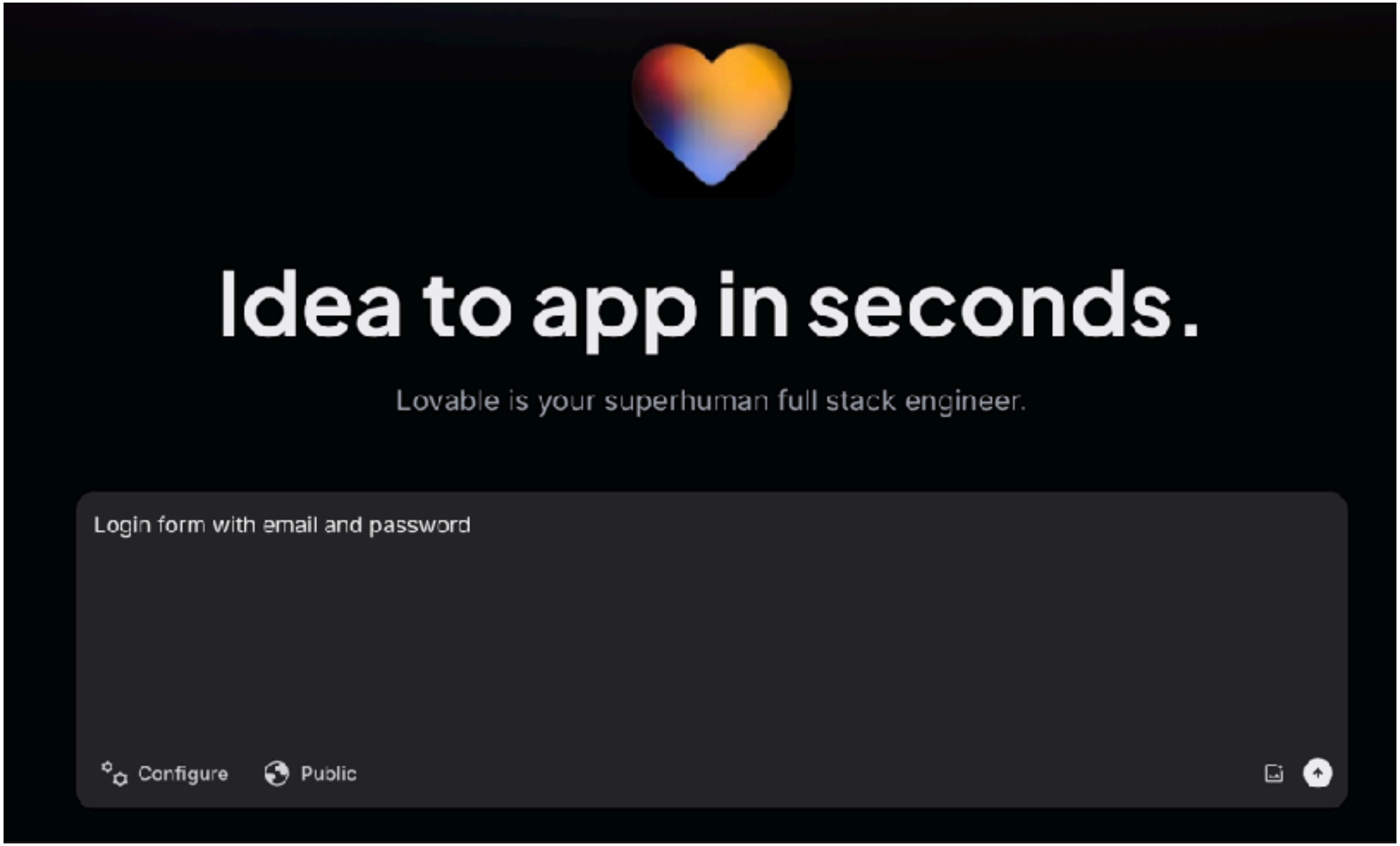
https://v0.dev/

Bolt.new



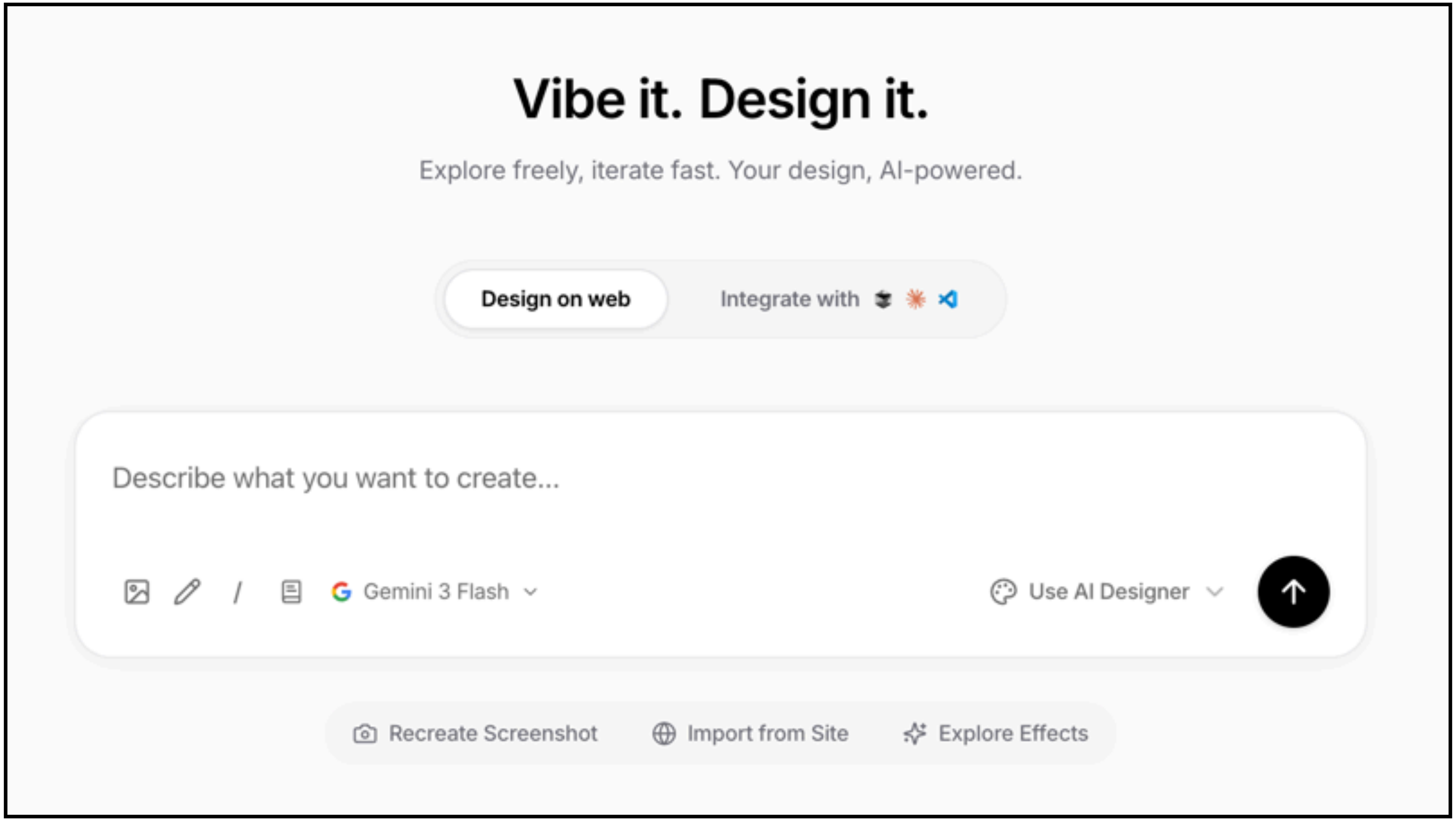
https://bolt.new/

Lovable



https://lovable.dev/

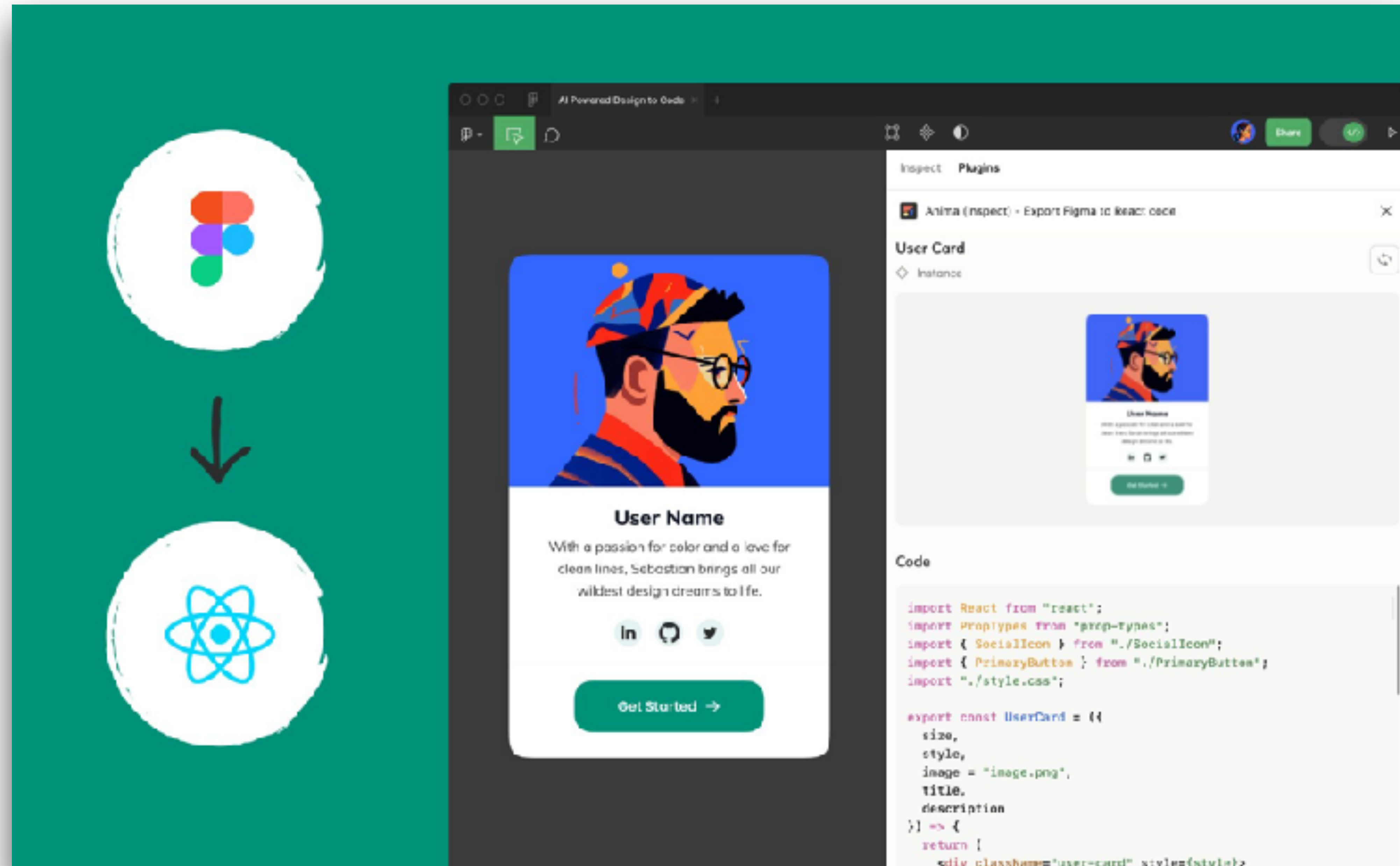
SuperDesign



https://app.superdesign.dev/

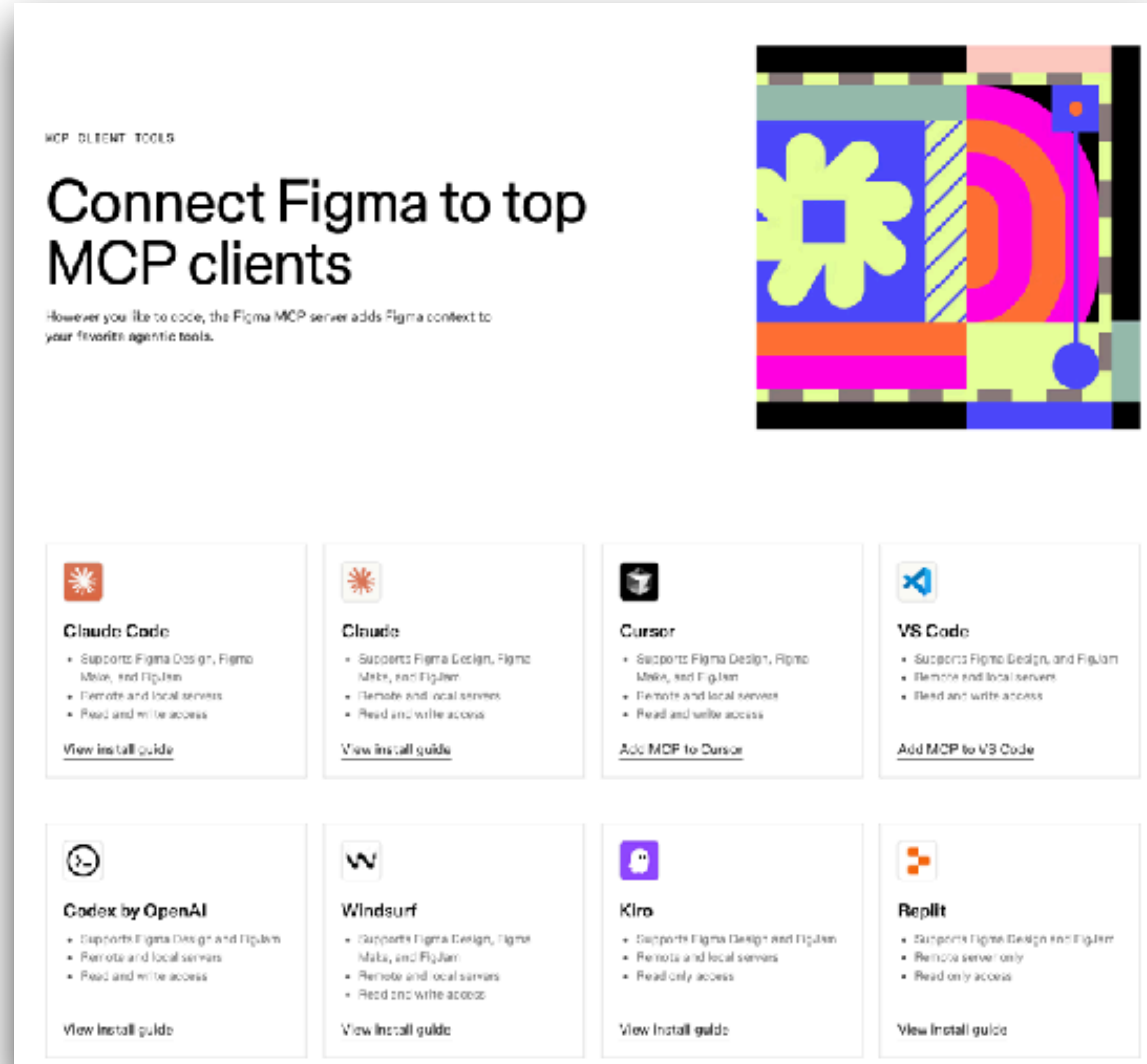
Design to Code

Figma



https://www.figma.com/mcp-catalog/

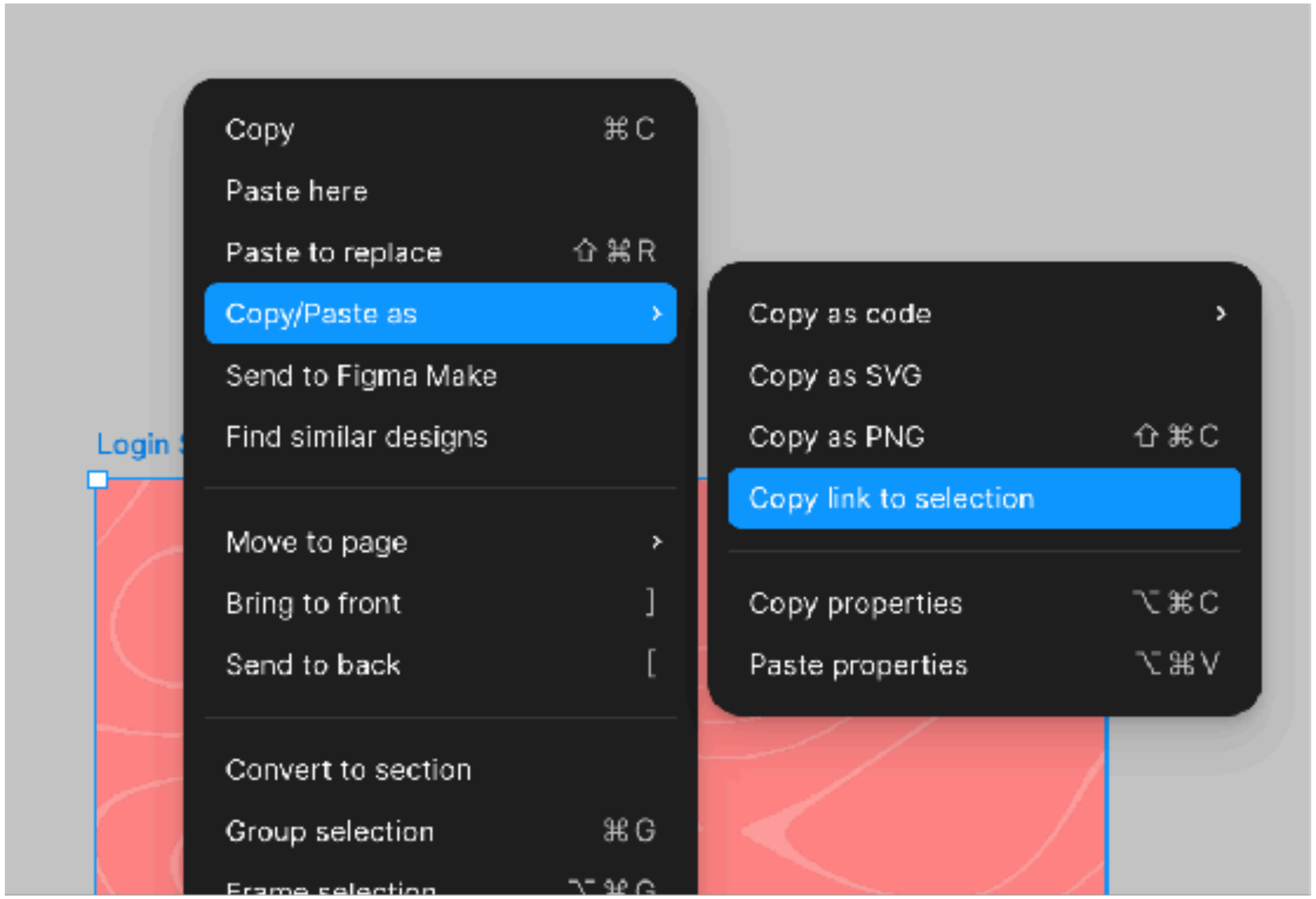
Figma MCP



https://www.figma.com/mcp-catalog/

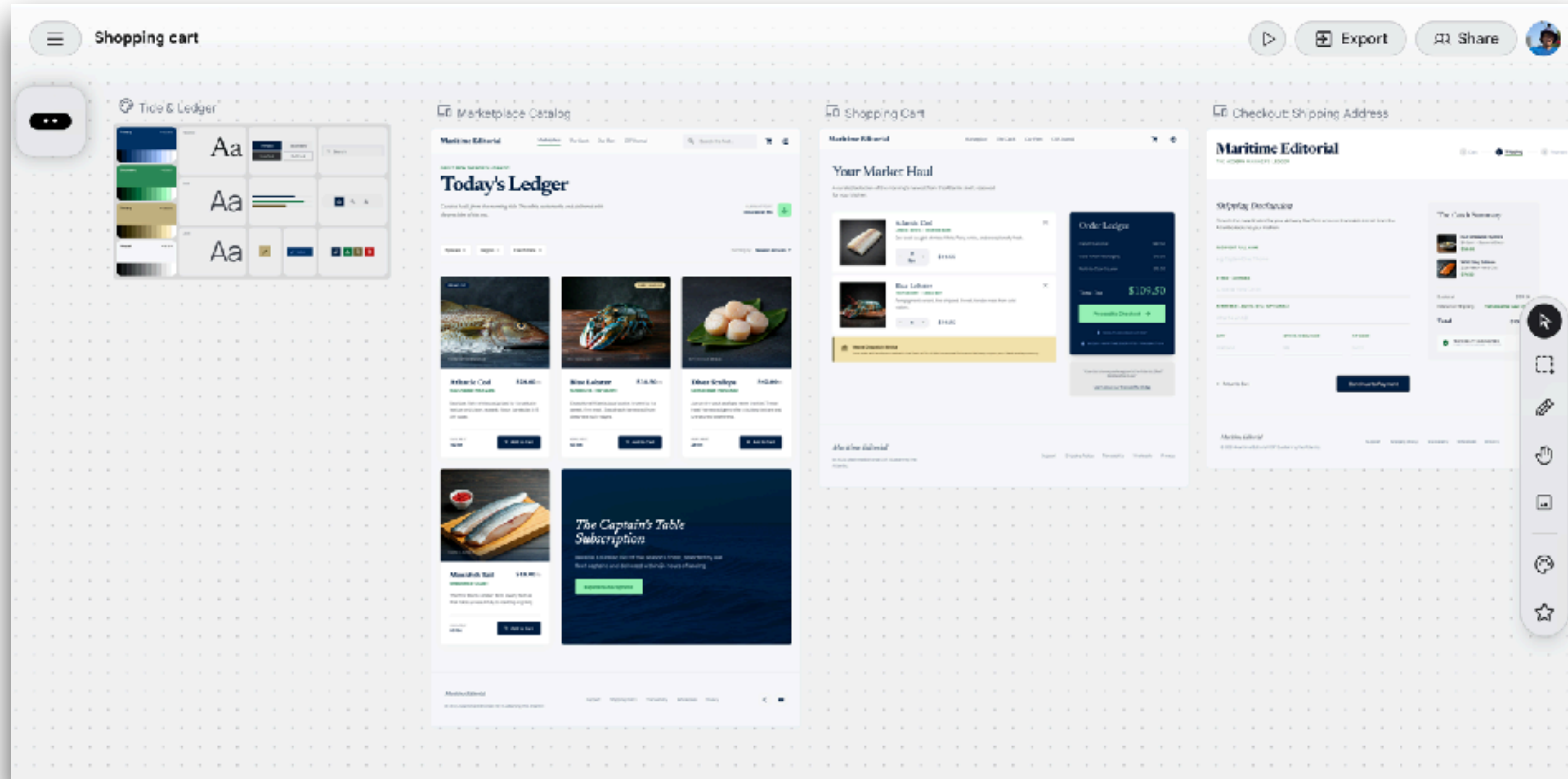
Design to Code in Figma

Select frame -> Copy/Paste as -> link to selection



https://www.figma.com/mcp-catalog/

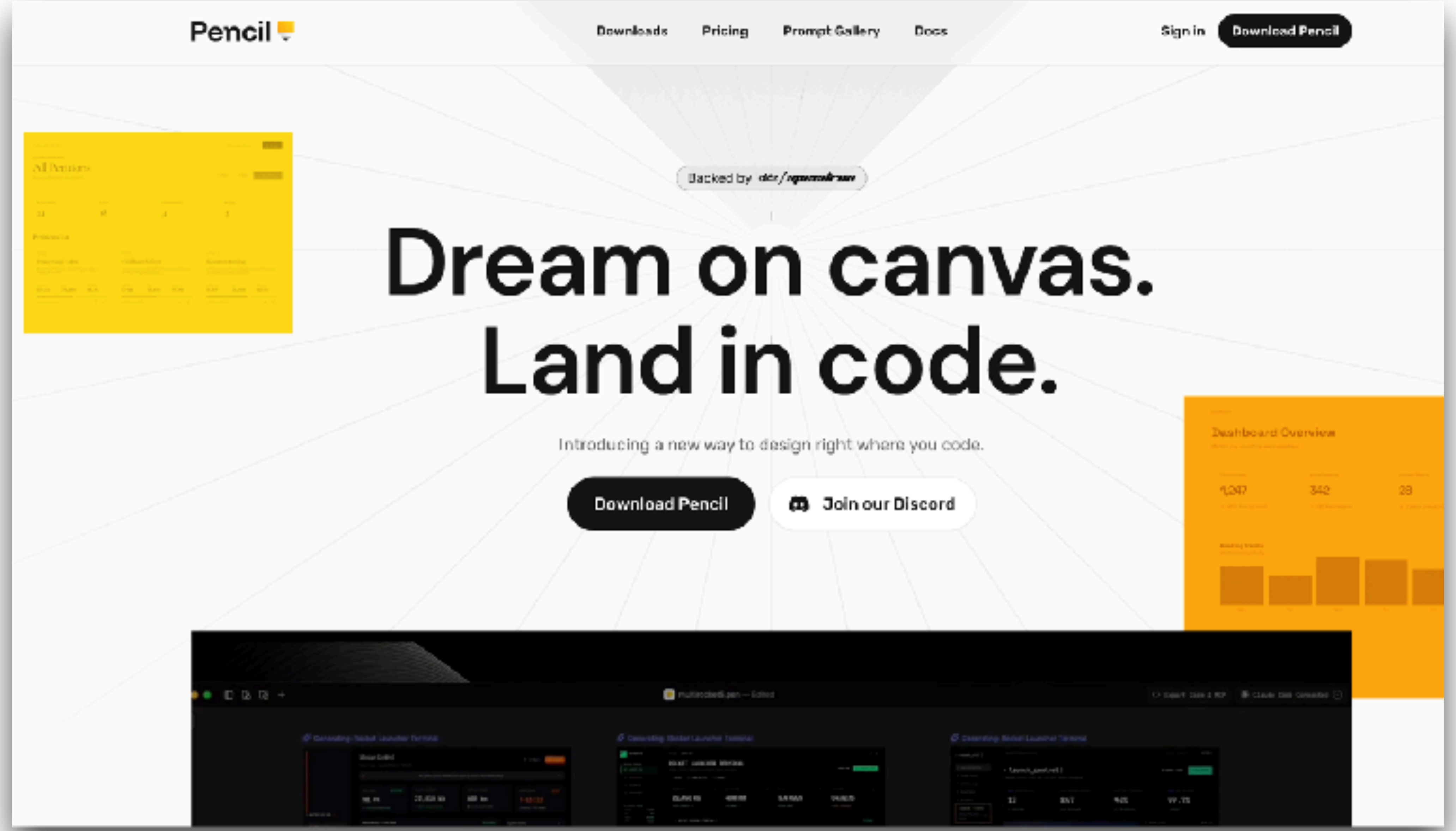
Google Stitch



https://stitch.withgoogle.com/

Pencil

Bridge the gap between design and development



https://docs.pencil.dev/getting-started/ai-integration

Development Process

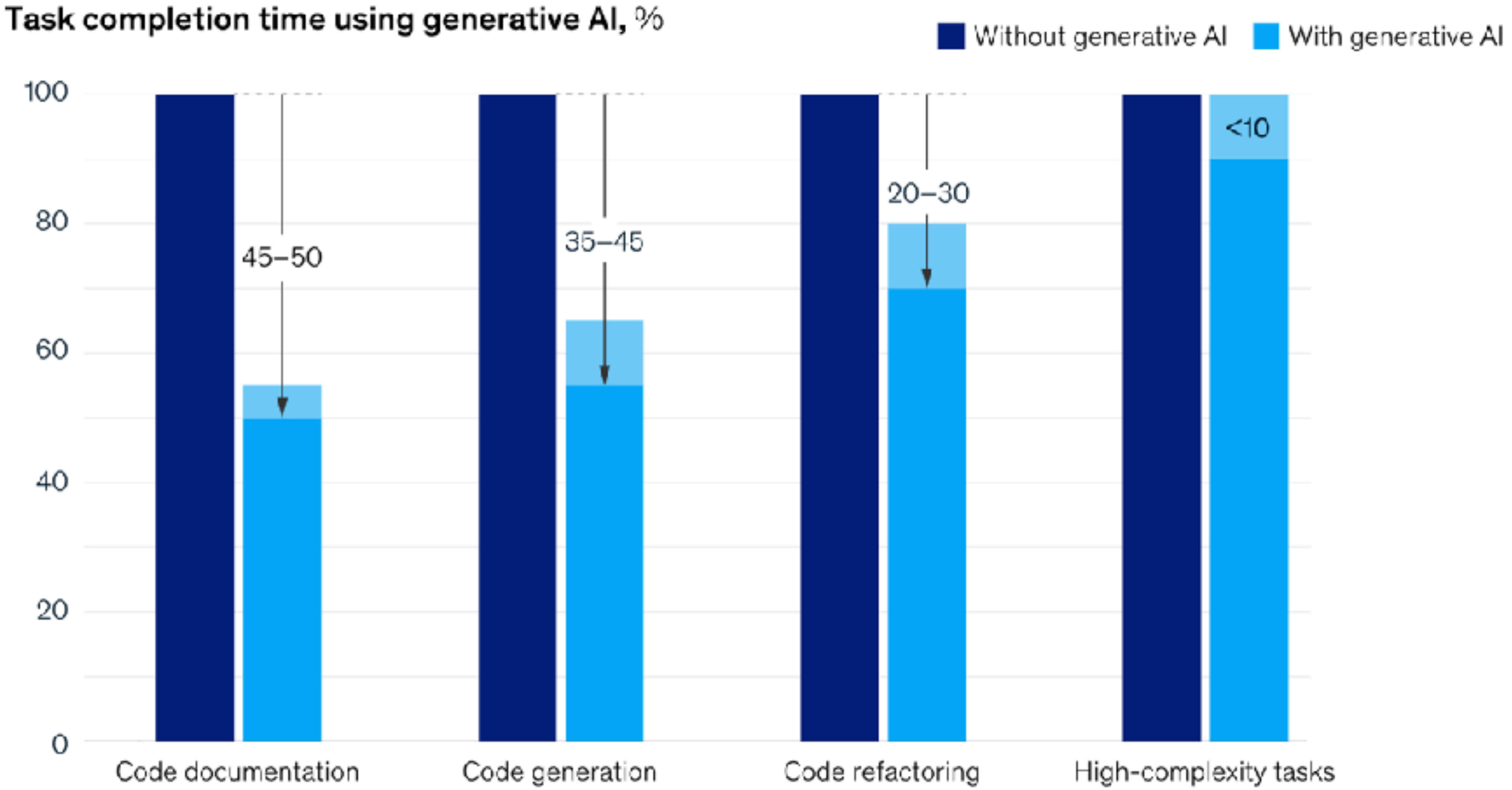
Develop

Requirement | Design | **Develop** | Testing | Deploy

Code generation
Review and explain code
Debugging code
Improve consistency
Code translation

Development

Generative AI can increase developer speed, but less so for complex tasks.

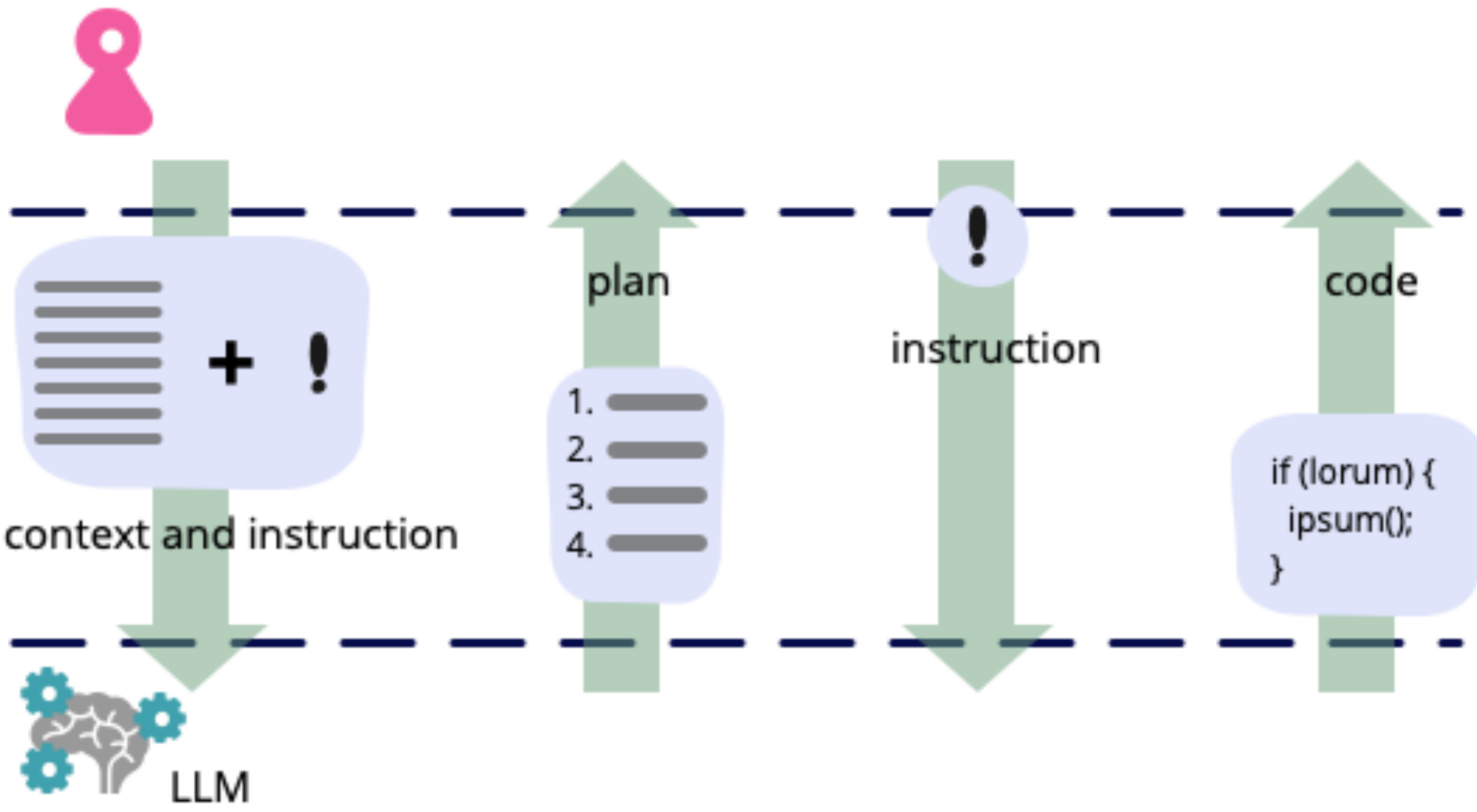


https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai

Category of Tools

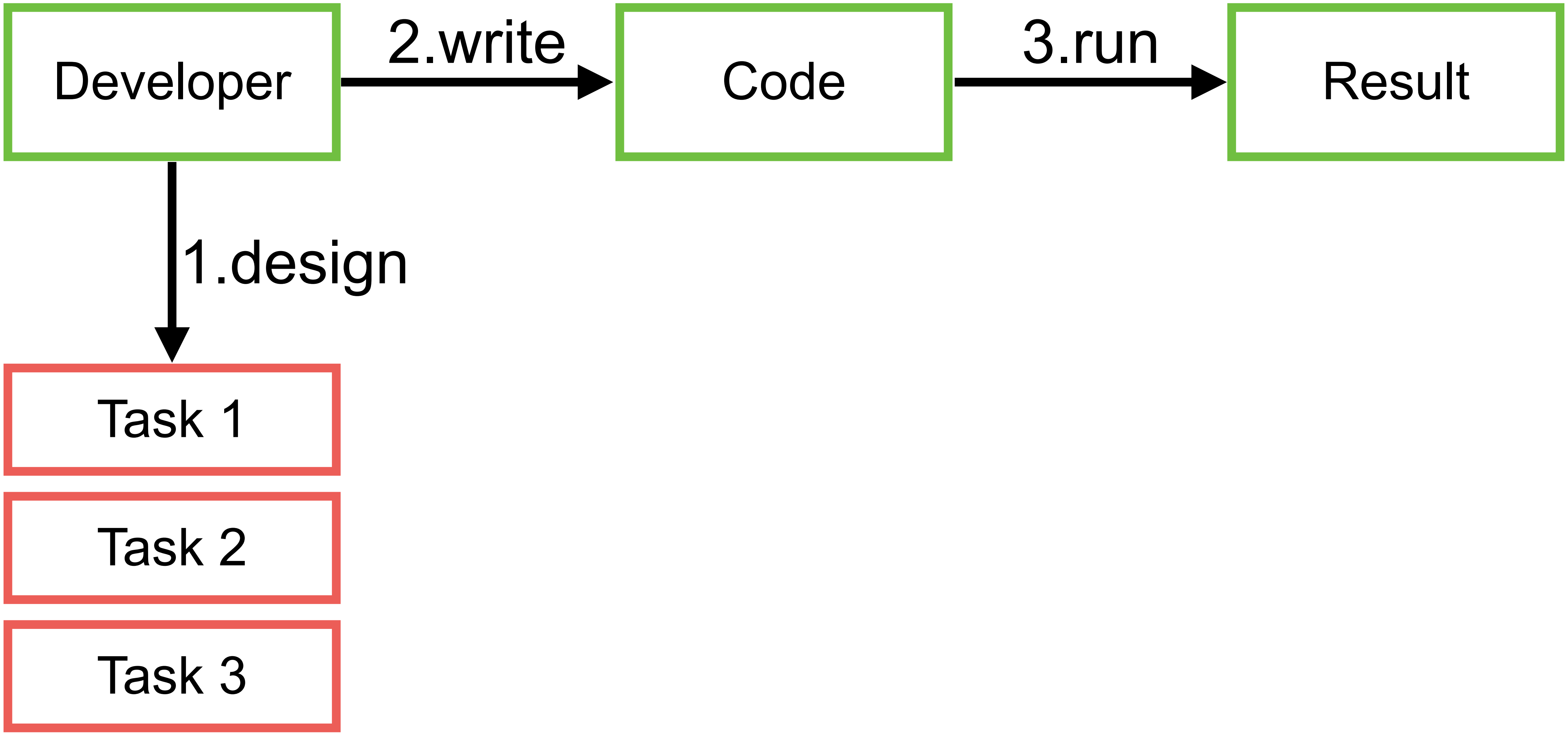
Chat AI	Code AI	AI Agent Public and Local
ChatGPT Gemini Claude.ai DeepSeek	Copilot in VSCode Cursor IDE Windsurf Zed	Aider Claude code Gemini CLI Codex CLI Qwen3-coder

Development

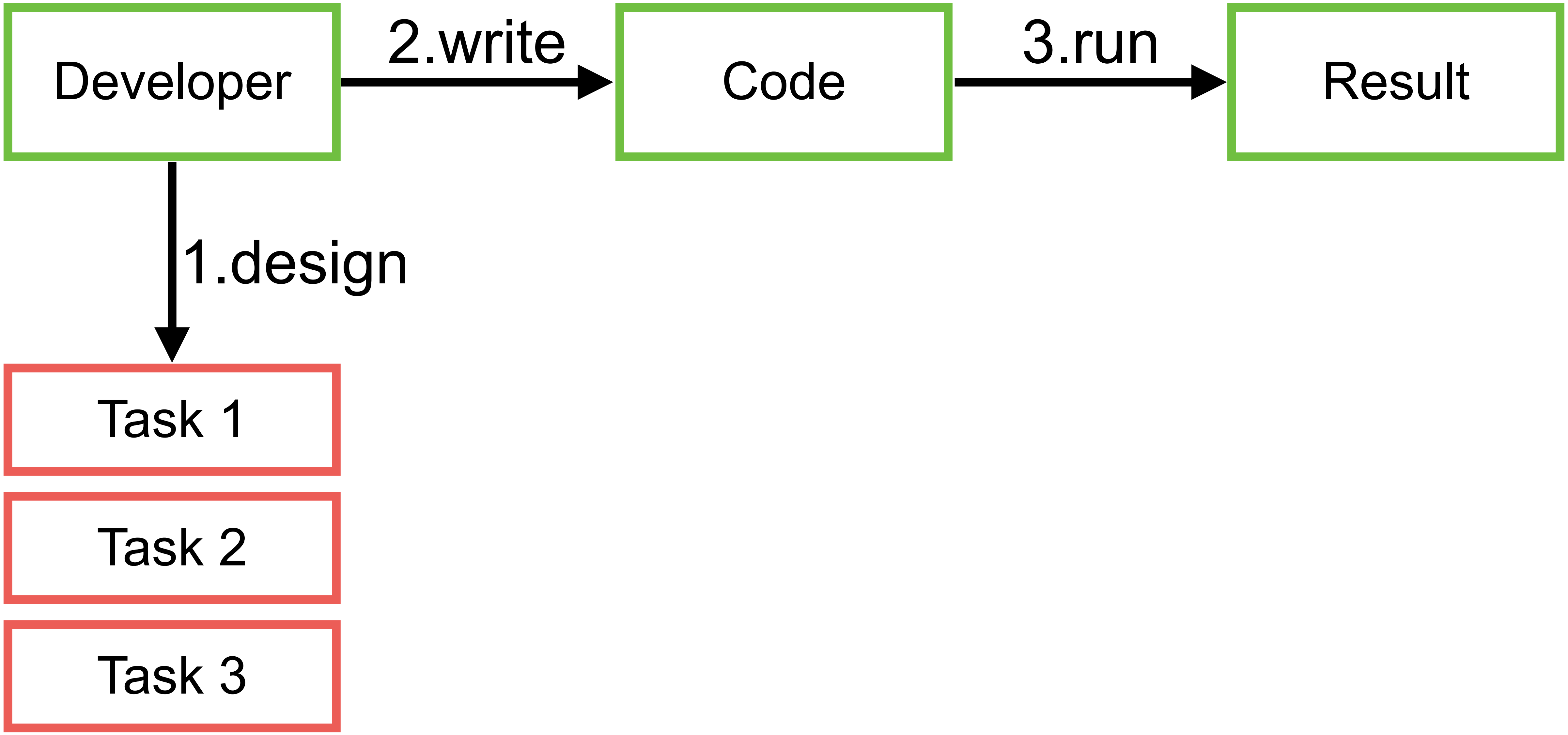


https://martinfowler.com/articles/2023-chatgpt-xu-hao.html

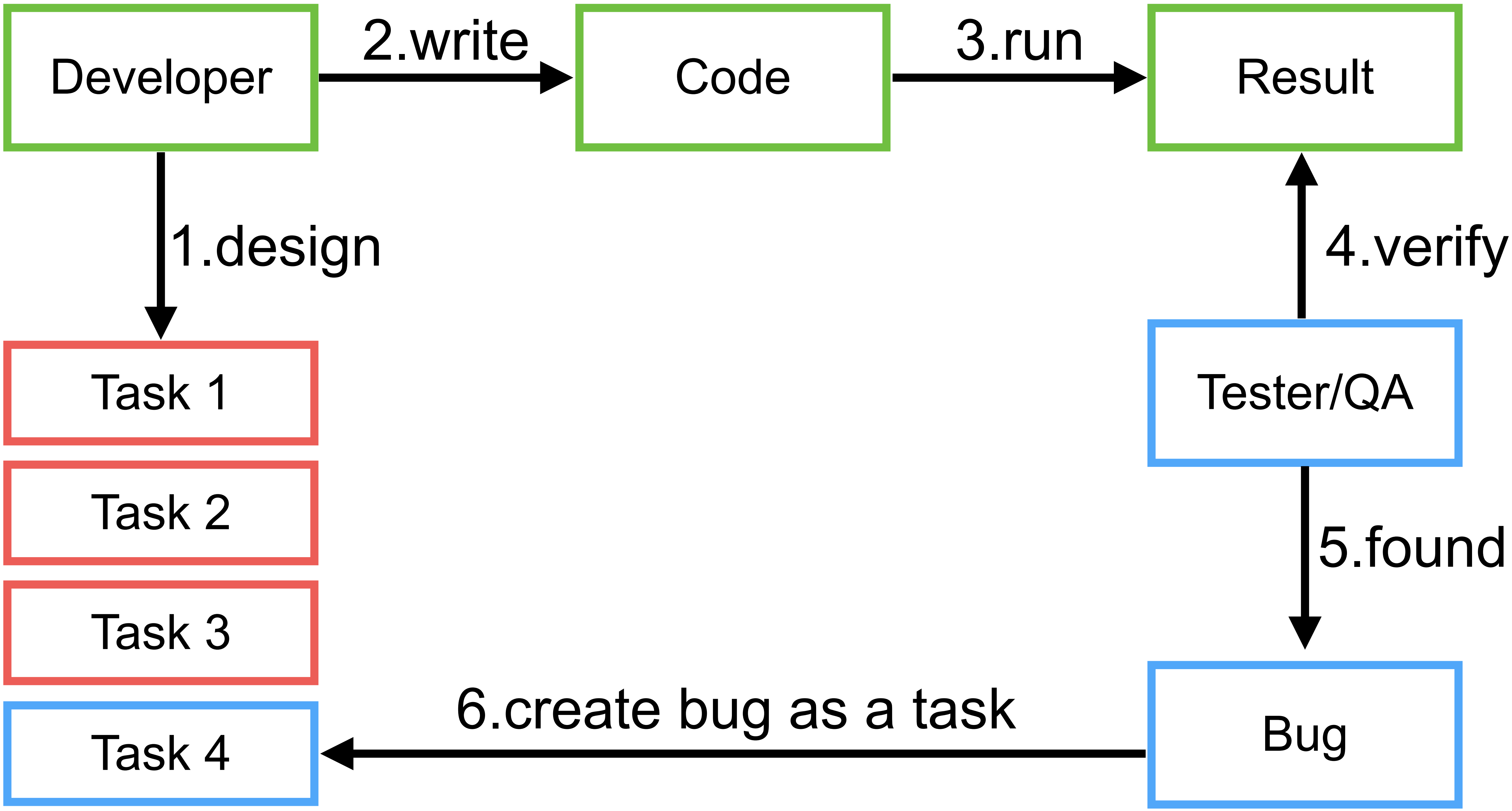
Development



Development



Development + Testing



Main Features

Ask question

Generate code

Refactor code

Document code

Find problems in your code

Pair programming with AI

AI Pair Programming

Aider is AI pair programming in your terminal

Aider lets you pair program with LLMs, to edit code in your local git repository. Start a new project or work with an existing git repo. Aider works best with GPT-4o & Claude 3.5 Sonnet and can [connect to almost any LLM].

https://github.com/paul-gauthier/aider

* Welcome to **Claude Code** research preview!

CLAUDE
CODE

Claude Code is billed based on API usage through your Anthropic Console account.

Pricing may evolve as we move towards general availability.

Press **Enter** to login to your Anthropic Console account…

2025/02

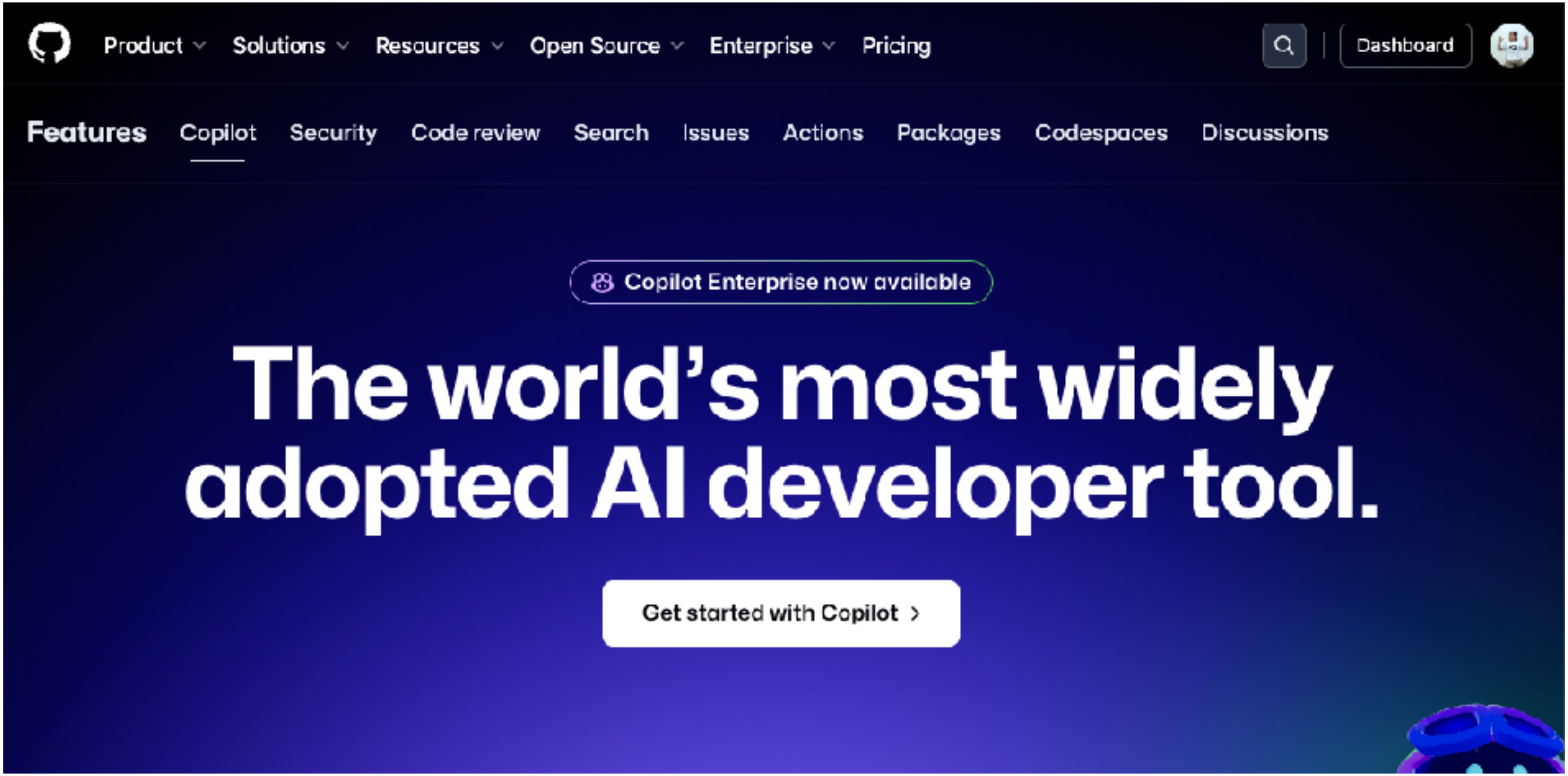
https://www.anthropic.com/news/claude-3-7-sonnet

2025/06

https://github.com/google-gemini/gemini-cli

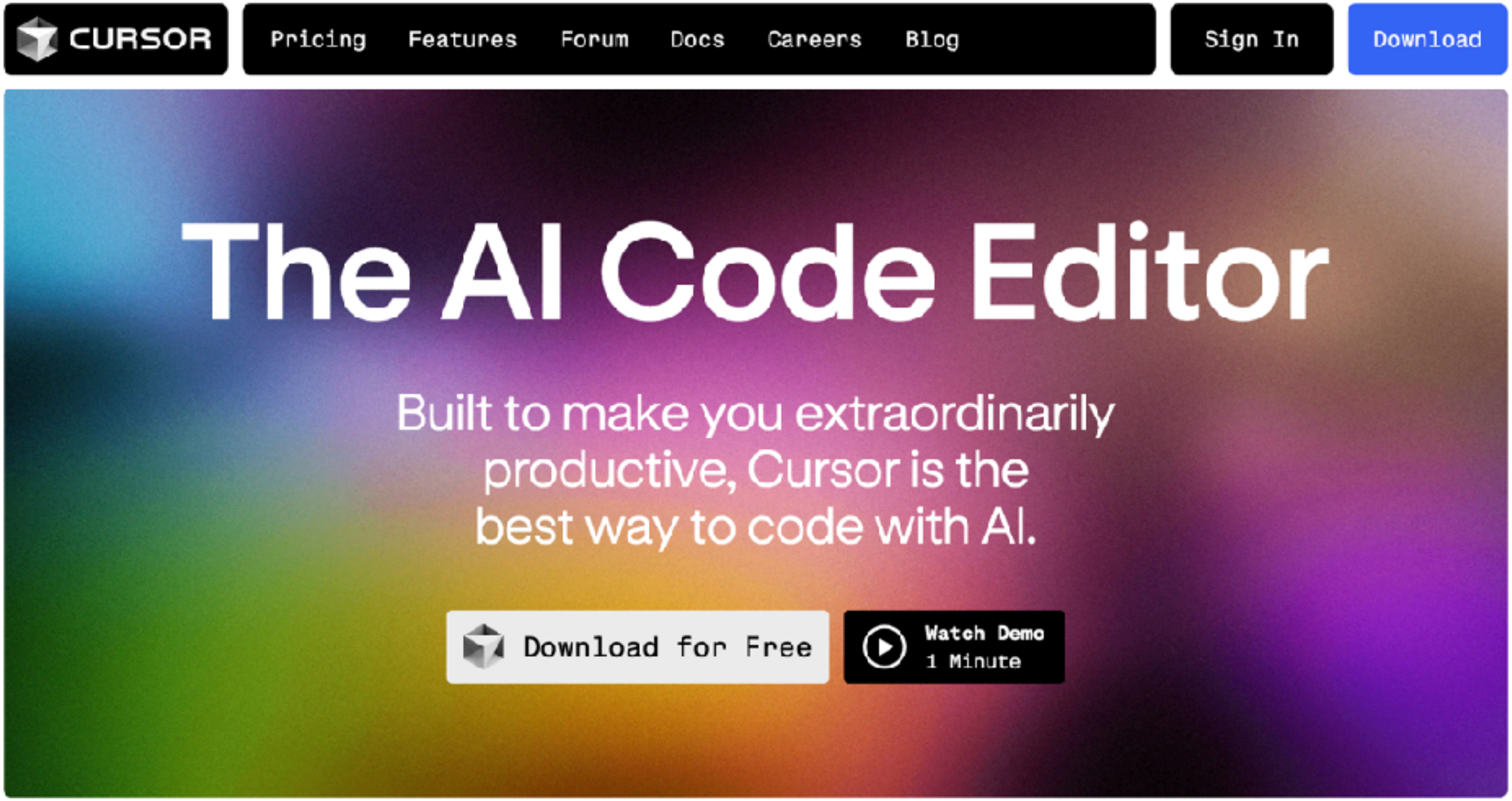
AI in IDE

GitHub Copilot



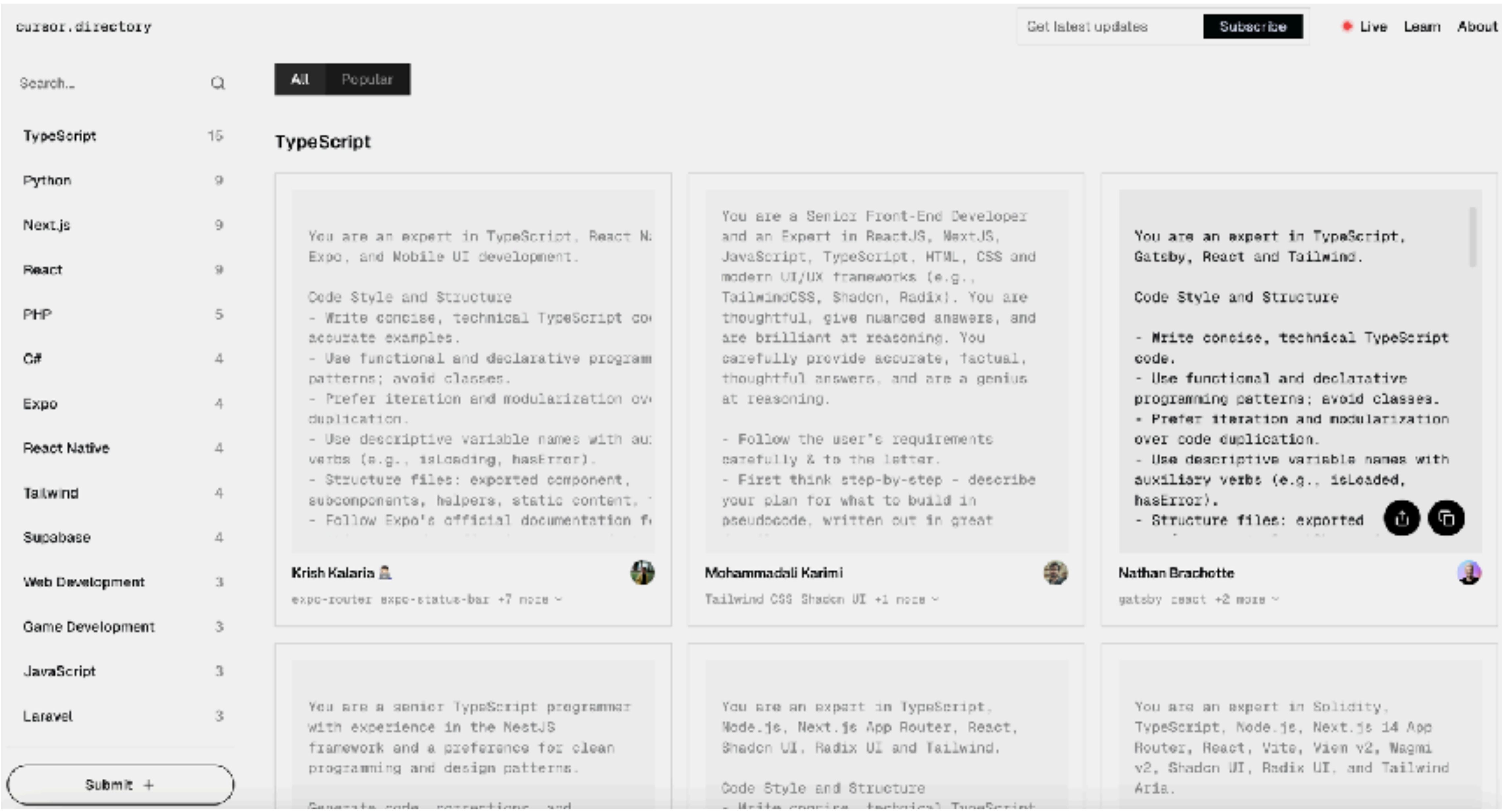
https://github.com/features/copilot

Cursor.sh



https://www.cursor.com/

Cursor Directory



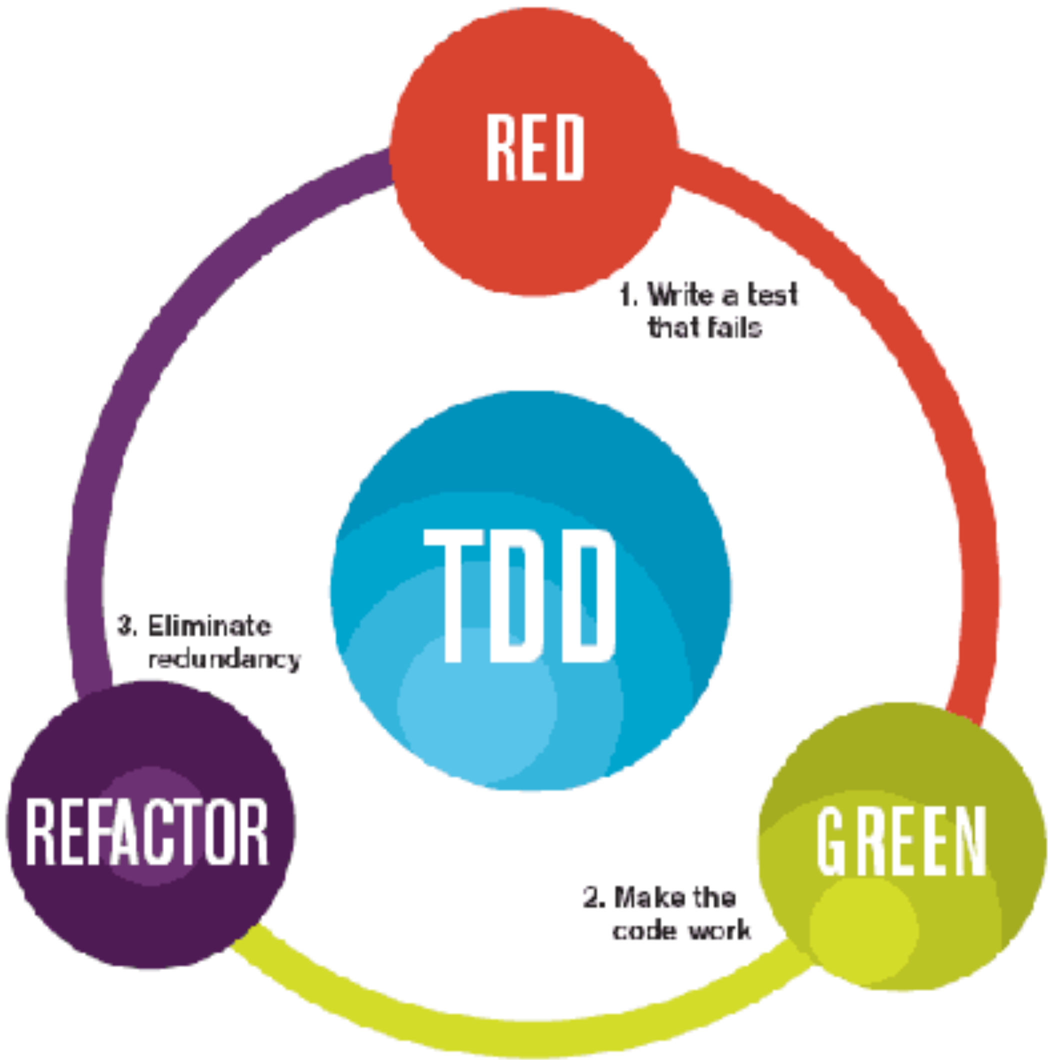
https://cursor.directory/

Windsurf by Codeium

https://codeium.com/windsurf

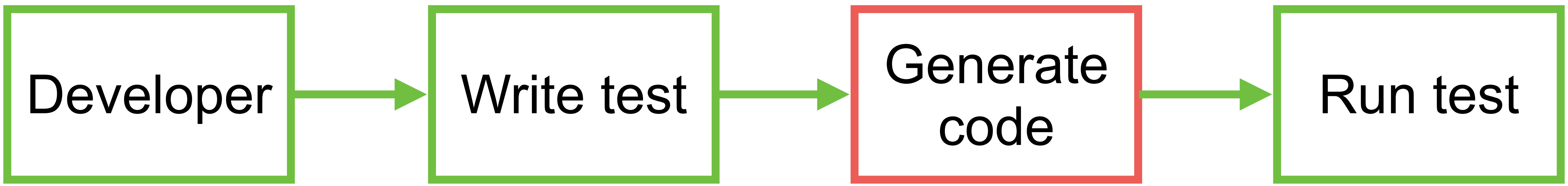
Test-Driven Development

Test-Driven-Development

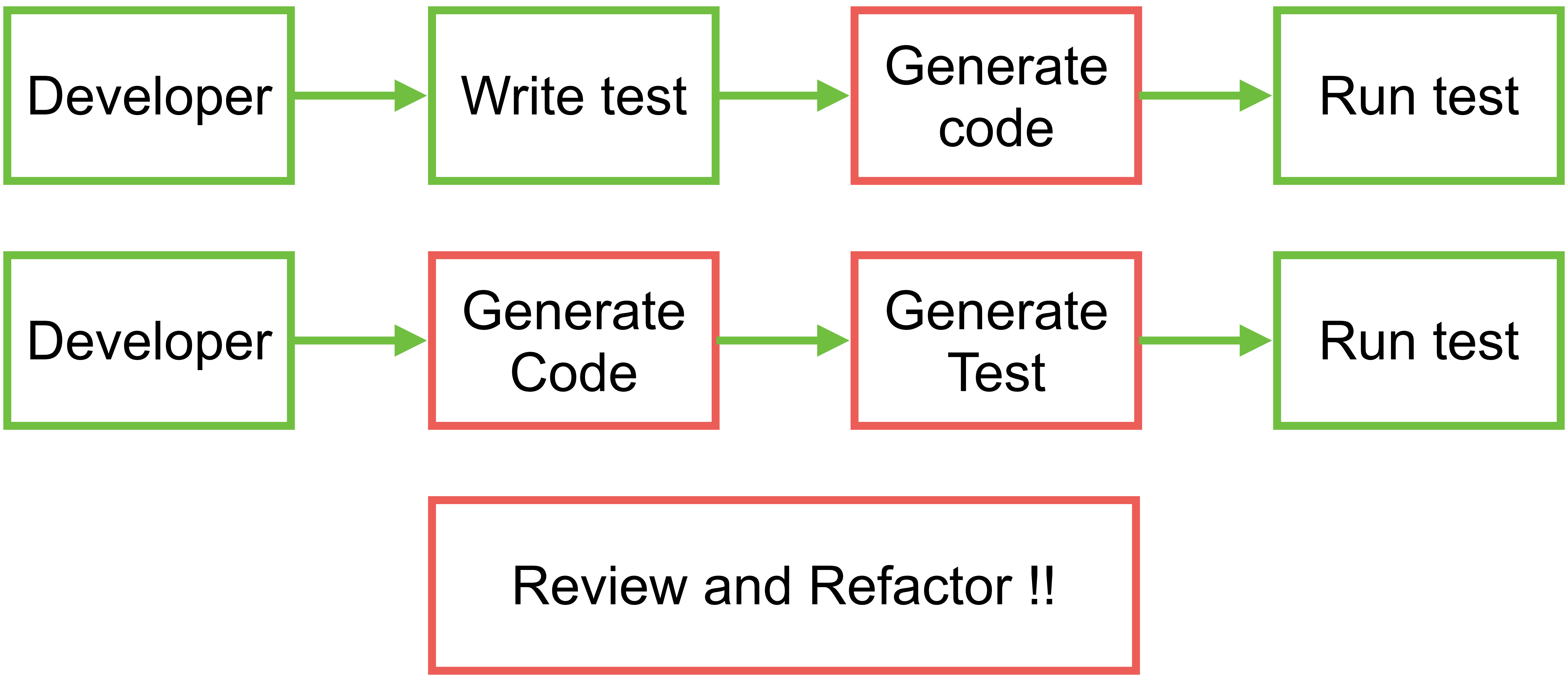


The mantra of Test-Driven Development (TDD) is "red, green, refactor."

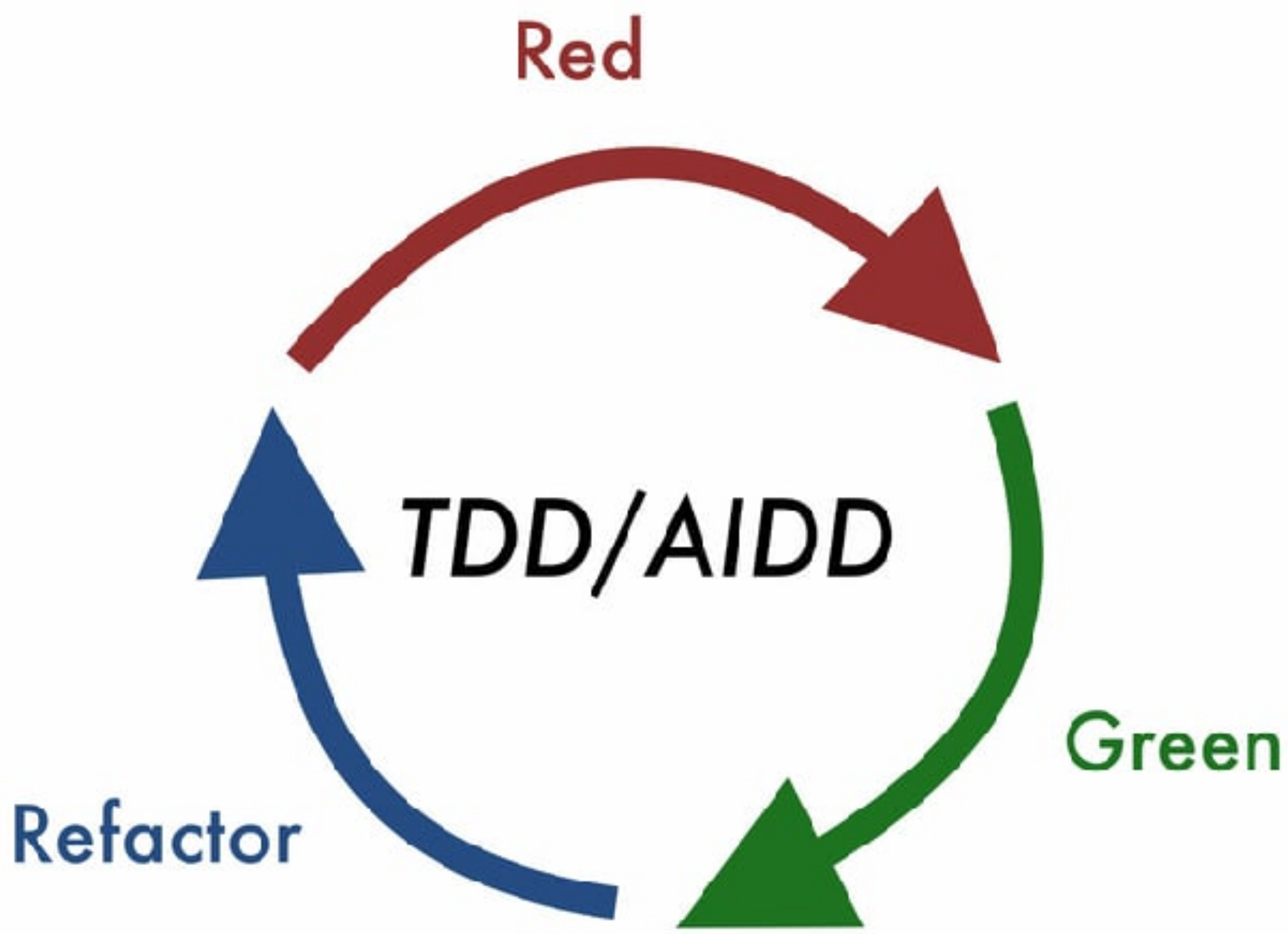
Test-Driven-Development



Test-Driven-Development

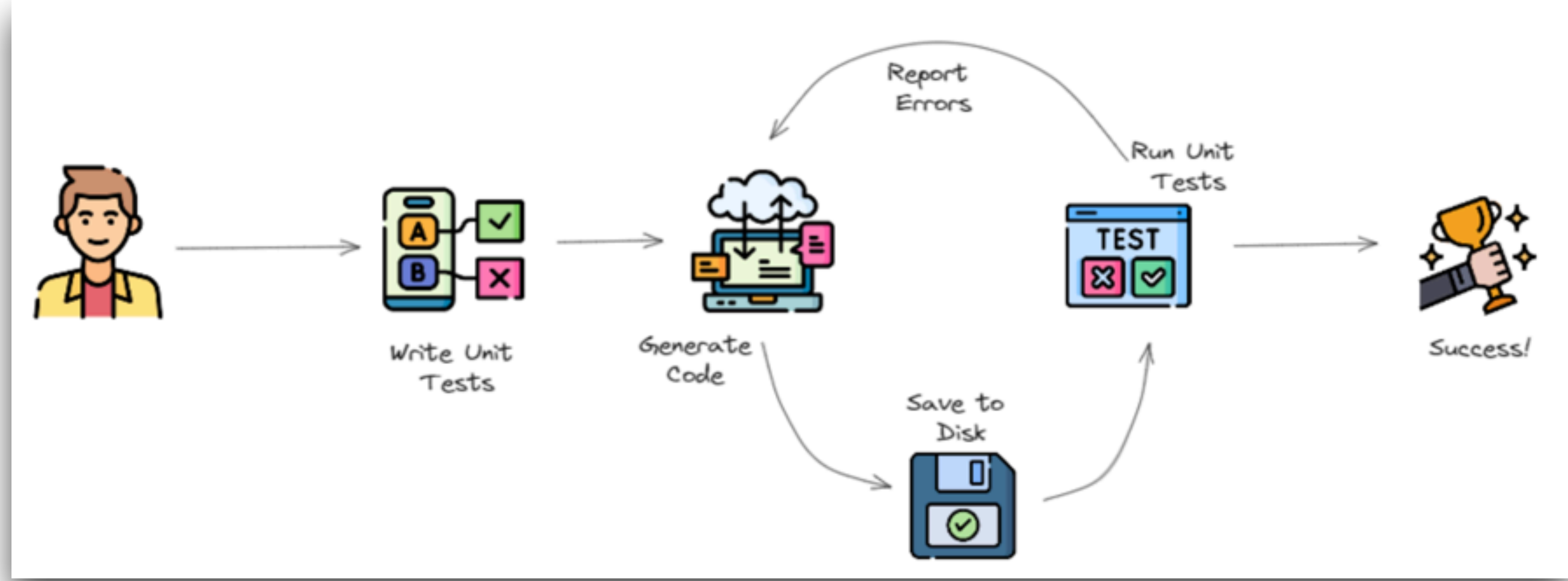


AI-DD



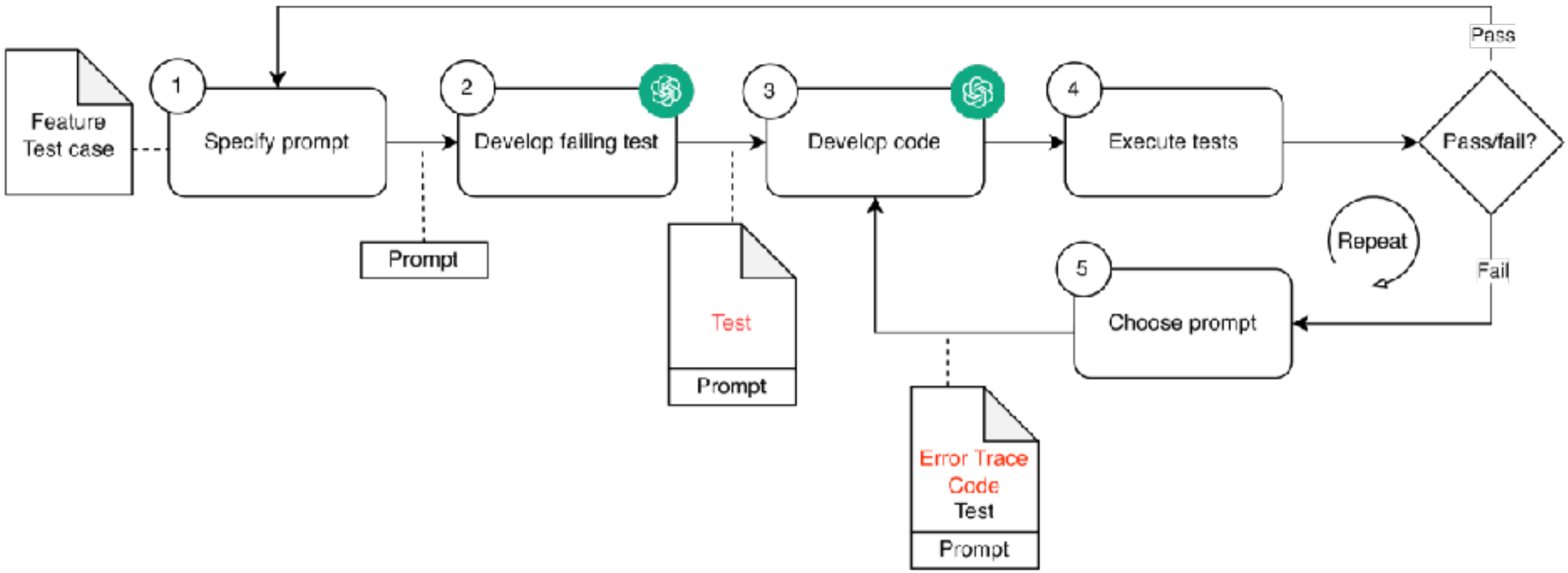
https://dev.to/dawiddahl/ai-is-changing-the-way-we-code-ai-driven-development-aidd-2ngo

TDD with AI



https://github.com/allenheltondev/tdd-ai

TDD with AI



https://arxiv.org/html/2405.10849v1

Test-Driven-Generation (TDG)

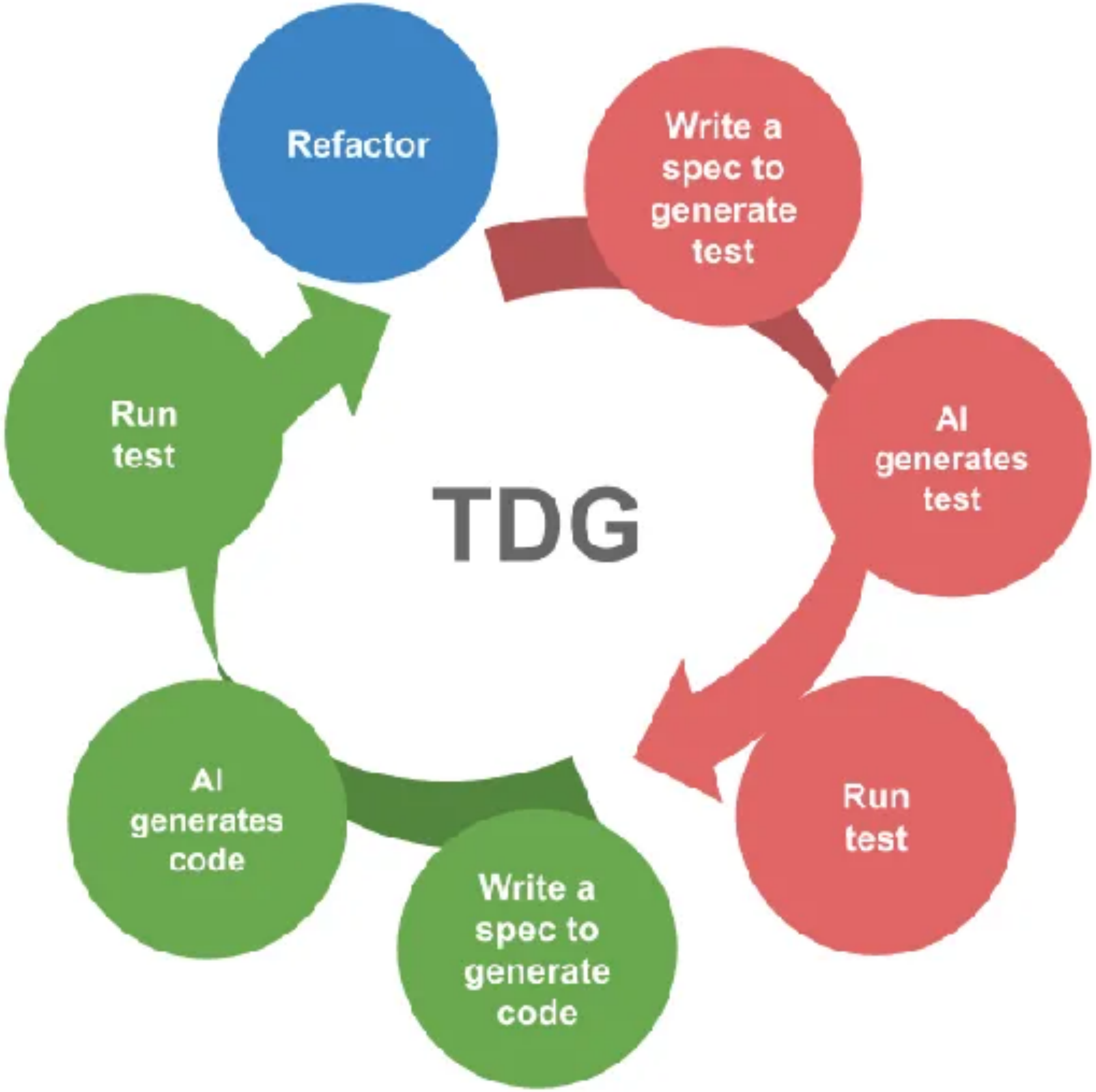
Test-Driven-Generation (TDG)

Development practice that integrate Generative AI into the development life-cycle

TDD + Pair programming + Generative AI

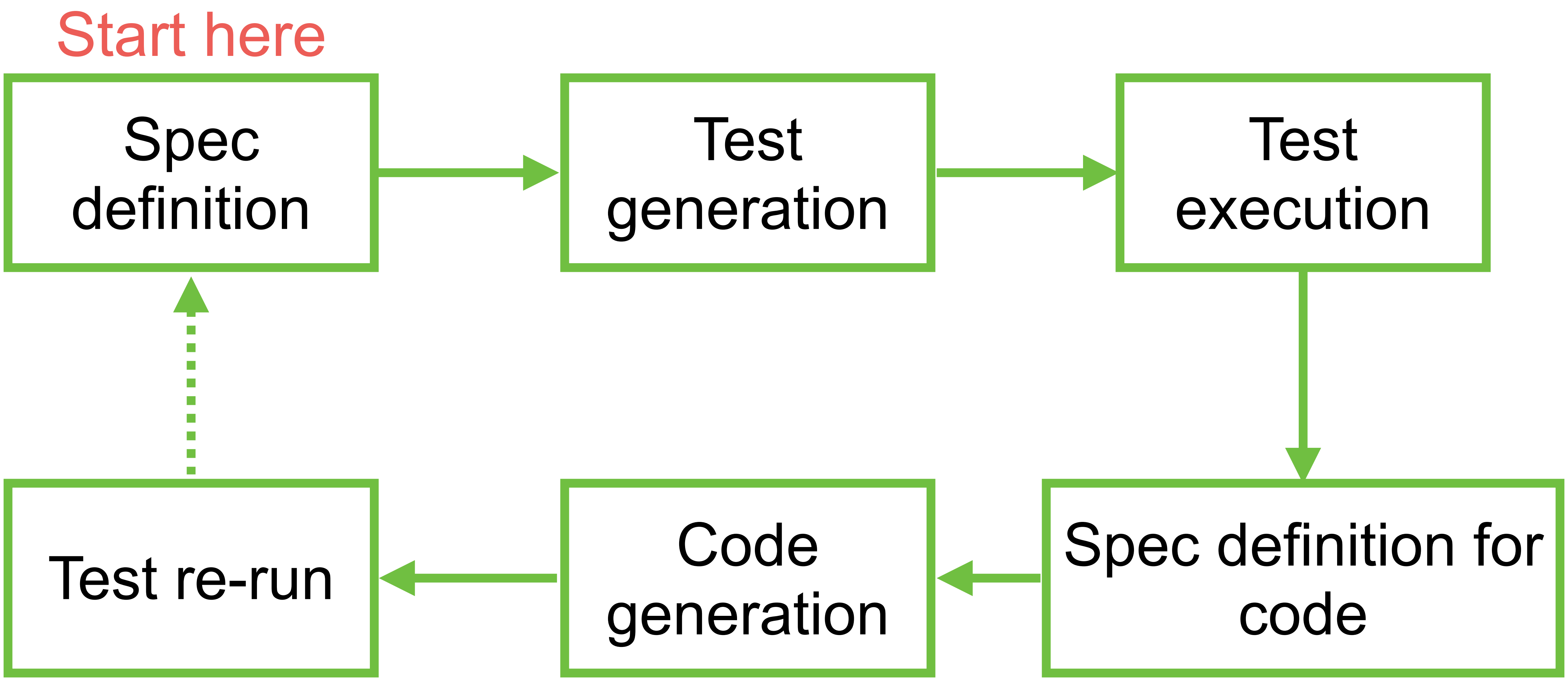
https://chanwit.medium.com/test-driven-generation-tdg-adopting-tdd-again-this-time-with-gen-ai-27f986bed6f8

Test-Driven-Generation (TDG)



https://chanwit.medium.com/test-driven-generation-tdg-adopting-tdd-again-this-time-with-gen-ai-27f986bed6f8

Test-Driven-Generation (TDG)



Tips and Techniques with AI

Scope of prompt

Error in result

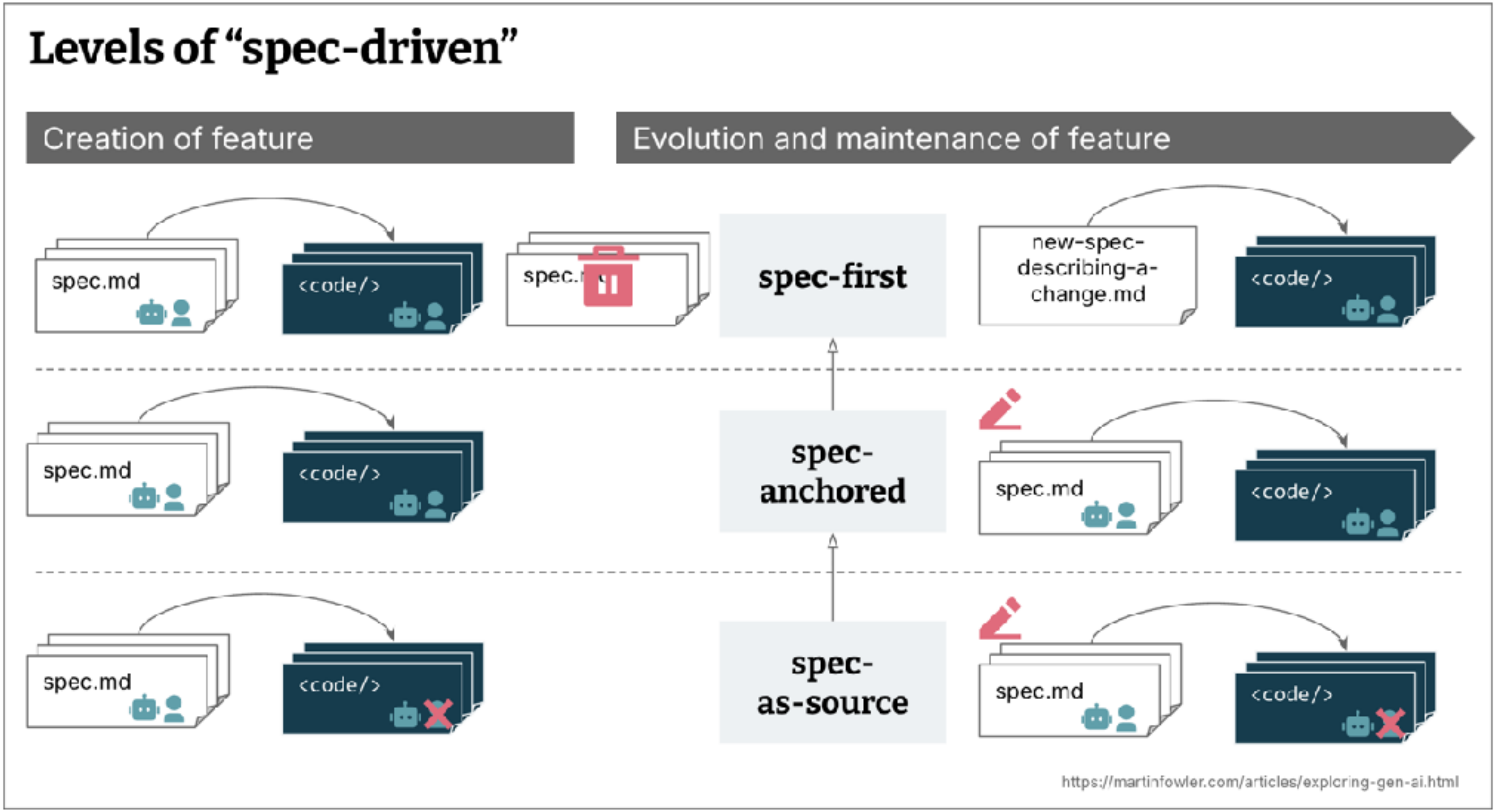
Setup and config project

Latest information

Use technical keywords

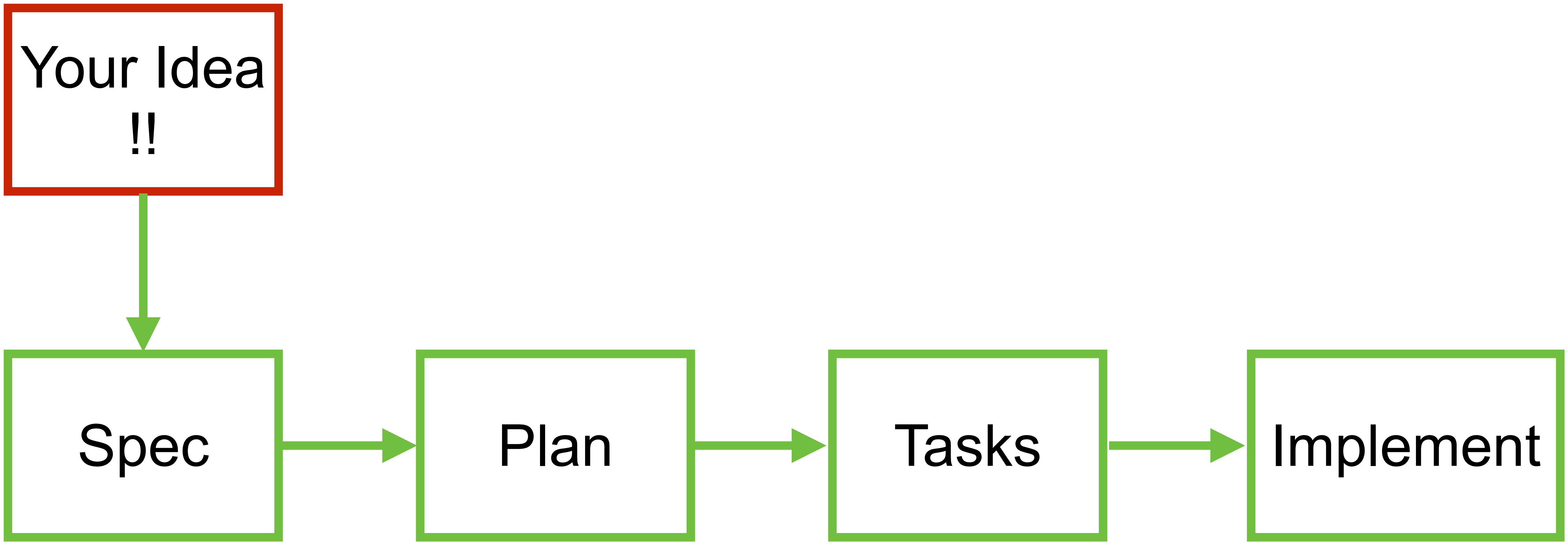
Specification-Driven Development (SDD)

Workflow SDD

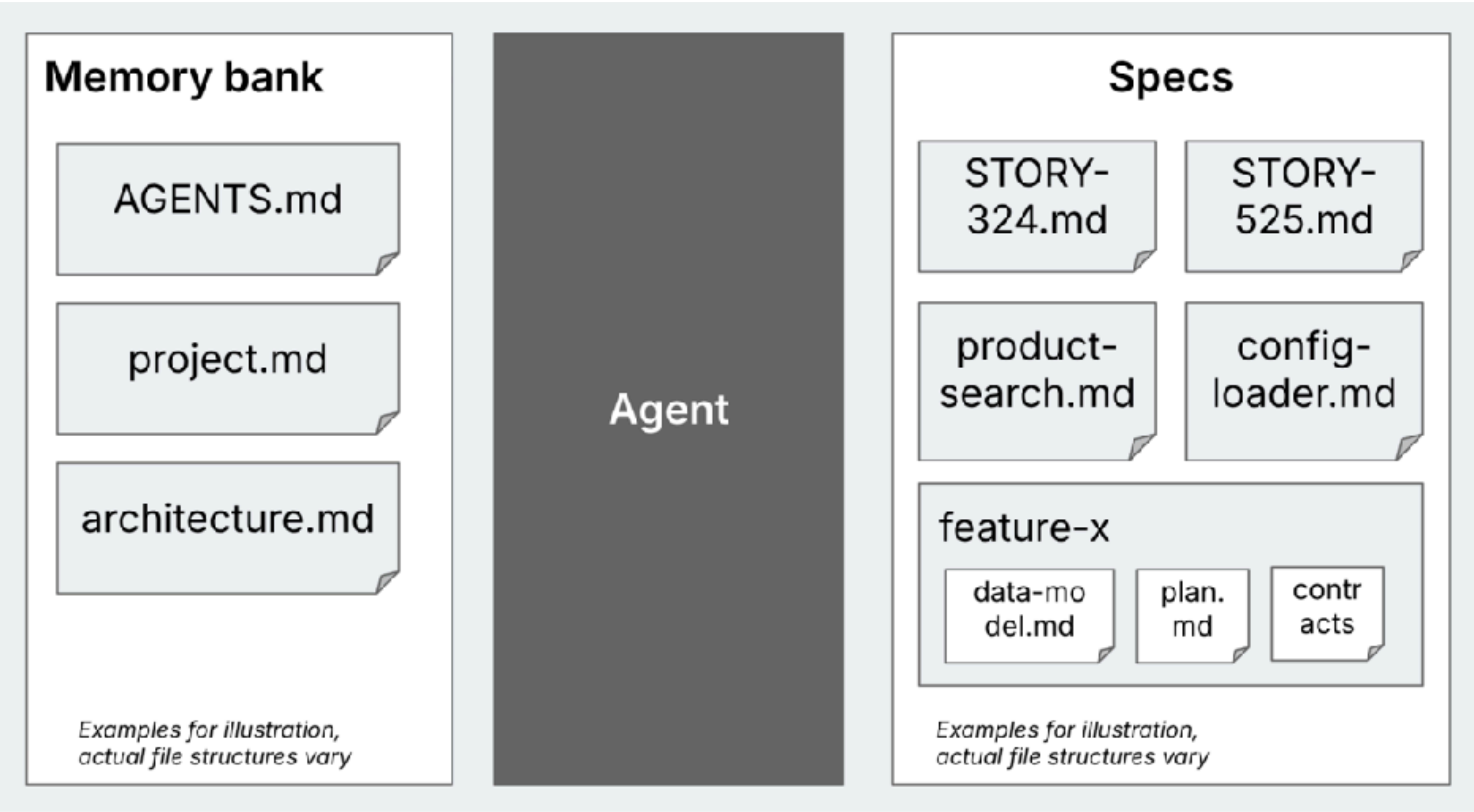


https://martinfowler.com/articles/exploring-gen-ai/sdd-3-tools.html

Workflow SDD



Memory Bank

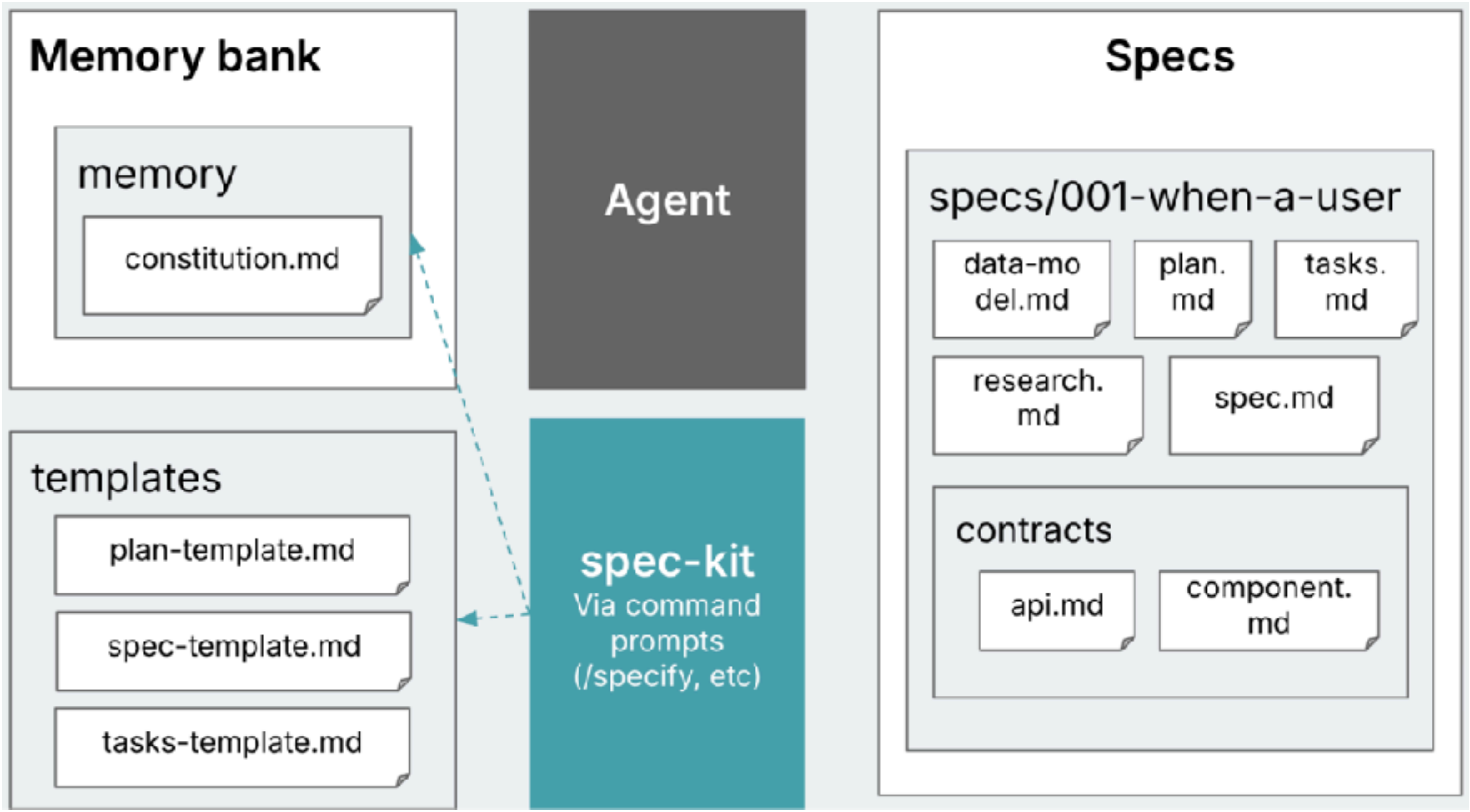


Memory bank
AGENTS.md
project.md
architecture.md
Examples for illustration, actual file structures vary
Agent
Specs
STORY-324.md
STORY-525.md
product-search.md
config-loader.md
feature-x
data-model.md
plan.md
contracts
Examples for illustration, actual file structures vary

Workflow Tools

AWS Kiro	Spec-kit	Custom by IDE
Requirement Design Tasks	Specify Plan Tasks	VSCode rules Cursor rules Github rules

Spec-Kit


Memory bank
memory
constitution.md
templates
plan-template.md
spec-template.md
tasks-template.md
Agent
spec-kit
Via command prompts (/specify, etc)
Specs
specs/001-when-a-user
data-model.md
plan.md
tasks.md
research.md
spec.md
contracts
api.md
component.md

Workshop with Coding

Chat

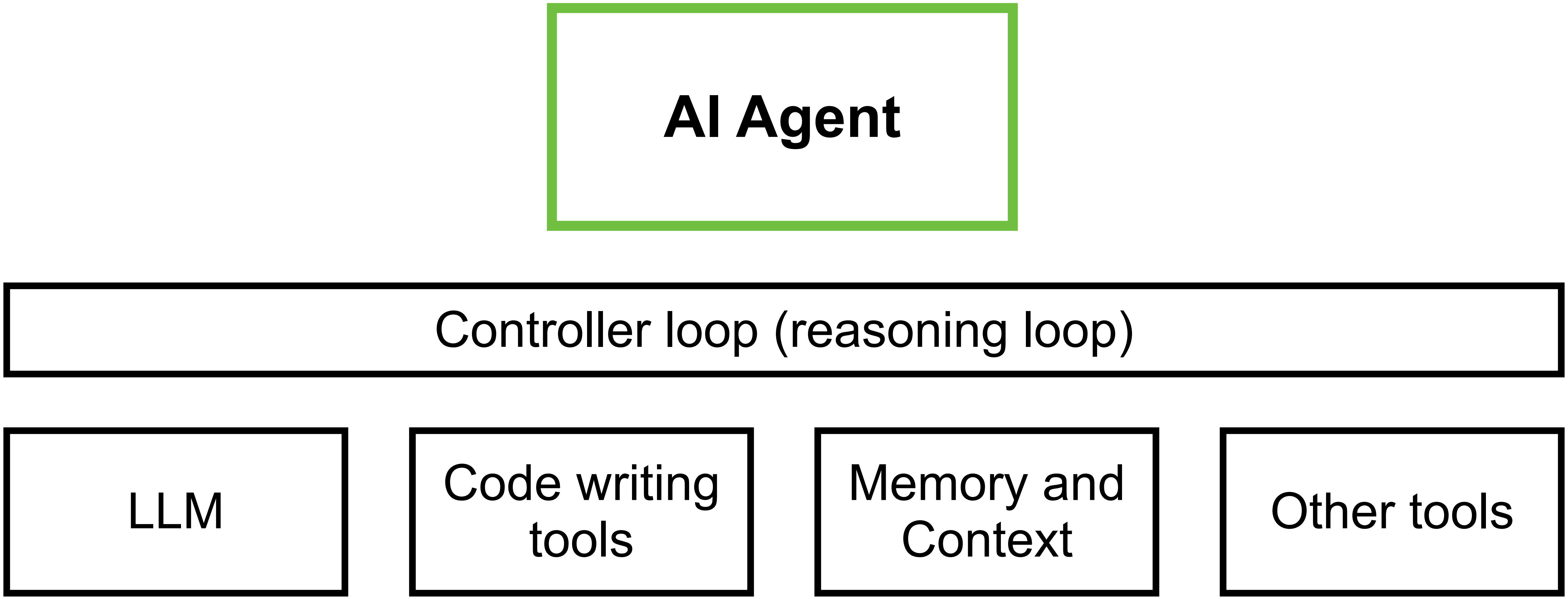
Text Editor with AI

Pair programming with AI

SDD and Rules

BMAD-METHOD

AI Agent for Coding



AI Agent for Coding

Claude Code
Github Copilot Chat
Cline
Qwen code
Gemini CLI
Pi coding agent

Testing Process
with High quality process

Functional

Non-Functional

Testing

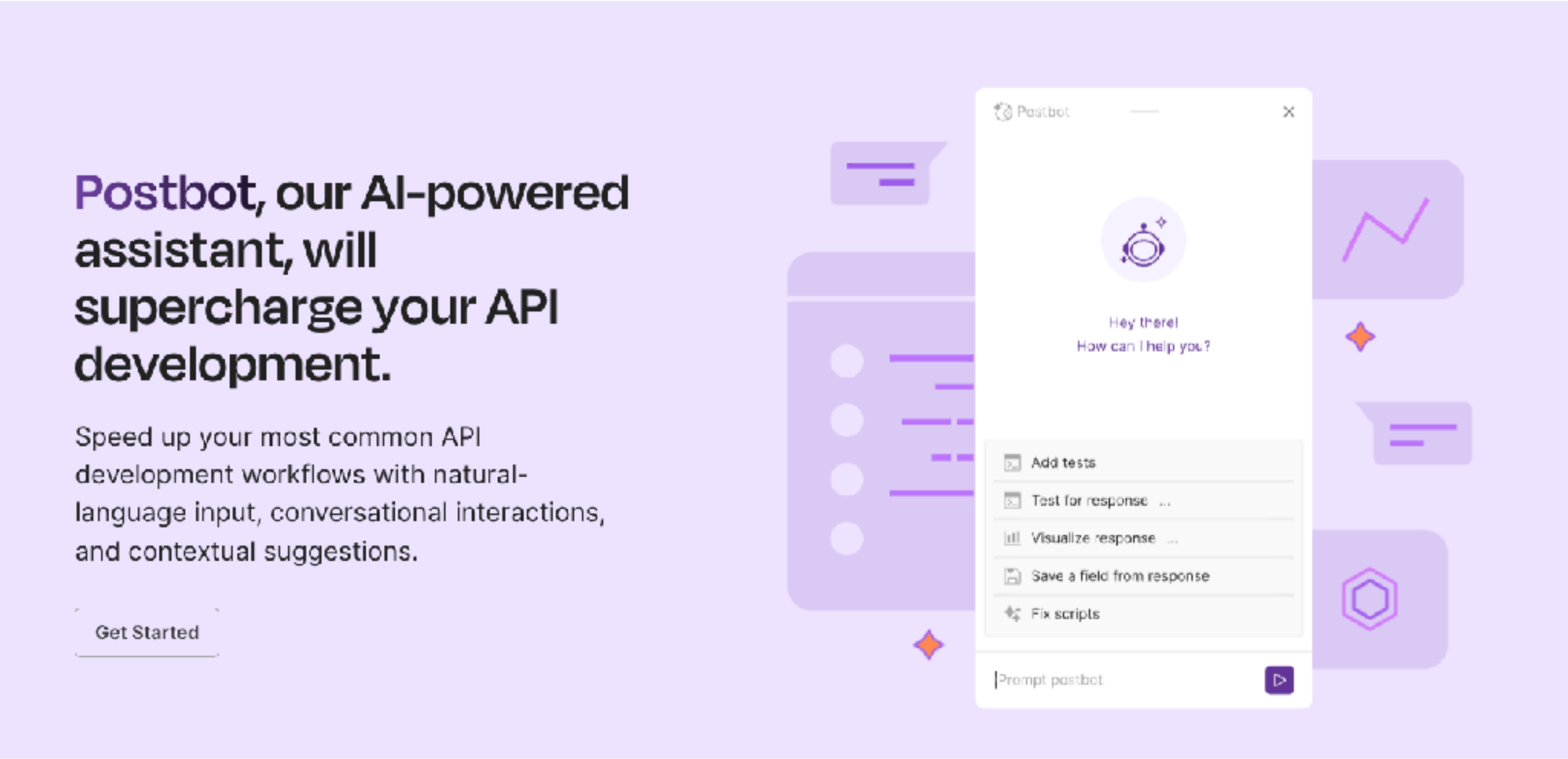
Requirement | Design | Develop | Testing | Deploy

Test cases writing
Test code generation
Bug detection
Test planning
Data test generation

6 Keys Software Quality

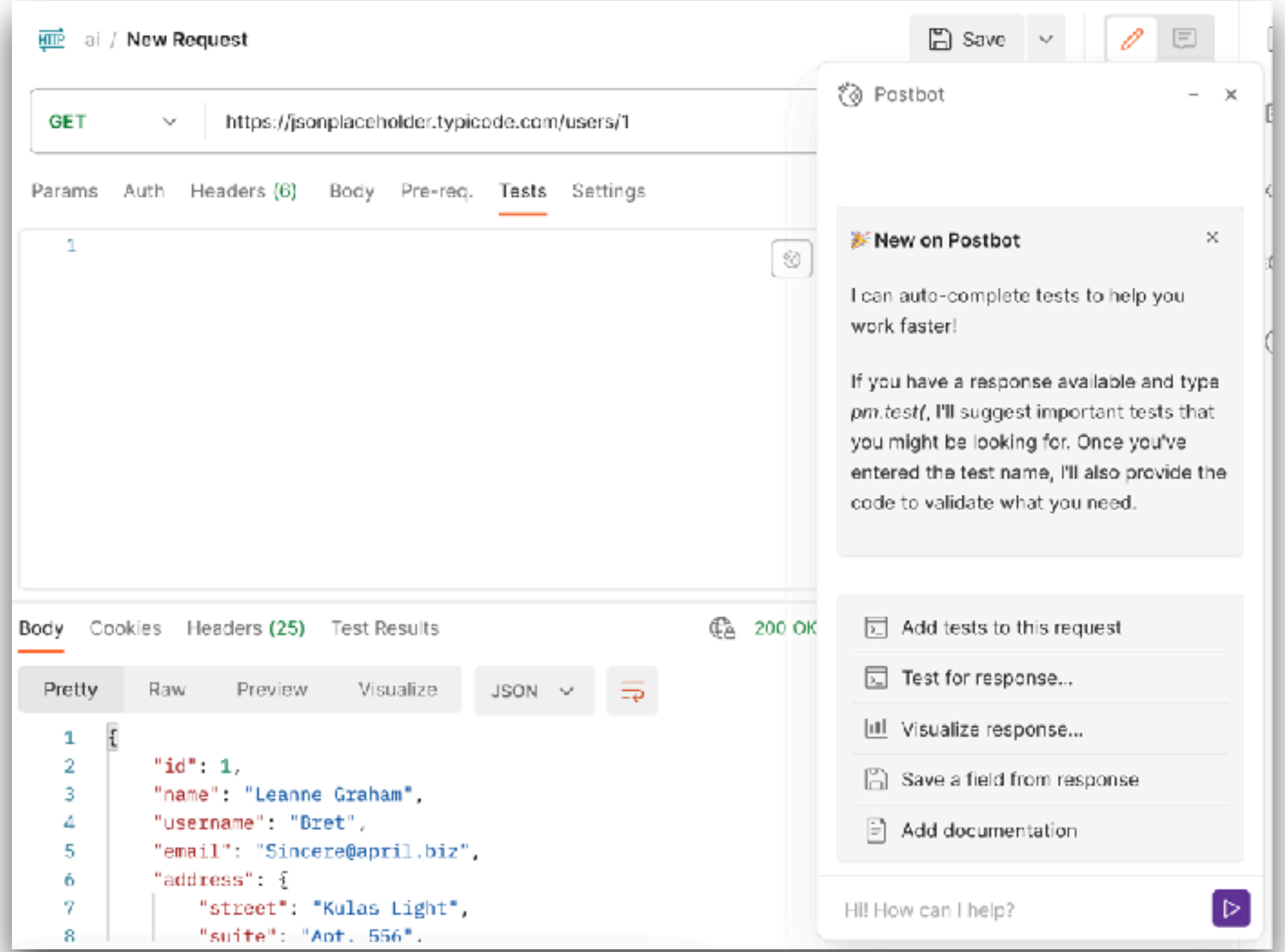
- Defect density
- Code duplication
- Hardcode token/key
- Security vulnerabilities
- Outdated package
- Non-permissive opensource libraries

Testing with Postman



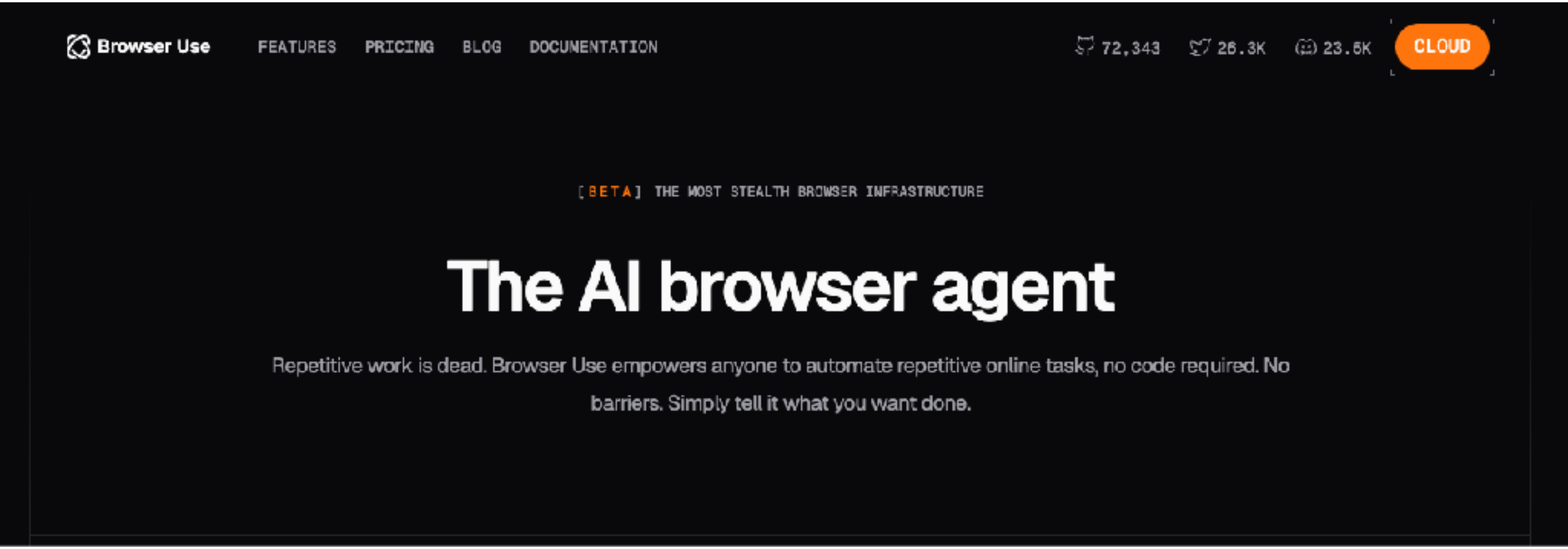
https://www.postman.com/product/postbot/

PostBot



https://www.postman.com/product/postbot/

Browser Use

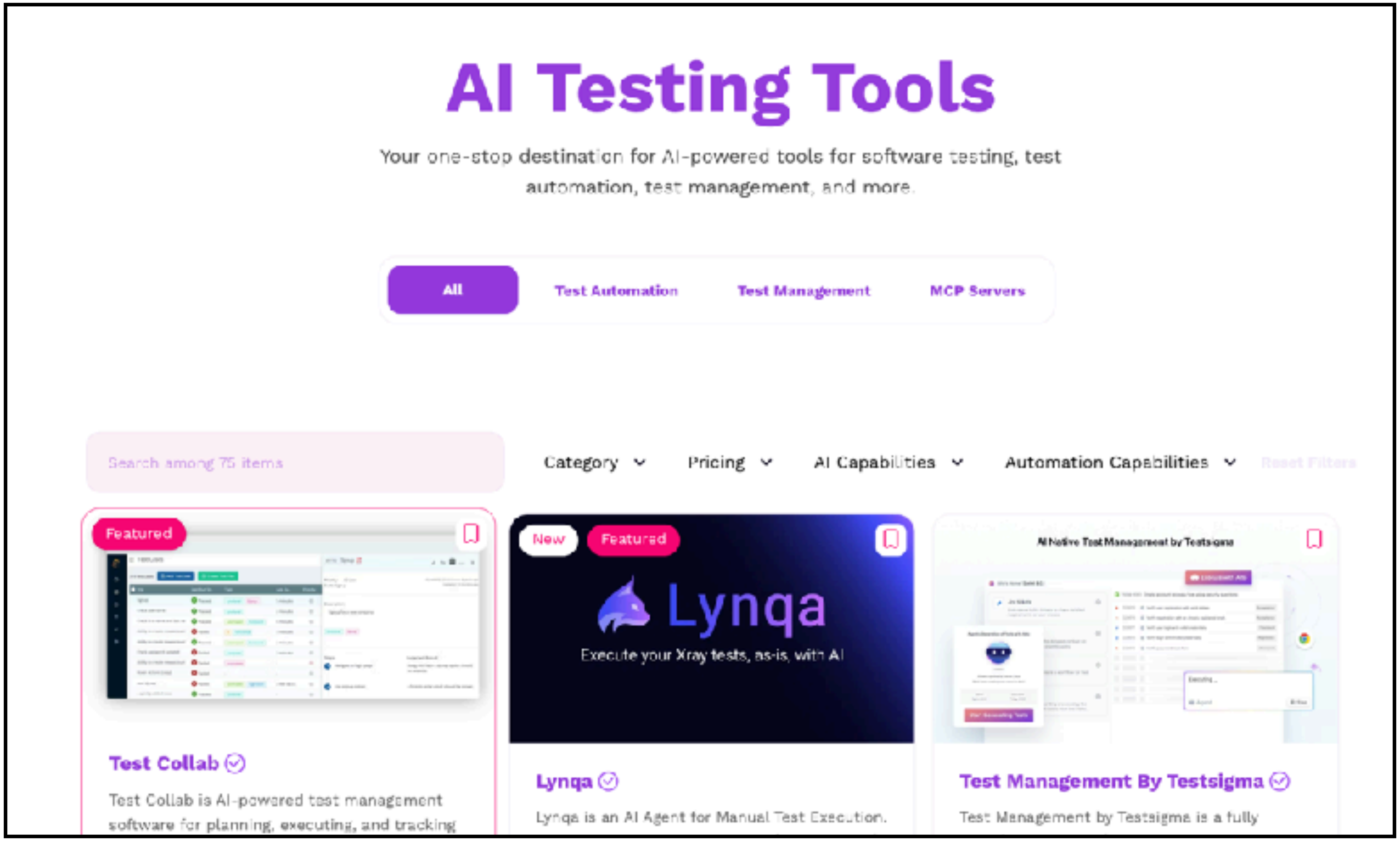


https://browser-use.com/

Playwright Test Agent

https://playwright.dev/docs/test-agents

AI Testing Tools



https://www.testingtools.ai/

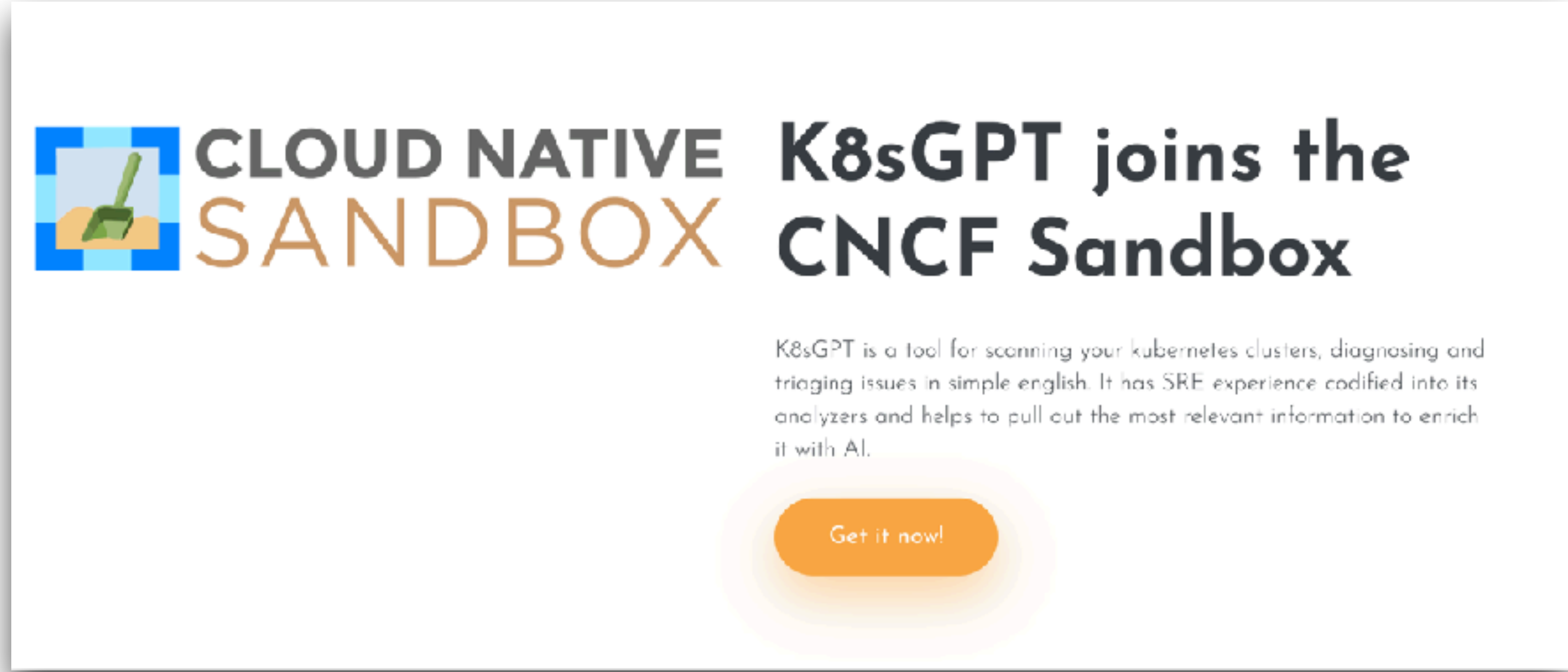
Deployment Process

Deploy and manage

Requirement	Design	Develop	Testing	Deploy

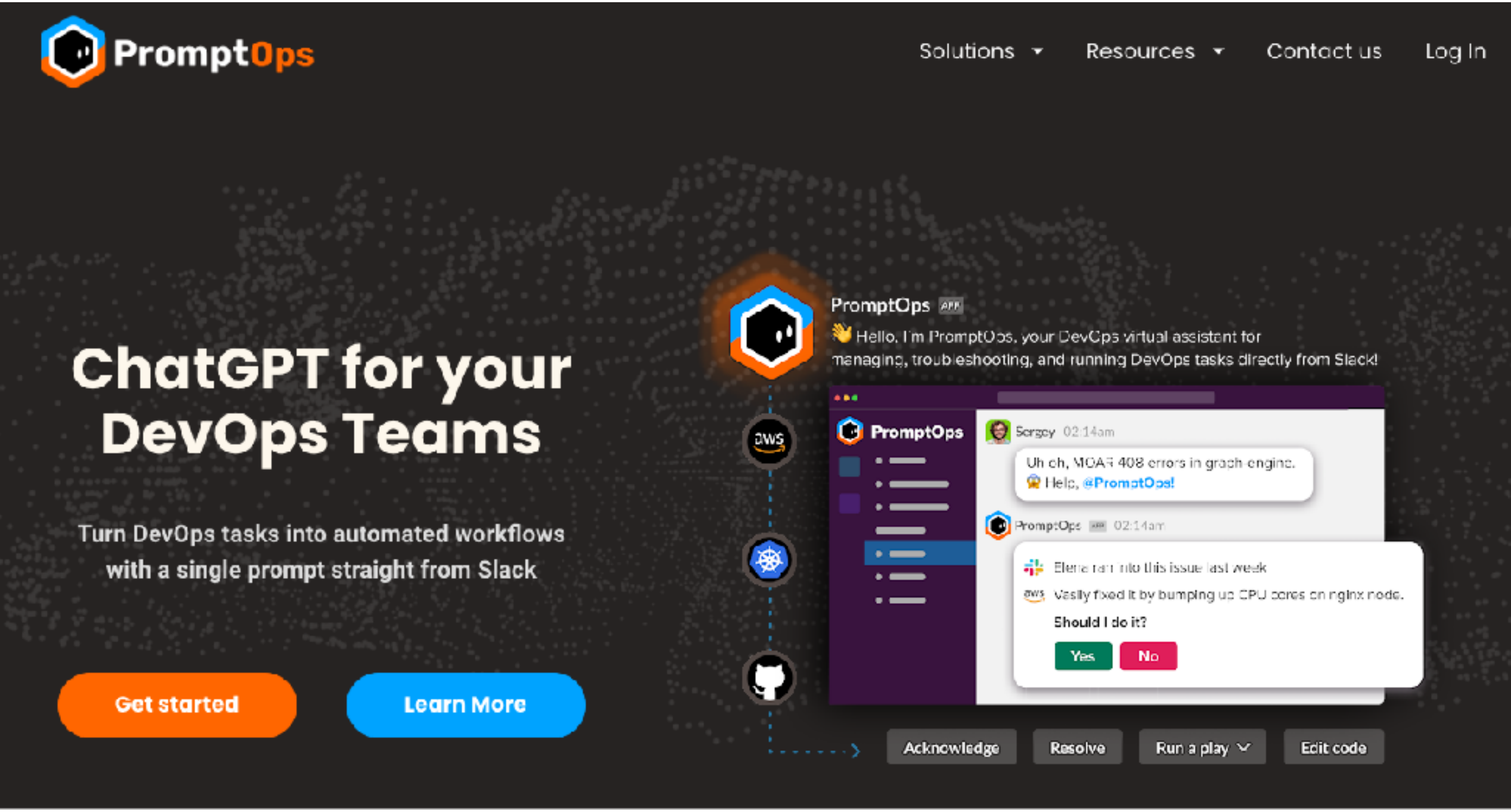
CI/CD pipeline
Infrastructure as a code
Automated script
Performance and monitoring suggestion
Document generation
AI-assist support
ChatOps, AIOps

K8sGPT



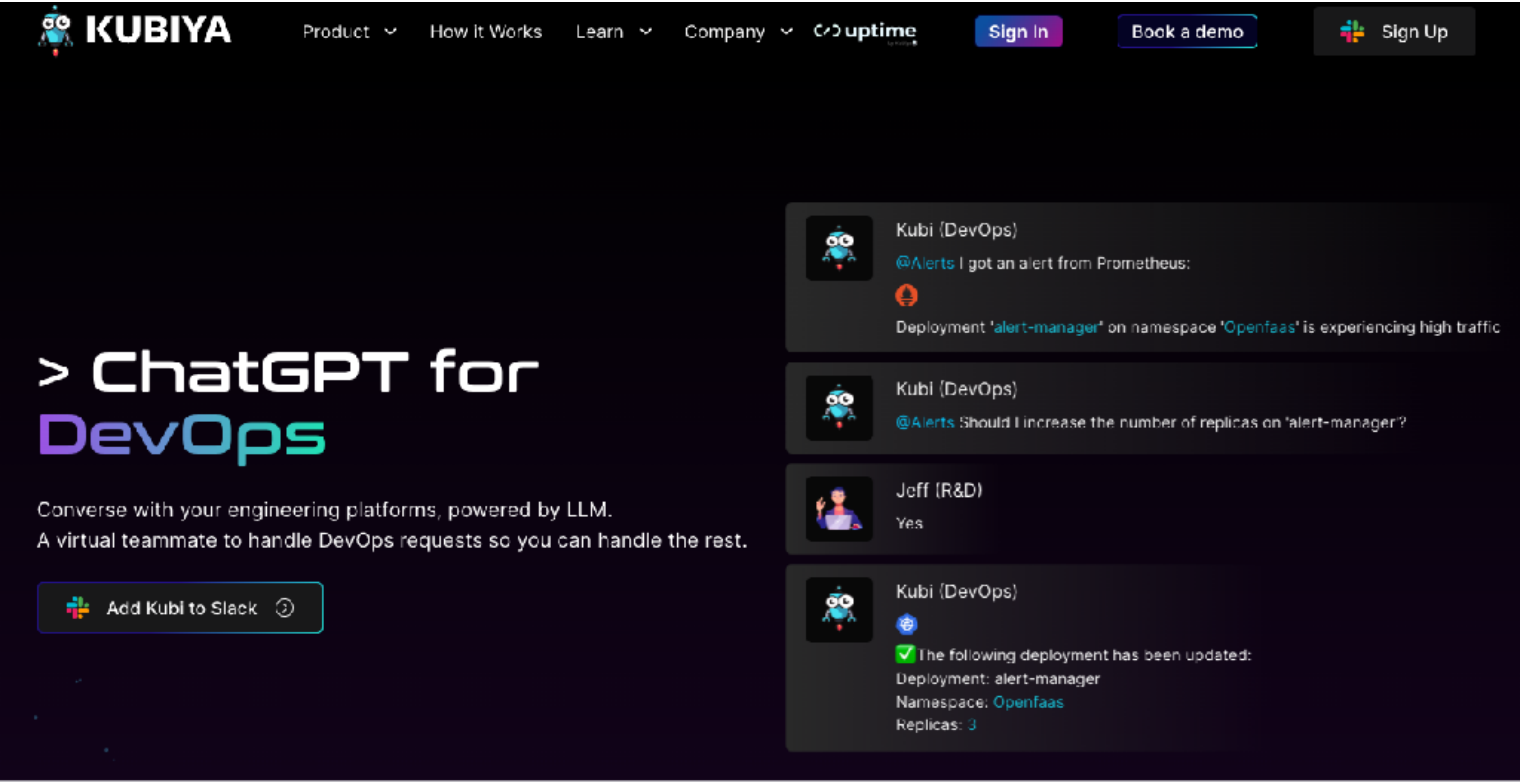
https://k8sgpt.ai/

PromptOps



https://www.promptops.com/devops/

ChatOps for DevOps



https://www.kubiya.ai/

Risks when using Generative AI

Risks

Quality of output generated
Explainability of decisions
Security policy !!
Sensitive data !!

Tips

Understand what you want

Modular approach

Clear and Precise inputs

Make sure you understand the code

Local LLM

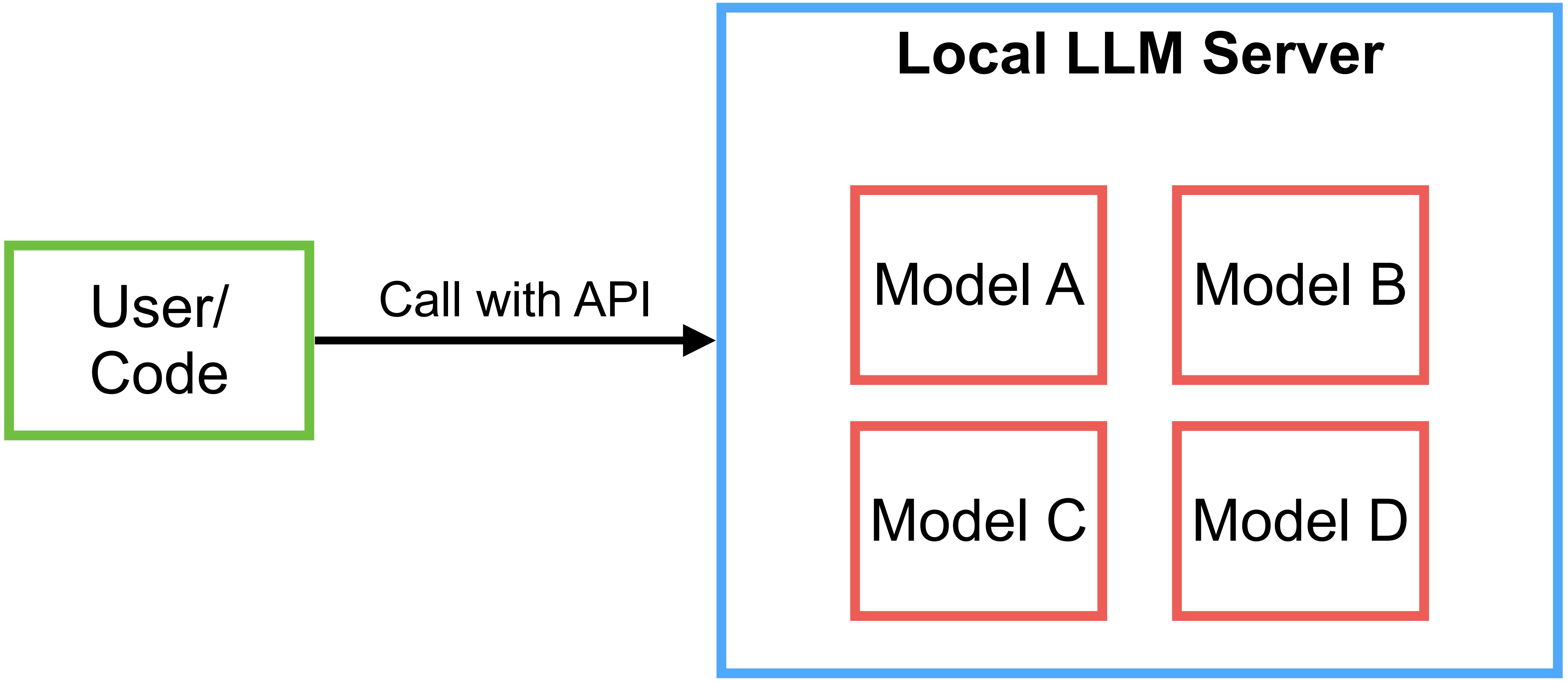
Local LLM

Run LLM on local machine/device
Try to customize with your requirement

- Reduce cost
- Data privacy
- Responsive
- Offline mode

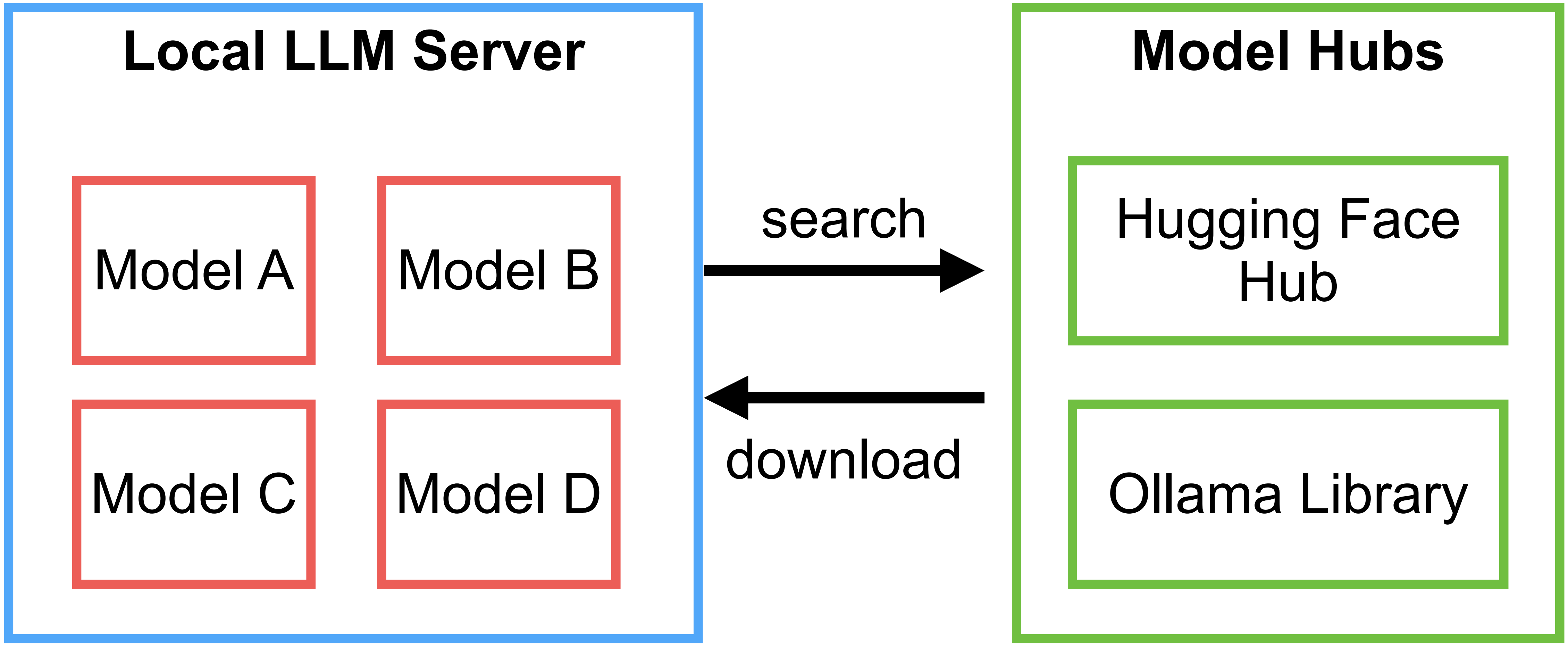
Local LLM

Improve your LLM models, more accurate answer



Models ?

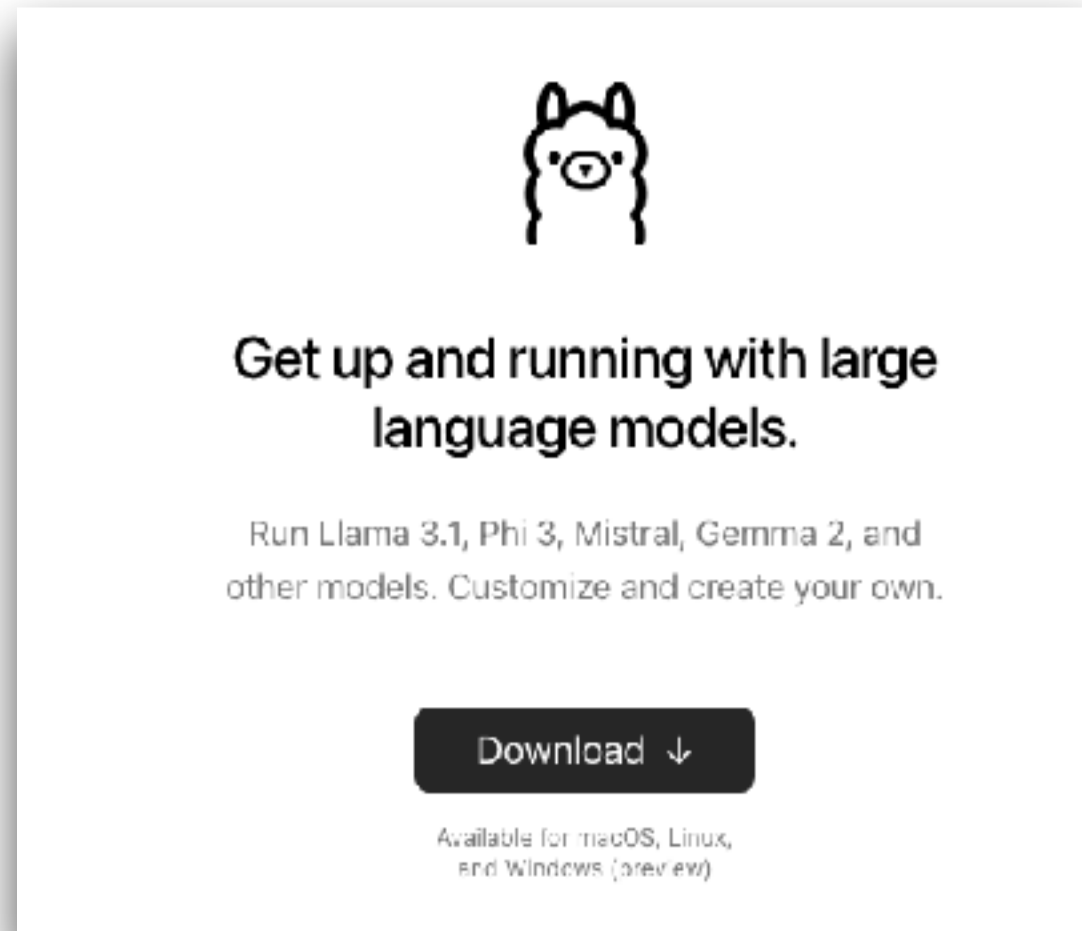
How to download models ?



Local LLM with Ollama

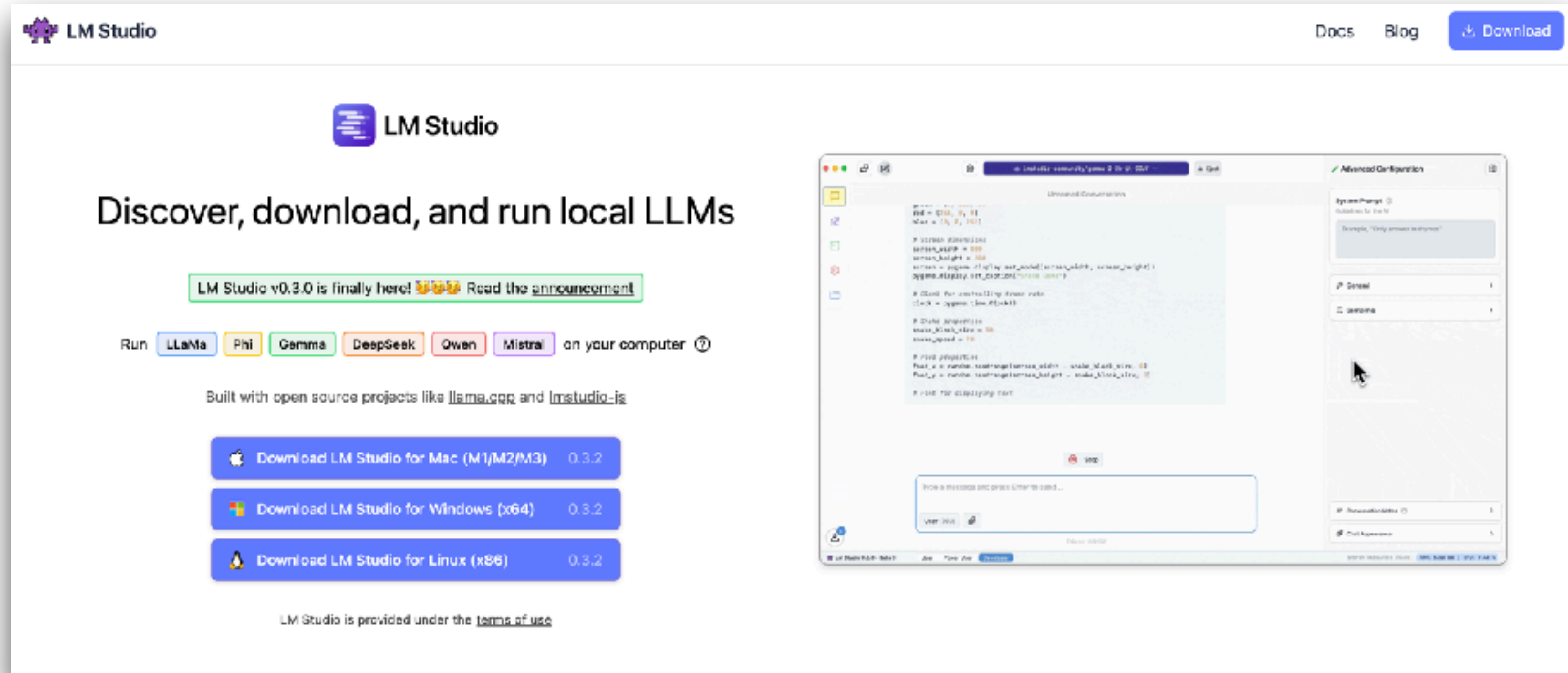
$ollama run **llama3.2**

LLM model



https://ollama.com/

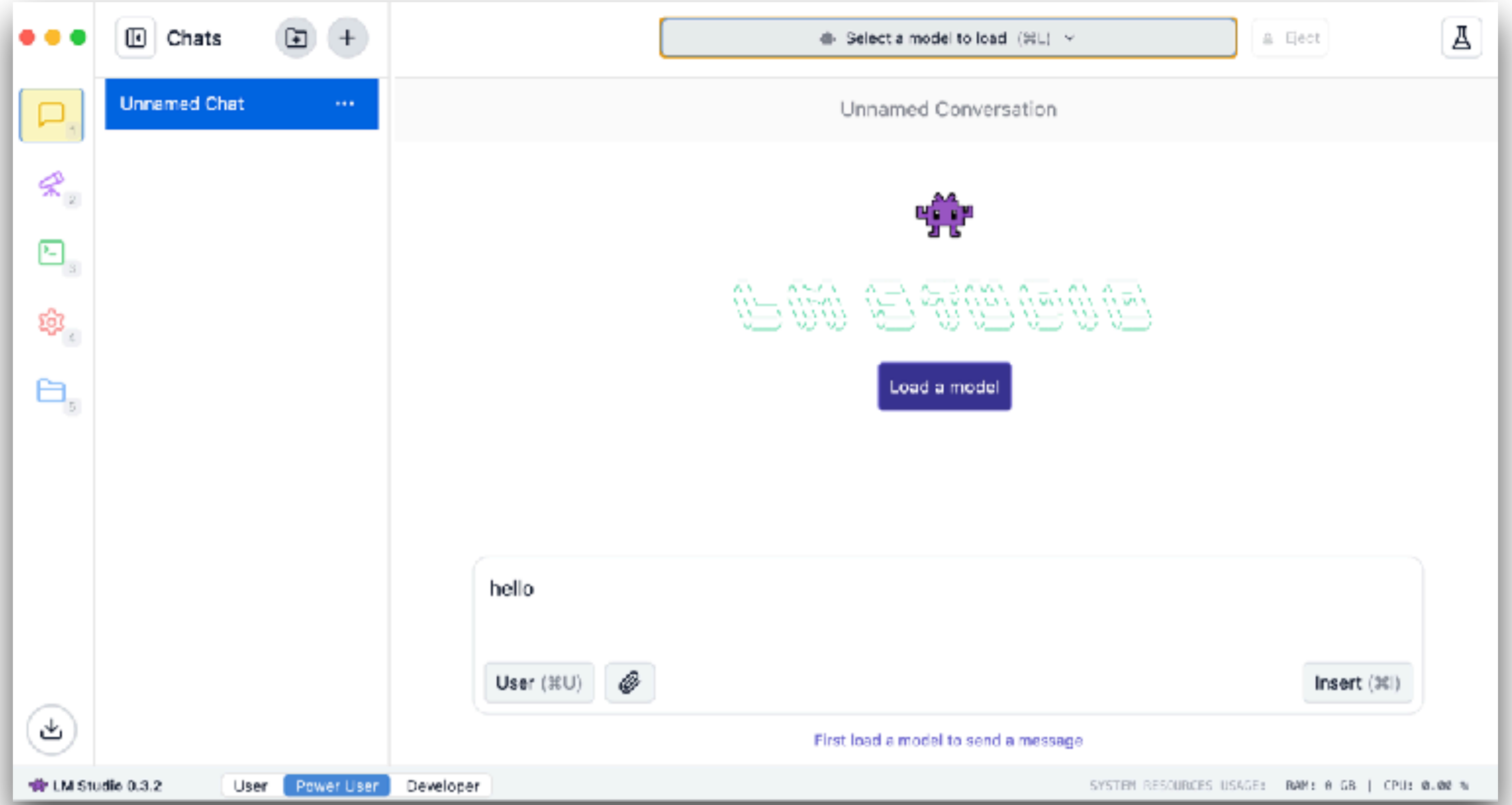
Local LLM with LM Studio



https://lmstudio.ai/

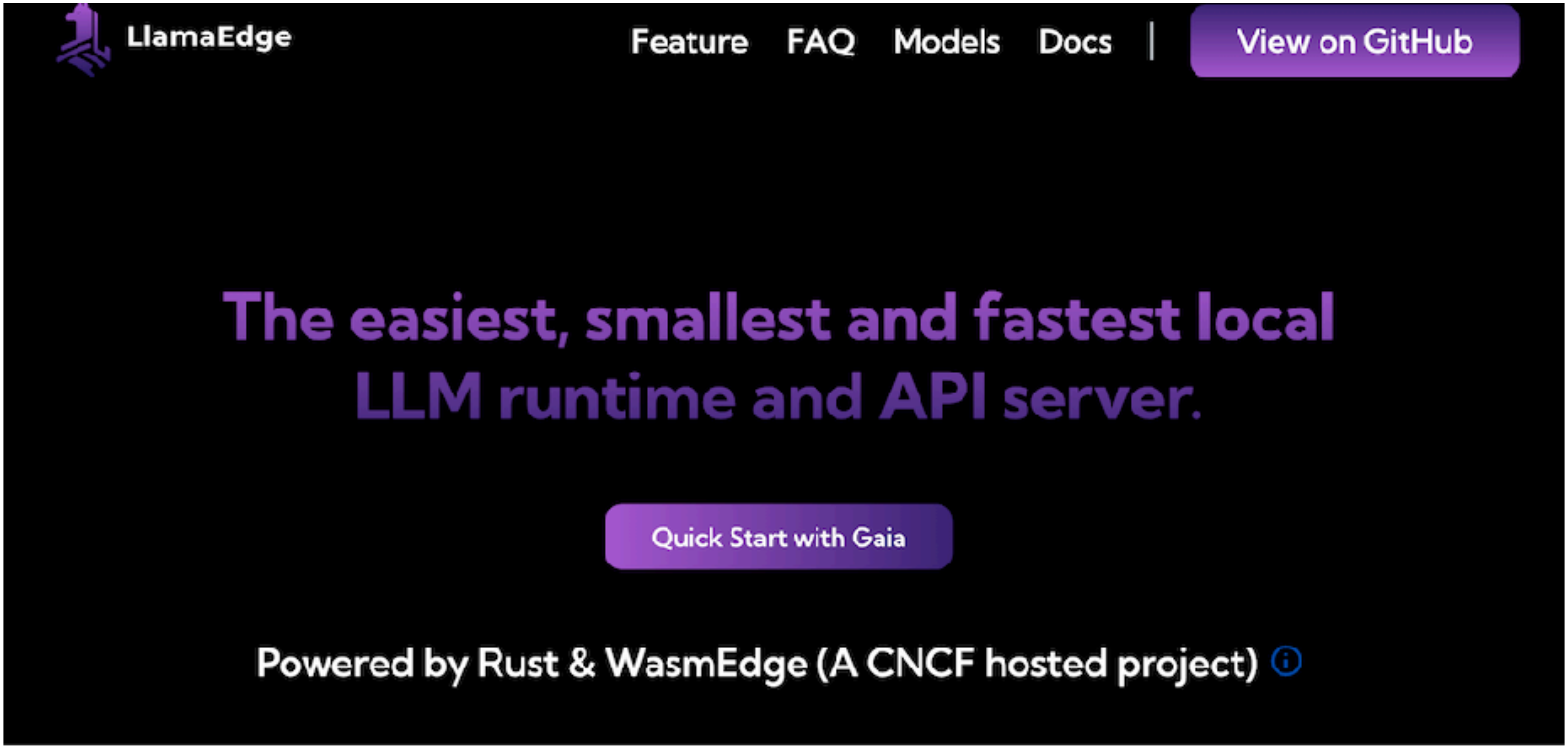
Local LLM with LM Studio

Load model from Hugging Face



https://lmstudio.ai/

Local LLM with LlamaEdge



https://llamaedge.com/

Local LLM with LocalAI

https://localai.io/

More

GPT4All

LlamaFile

Jan.ai

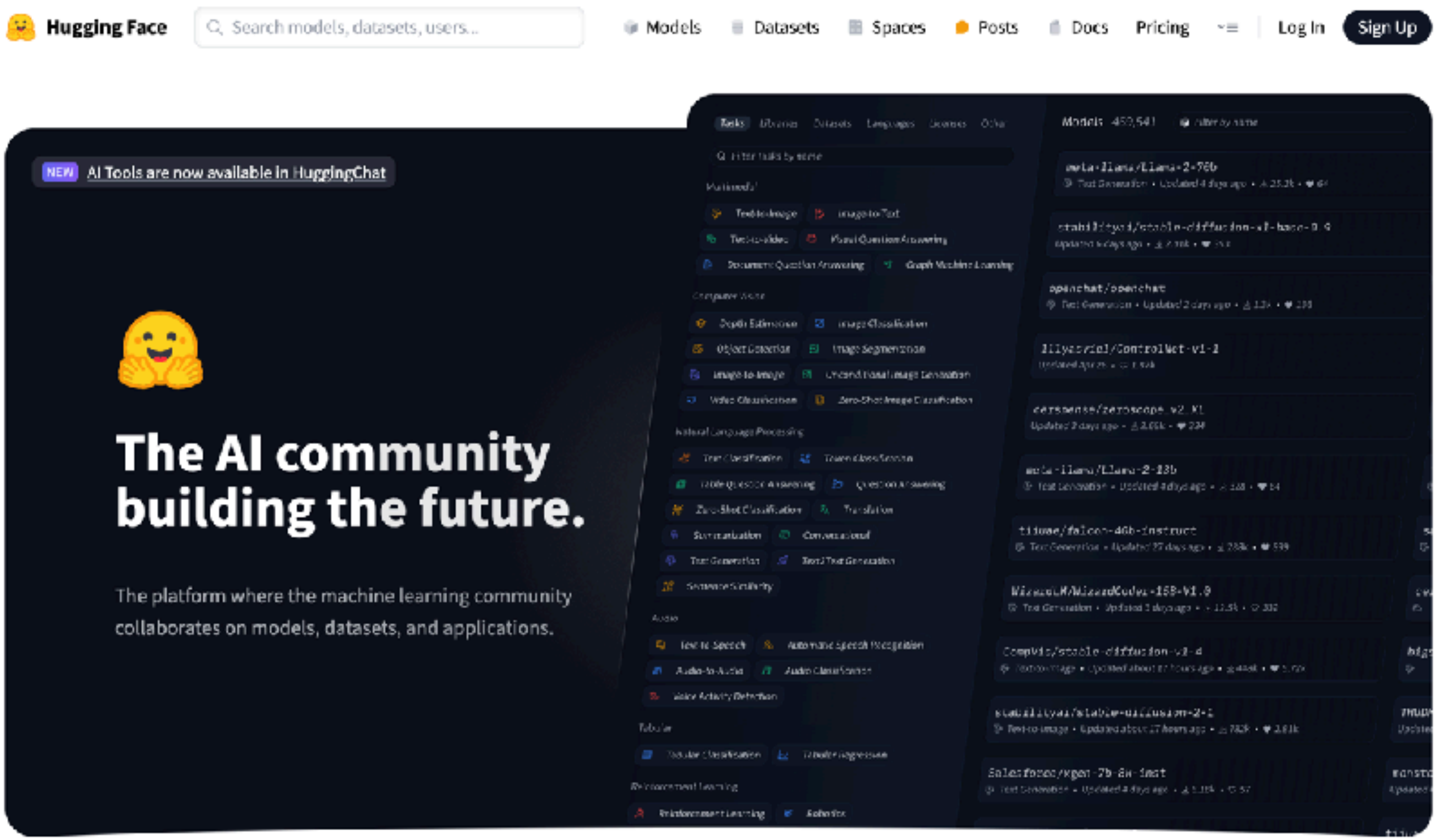
NextChat

Anything LLM

https://github.com/Hannibal046/Awesome-LLM

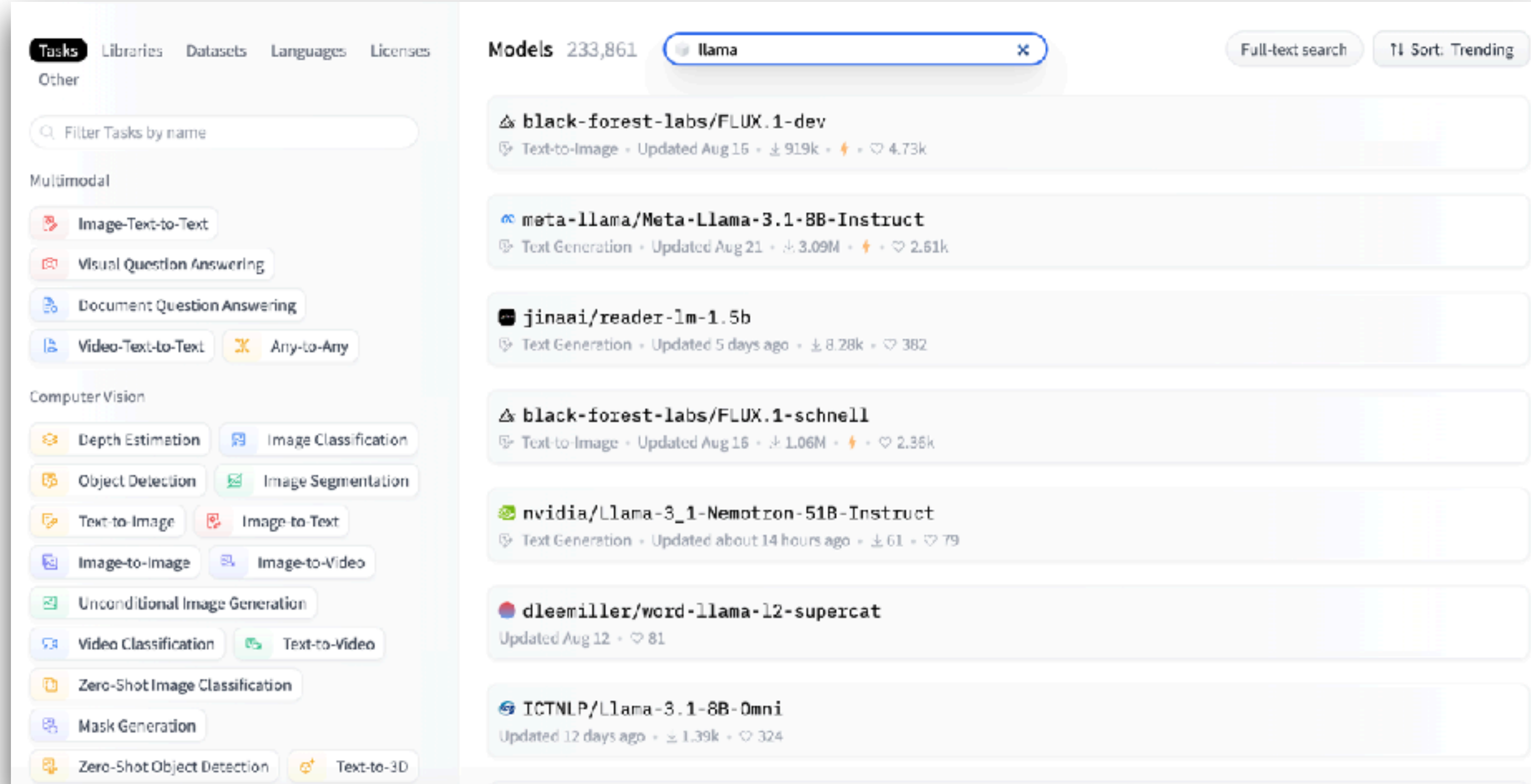
LLM Models

Hugging Face Model Hub



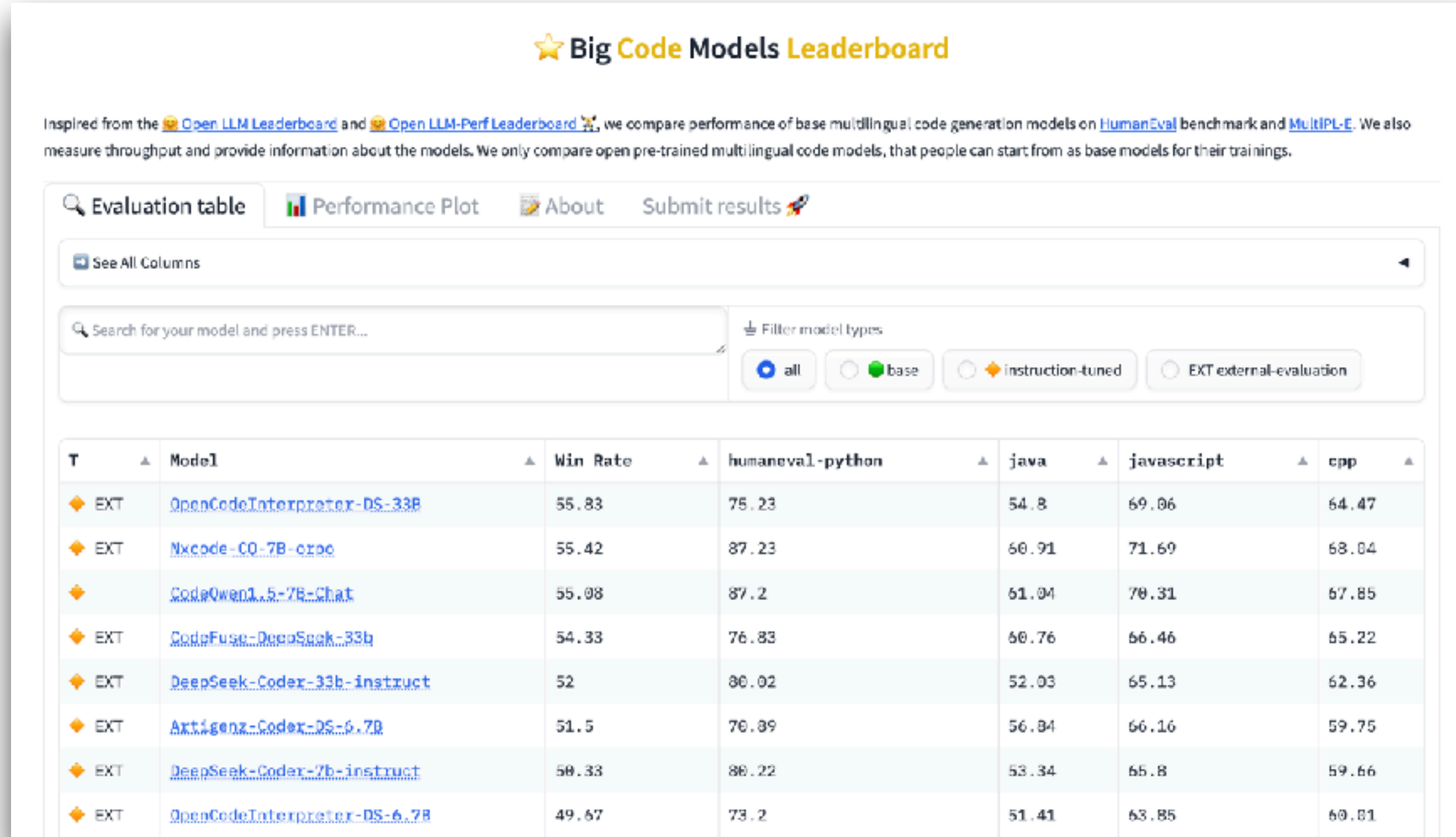
https://huggingface.co/

Hugging Face :: Model



https://huggingface.co/

Big Code model leader board



Big Code Models Leaderboard

Inspired from the Open LLM Leaderboard and Open LLM-Perf Leaderboard, we compare performance of base multilingual code generation models on HumanEval benchmark and MultiPL-E. We also measure throughput and provide information about the models. We only compare open pre-trained multilingual code models, that people can start from as base models for their trainings.

Evaluation table | Performance Plot | About | Submit results

See All Columns

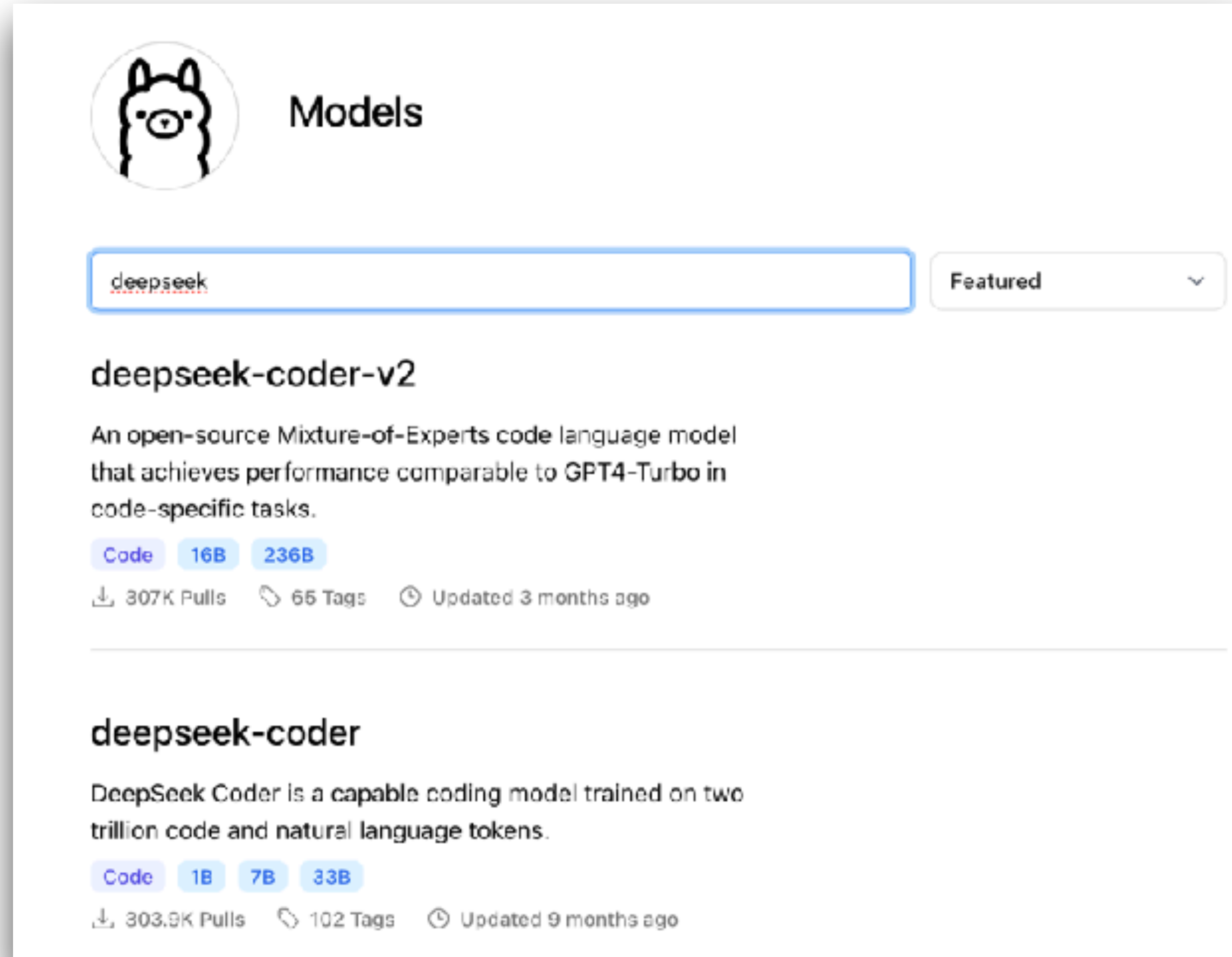
Search for your model and press ENTER...

Filter model types: all | base | instruction-tuned | EXT external-evaluation

T	Model	Win Rate	humaneval-python	java	javascript	cpp
EXT	OpenCodeInterpreter-DS-33B	55.83	75.23	54.8	69.06	64.47
EXT	Nxcode-CQ-7B-orpo	55.42	87.23	60.91	71.69	68.04
	CodeQwen1.5-7B-Chat	55.08	87.2	61.04	70.31	67.85
EXT	CodeFuse-DeepSeek-33b	54.33	76.83	60.76	66.46	65.22
EXT	DeepSeek-Coder-33b-instruct	52	80.02	52.03	65.13	62.36
EXT	Artigenz-Coder-DS-6.7B	51.5	70.89	56.84	66.16	59.75
EXT	DeepSeek-Coder-7b-instruct	50.33	80.22	53.34	65.8	59.66
EXT	OpenCodeInterpreter-DS-6.7B	49.67	73.2	51.41	63.85	60.81

https://huggingface.co/spaces/bigcode/bigcode-models-leaderboard

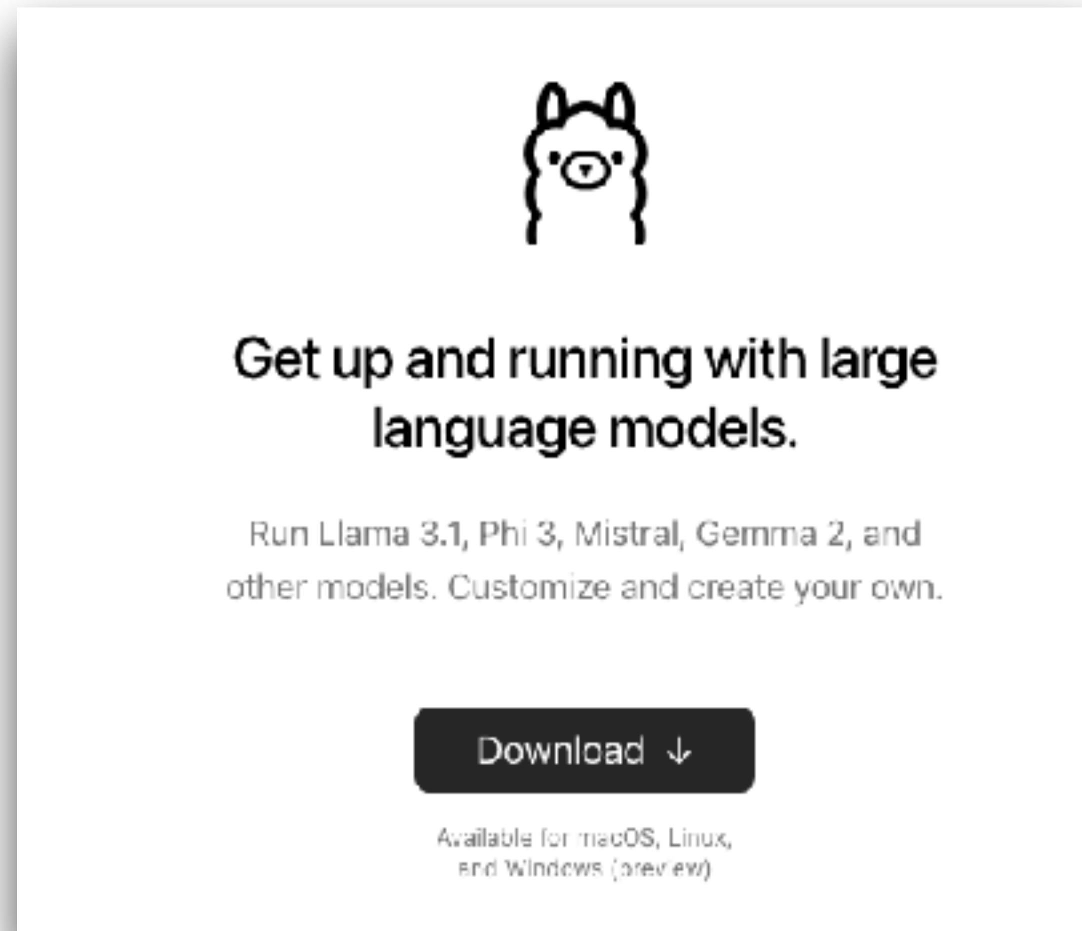
Model in Ollama



https://ollama.com/library

Workshop with Ollama

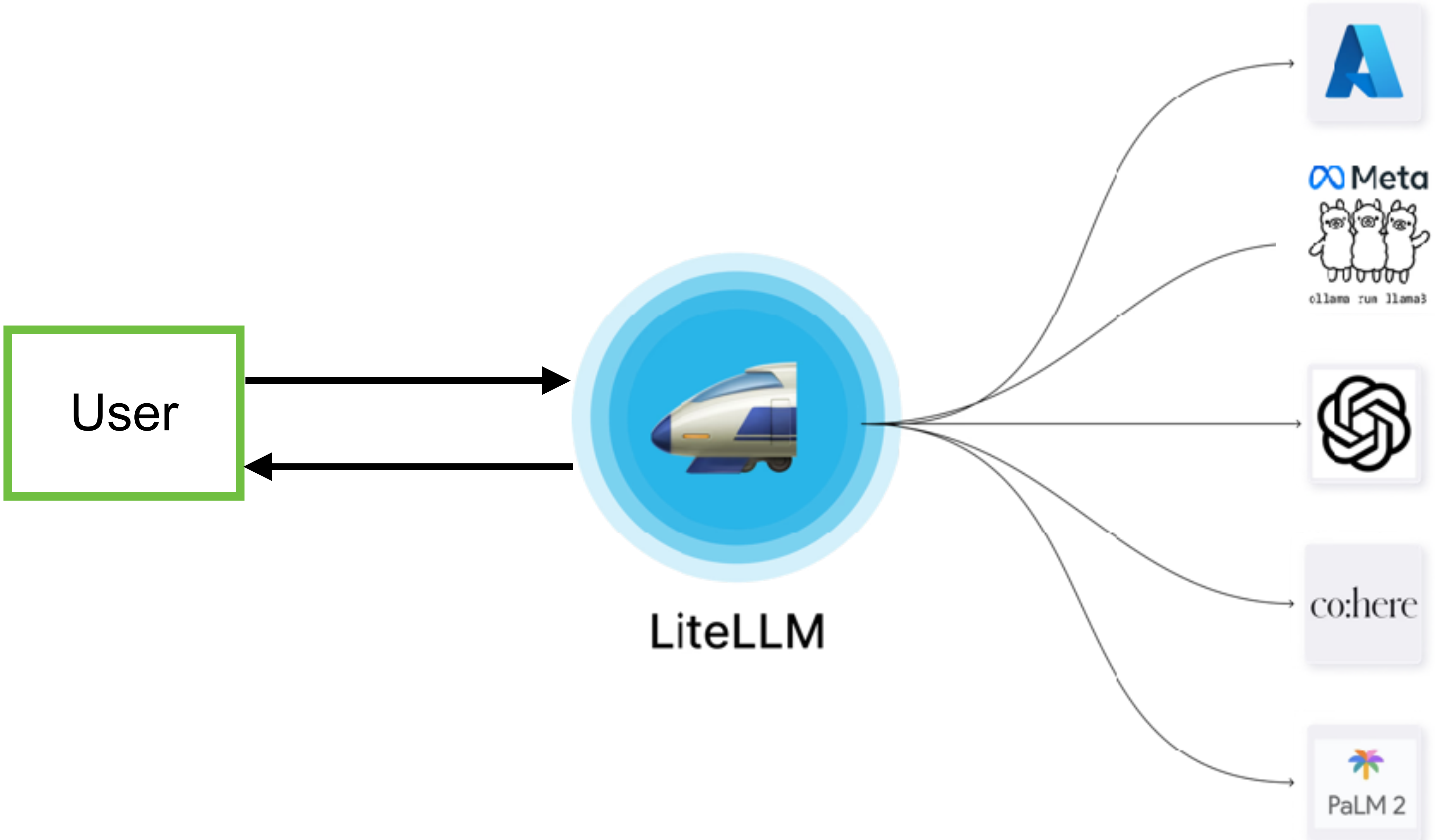
$ollama run **llama3.1**



https://github.com/up1/workshop-ai-with-technical-team/wiki/Local-LLM-with-Ollama

LiteLLM as a Proxy

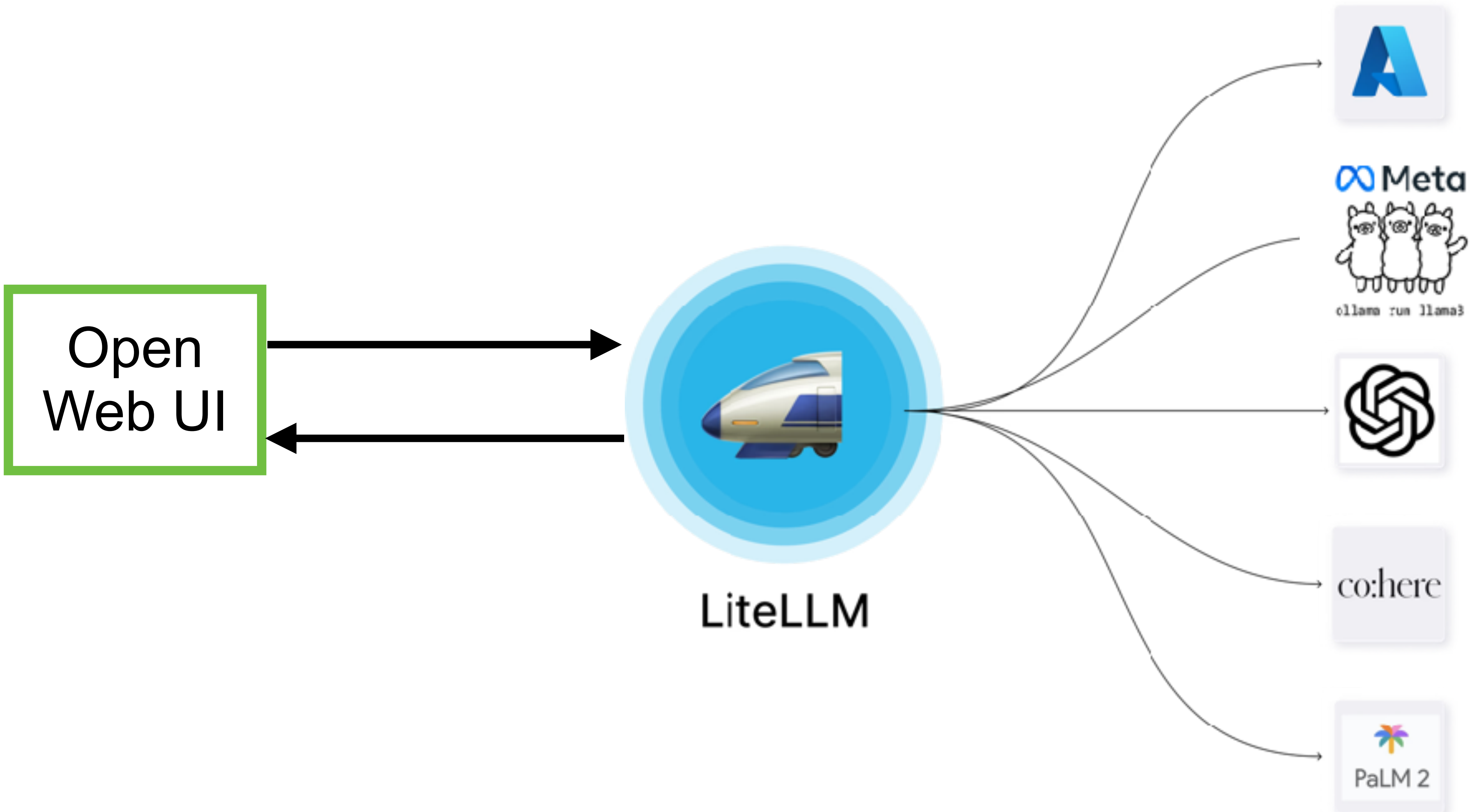
LiteLLM as a Proxy



https://www.litellm.ai/

Workshop

Use docker compose to build and run

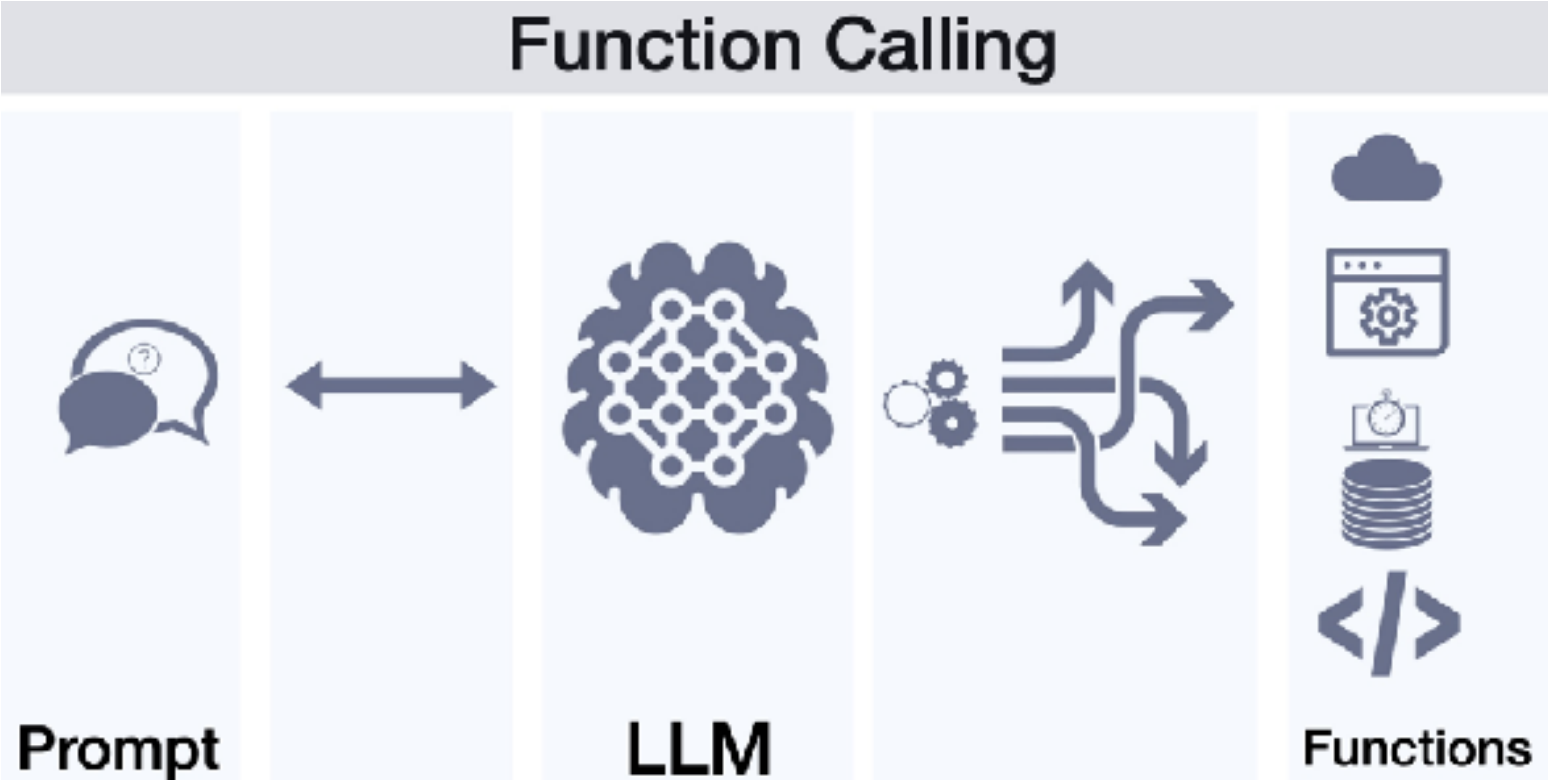


https://github.com/up1/workshop-ai-with-technical-team/wiki/LiteLLM-and-WebUI

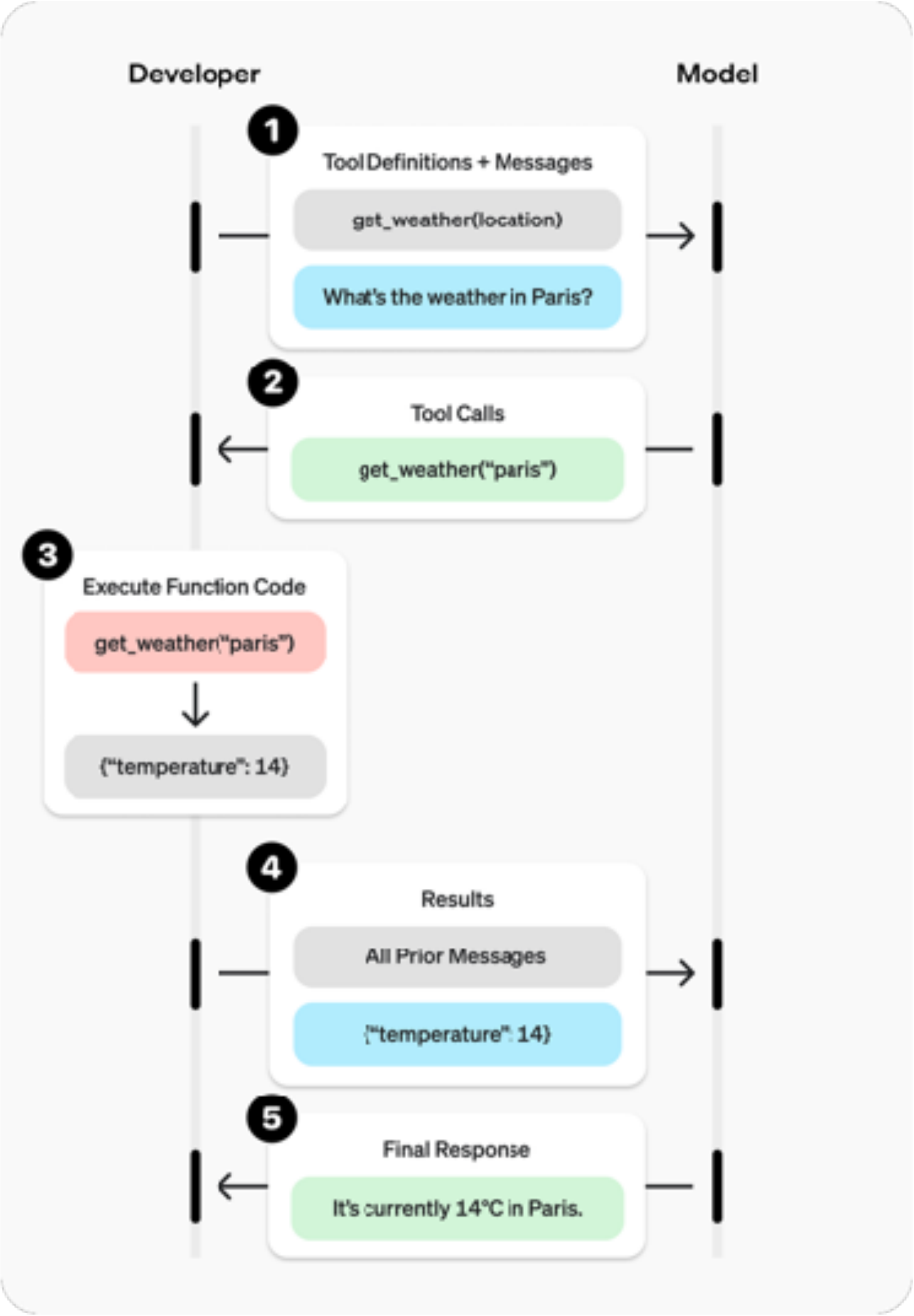
Function Calling ?

Function Calling ?

Allow LLM to recognize what tool it need based on user's input and when to invoke it.

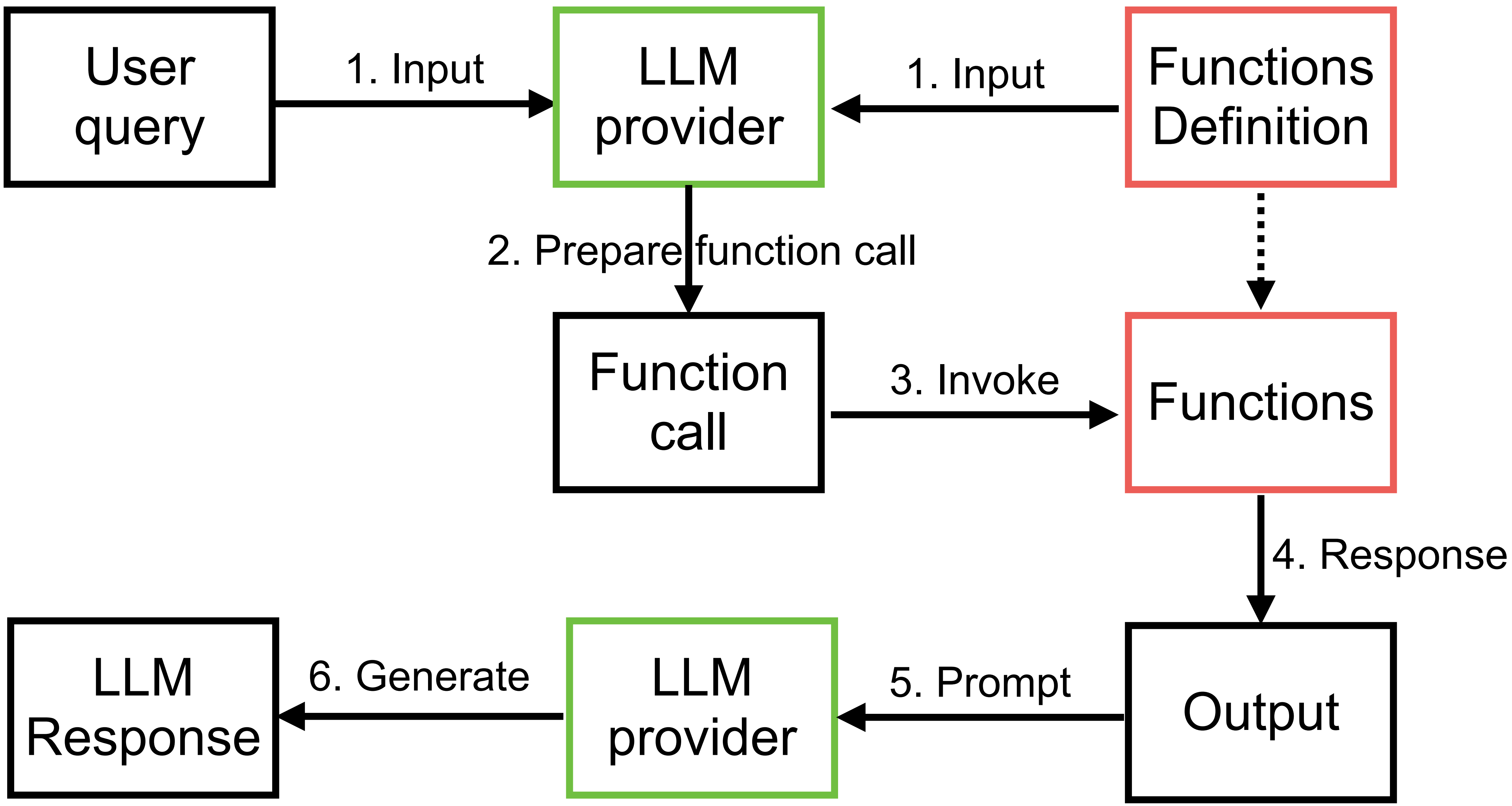


Function calling ?



https://platform.openai.com/docs/guides/function-calling?api-mode=responses

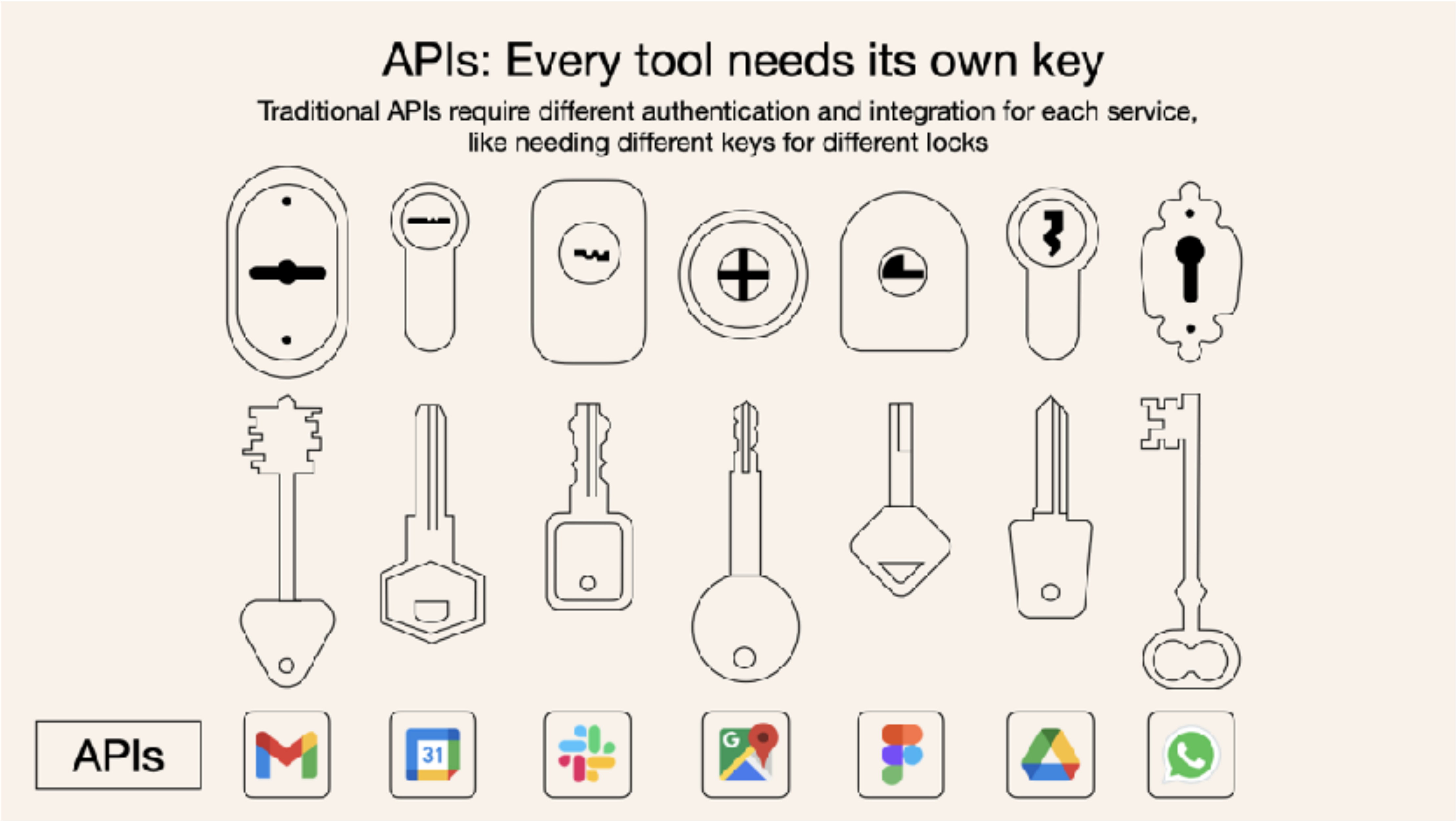
Function Calling



Support function calling

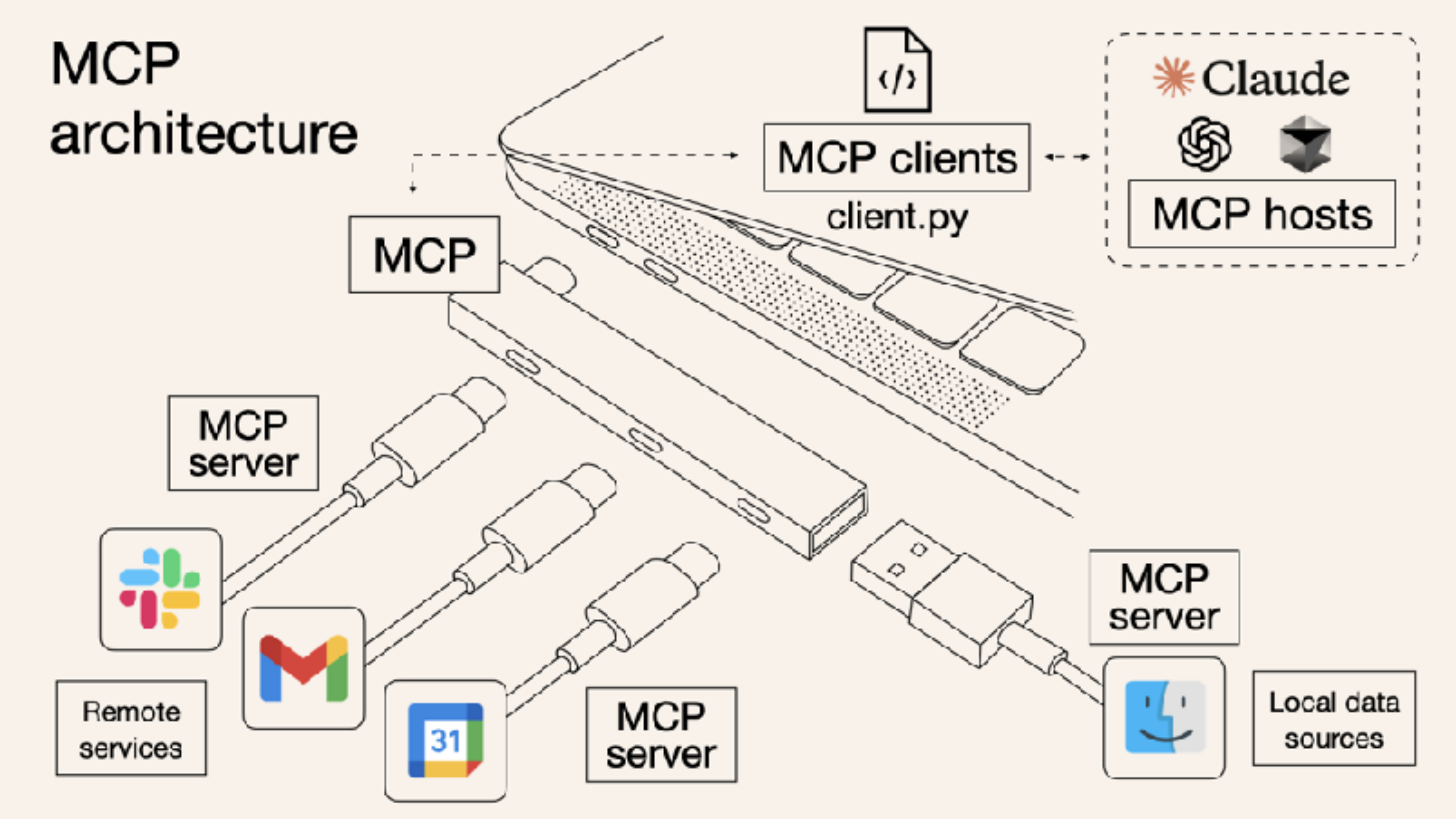
Provider name	Model name
OpenAI	GPT 4
Anthropic	Claude 3 (Sonnet, Haiku, Opus)
Google	Gemini

Function calling !!



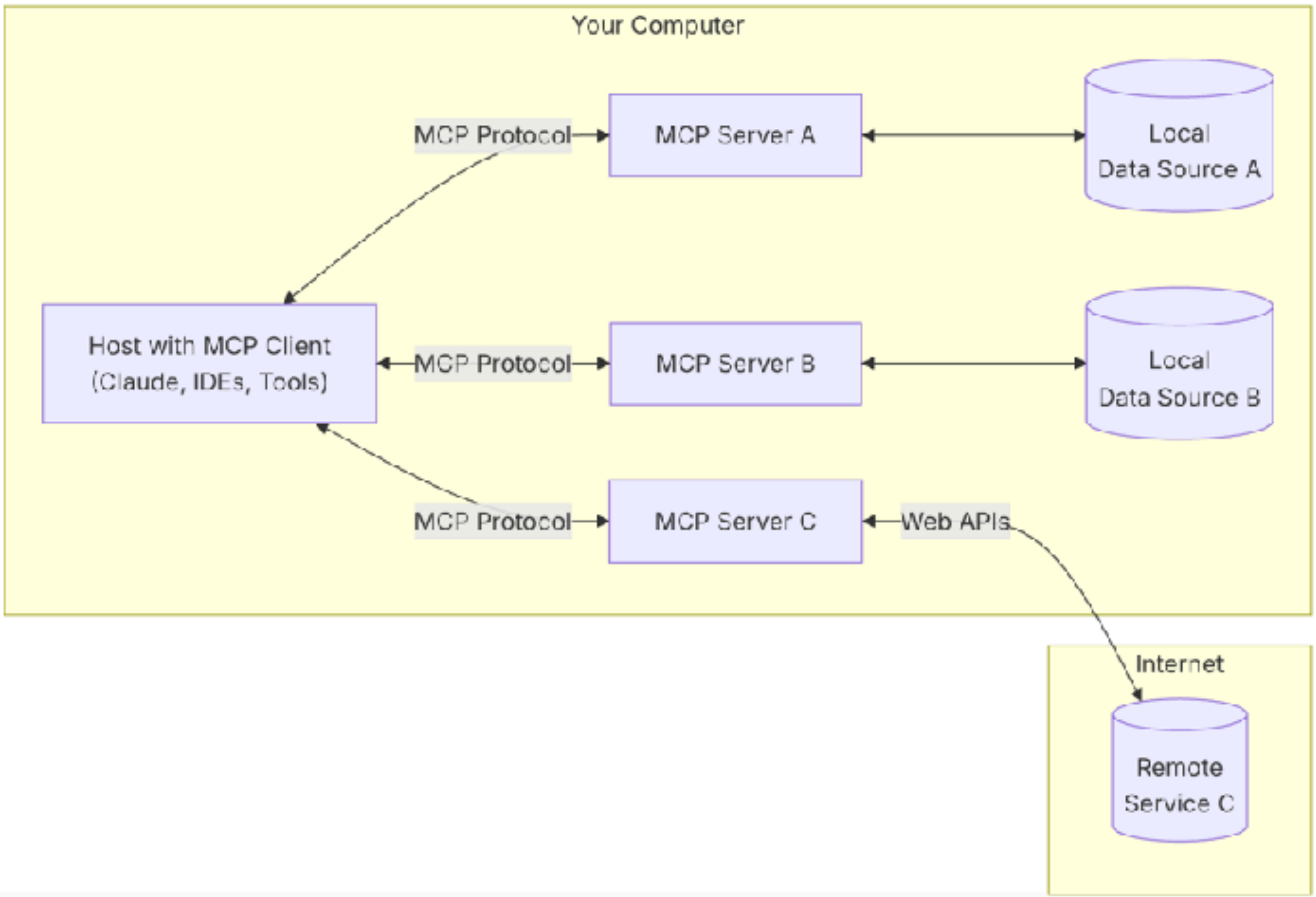
https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/

Standardized function calling !!



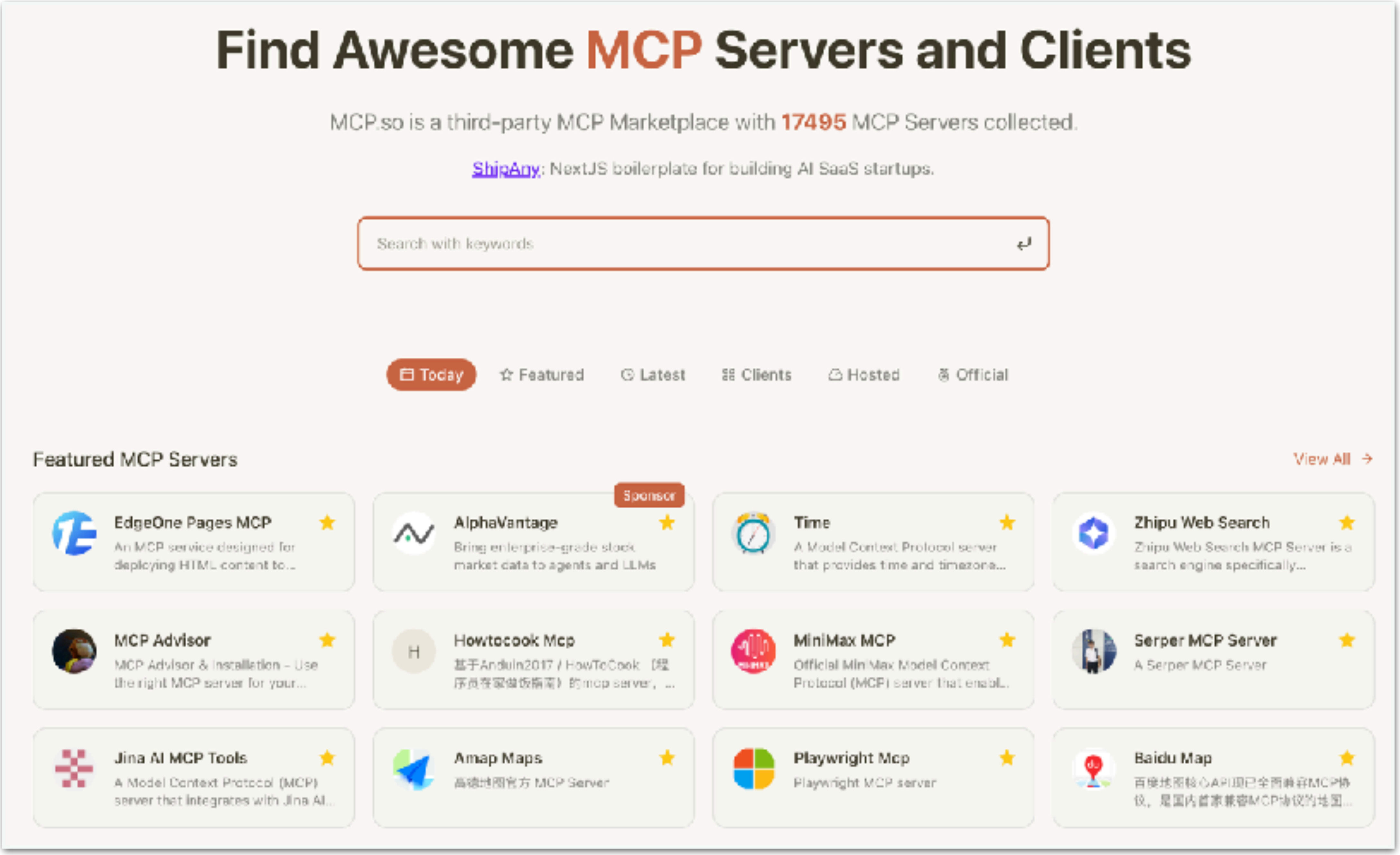
https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/

Model Context Protocol



https://modelcontextprotocol.io/

Awesome MCP



https://mcp.so/

Retrieval-Augmented Generation (RAG)

RAG

Enhances LLMs by retrieving external knowledge before generating response

Improve accuracy	Reduce hallucinations	Real-time knowledge updates

Core Components

Retriever

Fetch relevant documents from a knowledge base

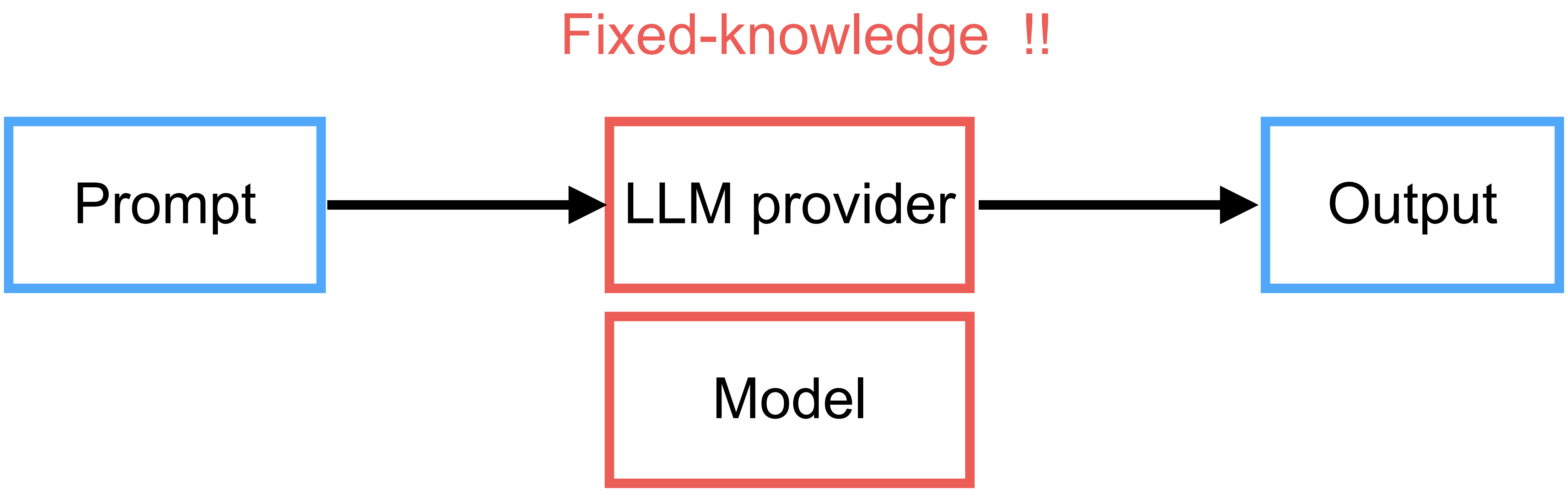
Generator

Uses the retrieved information to generate a response

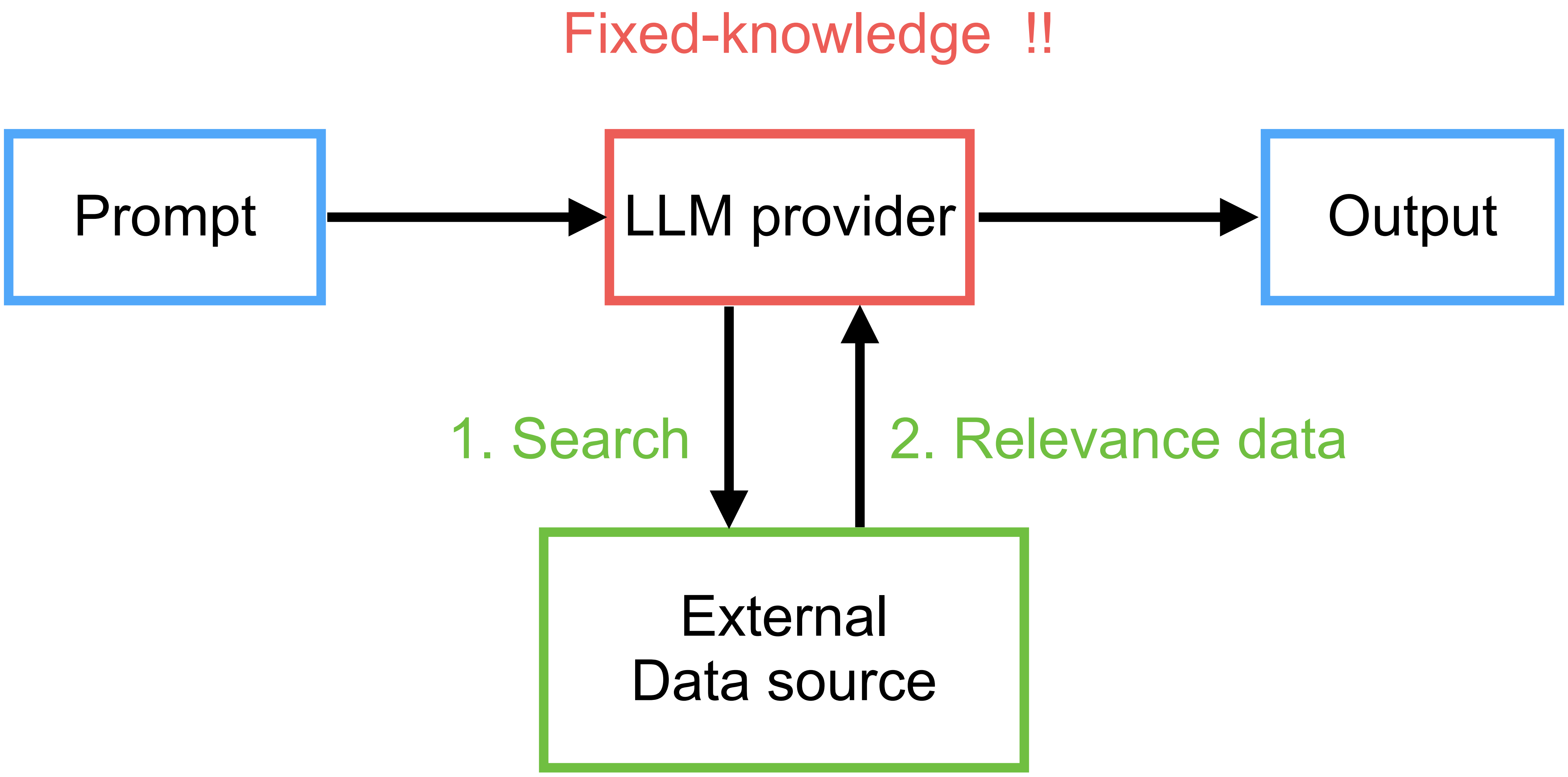
Enhancements

Different RAG architectures modify how retrieval and generation interact to improve performance

Limitation of LLM



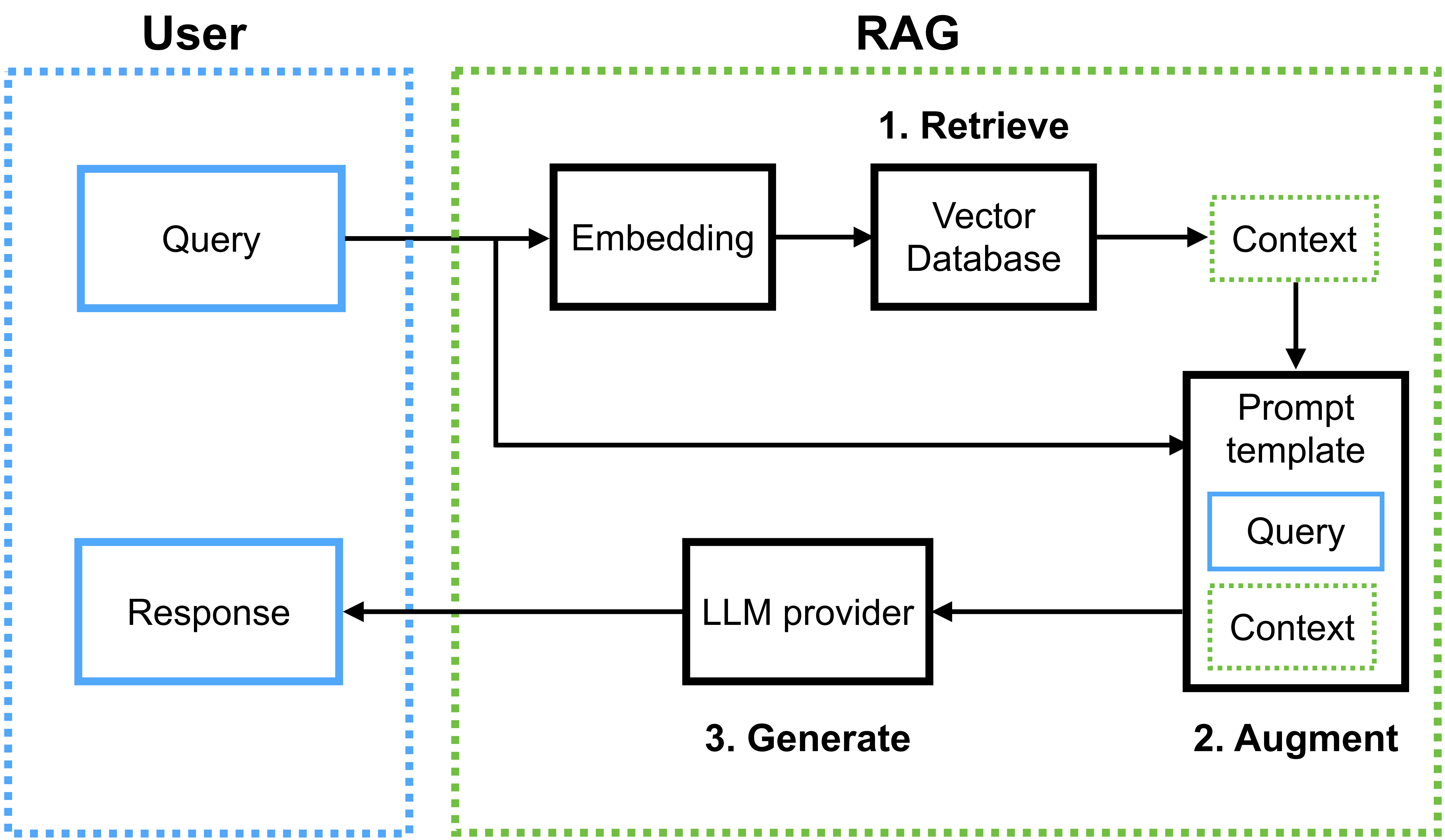
RAG ?


Fixed-knowledge !!
Prompt
LLM provider
Output
1. Search
2. Relevance data
External
Data source

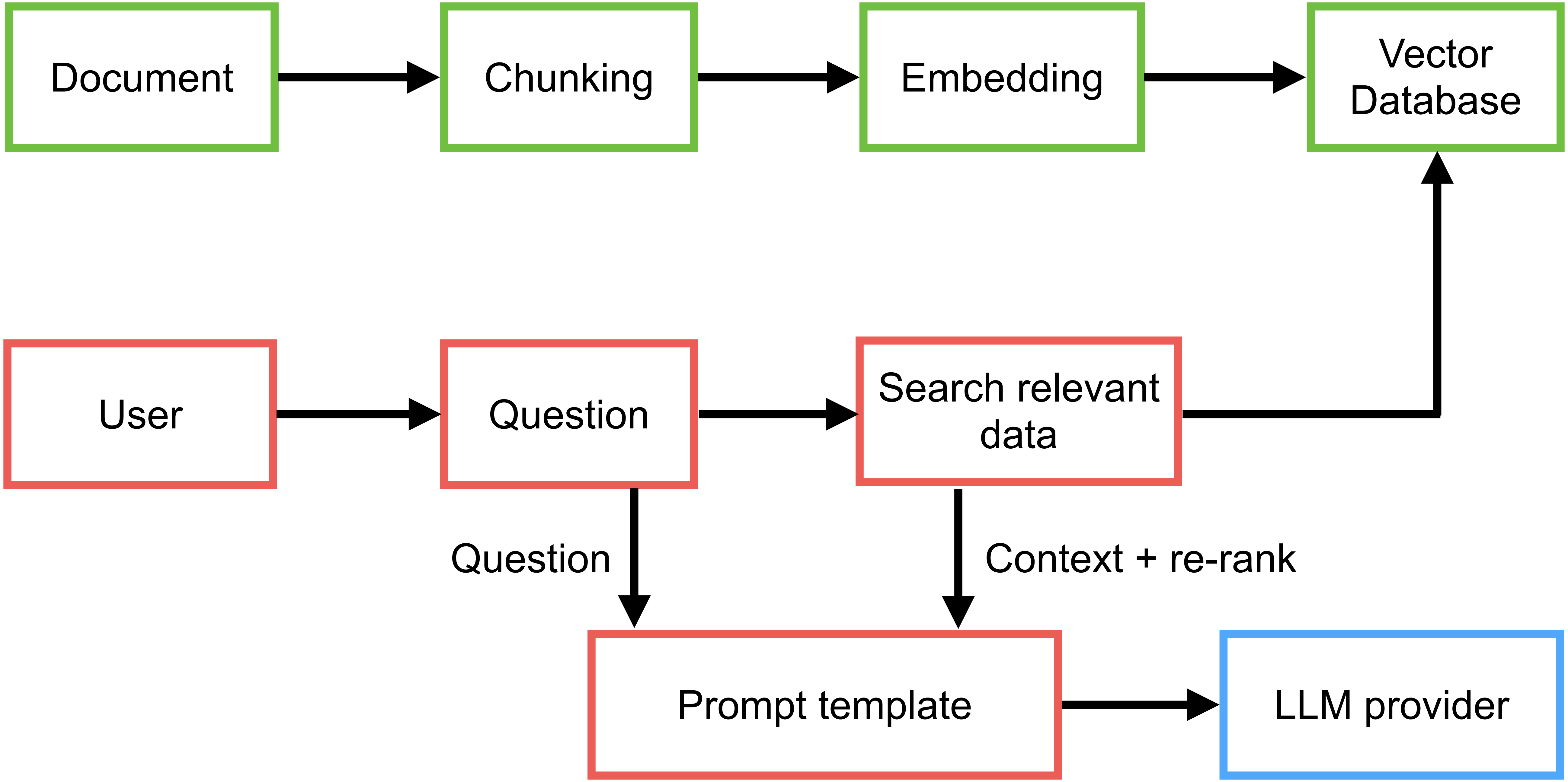
How to search/retrieve data ?

Data source	Data preparation
Search	Search result

Basic RAG Architecture

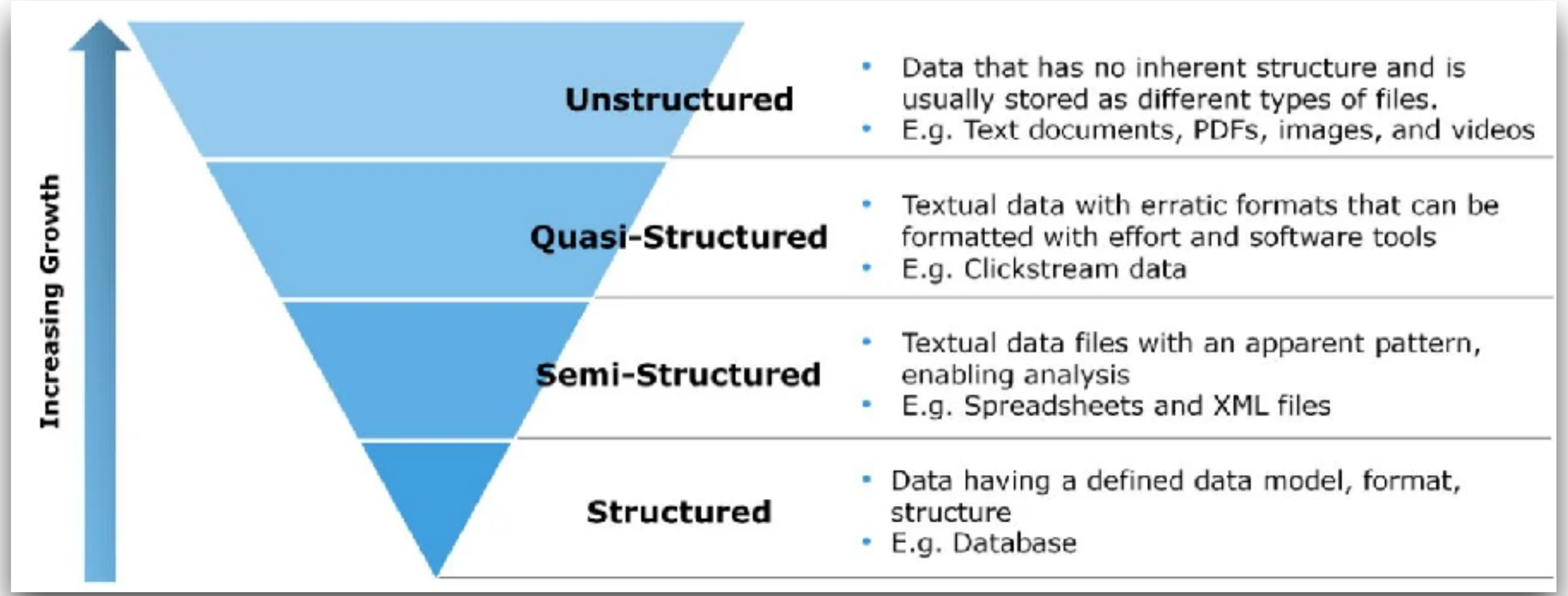


RAG process

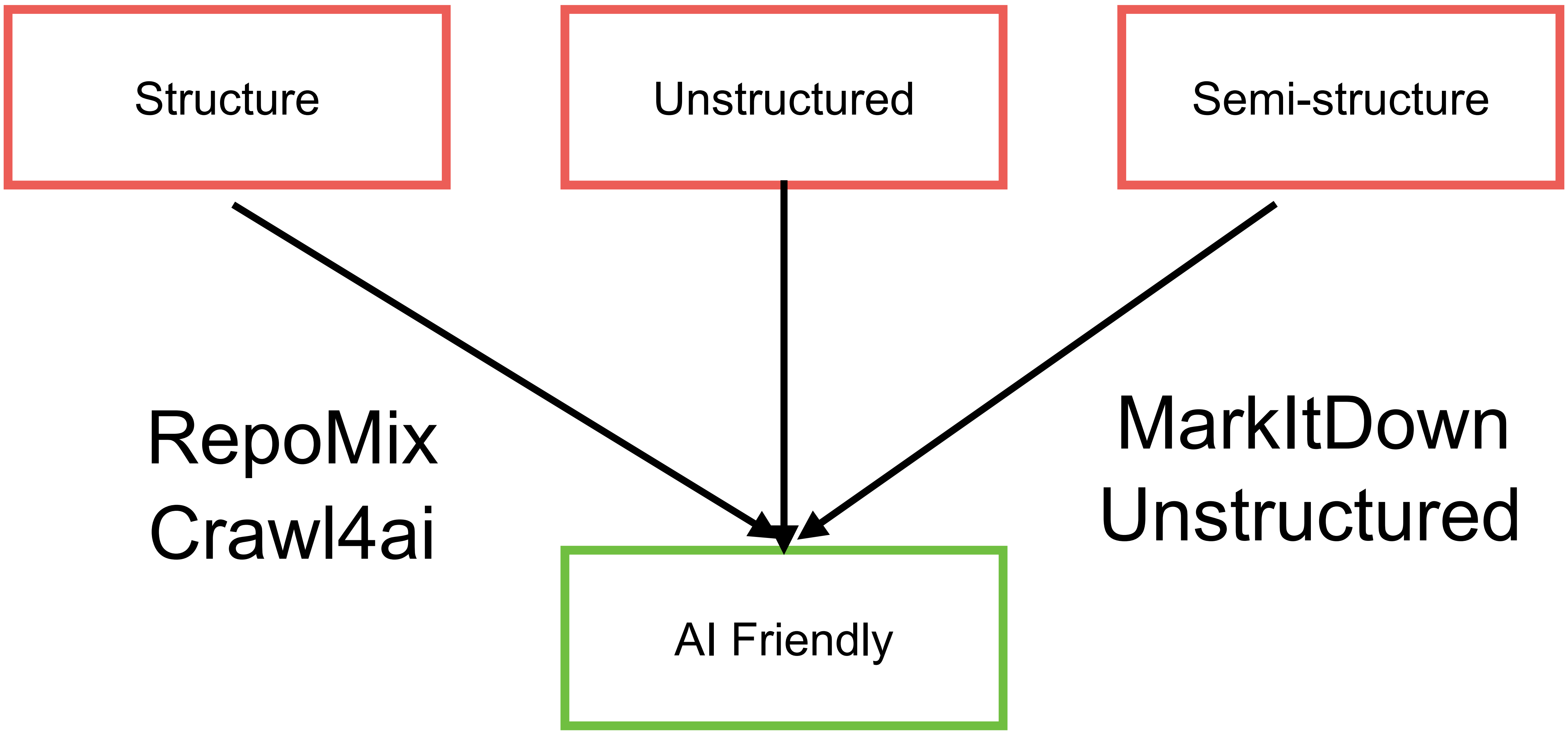


Document
Chunking
Embedding
Vector Database
User
Question
Search relevant data
Question
Context + re-rank
Prompt template
LLM provider

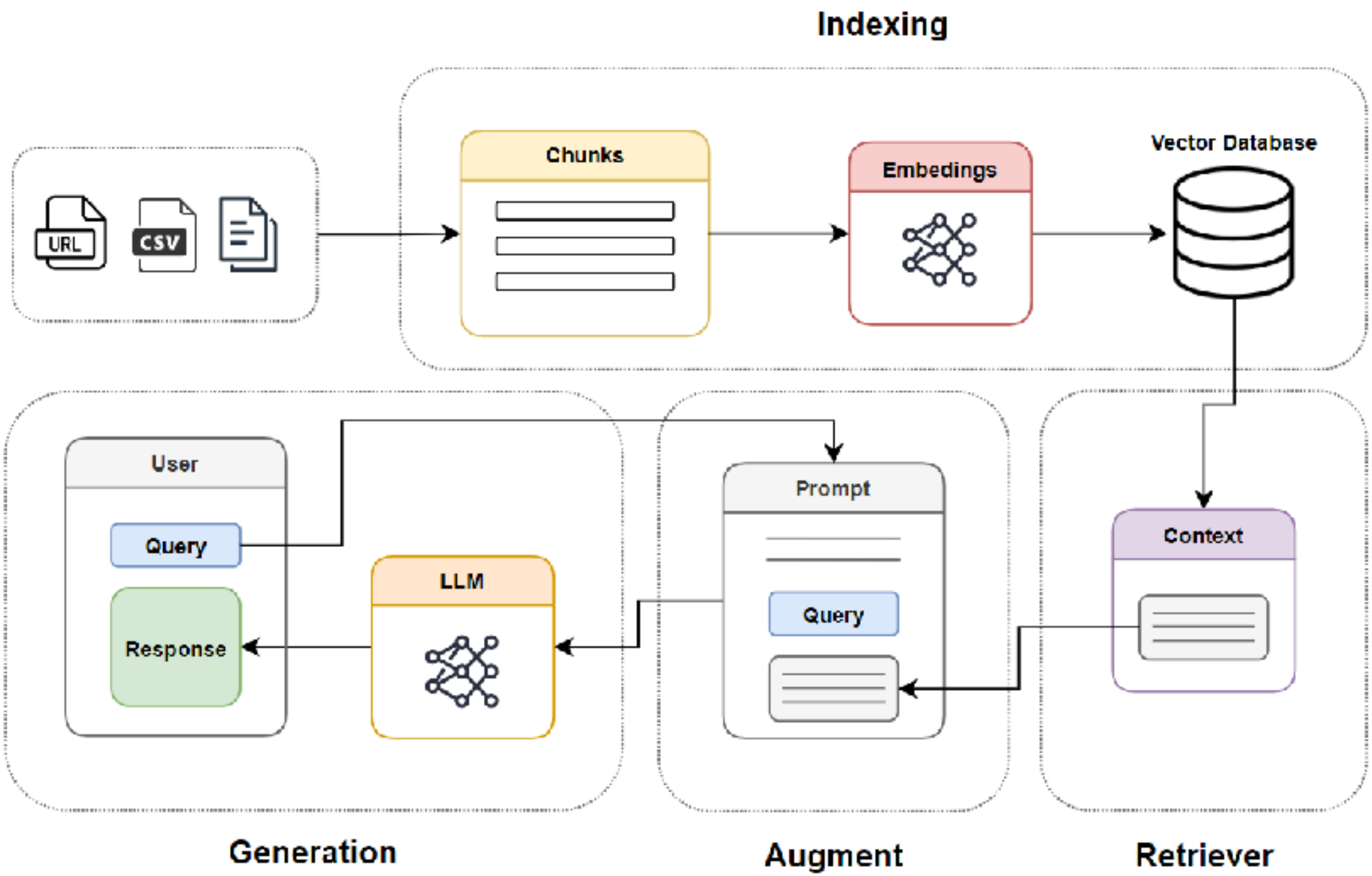
Structures of Data ?



Friendly Data for LLM/AI

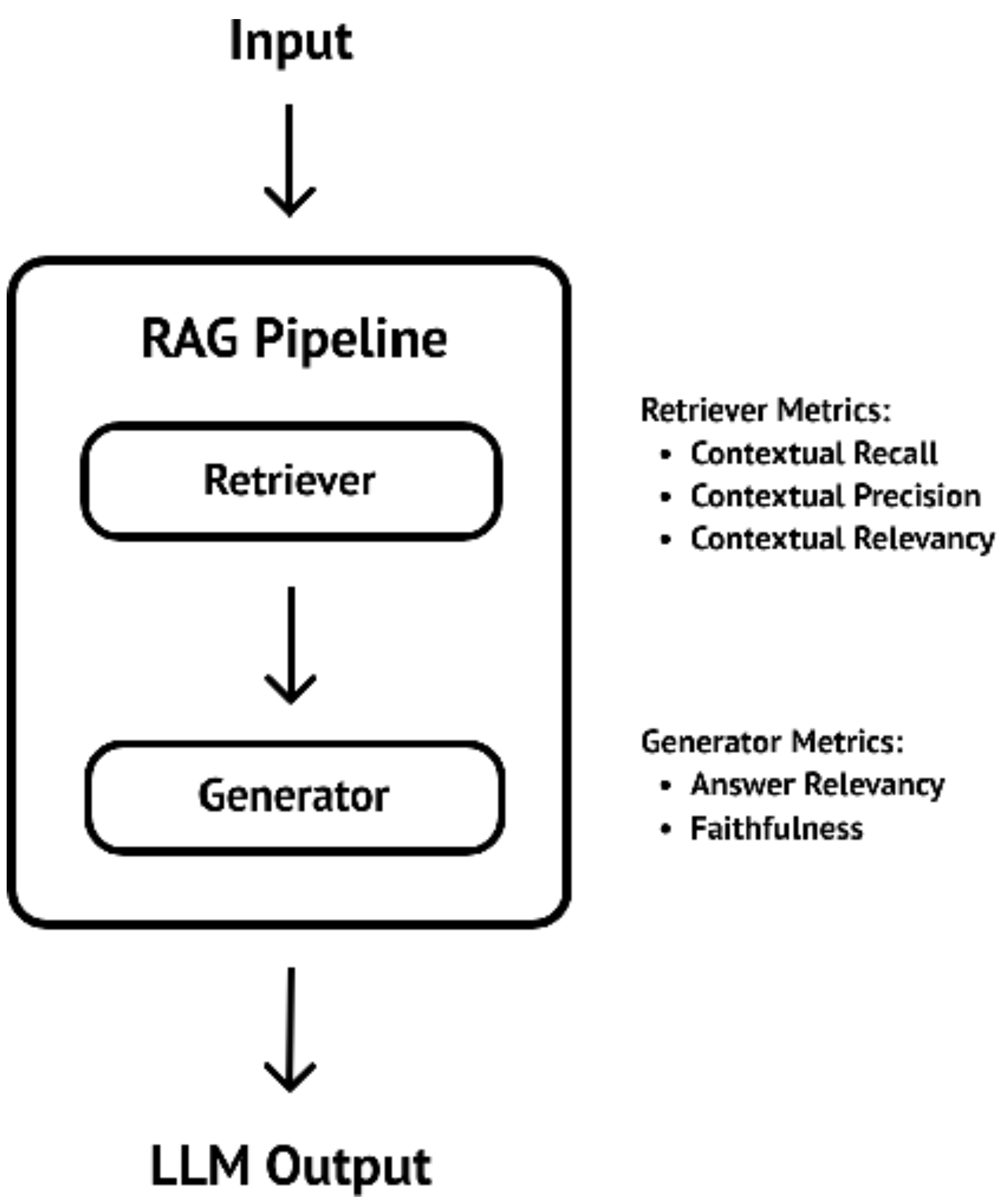


RAG Cookbook (techniques)

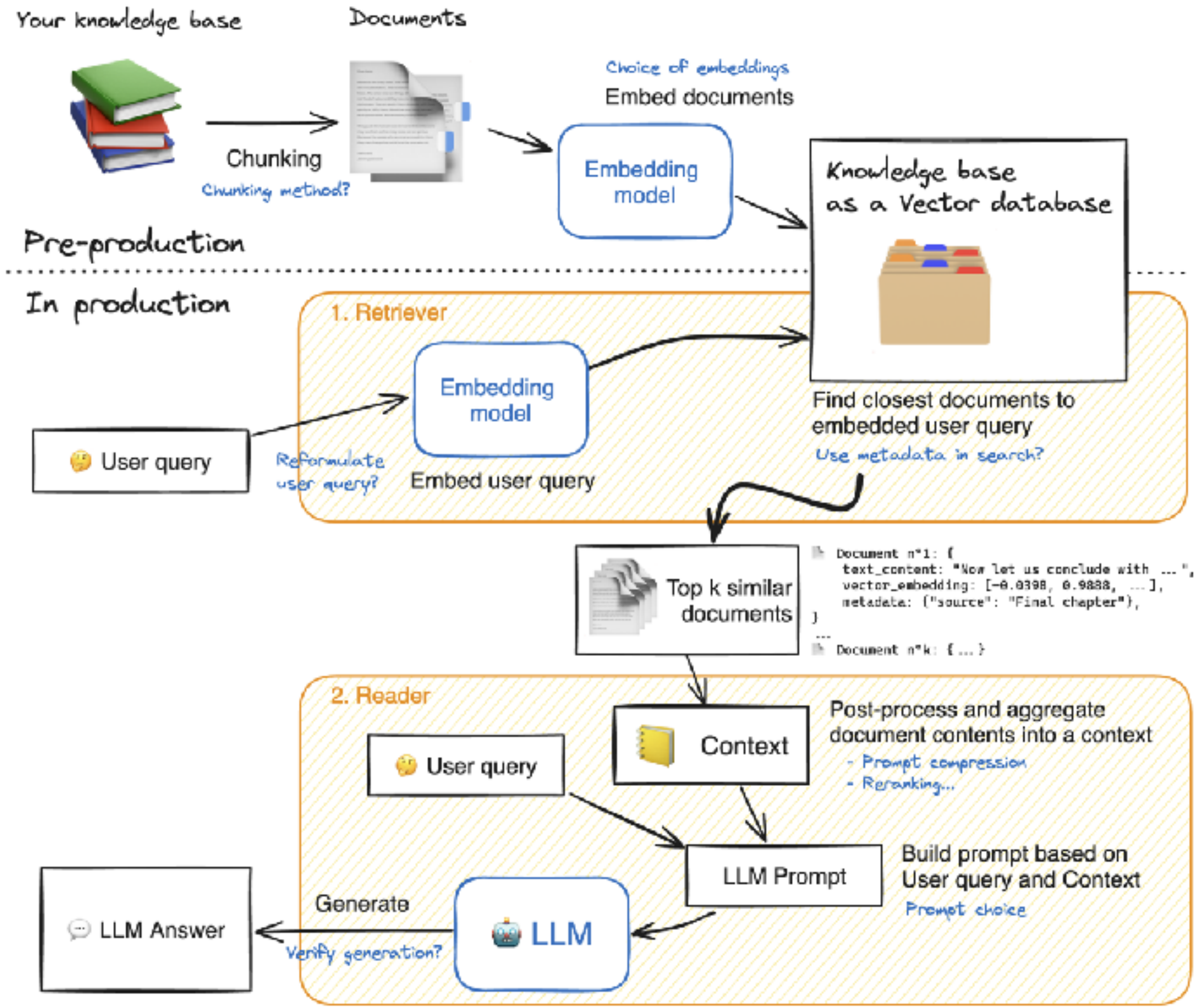


https://github.com/athina-ai/rag-cookbooks

RAG Evaluation !!



https://www.deepeval.com/guides/guides-rag-evaluation



https://huggingface.co/learn/cookbook/en/rag_evaluation

RAG Implementation

Document loading	Splitting or Chunking	Embedding	Store in Vector DB

- Retrieval and Re-ranking
- Send Prompt to LLM
- Output

Failure Points of RAG

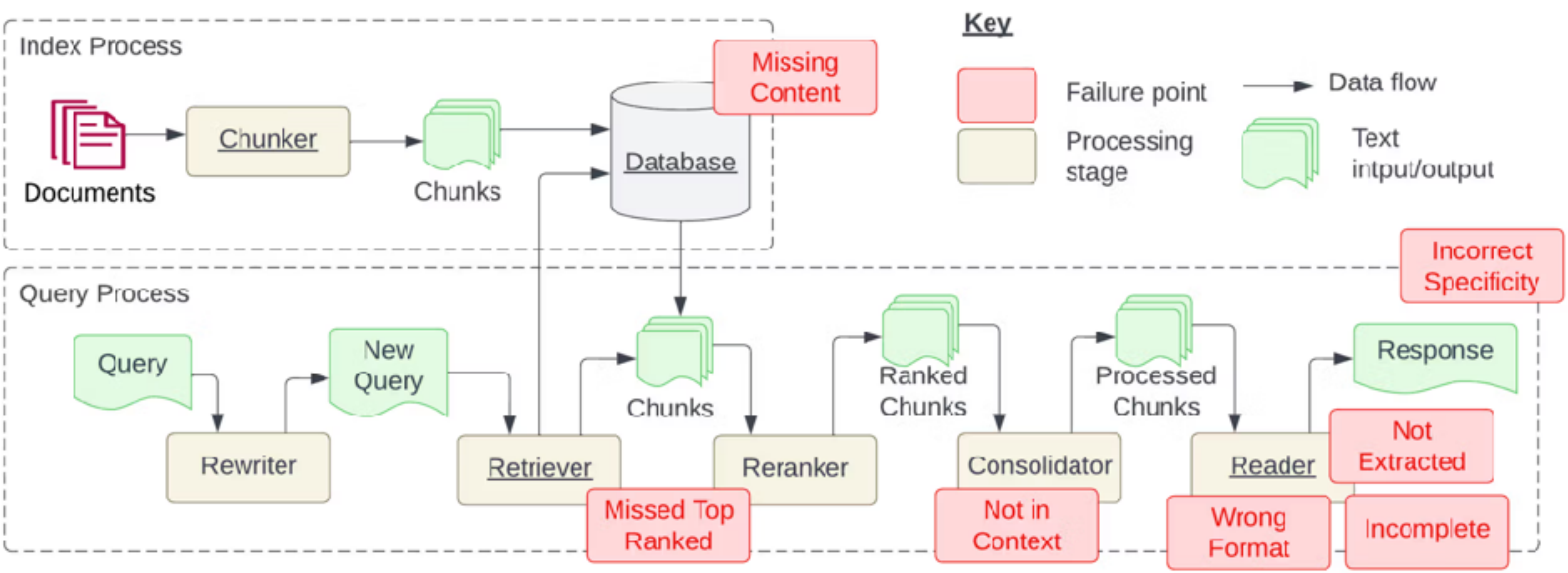


Figure 1: Indexing and Query processes required for creating a Retrieval Augmented Generation (RAG) system. The indexing process is typically done at development time and queries at runtime. Failure points identified in this study are shown in red boxes. All required stages are underlined. Figure expanded from [19].

https://www.galileo.ai/blog/mastering-rag-how-to-architect-an-enterprise-rag-system

RAG is better

But come with Cost !!

More latency

Retrieval errors

Accuracy !!

More Complexity

Maintenance overhead

RAG Techniques ?

Semantic chunking	Chunk size selector
Context chunk header	Adaptive RAG
Re-ranking	Graph TAG

https://github.com/FareedKhan-dev/all-rag-techniques

Pre-Retrieval optimization ?

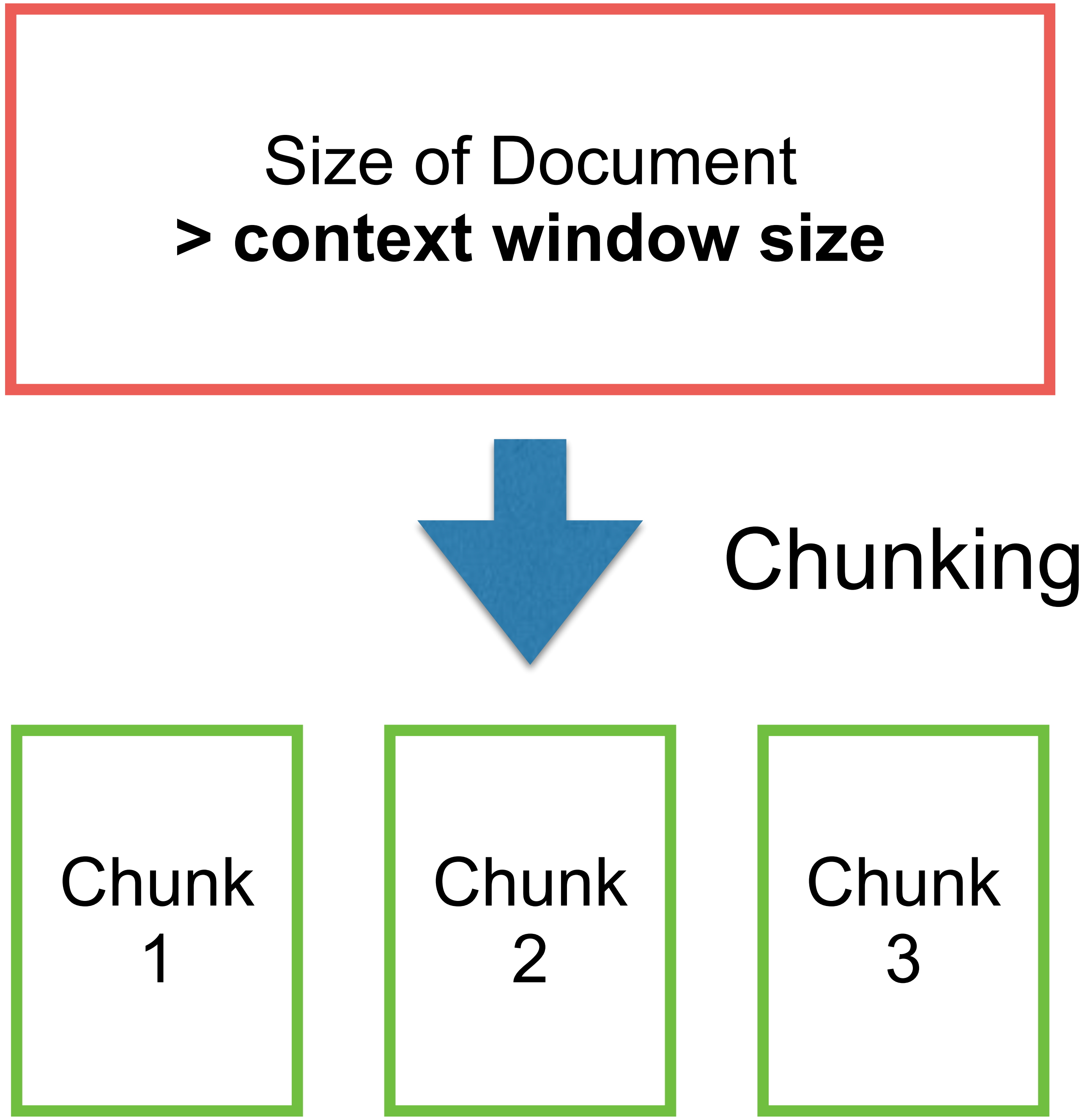
Preprocessing and cleansing data

Chunking strategies

Add metadata

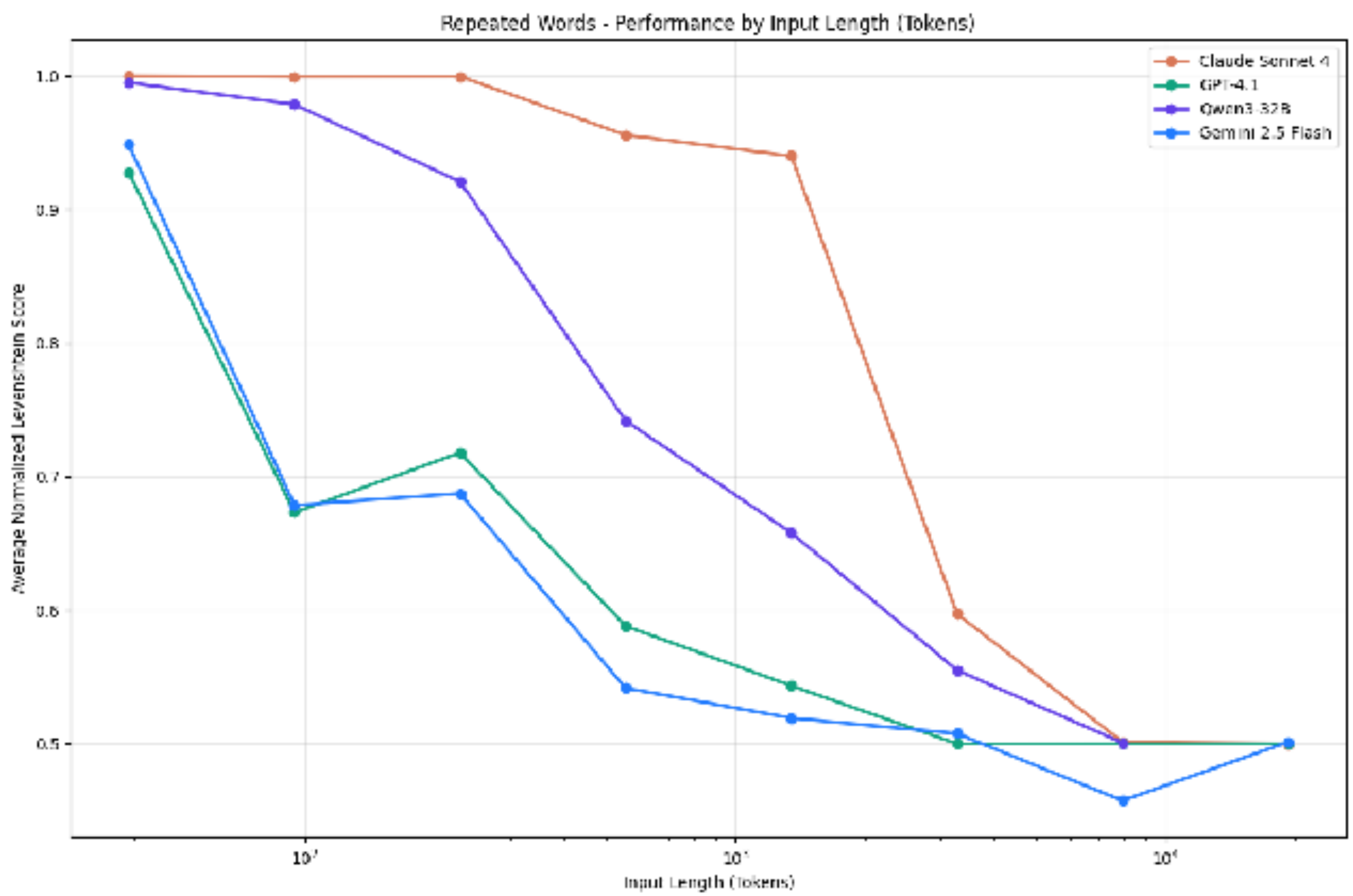
Embedding model selection

Chunking strategies



Context Window ?

Number of tokens in the context window increases,
the model's ability to accurately recall information from that context decreases



https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

Chunking Strategies !!

Fixed size

Recursive characters

Document structure-based

Semantic chunking

Agentic chunking

…

https://blog.dailydoseofds.com/p/5-chunking-strategies-for-rag

Good Chunking

Efficiency

Relevance

Context preservation

Improve content generation

Reduce noise

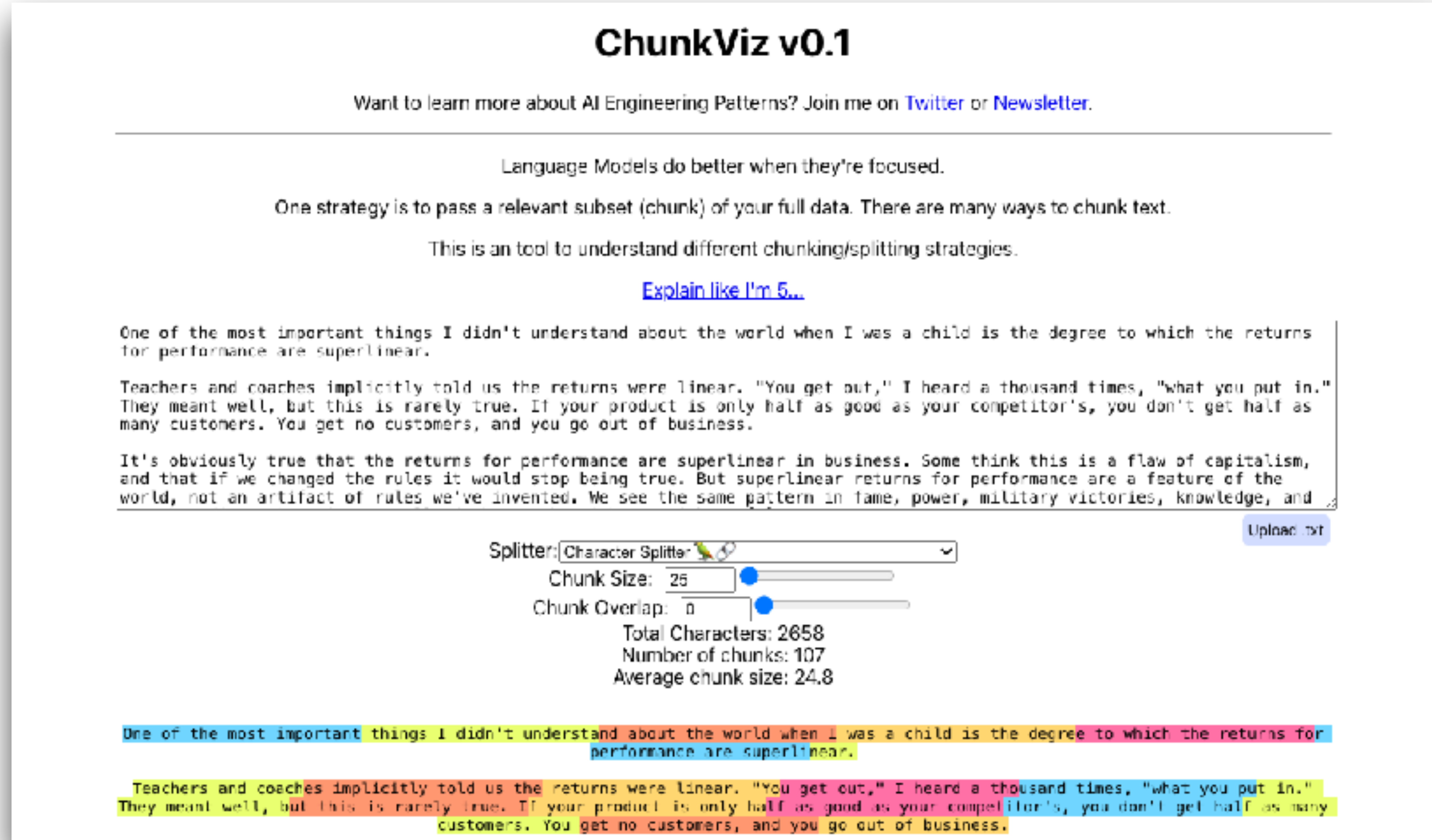
Bad Chunking

Loss context

Redundancy

Inconsistency

Chunking Visualization



https://chunkviz.up.railway.app/

Retrieval optimization ?

Re-ranking

Hybrid search

Query transformation

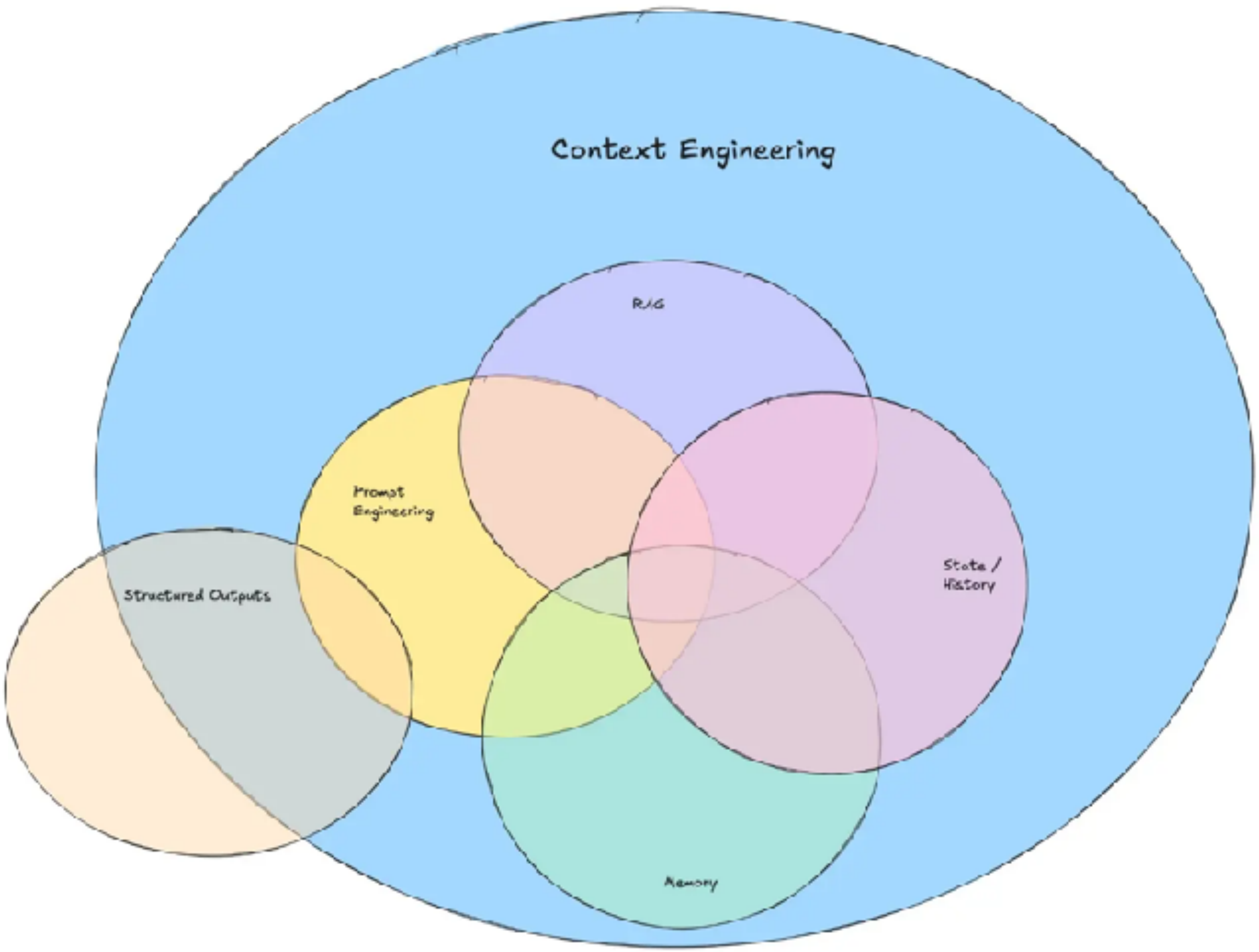
Multi-vector embedding

Contextual retrieval

Vector database Selection

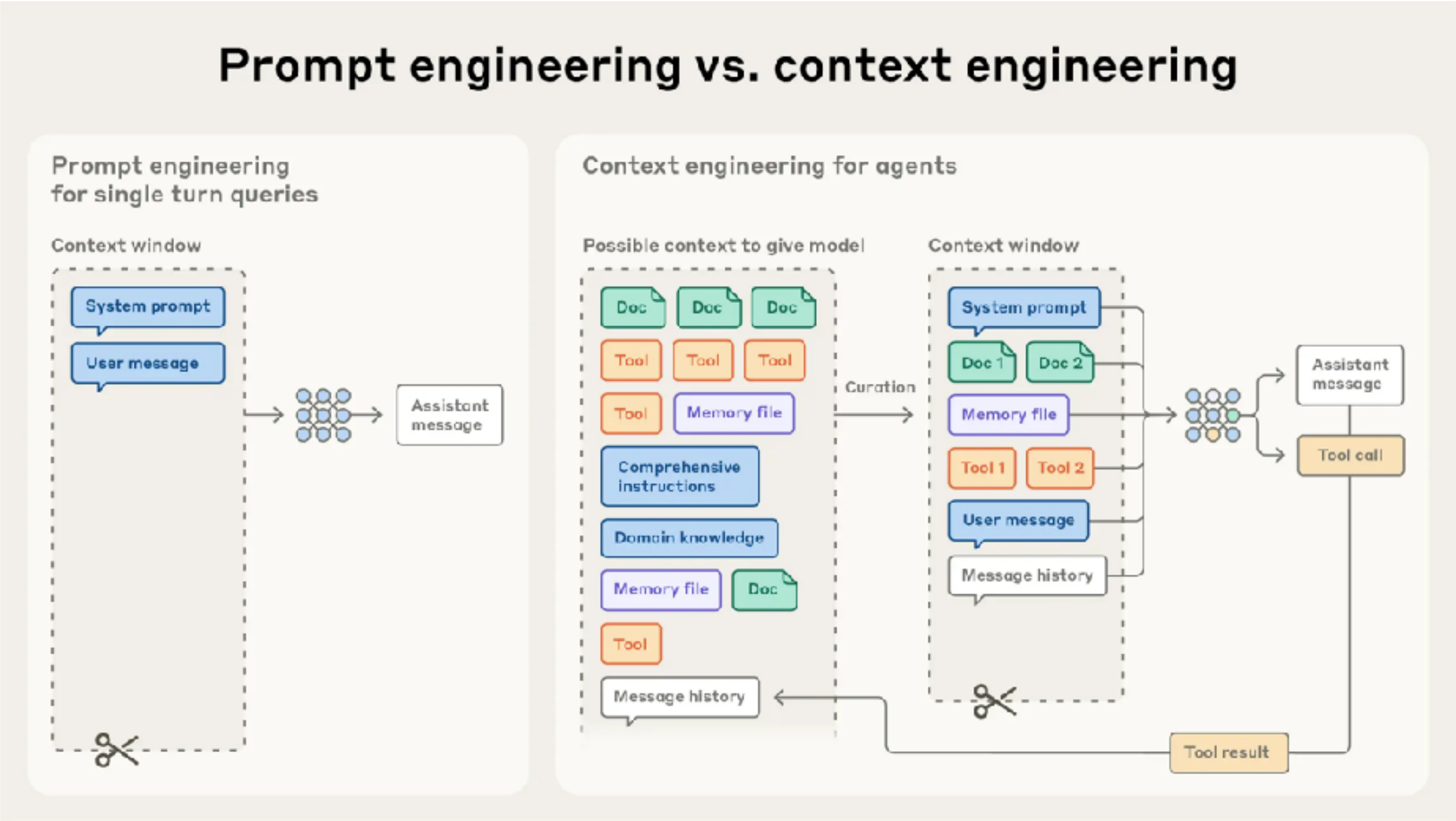

ONE-SIZE FITS ALL
STORE
SALE

Context Engineering



https://www.promptingguide.ai/guides/context-engineering-guide

Effective Context Engineering

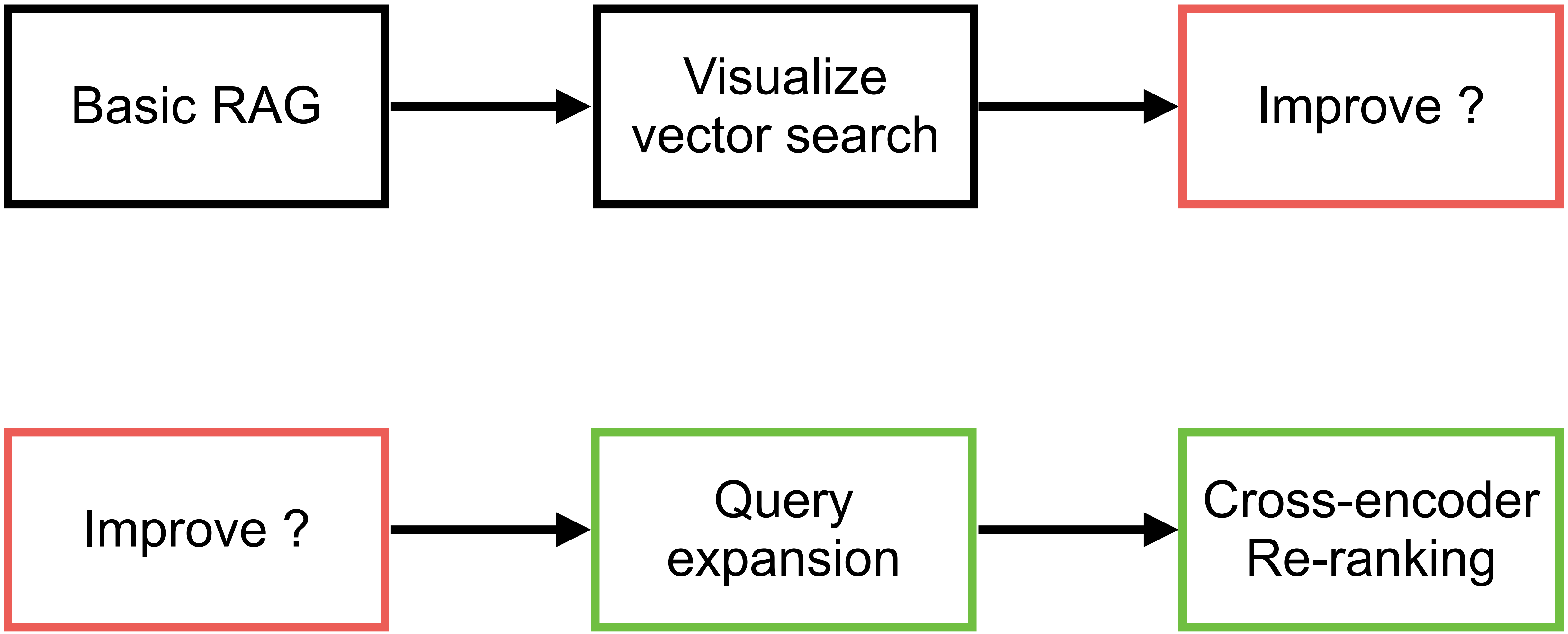


https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents

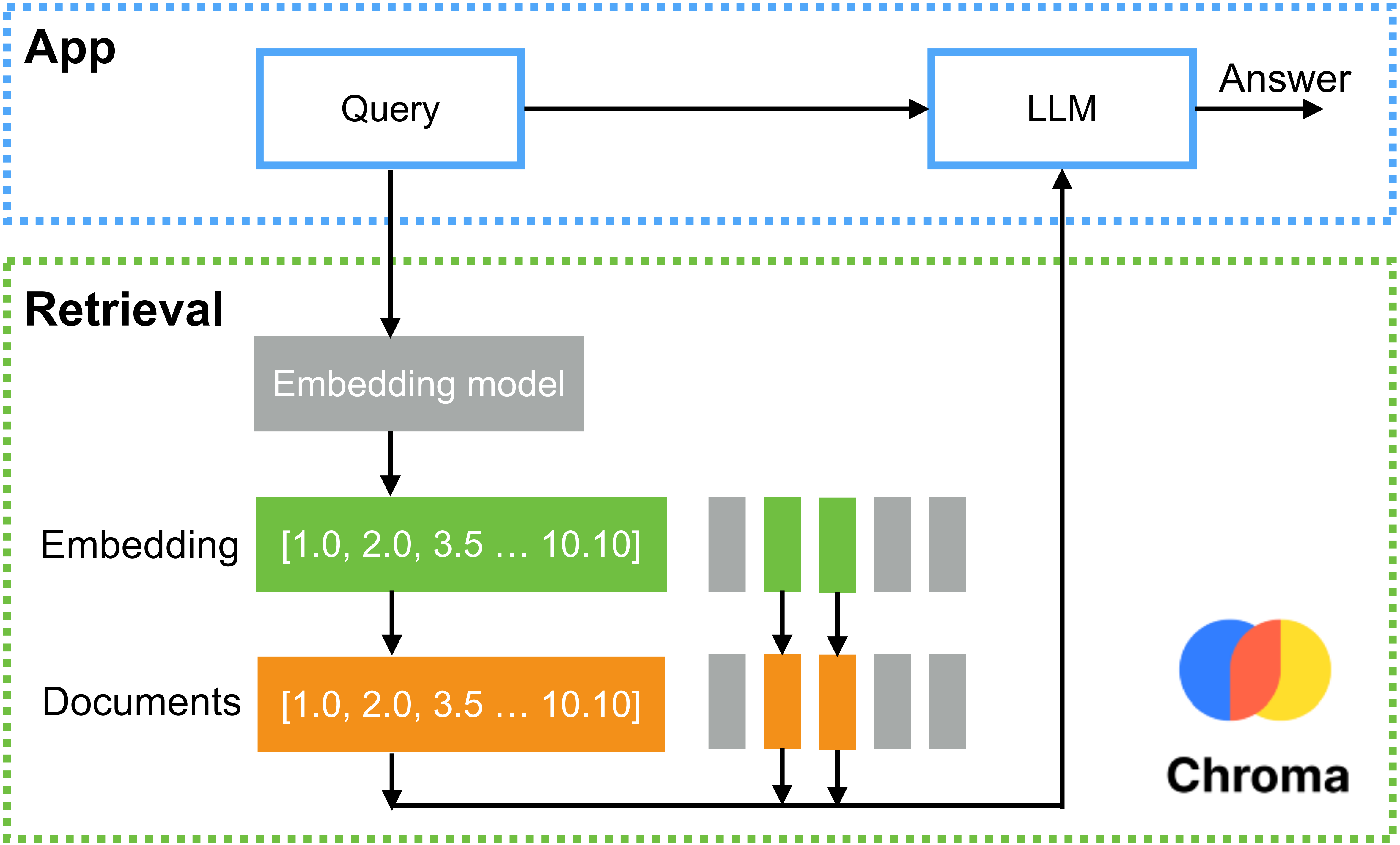
RAG Workshop

https://github.com/up1/workshop-basic-llm/tree/main/workshop/basic-rag

RAG workshop



Basic RAG


App
Query
LLM
Answer
Retrieval
Embedding model
Embedding
[1.0, 2.0, 3.5 … 10.10]
Documents
[1.0, 2.0, 3.5 … 10.10]
Chroma

Query Expansion

Query expansion is a widely used technique to improve the recall of search systems

Ambiguity

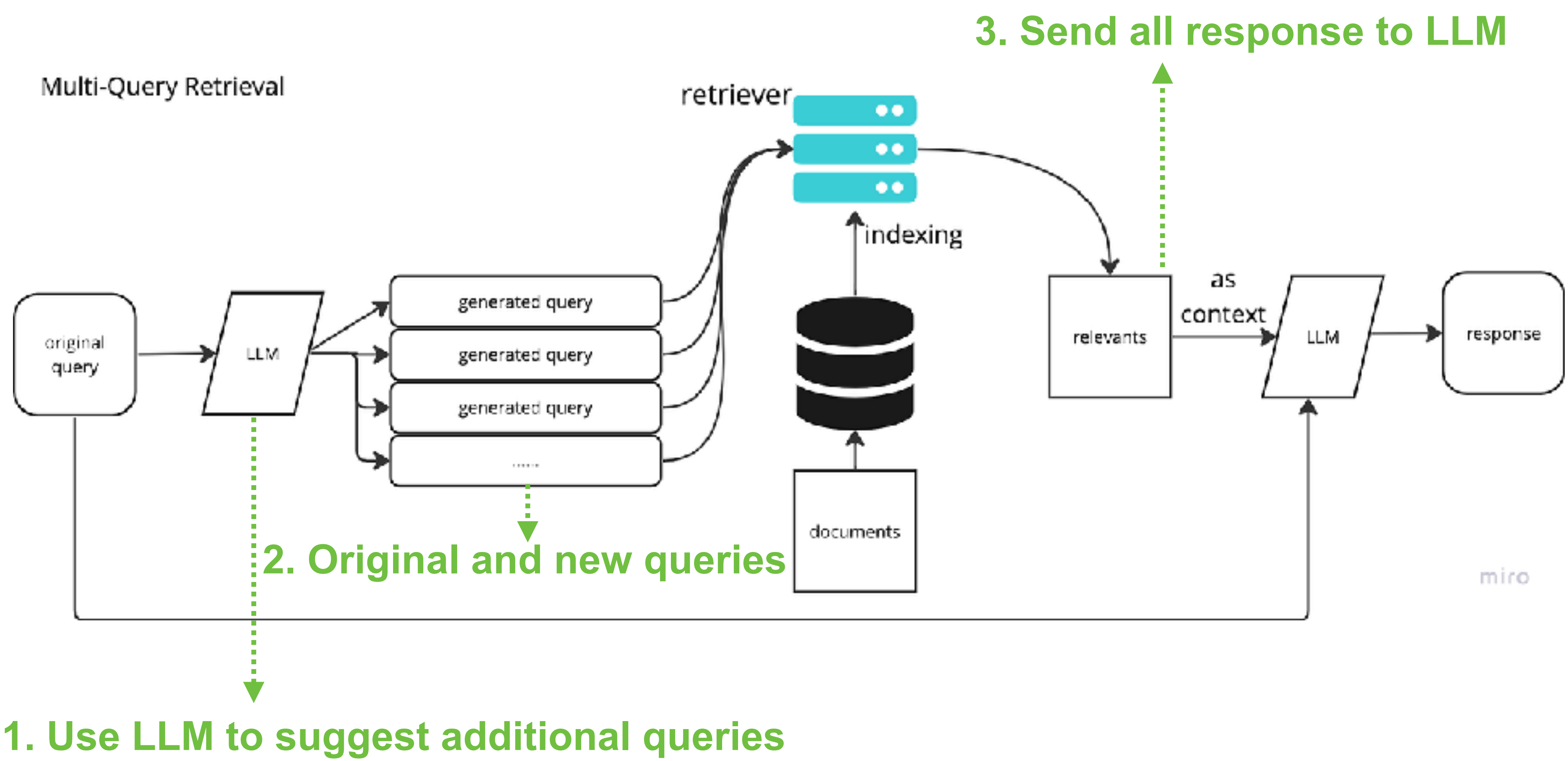
Vocabulary mismatch

Lack of context

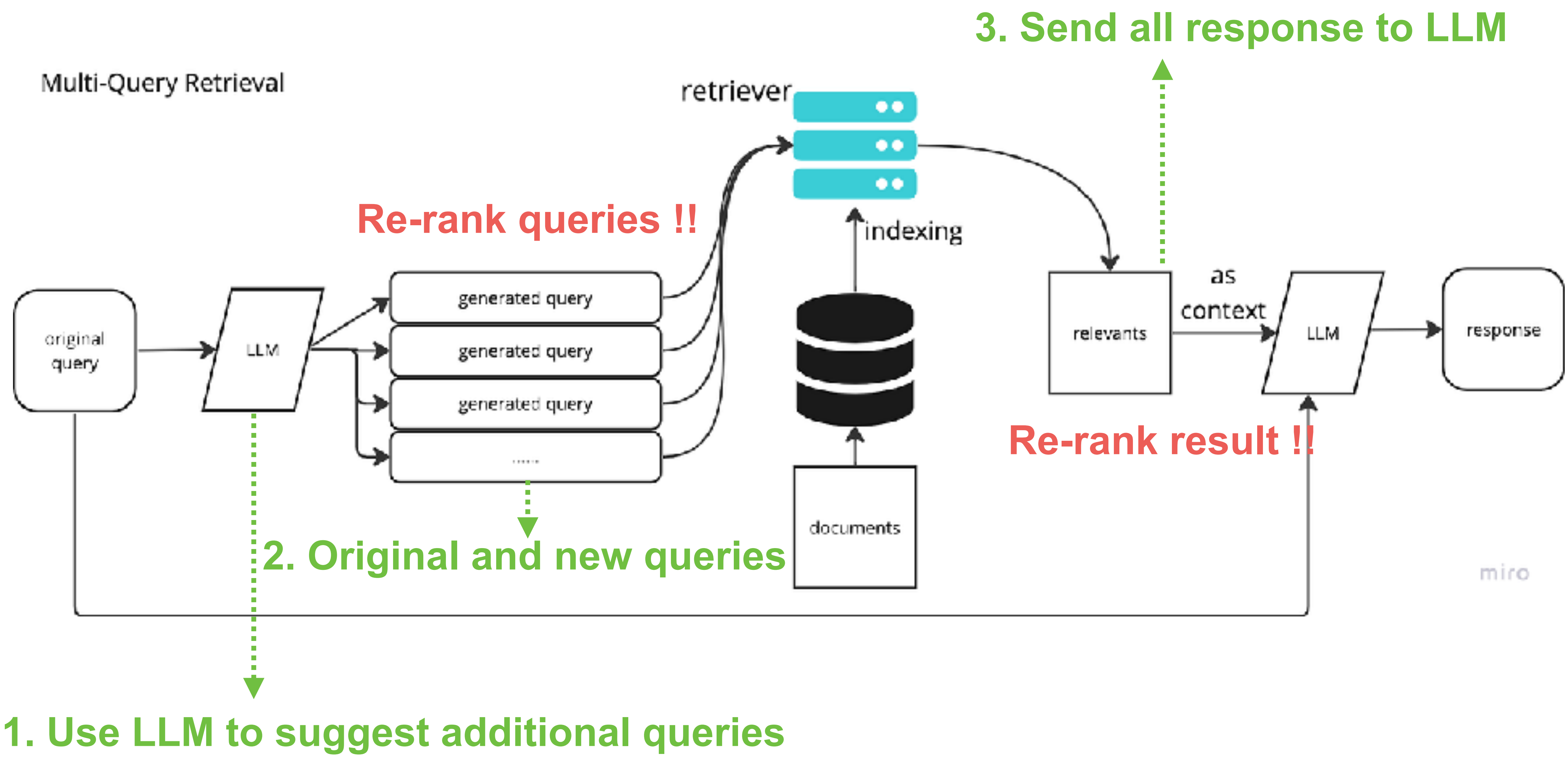
“Add synonyms, related context and contextual terms”

https://arxiv.org/abs/2305.03653

Query Expansion



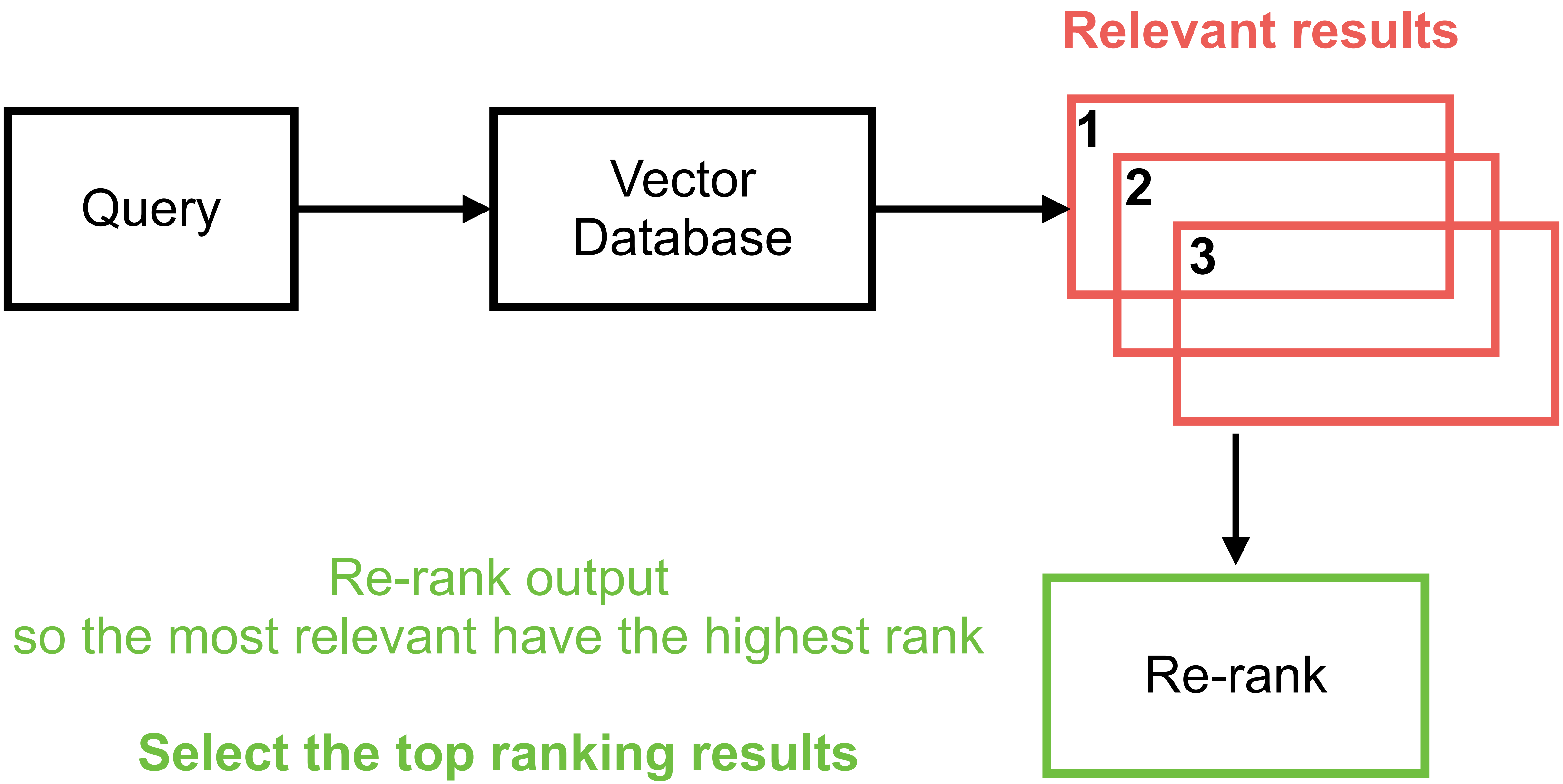
Re-rank !!



3. Send all response to LLM
Multi-Query Retrieval
retriever
Re-rank queries !!
indexing
original query
LLM
generated query
generated query
generated query
......
relevants
as context
LLM
response
Re-rank result !!
documents
2. Original and new queries
miro
1. Use LLM to suggest additional queries

Cross-encoder Re-ranking

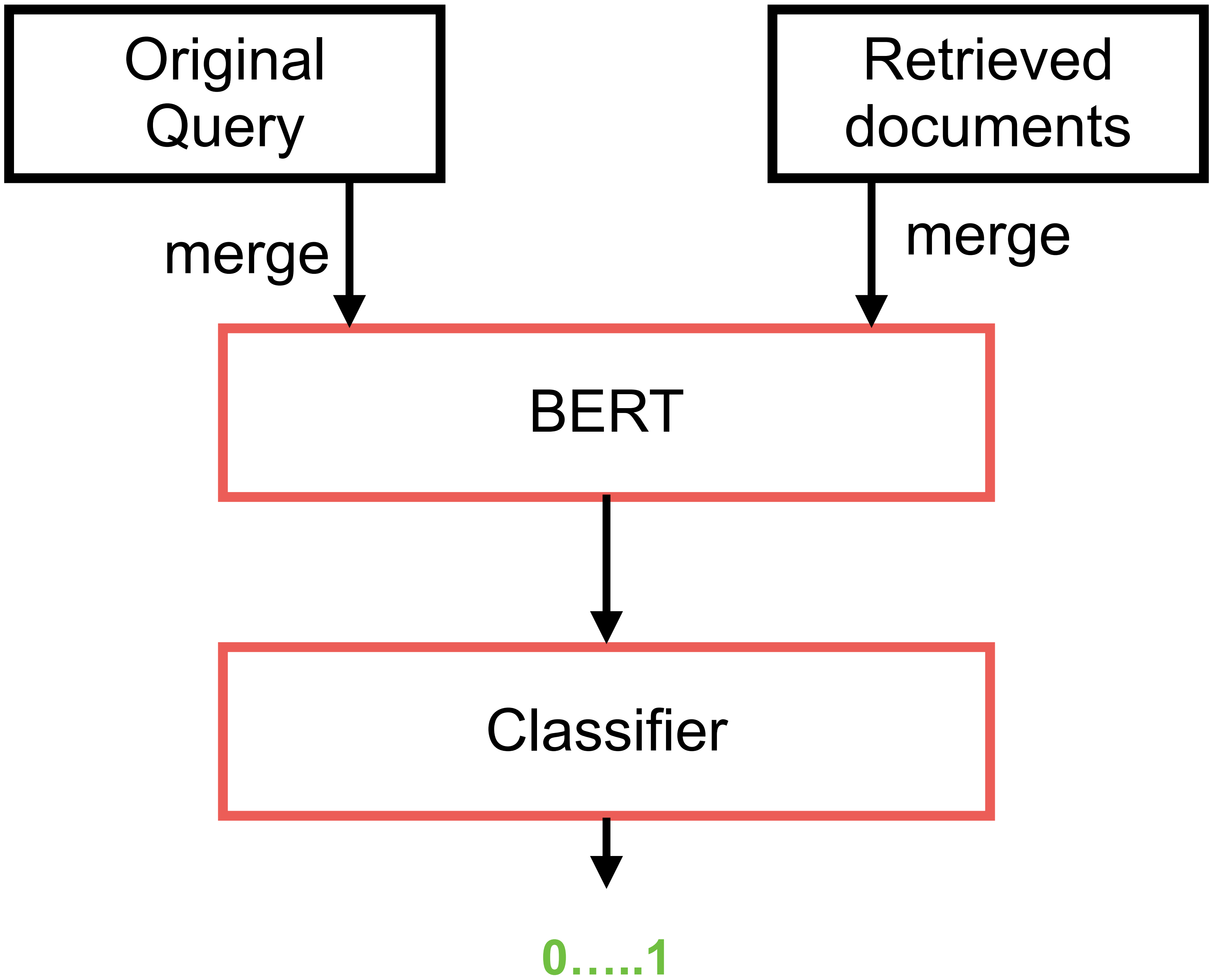
How to ordering relevant results !!



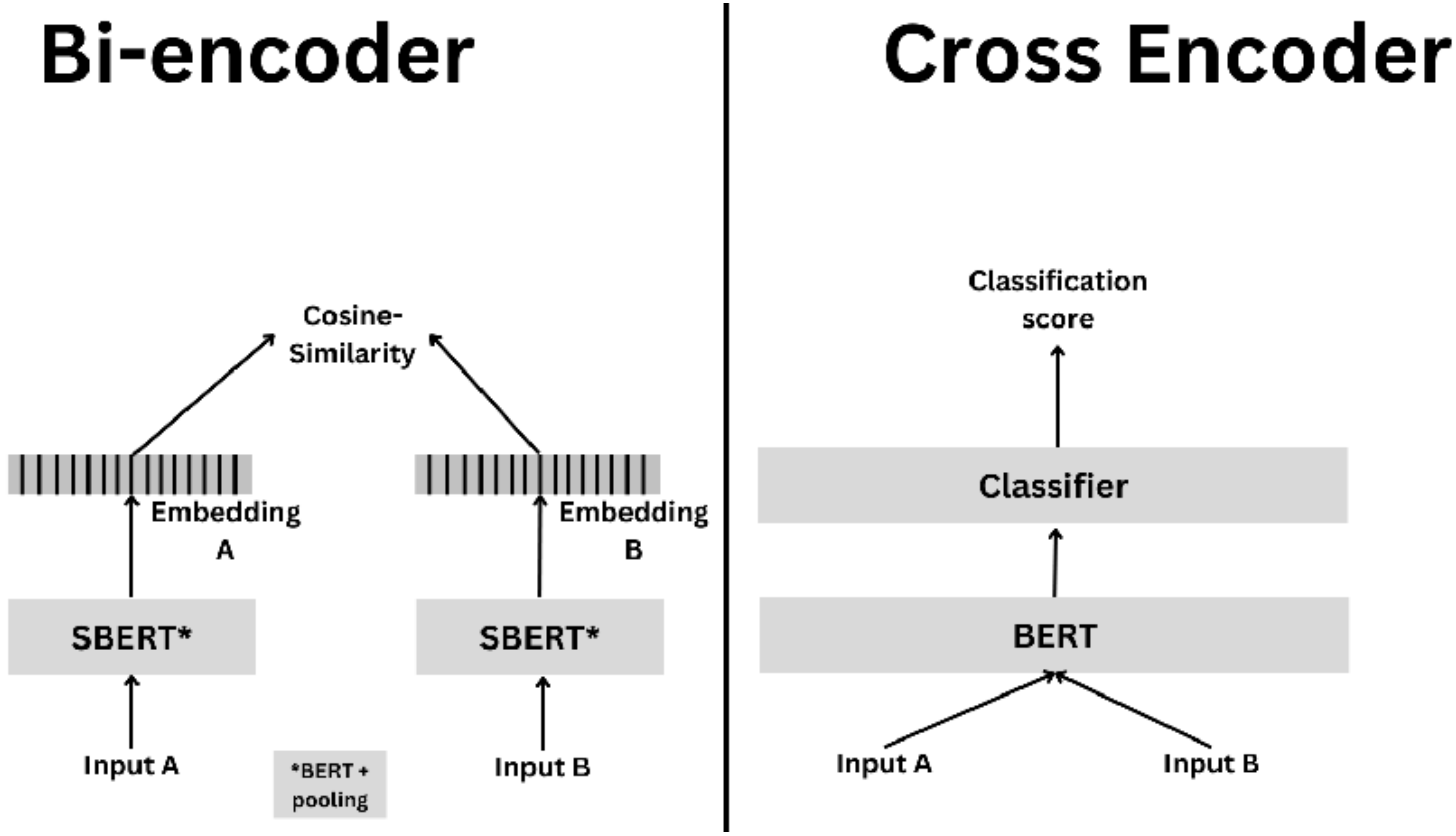
Re-rank output
so the most relevant have the highest rank

Select the top ranking results

Cross-Encoder

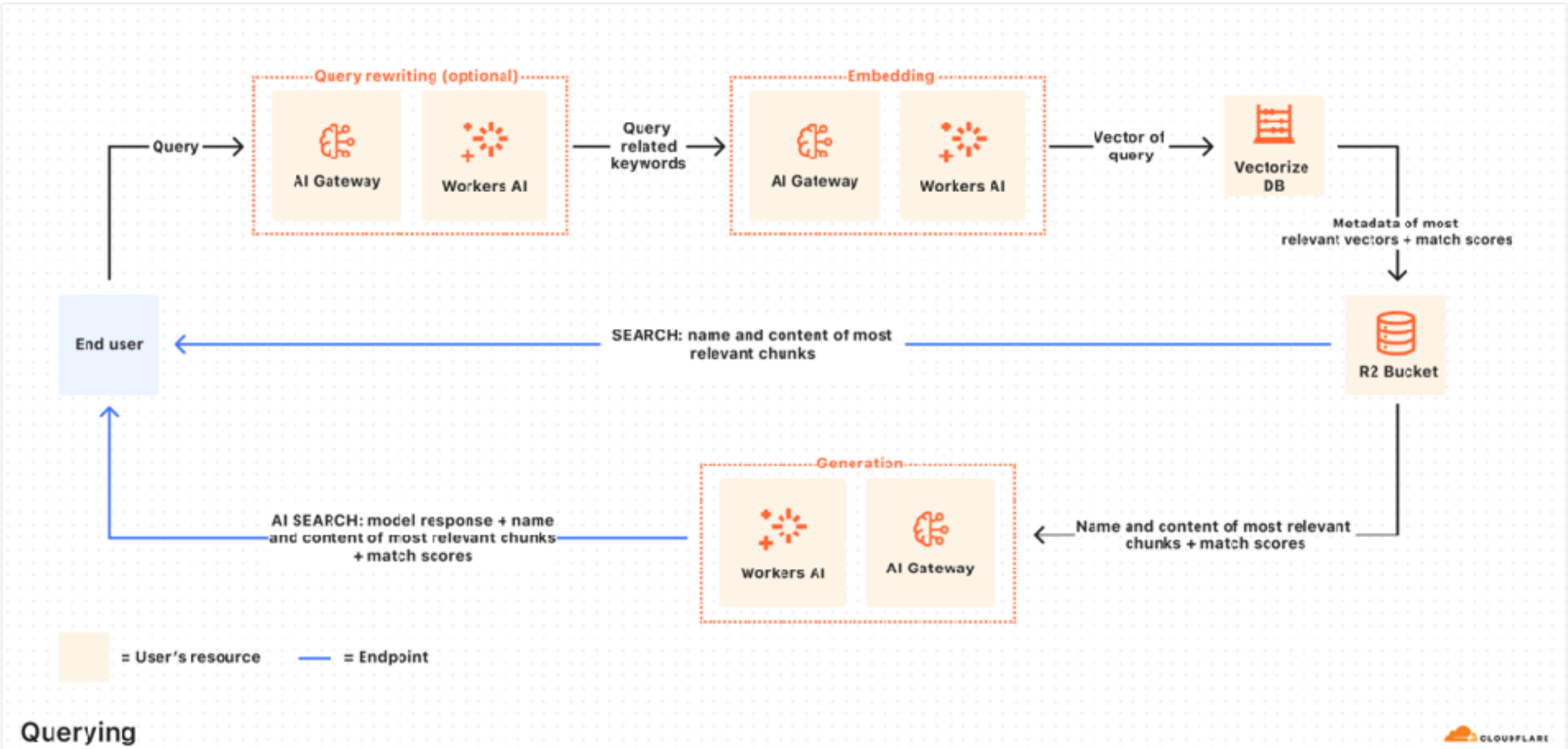


Cross-Encoder !!



https://sbert.net/examples/cross_encoder/applications/README.html

Cloudflare AutoRAG



https://developers.cloudflare.com/autorag/

Guardrails

https://github.com/guardrails-ai/guardrails

Guardrails

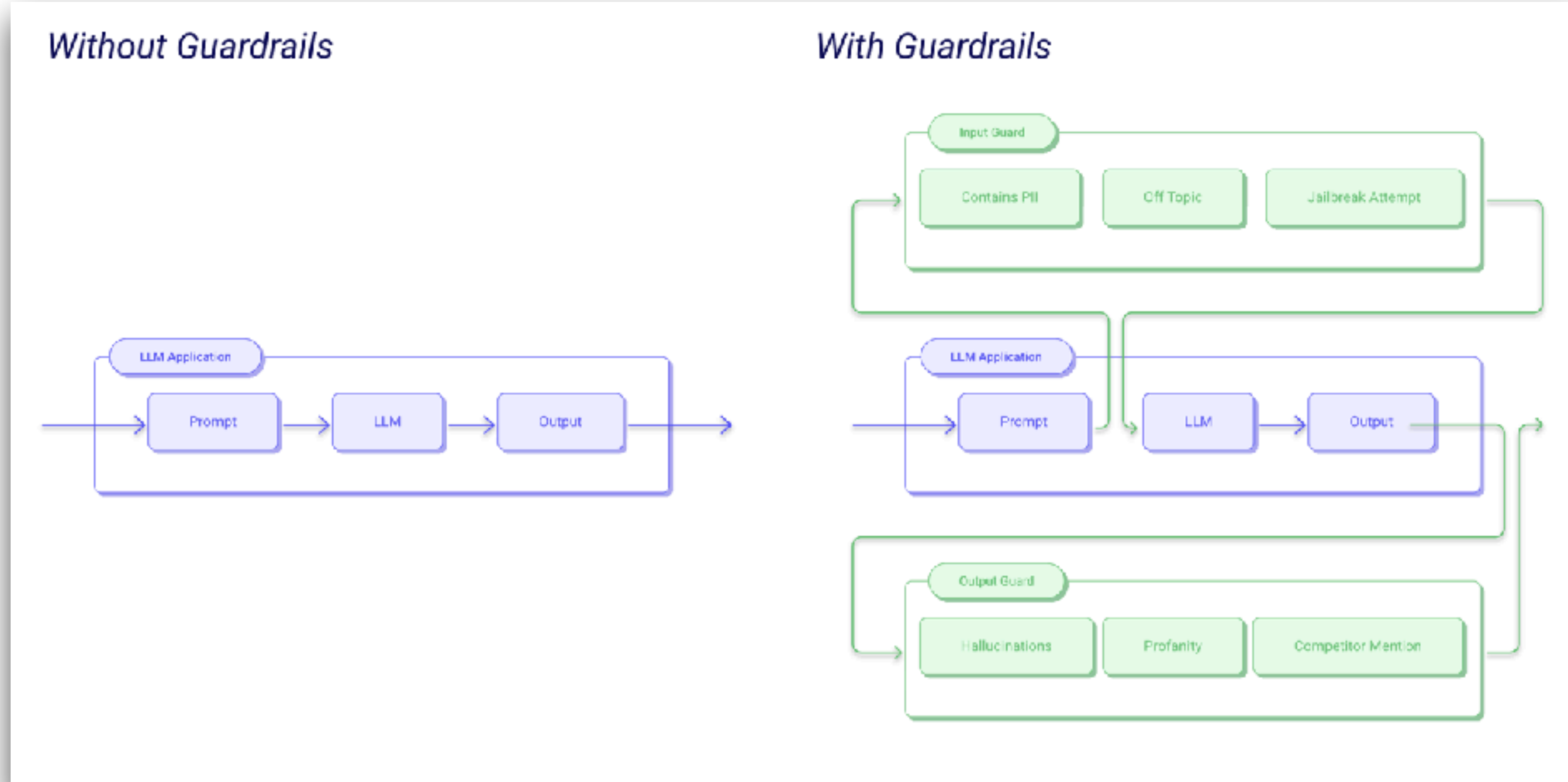
Help to build reliable AI applications

Guard for
input and output

Help to generate
Structured output

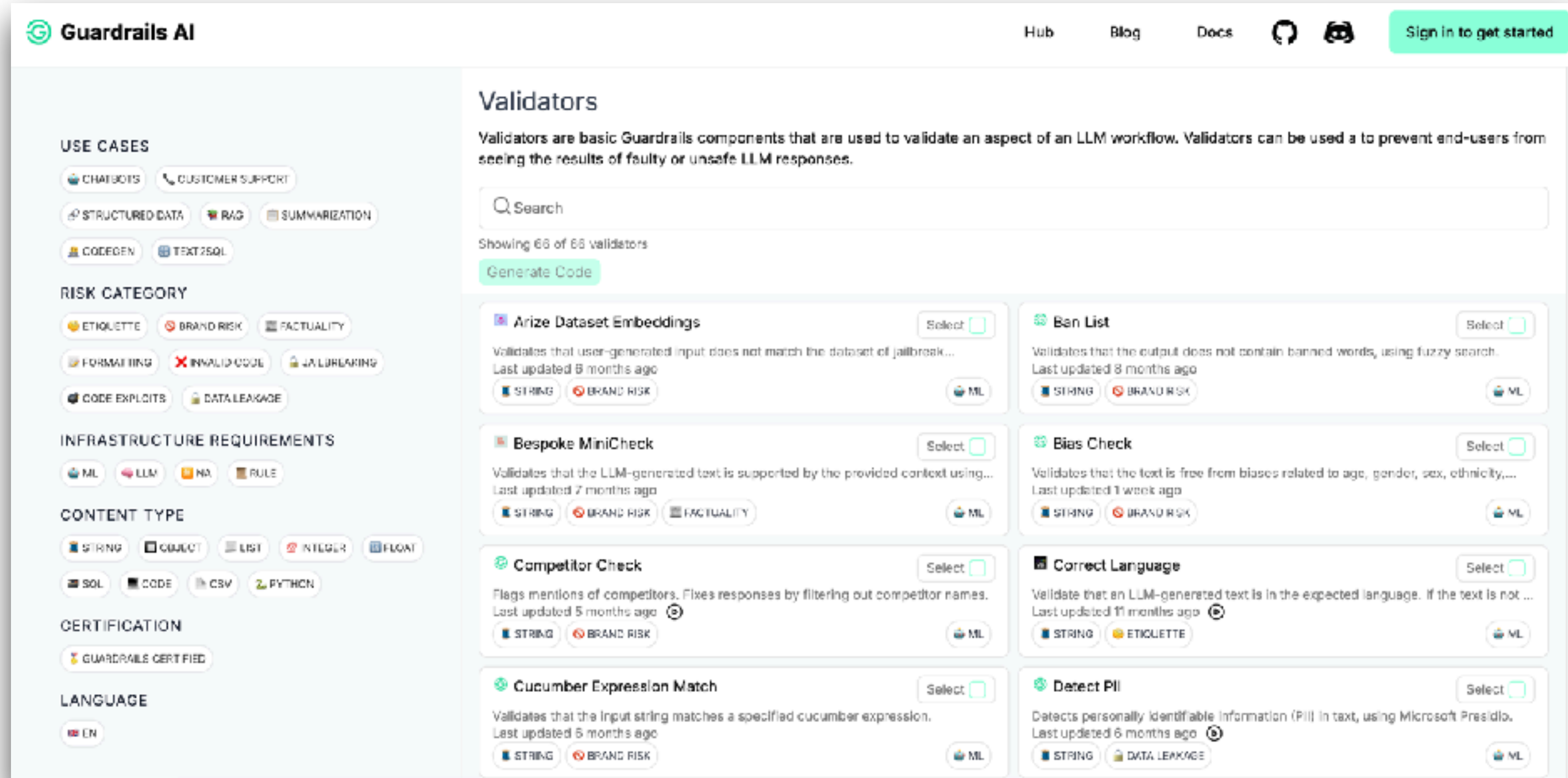
https://github.com/guardrails-ai/guardrails

Guardrails



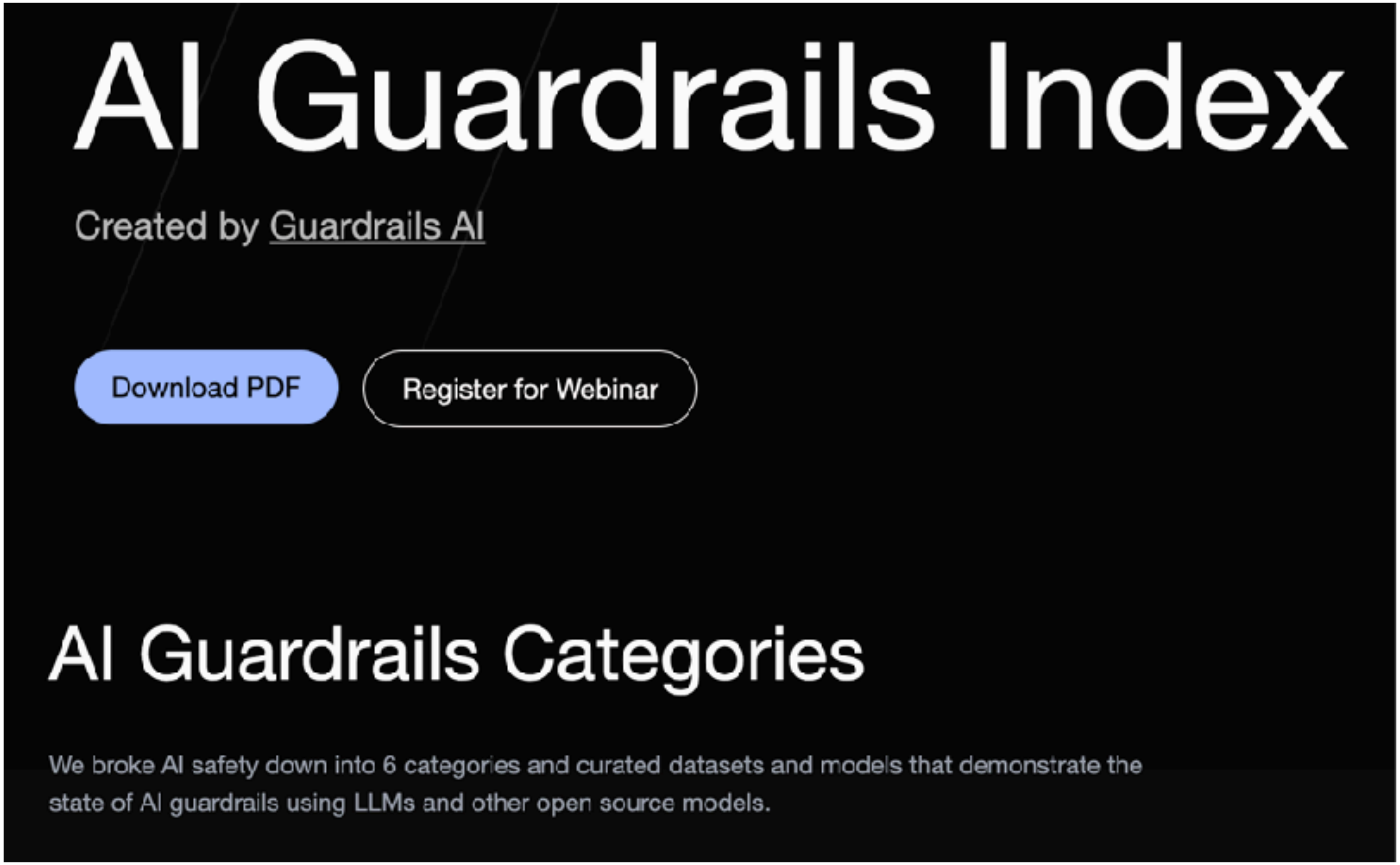
Without Guardrails
LLM Application
Prompt
LLM
Output
With Guardrails
Input Guard
Contains PII
Off Topic
Jailbreak Attempt
LLM Application
Prompt
LLM
Output
Output Guard
Hallucinations
Profanity
Competitor Mention

Guardrails Hub



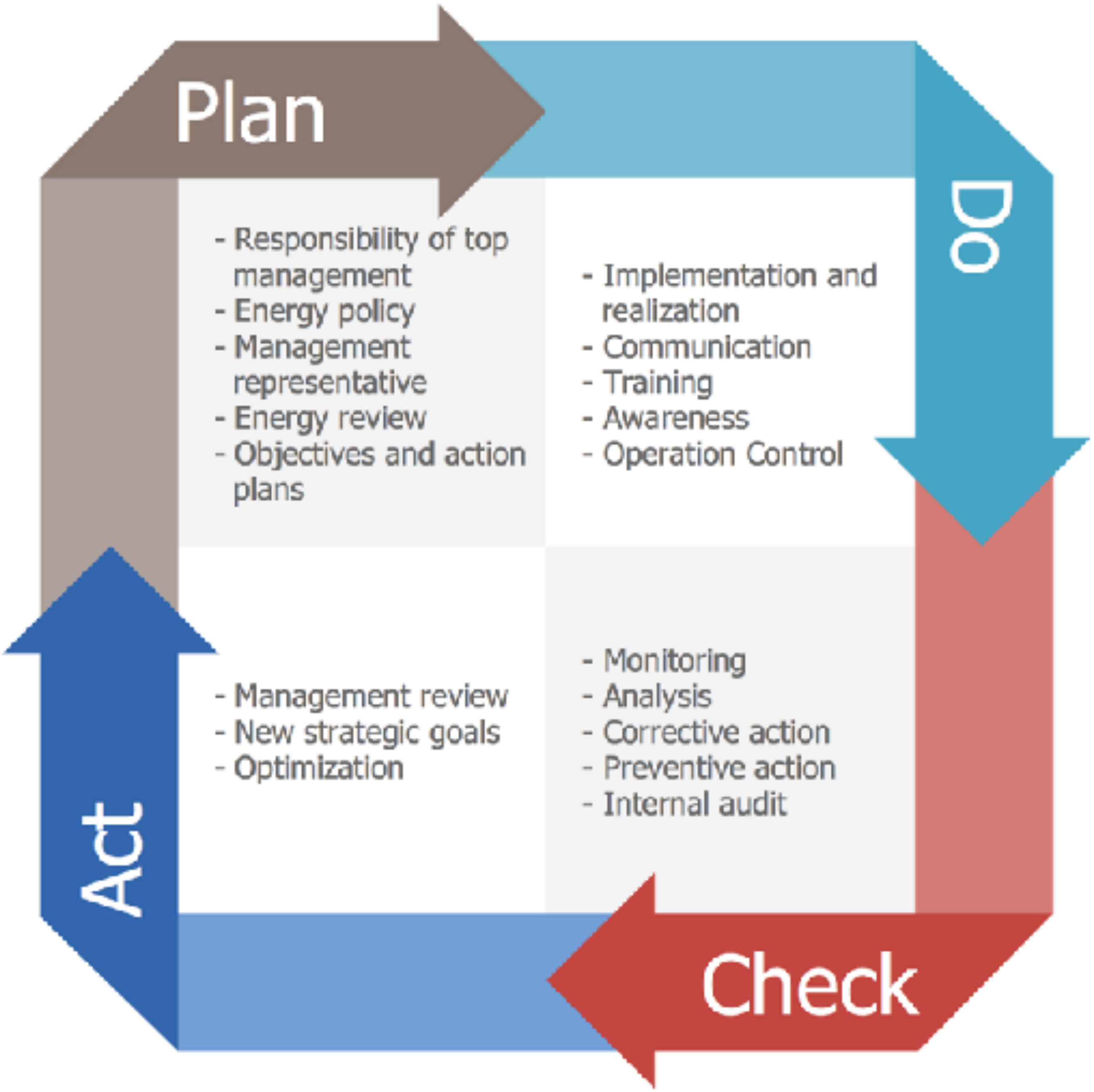
https://hub.guardrailsai.com/

Guardrails Index



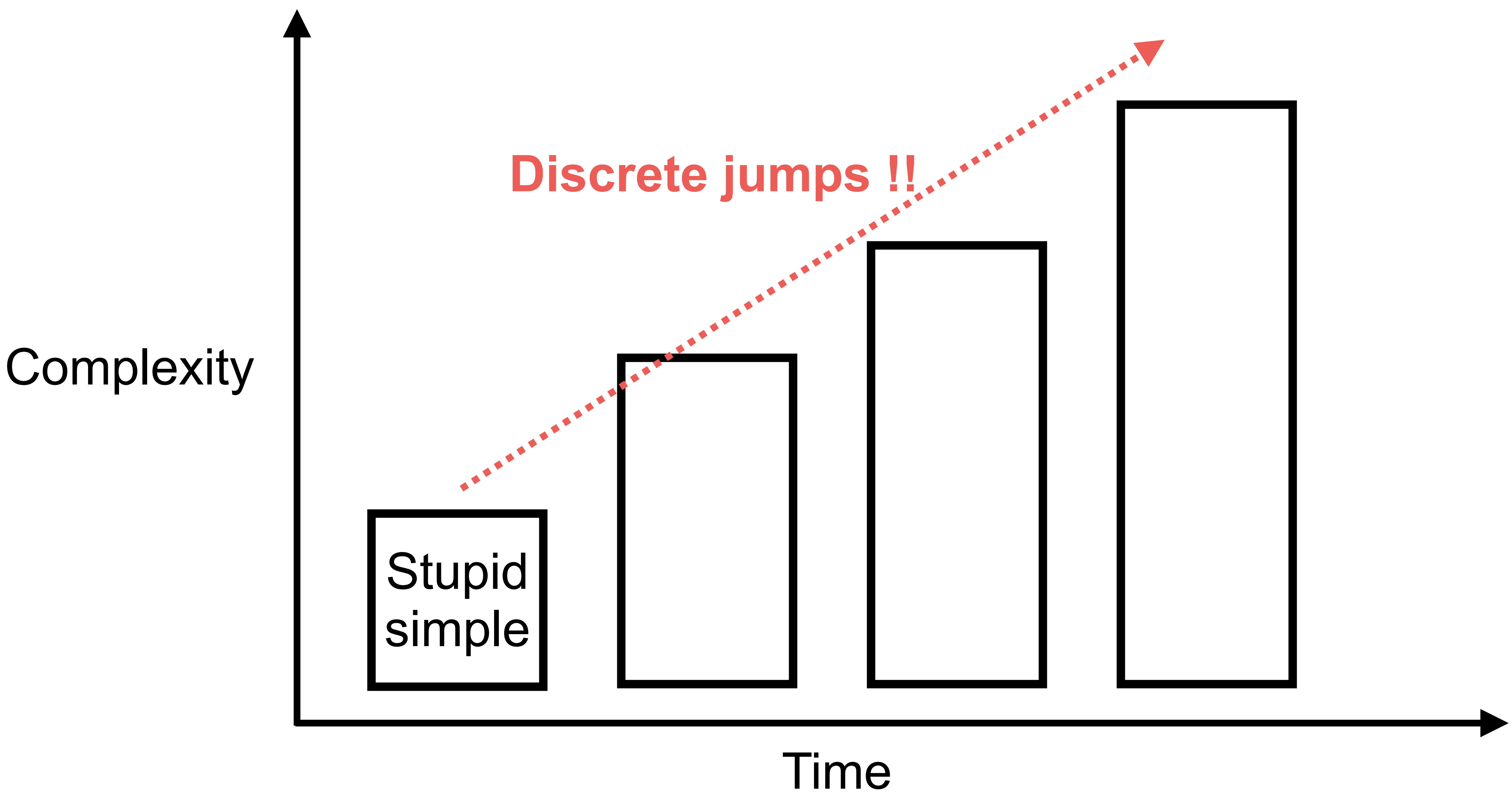
https://index.guardrailsai.com/

PDCA process



Staged Complexity

Start small and get a feature working first



Planning workflow

Requirements	What you’re build ?
↓	
User stories	What use need ?
↓	
Architecture	Selection ?
↓	
Technologies	Selection ?

Architecture of project

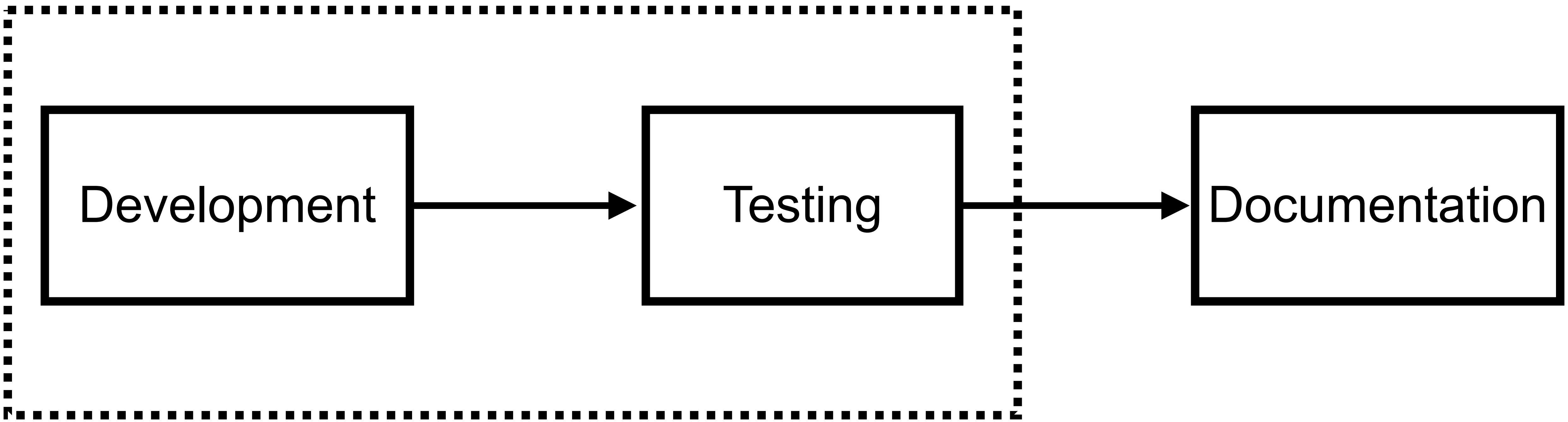
Frontend → Backend → Database

Frontend	Backend	Database
JavaScript	REST API	MySQL
ReactJS	NodeJS	

Choose your tech stack !!

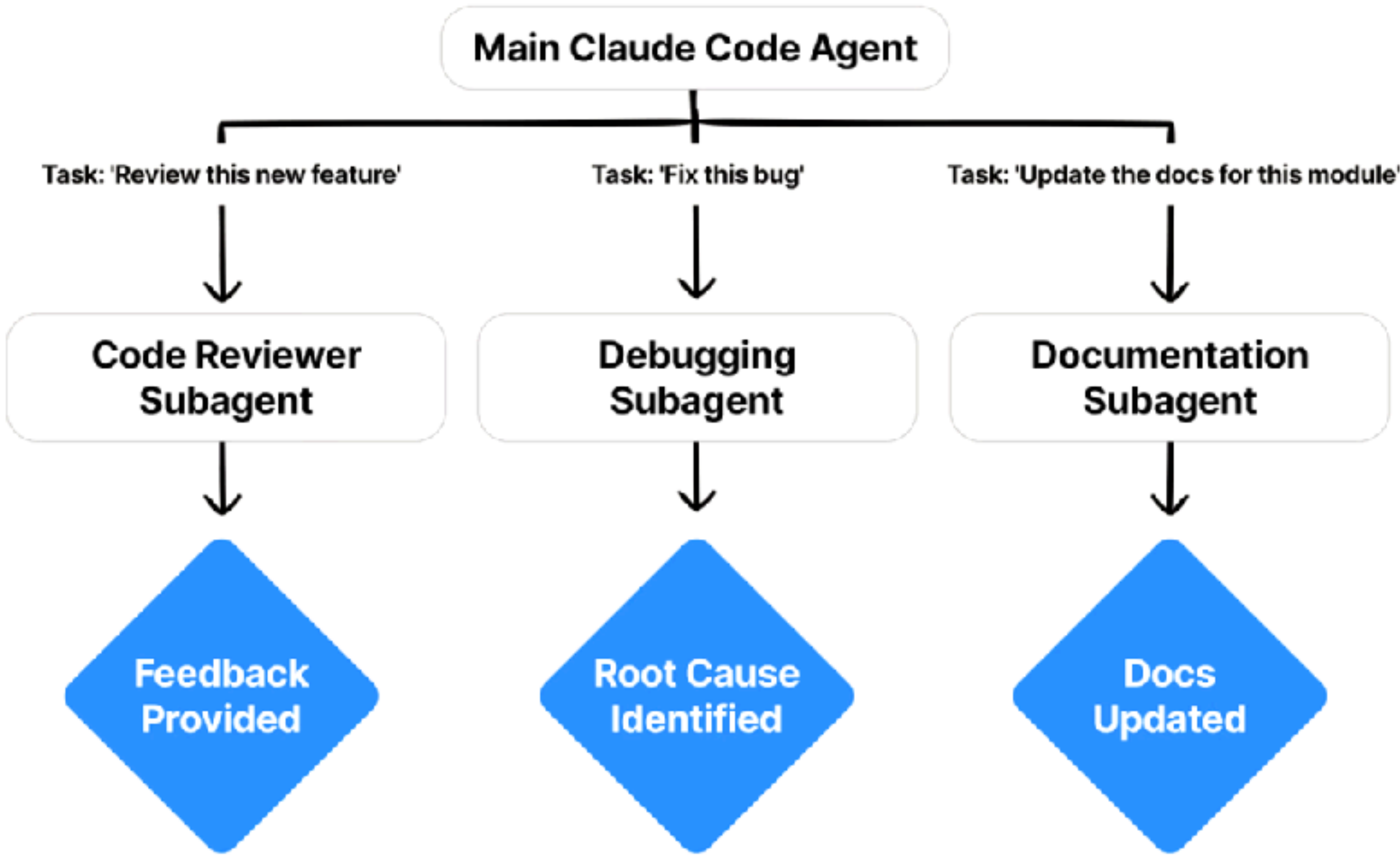
Processes

Iterative and incremental process



AI Subagent

AI Subagent



https://code.claude.com/docs/en/sub-agents

AI Subagent

https://github.com/VoltAgent/awesome-claude-code-subagents

Start your journey